PROOF OF THE BROUSTET–GONGYO CONJECTURE

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Abstract. We prove the Broustet–Gongyo conjecture. It asserts that any normal projective variety admitting a polarized endomorphism is of Calabi–Yau type, and we prove the conclusion more generally for normal projective varieties admitting an int-amplified endomorphism. For a polarized endomorphism of a normal projective variety, we also show that after replacing the endomorphism by an iterate, the variety together with the normalized ramification divisor of the endomorphism is a log canonical pair with $\mathbb{Q}$-linearly trivial log canonical divisor, as conjectured by Chang and Zhang.

Contents

1. Introduction 1
2. Preliminaries 8
3. The class group of an int-amplified variety 16
4. The canonical model of an int-amplified variety 23
5. A small model for the pushforwards of the ramification divisor 34
6. Polarized endomorphisms 40
7. Lc anti-canonical series 47
8. Leading divisors 51
9. Return from leading divisors 65
10. A dlt model on which the center has codimension equal to the rational rank 72
11. The residue model 80
12. Finite attainment 91
13. Proofs of the main theorems 98
References 100

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Y. Zhang asked whether, for a normal isolated singularity $X \ni x$ with a non-invertible finite endomorphism that is not étale in codimension one, there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ on X such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier and (X, Δ) is lc [Zha17, Question 1.6] (see also [BG17, Question 1.1]). We say that a normal variety X is *of lc type* if such a Δ exists. For surface singularities the answer is positive [Wah90, Theorem (a), (b), (c), p. 626]. [BdFF12, Theorems A(iii) and B] answered the question in the $\mathbb{Q}$-Gorenstein case: a non-invertible finite endomorphism forces the volume of $X \ni x$ in the sense of [BdFF12] to be zero, and a $\mathbb{Q}$-Gorenstein singularity with volume zero is lc. If X is a normal projective variety with an isolated singular point x, and $\varphi : X \to X$ is a finite endomorphism with $\varphi(x) = x$ and $\deg \varphi \geq 2$ that is étale in codimension one, then X is $\mathbb{Q}$-Gorenstein and lc [Zha17, Theorem 1.1].

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Outside the $\mathbb{Q}$-Gorenstein case, the volume of [BdFF12] does not detect lc type: there exists an affine cone over a smooth projective variety whose volume in the sense of [BdFF12] is zero, and which is not of lc type [Zha14, Theorems 1.2 and 1.3], [dFH09, Proposition 7.2].

Zhang asked the global version of the question [Zha17, Question 1.7], and Broustet and Gongyo proposed it as a conjecture [BG17, Conjecture 1.2]. A surjective endomorphism $f : X \to X$ of a projective variety X is *polarized* if $f^*H \sim qH$ for some ample Cartier divisor H on X and some integer $q > 1$ (see Definition 2.11). A normal projective variety X is *of Calabi–Yau type* if there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that (X, B) is lc and $K_X + B \sim_{\mathbb{Q}} 0$ (see Definition 2.3).

Conjecture 1.1 (Broustet–Gongyo)**.** *Let X be a normal projective variety admitting a polarized endomorphism. Then X is of Calabi–Yau type.*

We now explain what is known about the singularities of a variety X as in Conjecture 1.1. Broustet and Höring proved that if K_X is $\mathbb{Q}$-Cartier and X admits a polarized endomorphism f, then X is lc [BH14, Corollary 1.3], and that (X, D) is lc for any reduced divisor D with $f^{-1}(D) = D$ such that $K_X + D$ is $\mathbb{Q}$-Cartier [BH14, Corollary 3.3]. For a finite endomorphism f of degree $\deg f > 1$ of a normal variety X and a reduced divisor D with $f^{-1}(D) = D$ such that $K_X + D$ is $\mathbb{Q}$-Cartier, [BH14, Theorem 1.4] shows that, after replacing f by an iterate, any irreducible component Z of the locus where (X, D) is not lc satisfies $f^{-1}(Z) = Z$ and $\deg(f|_Z) = \deg f$.

A surjective endomorphism $h : X \to X$ is *int-amplified* if $h^*H - H$ is ample for some ample Cartier divisor H on X (see Definition 2.11). A polarized endomorphism f is int-amplified, since $f^*H - H \sim (q-1)H$ is ample. [Men20, Theorem 1.6] extends [BH14, Corollary 1.3] to int-amplified endomorphisms. An int-amplified endomorphism h is finite (see Lemma 2.12), and we let $R(h)$ be the ramification divisor of h, so that $K_X \sim h^*K_X + R(h)$ (see Definition-Proposition 2.10). Chang and Zhang extended [Men20, Theorem 1.6] to pairs: if $\Delta \geq 0$ is a $\mathbb{Q}$-divisor such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier and $\Delta + R(h) - h^*\Delta \geq 0$, then (X, Δ) is lc [CZ25, Theorem 1.8].

If K_X is not $\mathbb{Q}$-Cartier, lc singularities of X itself are not defined, and the question is whether X is of lc type. Consider the following three conditions on a normal variety X.

(a) X is of lc type. By [dFH09, Proposition 7.2], (a) is equivalent to lc in the sense of [dFH09, Definition 7.1]: there exists a positive integer m such that for any prime divisor E over X, the m-th log discrepancy of E, defined using $\mathcal{O}_X(-mK_X)$, is non-negative.

(b) For any prime divisor E over X, the supremum of the log discrepancies of E with respect to the pairs (X, B) is non-negative, where B runs over the $\mathbb{R}$-divisors $B \geq 0$ such that $K_X + B$ is $\mathbb{R}$-Cartier near the image of E on X [Has20, Definitions 4.1 and 4.2]. Here B may depend on E.

(c) X is *valuative lc* [Yos22, Definitions 4.1 and 4.3], a condition on each prime divisor E over X in terms of the orders along E of the sheaves $\mathcal{O}_X(mK_X)$, $m \geq 1$. By [Yos22, Theorem 4.4], (c) holds if and only if

$$\pi_*\mathcal{O}_{X_\pi}(m(K_{X_\pi} + E_\pi)) = \mathcal{O}_X(mK_X)$$

for any projective birational morphism $\pi : X_\pi \to X$ from a normal variety and any positive integer m, where E_π is the sum of the π-exceptional prime divisors.

Condition (a) implies (b), since the boundary Δ in (a) works for all prime divisors over X at once. Condition (b) implies (c) by [Has20, Proposition 4.7 and Theorem 4.9] and [BdFF12, Proposition 2.6], and (b) and (c) are not equivalent in general [Yos22, Remark 4.5] (cf. [BdFF12, Example 5.4]). Any normal projective variety with a polarized endomorphism satisfies (c) [Yos22, Theorem 1.4], and [Yos22, Corollary 7.5] proves (c) for normal projective varieties with an int-amplified endomorphism h, also for pairs (X, D), where D is a reduced divisor with $h^{-1}(D) = D$. However, there exist cones that satisfy (b) but not (a) (see [Yos22, Remark 4.5] and [Has20, Example 4.11], see also [Zha14, Theorems 1.2 and 1.3] and [dFH09, Proposition 7.2]). Thus estimates proved for each prime divisor over X separately do not imply the existence of one

boundary that works for all prime divisors over X at once, and [Yos22, Theorem 1.4] does not imply Conjecture 1.1.

On the anti-canonical side, let X be a normal projective variety with $\dim X = d$ admitting an int-amplified endomorphism. Then there exists a $\mathbb{Q}$-divisor $D \geq 0$ on X such that

$$(-K_X \cdot L_1 \cdots L_{d-1}) = (D \cdot L_1 \cdots L_{d-1})$$

for any Cartier divisors $L_1, \ldots, L_{d-1}$ on X [Men20, Theorem 1.5]. If moreover X is $\mathbb{Q}$-Gorenstein, then $-K_X$ is $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor [MZ22, Theorem 1.5].

Let X be a normal projective variety with a polarized endomorphism f. Conjecture 1.1 was proved in the following cases. Since f is int-amplified, results proved for int-amplified endomorphisms apply. Items (4) and (8) are such results, and (2) and (7) also include results on int-amplified endomorphisms that are étale in codimension one and on int-amplified endomorphisms h such that $\mathbb{C}(X)/h^*\mathbb{C}(X)$ is Galois, respectively.

(1) X is a surface or a $\mathbb{Q}$-factorial Mori dream space [BG17, Theorem 1.3]. Here Mori dream spaces are in the sense of Hu and Keel (see [BG17, Definition 3.1]), and an MMP can be run for any divisor on them [BG17, Remark 3.2].
(2) f is étale in codimension one [BG17, Proposition 4.1] (for $\mathbb{Q}$-Gorenstein X), [Yos22, Theorem 1.6], or more generally X admits an int-amplified endomorphism étale in codimension one [Yos22, Corollary 7.6], [Men20, Theorem 1.6], in which case X is lc and $K_X \sim_{\mathbb{Q}} 0$.
(3) X is not uniruled [NZ10, Theorem 3.2], in which case X has canonical singularities and $K_X \sim_{\mathbb{Q}} 0$.
(4) X is smooth and rationally connected. Then X is of Fano type, that is, there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ such that (X, Δ) is klt and $-(K_X + \Delta)$ is ample [Yos21, Corollary 1.4], and varieties of Fano type are of Calabi–Yau type.
(5) X is a smooth threefold [Men23, Theorem 1.2].
(6) X is a Gorenstein terminal threefold [CZ25, Theorem 1.2].
(7) X is smooth and admits an int-amplified endomorphism h such that the field extension $\mathbb{C}(X)/h^*\mathbb{C}(X)$ is Galois [JL26, Definition 2.1, Theorem 1.6].
(8) X is $\mathbb{Q}$-Gorenstein or of klt type, that is, there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ such that (X, Δ) is klt [Yos26, Corollary 1.3].

The strongest previous result is [Yos26, Corollary 1.3], and we now explain why it is limited to varieties that are $\mathbb{Q}$-Gorenstein or of klt type. Let h be an int-amplified endomorphism of X. [Yos26, Corollary 1.3] follows from [Yos26, Theorem 1.2], which starts from a pair (X, B) such that $B \geq 0$ is a $\mathbb{Q}$-divisor, $K_X + B$ is $\mathbb{Q}$-Cartier, and $R(h) + B - h^*B \geq 0$ [Yos26, Definition 3.1], and yields a $\mathbb{Q}$-divisor $B' \geq B$ such that (X, B') is lc and $K_X + B' \sim_{\mathbb{Q}} 0$. In the proof of [Yos26, Theorem 1.2], the obstruction to lc is located on divisorial valuations ord_E with $h_* \operatorname{ord}_E = e \operatorname{ord}_E$ for an integer $e > 1$, where $(h_* \operatorname{ord}_E)(\varphi) = \operatorname{ord}_E(h^*\varphi)$ for $\varphi \in \mathbb{C}(X)$ [Yos26, Theorem 5.1], and such divisors E are extracted on birational models of X to which an iterate of h lifts [Yos26, Proposition 6.5].

If X is $\mathbb{Q}$-Gorenstein, one takes $B = 0$. If X is of klt type, then there exists a small birational morphism $X' \to X$ such that X' is $\mathbb{Q}$-factorial and an iterate of h lifts to an endomorphism of X' [CZ25, Theorem A1.1] (cf. [MYY24, Theorem 1.2]), and [Yos26, Corollary 1.3] takes $B = 0$ on X'. If X is neither $\mathbb{Q}$-Gorenstein nor of klt type, no pair (X, B) satisfying the hypotheses of [Yos26, Theorem 1.2] is known in advance, and small birational morphisms as in [CZ25, Theorem A1.1] are known to exist only for varieties of klt type. Before this paper, Conjecture 1.1 was open for normal projective varieties that are neither $\mathbb{Q}$-Gorenstein nor of klt type, apart from the cases in the list above that do not assume either condition.

We now turn to the ramification divisor of a polarized endomorphism. Let X be a normal projective variety and $f : X \to X$ a *q-polarized* endomorphism, that is, $f^*H \sim qH$ for an ample Cartier divisor H on X (see Definition 2.11), and let $R(f)$ be the ramification divisor of f (see

Definition-Proposition 2.10). We call $\frac{1}{q-1}R(f)$ the *normalized ramification divisor* of f. For any positive integer s, the iterate f^s is q^s-polarized (see Lemma 2.12). [Men23, Conjecture 1.3] attributes the following conjecture to Gongyo.

Conjecture 1.2 (Gongyo)**.** *Let X be a smooth projective variety and $f : X \to X$ a q-polarized endomorphism. Then there exists a positive integer s such that $(X, \frac{1}{q^s-1}R(f^s))$ is lc.*

If X is $\mathbb{Q}$-Gorenstein, then there exists a positive integer s such that $(X, \frac{1}{q^{ms}-1}R(f^{ms}))$ is lc for any positive integer m [Yos26, Theorem 1.4]. Since a smooth variety is $\mathbb{Q}$-Gorenstein, Conjecture 1.2 holds. For surfaces, Conjecture 1.2 was also proved for $\rho(X) > 1$ [Men23, Theorem 4.1], for $\mathbb{P}^2$ with a curve C such that $f^{-1}(C) = C$ [Men23, Theorem 4.2], and for $\mathbb{P}^2$ [LM26, Theorem 1.3].

Chang and Zhang proposed a stronger conjecture [CZ25, Conjecture 1.5] for pairs (X, Δ), where $\Delta \geq 0$ is a $\mathbb{Q}$-divisor with $\Delta + R(f) - f^*\Delta \geq 0$, and $K_X + \Delta$ is not assumed $\mathbb{Q}$-Cartier [CZ25, Definition 1.4]. For $\Delta = 0$ and a normal projective variety X, [CZ25, Conjecture 1.5] predicts that, after replacing f by an iterate f^s and q by q^s, the $\mathbb{Q}$-divisor $K_X + \frac{1}{q-1}R(f)$ is $\mathbb{Q}$-Cartier,

$$K_X + \frac{1}{q-1}R(f) \equiv 0,$$

and $(X, \frac{1}{q-1}R(f))$ is lc. Since a projective lc pair (X, B) with $K_X + B \equiv 0$ satisfies $K_X + B \sim_{\mathbb{Q}} 0$ [Gon13, Theorem 1.2], the prediction is equivalent to the statement that, after replacing f by an iterate, $(X, \frac{1}{q-1}R(f))$ is lc and $K_X + \frac{1}{q-1}R(f) \sim_{\mathbb{Q}} 0$ (cf. [CZ25, Remark 1.6]).

The following cases were known. After replacing f by an iterate, the coefficients of $\frac{1}{q-1}R(f)$ are at most 1 [Men23, Theorem 3.1], and [CZ25, Theorem 1.7] proves the version for pairs, which implies [CZ25, Conjecture 1.5] for curves. [CZ25, Theorem 1.16] proves [CZ25, Conjecture 1.5] for pairs (X, Δ) with $\Delta \geq 0$ and $\Delta + R(f) - f^*\Delta \geq 0$ on normal projective surfaces X with $\rho(X) \geq 2$, or with a prime divisor P such that $f^{-s}(P) = P$ for some positive integer s. If f is a Galois cover of $\mathbb{P}^d$, then $(\mathbb{P}^d, \frac{1}{q-1}R(f))$ is lc without passing to an iterate [LZ26, Theorem 1.4]. For klt pairs, [CZ25, Conjecture 1.5] reduces to the case of Picard number one [CZ25, Theorem 1.13]. If X is $\mathbb{Q}$-factorial or of klt type and $\dim X \leq 3$, then after replacing f by an iterate, $f^*K_X - qK_X$ is $\mathbb{Q}$-Cartier and $f^*K_X - qK_X \equiv 0$ [CZ25, Theorem 1.15].

For the positive integer s of [Yos26, Theorem 1.4] and any positive integer m, [Yos26, Theorem 1.4] does not prove that

$$K_X + \frac{1}{q^{ms}-1}R(f^{ms}) \equiv 0.$$

For a pair (X, B) such that $B \geq 0$ is a $\mathbb{Q}$-divisor, $K_X + B$ is $\mathbb{Q}$-Cartier, $R(f) + B - f^*B \geq 0$, and $f^*(K_X + B) \equiv q(K_X + B)$, [Yos26, Corollary 7.6] shows that, after replacing f by an iterate and q by the corresponding power, the $\mathbb{Q}$-divisor

$$B' = B + \sum_{j=1}^{N} (\deg f)^{-j} (f^j)_* \big(R(f) + B - f^*B\big) + \frac{q^{1-N}}{q-1} (\deg f)^{-1} f_* \big(R(f) + B - f^*B\big)$$

satisfies $B' \geq B$, (X, B') is lc, and $K_X + B' \sim_{\mathbb{Q}} 0$ for all sufficiently large N.

Both [Yos26, Theorem 1.4] and the version for pairs [Yos26, Theorem 8.1] start from a pair (X, B) with $K_X + B$ $\mathbb{Q}$-Cartier, which for $B = 0$ means that K_X is $\mathbb{Q}$-Cartier. By Definition-Proposition 2.10(1), $K_X \sim f^*K_X + R(f)$, and hence

$$K_X + \frac{1}{q-1}R(f) \sim_{\mathbb{Q}} \frac{1}{q-1}\big(qK_X - f^*K_X\big).$$

Thus $K_X + \frac{1}{q-1}R(f) \sim_{\mathbb{Q}} 0$ if and only if $f^*K_X \sim_{\mathbb{Q}} qK_X$ (cf. [CZ25, Remark 1.6]). In particular, [CZ25, Conjecture 1.5] for $\Delta = 0$ predicts that $f^*K_X \sim_{\mathbb{Q}} qK_X$ after replacing f by an iterate

f^s and q by q^s. If X is not $\mathbb{Q}$-Gorenstein, then $K_X + \frac{1}{q-1}R(f)$ is not known to be $\mathbb{Q}$-Cartier in advance.

For pairs that are already log Calabi–Yau, polarized endomorphisms are understood structurally: dlt log Calabi–Yau pairs with a polarized endomorphism are finite quotients of toric log Calabi–Yau fibrations over abelian varieties [MYY24, Theorem 1.1], and, in the Fano-type case with reduced boundary, finite quotients of toric log Calabi–Yau pairs [MYY25, Theorem 1].

The goal of this paper is to prove Conjecture 1.1 for all normal projective varieties, in the more general form for int-amplified endomorphisms, and to show that for a q-polarized endomorphism f of a normal projective variety, the normalized ramification divisors of iterates of f are the boundaries predicted by [CZ25, Conjecture 1.5] for $\Delta = 0$. For an int-amplified endomorphism h, we start from the $\mathbb{Q}$-divisors used in [Yos26], partial sums of the $\mathbb{Q}$-divisors $(\deg h)^{-j}(h^j)_*R(h)$ plus a finite correction (cf. [Yos26, Definition 3.7, Lemma 3.8]), and the boundary that we construct is defined by a general linear combination of products of sections whose divisors are positive integer multiples of such $\mathbb{Q}$-divisors.

However, we use h only on X: through relations satisfied by the operator $(\deg h)^{-1}h_*$ on the group of $\mathbb{Q}$-divisors on X modulo $\mathbb{Q}$-linear equivalence, and through the behavior of log discrepancies on X under h. On a small birational model of X to which no endomorphism is lifted, we then prove that the boundary defined by such a general linear combination is lc. The divisorial valuations ord_E with $h_*\,\mathrm{ord}_E = e\,\mathrm{ord}_E$ for an integer $e > 1$ used in [Yos26] are replaced by valuations singled out by a condition on a sequence of $\mathbb{Q}$-divisors on a small birational model of X, a condition in which no endomorphism appears, and the extraction on models to which an iterate of h lifts is replaced by induction on dimension.

Main results. Our first main theorem concerns int-amplified endomorphisms in the sense of [Men20, Section 1].

Theorem 1.3. *Let X be a normal projective variety admitting an int-amplified endomorphism. Then X is of Calabi–Yau type, that is, there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that (X, B) is lc and $K_X + B \sim_{\mathbb{Q}} 0$.*

Since polarized endomorphisms are int-amplified (Lemma 2.12), Theorem 1.3 implies the conjecture of Broustet and Gongyo.

Corollary 1.4. *Conjecture 1.1 holds.*

For polarized endomorphisms, the boundary in Theorem 1.3 can be taken to be a normalized ramification divisor.

Theorem 1.5. *Let X be a normal projective variety and $f : X \to X$ a q-polarized endomorphism. Then there exists a positive integer s such that for any positive integer m, the $\mathbb{Q}$-divisor*

$$B_m := \frac{1}{q^{ms} - 1}R(f^{ms})$$

satisfies $B_m \geq 0$, (X, B_m) is lc, and $K_X + B_m \sim_{\mathbb{Q}} 0$.

In particular, the case $\Delta = 0$ of [CZ25, Conjecture 1.5] holds.

Theorem 1.5 also controls the canonical divisor K_X, which is not assumed to be $\mathbb{Q}$-Cartier. Indeed, by Theorem 1.5 with $m = 1$ and Definition-Proposition 2.10(1), we have $(f^s)^*K_X \sim_{\mathbb{Q}} q^sK_X$ (see the proof of Corollary 1.6). By Construction-Proposition 4.2, the $\mathcal{O}_X$-algebra $\bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ is finitely generated, and we obtain the following lift of f^s.

Corollary 1.6. *Let X be a normal projective variety and $f : X \to X$ a q-polarized endomorphism. Then there exists a positive integer s such that $(f^s)^*K_X \sim_{\mathbb{Q}} q^sK_X$. Moreover, the $\mathcal{O}_X$-algebra $\bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ is finitely generated, and there exists a q^s-polarized endomorphism g of $X' := \mathrm{Proj}_X \bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ such that $\pi \circ g = f^s \circ \pi$, where $\pi : X' \to X$ is the structure morphism.*

Remark 1.7. Unlike in Theorem 1.5, for an int-amplified endomorphism h, the boundary in Theorem 1.3 cannot always be taken to be of the form $cR(h^n)$ with $c \in \mathbb{Q}$ and $n \geq 1$. For example, let h be the endomorphism of $\mathbb{P}^1 \times \mathbb{P}^1$ defined by

$$h([x_0 : x_1], [y_0 : y_1]) = ([x_0^2 : x_1^2], [y_0^3 : y_1^3]).$$

Then h is int-amplified, since $h^*\mathcal{O}(1,1) \otimes \mathcal{O}(-1,-1) \cong \mathcal{O}(1,2)$ is ample. For any positive integer n, we have

$$R(h^n) = (2^n - 1)(P_0 + P_1) + (3^n - 1)(P_0' + P_1'),$$

where $P_i := \{x_i = 0\}$ and $P_i' := \{y_i = 0\}$. Since $K_{\mathbb{P}^1 \times \mathbb{P}^1} \sim -P_0 - P_1 - P_0' - P_1'$ and $2^n \neq 3^n$, there exists no rational number c such that $K_{\mathbb{P}^1 \times \mathbb{P}^1} + cR(h^n) \sim_{\mathbb{Q}} 0$.

Sketch of the proof. For an int-amplified endomorphism h of X, Theorem 1.3 follows from three results on the $\mathbb{Q}$-divisors $D_j = (\deg h)^{-j}(h^j)_* R(h)$, $j \geq 1$.

(i) Construction-Proposition 3.1: for some $\ell \geq 1$ and $a_{j,N} \in \mathbb{Q}$ with $\lim_{N \to +\infty} a_{j,N} = 0$,

$$K_X + \sum_{j=1}^{N} D_j + \sum_{j=1}^{\ell} a_{j,N} D_j \sim_{\mathbb{Q}} 0 \quad \text{for any } N \geq \ell.$$

(ii) Construction-Proposition 5.2: on a small birational model X' with $(X', 0)$ lc, the strict transforms D_j' of the D_j are $\mathbb{Q}$-Cartier, $(X', \sum_{j \leq N} D_j')$ is lc for any $N \geq 1$, and the relation in (i) holds with $K_{X'}$ and the D_j'; that is, $(D_j')_{j \geq 1}$ is an lc anti-canonical series on $(X', 0)$ (Definition 7.1). It uses Theorems 4.1 and 4.4 and Proposition 4.5.

(iii) Theorem 12.1: X' carries a $\mathbb{Q}$-divisor $B' \geq 0$ with (X', B') lc and $K_{X'} + B' \sim_{\mathbb{Q}} 0$, which descends to X by Lemma 2.19.

Only (i) and (ii) use h, and Theorem 12.1, used in (iii), is the main new input. Theorem 1.5, which implies Corollary 1.4 without Theorem 12.1, follows from Construction-Proposition 6.2 and Lemma 6.4, which reduce it to the $\mathbb{Q}$-Gorenstein case (Proposition 6.3, whose lc part is [Yos26, Theorem 8.1]).

By Lemma 12.3, the corrected boundaries $G_N = \sum_{j \leq N} D_j' + \sum_{j \leq m} c_{j,N} D_j'$ for some $m \geq 1$, where $c_{j,N} \to 0$ as $N \to +\infty$ and $K_{X'} + G_N \sim_{\mathbb{Q}} 0$ (Definition 7.1), fail, for large N depending on the point, to be lc only near the centers of stationary valuations, that is, quasi-monomial valuations v with $\sum_{j \geq 1} v(D_j') = a(v, X', 0)$.

The key inductive step is as follows. Let a corrected boundary G satisfy $a(v, X', G) = 0$ at a stationary valuation v of rational rank r. On a log resolution on which v is monomial in r components $E_1, \dots, E_r$, a component S of $E_1 \cap \cdots \cap E_r$ (of codimension r) carries the leading divisor of G, which is $\frac{1}{l}$ times the divisor on S of the coefficient of the lowest-weight monomial in a local equation of the pullback of lG for a positive integer l such that lG is Cartier. If the leading divisor of G defines a sub-lc sub-pair on S, then (X', G) is lc near the center of v (Construction-Proposition 8.1, Proposition 9.1), and Construction-Proposition 11.4 reduces the sub-lc property of the leading divisor of G on S to Theorem 12.1 in dimension $\dim X - r$. We only show that B' exists, as a general member of a corrected system (Definition 7.3).

The new ideas behind (i)–(iii) and Theorem 1.5 are as follows.

The relation in (i) and the bound in (ii). We need a $\mathbb{Q}$-linear relation in (i), not only a numerical one as in [Yos26, Lemma 3.8], because the proof of Theorem 12.1 multiplies and spans sections defining corrected boundaries (Definition 7.3). The $\mathbb{Q}$-vector space $\mathrm{Cl}(X)_{\mathbb{Q}}$ of $\mathbb{Q}$-divisors on X modulo $\mathbb{Q}$-linear equivalence may have infinite dimension, since it contains the image of Pic^0 of a resolution of X. Construction-Proposition 3.1 finds annihilating polynomials of $(\deg h)^{-1} h_*$ on that image, by [Men20, Theorem 1.8(2)], and on the quotient, by the contraction of Lemma 3.4, and their product annihilates $(\deg h)^{-1} h_*$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$.

For the partial sums $\sum_{j \leq N} D_j$, [Yos26, Lemma 3.3, (3.6)] compares log discrepancies through h on a pair, but $K_X + \sum_{j \leq N} D_j$ need not be $\mathbb{Q}$-Cartier. Construction-Proposition 5.2 bounds the partial sums above by the anti-canonical boundaries $\sum_{j \leq k} D_j + (\deg h)^{-k} (h^k)_* \Delta_0$, $k \geq N$,

for $\Delta_0 := \sum_{j\le N_*} D_j + \sum_{j\le \ell} a_{j,N_*} D_j$, where $N_* \ge \ell$ is fixed with $1 + a_{j,N_*} > 0$ for any $j \le \ell$, so that $\Delta_0 \ge 0$ and $K_X + \Delta_0 \sim_{\mathbb{Q}} 0$ by (i). Their log discrepancies satisfy a weighted-average identity along h and are bounded below by quantities tending to 0 by Lemma 3.4.

Lc centers of the canonical model. Since by (i) the classes of $D_1, \dots, D_\ell$ span those of all D_j, Lemma 5.1 needs only ℓ applications of Theorem 2.18 over the canonical model X_0 of X (denoted X' in Construction-Proposition 4.2 and Corollary 1.6), given by Theorem 4.1 and [Has20, Theorems 4.17 and 4.19], provided that no lc center of $(X_0, 0)$ is contained in the strict transform of a D_j. Proposition 4.5 replaces the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $K_X + B$ of [Yos26, Proposition 3.4] by the $\boldsymbol{b}$-divisor $\mathbf{K}_X + \mathbf{E}_X - \mathrm{Env}_X(K_X)$ (Definitions 2.14 and 2.15), whose coefficients at exceptional divisors are the log discrepancies of $(X_0, 0)$ and which does not increase under finite pullback (cf. [Yos22, Proposition 4.11]). It also uses Theorem 4.4, whose difficulty is uniformity: the boundaries of [Has20, Theorem 4.8] with log discrepancy at least $-1/i$ at an lc place have classes depending on i. Their strict transforms B_i satisfy $(B_i \cdot C) = -(K_{X_0} \cdot C) < 0$ on any curve C contracted by $X_0 \to X$, so Lemma 4.3 bounds the order of B_i at a point of C below by a positive number independent of i.

Finite attainment. On X' no endomorphism is available to extract, as in [Yos26, Proposition 6.5], the valuations at which the corrected boundaries fail to be lc. The stationary valuations of Lemma 12.3, found by [Xu20, Theorem 1.1], are quasi-monomial (Definition 2.7) and may have irrational weights. For a stationary valuation v, Proposition 9.1 extracts by [Mor20, Theorem 1] the divisorial valuation of an integral weight vector $\mathbf{e}$ near the ray of the weights of v, and uses inversion of adjunction (Theorem 2.21) on the extracted divisor. [Mor20, Theorem 1] needs $a(w, X', \sum_{j\le N} D'_j) < 1$ for some N, where w is the divisorial valuation with weights $\mathbf{e}$. As $N \to +\infty$, $a(w, X', \sum_{j\le N} D'_j)$ decreases to a function of $\mathbf{e}$ that is homogeneous of degree one and vanishes at the weights of v (Lemma 8.5), so approximating the direction of the weights is not enough. We choose $\mathbf{e}$ by simultaneous approximation [Cas57] (Construction-Proposition 9.3).

For the induction, the boundary of the sub-pair on the stratum S of Construction-Proposition 8.1 may have negative coefficients, and the center of v may have codimension larger than the rational rank r of v. Construction-Proposition 10.1 adds a multiple of a partial sum to the boundary and runs an MMP ([OX12, Theorem 1.1], [Bir12, Theorem 3.4]) to reach codimension r. Then the sub-pairs on S push forward to pairs on the normalization of the center of v in the residue field of v (Definition-Proposition 11.2), which is pseudo-lc (Definition 2.17, Lemma 11.3). Construction-Proposition 11.4 adds finitely many moment rows (Definition 7.1), which remove the prime divisors with a negative coefficient that the leading divisors of the partial sums cancel only in the limit, and constructs a residue model of dimension $\dim X - r$. Since the vanishing of the log discrepancy at v is itself a moment row (Lemma 8.5) and rows pass from each pair to its residue model, Theorem 12.1 allows finitely many moment rows.

Polarized endomorphisms. Lemma 6.1 lifts f^s to the canonical model of X once the strict transform of $(f^s)^* K_X$ is relatively ample, but this is known only after Theorem 1.5 (Corollary 1.6). Construction-Proposition 6.2 constructs one small birational model of X on which the strict transforms of all $\mathbb{Q}$-divisors with classes in the span W of K_X, an ample H with $f^*H \sim qH$, and all $R(f^j)$ are $\mathbb{Q}$-Cartier, where $\dim W < +\infty$ and $f^*W = W$ by Construction-Proposition 3.1. Since $q^{-1}f^*$ and its inverse $q^{1-\dim X} f_*$ both map into itself a cone without lines in the numerical image of W, and $q^{-1}f^*$ fixes the interior point given by H, some positive power of $q^{-1}f^*$ is close to the identity (cf. [MZ18, Proposition 2.9]). Thus some $(f^a)^*$ keeps an ample class ample, and Lemma 6.1 lifts f^a.

The analogue for h needs a real number $c > 0$ such that the operators $(c^{-1}h^*)^k$, $k \in \mathbb{Z}$, are uniformly bounded on the numerical image of the corresponding space W. No such c exists for the endomorphism of Remark 1.7, whose pullback has eigenvalues 2 and 3 on $N^1(\mathbb{P}^1 \times \mathbb{P}^1)_{\mathbb{R}}$, which is the numerical image of W there. In general we do not know whether an iterate of h lifts to a small $\mathbb{Q}$-Gorenstein model.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce quasi-monomial valuations, log canonical thresholds of graded sequences, nef envelopes, and pseudo-lc varieties in the sense of Hashizume, and collect basic facts on int-amplified and polarized endomorphisms.

In Section 3, for an int-amplified endomorphism h of a normal projective variety X, we show that the operator $(\deg h)^{-1}h_*$ on the group $\mathrm{Cl}(X)_{\mathbb{Q}}$ of $\mathbb{Q}$-divisors on X modulo $\mathbb{Q}$-linear equivalence is annihilated by a polynomial whose roots have absolute value less than 1, and we deduce an expression of $-K_X$ in terms of the $\mathbb{Q}$-divisors $(\deg h)^{-j}(h^j)_*R(h)$, $j \geq 1$.

In Section 4 we show that a normal projective variety admitting an int-amplified endomorphism is pseudo-lc, so that the $\mathcal{O}_X$-algebra $\bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ is finitely generated [Has20, Theorems 4.17 and 4.19], and we study the canonical model $\operatorname{Proj}_X \bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ of X (see Construction-Proposition 4.2) and the lc centers of the canonical model.

Section 5 constructs a small birational model of X on which the strict transforms of the $\mathbb{Q}$-divisors $(\deg h)^{-j}(h^j)_*R(h)$, $j \geq 1$, are $\mathbb{Q}$-Cartier with lc partial sums, for an int-amplified endomorphism h of X.

In Section 6, for a q-polarized endomorphism f of X, we construct a small birational model of X with $\mathbb{Q}$-Cartier canonical divisor to which an iterate of f lifts, and we study the normalized ramification divisors $\frac{1}{q^n-1}R(f^n)$.

In Section 7 we introduce lc anti-canonical series, correction families, moment rows, and stationary valuations (see Definition 7.1), and corrected systems (see Definition 7.3).

In Section 8 we introduce adapted carriers and leading divisors of a quasi-monomial valuation (see Construction-Proposition 8.1), and we study the leading exponents and the leading divisors of an lc anti-canonical series along an adapted carrier of a stationary valuation (see Lemmas 8.5 and 8.6).

In Section 9 we show by inversion of adjunction that if v is a quasi-monomial valuation that is stationary for an lc anti-canonical series on (X, B) (Definition 7.1), the log discrepancy of $(X, B + G)$ at v is zero, and the sub-pair formed by the leading divisor of G on the stratum of an adapted carrier of v is sub-lc, then $(X, B + G)$ is lc on a neighborhood of the center of v (see Proposition 9.1).

In Section 10, for a quasi-monomial valuation v that is stationary for an lc anti-canonical series on (X, B), we construct a $\mathbb{Q}$-factorial dlt model on which the center of v has codimension equal to the rational rank of v (see Construction-Proposition 10.1).

In Section 11 we construct the residue model (see Construction-Proposition 11.4), a pair of lower dimension carrying an lc anti-canonical series.

In Section 12 we argue by induction on dimension and combine the results of Sections 7–11 to prove Theorem 12.1: for a projective lc pair (X, B) such that B is a $\mathbb{Q}$-divisor and (X, B) carries an lc anti-canonical series, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on X such that $(X, B + G)$ is lc and $K_X + B + G \sim_{\mathbb{Q}} 0$.

In Section 13 we prove Theorems 1.3 and 1.5 and Corollaries 1.4 and 1.6.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that

(X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c\,\mathrm{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let (X, B) and (X', B') be sub-pairs. We say that (X, B) and (X', B') are *crepant birational* if there exist a normal variety W and projective birational morphisms $g : W \to X$ and $g' : W \to X'$ such that $g^*(K_X + B) = g'^*(K_{X'} + B')$. For morphisms $p : X \to Z$ and $p' : X' \to Z$ to a variety Z, we say that (X, B) and (X', B') are *crepant birational over* Z if there exist a normal variety W and projective birational morphisms $g : W \to X$ and $g' : W \to X'$ such that $g^*(K_X + B) = g'^*(K_{X'} + B')$ and $p \circ g = p' \circ g'$.

We recall the following definition of Broustet and Gongyo [BG17, Definition 2.3].

Definition 2.3. Let X be a normal projective variety. We say that X is *of Calabi–Yau type* if there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that (X, B) is lc and $K_X + B \sim_{\mathbb{Q}} 0$.

Definition 2.4. Let (X, B) be a sub-pair. For a prime divisor E over X, we denote by $\mathrm{center}_X E$ the image of E in X. The *non-lc locus* $\mathrm{Nlc}(X, B)$ of (X, B) is the union of $\mathrm{center}_X E$ for all prime divisors E over X such that $a(E, X, B) < 0$. The *non-klt locus* $\mathrm{Nklt}(X, B)$ of (X, B) is the union of $\mathrm{center}_X E$ for all prime divisors E over X such that $a(E, X, B) \leq 0$.

Lemma 2.5. *Let X be a normal variety and D a $\mathbb{Q}$-divisor on X. Assume that D is $\mathbb{R}$-Cartier. Then D is $\mathbb{Q}$-Cartier.*

Proof. Since X is quasi-compact, there exist finitely many affine open subsets $U_1, \dots, U_n$ covering X such that, for any k, $D|_{U_k} = \sum_i r_{k,i} C_{k,i}$ for finitely many real numbers $r_{k,i}$ and Cartier divisors $C_{k,i}$ on U_k. Fix k. Comparing the coefficients at the finitely many prime divisors in $\mathrm{Supp}\, D|_{U_k} \cup \bigcup_i \mathrm{Supp}\, C_{k,i}$, the vector $(r_{k,i})_i$ is a real solution of a system of linear equations with rational coefficients. Since the solvability of such a system is determined by the ranks of the coefficient matrix and the augmented matrix, the system also has a rational solution $(s_{k,i})_i$. Thus $D|_{U_k} = \sum_i s_{k,i} C_{k,i}$, and $l_k D|_{U_k}$ is Cartier for a common denominator l_k of the $s_{k,i}$. Therefore lD is Cartier for $l := \mathrm{lcm}(l_1, \dots, l_n)$. □

2.2. **Valuations.** We follow [JM12, Sections 1.2 and 3] for valuations.

Definition 2.6. Let X be a normal variety. A *real valuation* of $\mathbb{C}(X)$ is a map $v : \mathbb{C}(X)^* \to \mathbb{R}$ such that $v(\phi\psi) = v(\phi) + v(\psi)$, $v(\phi + \psi) \geq \min\{v(\phi), v(\psi)\}$, and $v(\mathbb{C}^*) = 0$, with the convention $v(0) = +\infty$. A scheme point $\xi \in X$ is the *center* of v on X if $v \geq 0$ on $\mathcal{O}_{X,\xi}$ and $v > 0$ on the maximal ideal of $\mathcal{O}_{X,\xi}$, and $\mathrm{center}_X v$ denotes the closure of ξ. If v has a center ξ on X and D is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, then $v(D) := \frac{1}{l} v(\phi)$, where l is a positive integer such that lD is Cartier near ξ and ϕ is a local equation of lD at ξ. The *residue field* k_v of v is the residue field of the valuation ring of v. The *rational rank* of v is $\dim_{\mathbb{Q}}(v(\mathbb{C}(X)^*) \otimes_{\mathbb{Z}} \mathbb{Q})$.

Definition 2.7. Let X be a normal variety, $h : X' \to X$ a projective birational morphism from a smooth variety X', $E_1, \dots, E_r$ distinct components of a reduced simple normal crossing divisor on X', η the generic point of an irreducible component of $E_1 \cap \cdots \cap E_r$, and $\gamma \in \mathbb{R}^r_{\geq 0} \setminus \{0\}$. Let $z_1, \dots, z_r$ be local equations of $E_1, \dots, E_r$ at η, and identify $\widehat{\mathcal{O}}_{X',\eta}$ with $\kappa[[z_1, \dots, z_r]]$ for a coefficient field κ. The *monomial valuation* v_γ with weights γ in $E_1, \dots, E_r$ at η is defined by

$$v_\gamma\Big(\sum_{\beta \in \mathbb{N}^r} c_\beta z^\beta\Big) := \min\Big\{\sum_{i=1}^r \gamma_i \beta_i \;\Big|\; c_\beta \neq 0\Big\}.$$

A real valuation of $\mathbb{C}(X)$ is *quasi-monomial* if it is equal to v_γ for some such data.

Definition-Proposition 2.8. Let (X, B) be a sub-pair and v a real valuation of $\mathbb{C}(X)$ with a center on X. Let $h : X' \to X$ be a log resolution of (X, B), write $K_{X'} + B' := h^*(K_X + B)$, and let $A_{X'}$ be the log discrepancy function of the smooth variety X' in the sense of [JM12, Sections 5.1 and 5.2]. We define

$$a(v, X, B) := A_{X'}(v) - v(B') \in \mathbb{R} \cup \{+\infty\},$$

where $v(B') := \sum_F (\mathrm{mult}_F B')v(F)$ and F runs over the prime divisors on X'. Then $a(v, X, B)$ does not depend on the choice of h, and $a(v, X, B)$ coincides with the log discrepancy of Definition 2.1 if $v = c\,\mathrm{ord}_E$ for a prime divisor E over X and a real number $c > 0$. Moreover, if $v = v_\gamma$ is the monomial valuation with weights γ in components $E_1, \dots, E_r$ of a reduced simple normal crossing divisor on X' containing $\mathrm{Exc}(h) \cup \mathrm{Supp}\, B'$, at the generic point η of an irreducible component of $E_1 \cap \cdots \cap E_r$, in the sense of Definition 2.7, then $a(v, X, B) = \sum_{i=1}^r \gamma_i a(E_i, X, B)$. For any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on X, we have $a(v, X, B + L) = a(v, X, B) - v(L)$. If (X, B) is sub-lc, then $a(v, X, B) \geq 0$.

Proof. Since h is proper, v has a center on X' by the valuative criterion of properness. Since X' is smooth, any prime divisor F on X' is Cartier, so $v(F)$ is defined. If B' is a $\mathbb{Q}$-divisor, then $v(B')$ coincides with the value of Definition 2.6, since a local equation of a multiple of B' is a product of powers of local equations of the components of B'.

We show that $a(v, X, B)$ does not depend on h. Any two log resolutions of (X, B) are dominated by a third one. Thus it suffices to compare h with $h \circ g$, where $g : X'' \to X'$ is a projective birational morphism from a smooth variety X'' such that $h \circ g$ is a log resolution of (X, B). Write $K_{X''} + B'' := (h \circ g)^*(K_X + B)$ and $K_{X''/X'} := K_{X''} - g^* K_{X'}$. By [JM12, Remark 5.6],

$$A_{X'}(v) = A_{X''}(v) + v(K_{X''/X'}).$$

On the other hand, $B'' = g^*B' - K_{X''/X'}$. Since $v(g^*F) = v(F)$ for any prime divisor F on X', we have

$$v(B'') = v(B') - v(K_{X''/X'}).$$

Since $v(K_{X''/X'}) \in \mathbb{R}$, we obtain

$$A_{X''}(v) - v(B'') = A_{X'}(v) - v(K_{X''/X'}) - v(B') + v(K_{X''/X'}) = A_{X'}(v) - v(B'),$$

where both sides are $+\infty$ when $A_{X'}(v) = +\infty$.

We prove the formula for $v = v_\gamma$. By [JM12, Proposition 5.1(ii)] and [JM12, (5.1)], applied to X' and the given simple normal crossing divisor,

$$A_{X'}(v_\gamma) = \sum_{i=1}^r \gamma_i A_{X'}(\mathrm{ord}_{E_i}) = \sum_{i=1}^r \gamma_i.$$

Let F be a component of the given simple normal crossing divisor different from $E_1, \dots, E_r$. Then $\eta \notin F$, since $\mathrm{codim}(E_1 \cap \cdots \cap E_r \cap F) = r + 1$. Thus $v_\gamma(F) = 0$. Moreover, $v_\gamma(E_i) = \gamma_i$ for any i. Since $\mathrm{Supp}\, B'$ is contained in the given simple normal crossing divisor, $v_\gamma(B') = \sum_i \gamma_i \,\mathrm{mult}_{E_i} B'$. Subtracting and using $a(E_i, X, B) = 1 - \mathrm{mult}_{E_i} B'$, we obtain $a(v_\gamma, X, B) = \sum_i \gamma_i a(E_i, X, B)$.

Let $v = c\,\mathrm{ord}_E$. Since $a(v, X, B)$ does not depend on h, we may assume that E is a prime divisor on X'. Since E is a prime divisor on the smooth variety X', $A_{X'}(\mathrm{ord}_E) = 1$ [JM12, Section 1.3], so $A_{X'}(c\,\mathrm{ord}_E) = cA_{X'}(\mathrm{ord}_E) = c$ by [JM12, Proposition 5.1 and (5.1)]. Moreover, $v(F) = c\,\mathrm{mult}_E F$ for any prime divisor F on X', so $v(B') = c\,\mathrm{mult}_E B'$. Thus

$$a(c\,\mathrm{ord}_E, X, B) = c - c\,\mathrm{mult}_E B' = ca(E, X, B).$$

Let L be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Since $a(v, X, B)$ and $a(v, X, B + L)$ do not depend on the choice of the log resolution, we may assume that h is also a log resolution of $(X, B + L)$ [KM98, Theorem 0.2]. Then $K_{X'} + B' + h^*L = h^*(K_X + B + L)$. Moreover, $v(h^*L) = v(L)$,

because a local equation of a Cartier multiple lL at the center of v on X pulls back to a local equation of lh^*L at the center of v on X'. Thus

$$a(v, X, B + L) = A_{X'}(v) - v(B') - v(h^*L) = a(v, X, B) - v(L),$$

where both sides are $+\infty$ when $A_{X'}(v) = +\infty$.

Finally, assume that (X, B) is sub-lc, and let $F_1, \dots, F_k$ be the components of $\operatorname{Supp} B'$. Since h is a log resolution of (X, B), $F_1 + \cdots + F_k$ is a reduced simple normal crossing divisor on X'. The retraction $r(v)$ of v to the monomial valuations in $F_1, \dots, F_k$ satisfies $r(v)(F_i) = v(F_i)$ for any i, $A_{X'}(r(v)) = \sum_i v(F_i)$, and $A_{X'}(r(v)) \leq A_{X'}(v)$ [JM12, Section 4.3, (5.1), and Corollary 5.4]. Since (X, B) is sub-lc, $\operatorname{mult}_{F_i} B' \leq 1$ for any i. Since $v(F_i) \geq 0$ for any i, we obtain $v(B') \leq \sum_i v(F_i) \leq A_{X'}(v)$, that is, $a(v, X, B) \geq 0$. □

We use the conventions of [Xu20, Section 2] for log canonical thresholds of graded sequences of ideals, and we require in the following definition that (X, B) is sub-klt on a neighborhood of ξ because [Xu20, Theorem 1.1] concerns klt germs, and we only use log canonical thresholds of graded sequences of ideals near points where (X, B) is klt (Lemma 12.3). At points of $\operatorname{Nklt}(X, B)$, Lemma 12.3 uses lc places instead of log canonical thresholds.

Definition 2.9. Let (X, B) be a sub-pair, $\xi \in X$ a scheme point such that (X, B) is sub-klt on a neighborhood of ξ, and $\mathfrak{a}_\bullet = (\mathfrak{a}_l)_{l \geq 1}$ a *graded sequence of ideals* of $\mathcal{O}_{X,\xi}$, that is, a sequence of non-zero ideals of $\mathcal{O}_{X,\xi}$ such that $\mathfrak{a}_l \mathfrak{a}_{l'} \subset \mathfrak{a}_{l+l'}$ for any $l, l' \geq 1$. For a real valuation v of $\mathbb{C}(X)$ that has a center on X with $\xi \in \operatorname{center}_X v$, and a non-zero ideal $\mathfrak{a} \subset \mathcal{O}_{X,\xi}$, we write $v(\mathfrak{a}) := \min\{v(\phi) \mid \phi \in \mathfrak{a}\}$, and we define

$$v(\mathfrak{a}_\bullet) := \lim_{l \to +\infty} \frac{1}{l} v(\mathfrak{a}_l) = \inf_{l \geq 1} \frac{1}{l} v(\mathfrak{a}_l),$$

where the limit exists and is equal to the infimum by [JM12, Lemma 2.3], since $v(\mathfrak{a}_{l+l'}) \leq v(\mathfrak{a}_l) + v(\mathfrak{a}_{l'})$ for any $l, l' \geq 1$. The *log canonical threshold* of $\mathfrak{a}_\bullet$ with respect to (X, B) is

$$\operatorname{lct}(X, B; \mathfrak{a}_\bullet) := \inf_v \frac{a(v, X, B)}{v(\mathfrak{a}_\bullet)},$$

where v runs over all real valuations of $\mathbb{C}(X)$ that have a center on X with $\xi \in \operatorname{center}_X v$ and such that $a(v, X, B) < +\infty$ and $v(\mathfrak{a}_\bullet) > 0$. We say that v *computes* $\operatorname{lct}(X, B; \mathfrak{a}_\bullet)$ if $a(v, X, B) = \operatorname{lct}(X, B; \mathfrak{a}_\bullet) \cdot v(\mathfrak{a}_\bullet)$.

2.3. **Finite morphisms.** For a normal variety X, we denote by $\operatorname{Cl}(X)_{\mathbb{Q}}$ the $\mathbb{Q}$-vector space of $\mathbb{Q}$-divisors on X modulo $\mathbb{Q}$-linear equivalence, and by $[D] \in \operatorname{Cl}(X)_{\mathbb{Q}}$ the class of a $\mathbb{Q}$-divisor D on X.

Definition-Proposition 2.10. Let $f : X \to Y$ be a finite surjective morphism between normal varieties. For a prime divisor E on X, let e_E be the ramification index of f along E. For a $\mathbb{Q}$-divisor D on Y, let f^*D be the $\mathbb{Q}$-divisor on X whose coefficient at any prime divisor E on X is e_E times the coefficient of $f(E)$ in D, and let f_* be the pushforward of cycles. The *ramification divisor* of f is

$$R(f) := \sum_E (e_E - 1)E,$$

where E runs over the prime divisors on X. Then the following hold.

(1) $R(f) \geq 0$ and $K_X = f^*K_Y + R(f)$, where $K_Y = \operatorname{div}(\omega)$ for a non-zero rational differential form ω of top degree on Y and $K_X := \operatorname{div}(f^*\omega)$ (if $Y = X$, then $\operatorname{div}(\omega)$ and $\operatorname{div}(f^*\omega)$ are two canonical divisors of X, which differ in general), and in particular $K_X \sim f^*K_X + R(f)$ if $Y = X$.

(2) f^* and f_* preserve $\mathbb{Q}$-linear equivalence, and $f_* f^* D = (\deg f)D$ for any $\mathbb{Q}$-divisor D on Y.

(3) For any finite surjective morphism $g : Y \to Y'$ between normal varieties, $R(g \circ f) = R(f) + f^*R(g)$.

(4) Let (Y, B) be a sub-pair such that B is a $\mathbb{Q}$-divisor, and write $B_X := f^*B - R(f)$. Then (X, B_X) is a sub-pair with $K_X + B_X = f^*(K_Y + B)$, and $a(E, X, B_X) = ea(E_Y, Y, B)$ for any prime divisor E over X, where E_Y is the prime divisor over Y and e is the positive integer such that the restriction of ord_E to $\mathbb{C}(Y)$ is $e \operatorname{ord}_{E_Y}$.

Proof. Since f is finite and surjective, $f(E)$ is a prime divisor on Y for any prime divisor E on X. By the definition of e_E,

$$\operatorname{ord}_E(\phi \circ f) = e_E \operatorname{ord}_{f(E)}(\phi)$$

for any $\phi \in \mathbb{C}(Y)^*$, so $f^* \operatorname{div}(\phi) = \operatorname{div}(\phi \circ f)$.

(1) Since $e_E \geq 1$, $R(f) \geq 0$. The equality $K_X = f^*K_Y + R(f)$ is the Hurwitz formula, which we check at a prime divisor E on X. Let t and y be uniformizers of $\mathcal{O}_{X,E}$ and $\mathcal{O}_{Y,f(E)}$. Then $y = ut^{e_E}$ for a unit u of $\mathcal{O}_{X,E}$, and since the characteristic is zero, the coefficients of K_X and f^*K_Y at E differ by $e_E - 1$ (cf. [KM98, Proposition 5.20]). If $Y = X$, then $K_Y \sim K_X$, so $f^*K_Y \sim f^*K_X$ by $f^* \operatorname{div}(\phi) = \operatorname{div}(\phi \circ f)$, and $K_X \sim f^*K_X + R(f)$.

(2) The equality $f^* \operatorname{div}(\phi) = \operatorname{div}(\phi \circ f)$ shows that f^* preserves $\mathbb{Q}$-linear equivalence. For any $\psi \in \mathbb{C}(X)^*$, $f_* \operatorname{div}(\psi)$ is the divisor of the norm of ψ from $\mathbb{C}(X)$ to $\mathbb{C}(Y)$ [Ful98, Proposition 1.4(b)]. Thus f_* preserves $\mathbb{Q}$-linear equivalence. Let P be a prime divisor on Y. The prime divisors E on X with $f(E) = P$ correspond to the maximal ideals of the integral closure of the discrete valuation ring $\mathcal{O}_{Y,P}$ in $\mathbb{C}(X)$. By [Ser79, Chapter I, §4, Proposition 10],

$$f_*f^*P = \sum_{f(E)=P} e_E[\mathbb{C}(E) : \mathbb{C}(P)]P = (\deg f)P.$$

By linearity, $f_*f^*D = (\deg f)D$ for any $\mathbb{Q}$-divisor D on Y.

(3) Let E be a prime divisor on X and e' the ramification index of g along $f(E)$. For any $\phi \in \mathbb{C}(Y')^*$,

$$\operatorname{ord}_E(\phi \circ g \circ f) = e_E \operatorname{ord}_{f(E)}(\phi \circ g) = e_E e' \operatorname{ord}_{g(f(E))}(\phi),$$

so the ramification index of $g \circ f$ along E is $e_E e'$. Thus the coefficient of E in $R(g \circ f)$ is

$$e_E e' - 1 = (e_E - 1) + e_E(e' - 1),$$

which is the coefficient of E in $R(f) + f^*R(g)$.

(4) Since B is a $\mathbb{Q}$-divisor and $K_Y + B$ is $\mathbb{R}$-Cartier, $K_Y + B$ is $\mathbb{Q}$-Cartier by Lemma 2.5. Let D be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y and E a prime divisor on X. Choose a positive integer l such that lD is given by a local equation ϕ near the generic point of $f(E)$. Then $\operatorname{ord}_E(\phi \circ f) = e_E \operatorname{ord}_{f(E)}(\phi)$, so f^*D defined above is equal to the pullback of D as a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. By (1),

$$K_X + B_X = f^*K_Y + R(f) + f^*B - R(f) = f^*(K_Y + B),$$

and $K_X + B_X$ is $\mathbb{Q}$-Cartier. Since f is finite and Y is quasi-projective, X is quasi-projective. Thus (X, B_X) is a sub-pair.

Let E be a prime divisor over X and w the restriction of ord_E to $\mathbb{C}(Y)$. Since ord_E has a center on X and f is finite, w has a center on Y. Since $\mathbb{C}(X)$ is a finite extension of $\mathbb{C}(Y)$, $w(\mathbb{C}(Y)^*) = e\mathbb{Z}$ for a positive integer e, and the residue field k_w has transcendence degree $\dim Y - 1$ over $\mathbb{C}$. Hence $w = e \operatorname{ord}_{E_Y}$ for a prime divisor E_Y over Y [KM98, Lemma 2.45]. Let $h_Y : Y' \to Y$ be a log resolution of (Y, B) such that E_Y is a prime divisor on Y', let X' be the normalization of Y' in $\mathbb{C}(X)$, and let $f' : X' \to Y'$ be the induced finite morphism. By the universal property of the normalization, there exists a projective birational morphism $h_X : X' \to X$ such that $f \circ h_X = h_Y \circ f'$. Let ζ be the center of ord_E on X'. Then $f'(\zeta)$ is the generic point of E_Y, and f' is finite, so ζ is the generic point of a prime divisor on X'. Thus $\mathcal{O}_{X',\zeta}$ is a discrete valuation ring dominated by the valuation ring of ord_E, so E is a prime divisor on X', and the ramification index of f' along E is e.

Write $K_{Y'} + B_{Y'} := h_Y^*(K_Y + B)$ and $K_{X'} + B_{X'} := h_X^*(K_X + B_X)$. Then

$$K_{X'} + B_{X'} = h_X^* f^*(K_Y + B) = f'^* h_Y^*(K_Y + B) = f'^*(K_{Y'} + B_{Y'}).$$

By (1) applied to f', $B_{X'} = f'^*B_{Y'} - R(f')$, so

$$\operatorname{mult}_E B_{X'} = e \operatorname{mult}_{E_Y} B_{Y'} - (e-1).$$

Since log discrepancies may be computed on any normal variety projective and birational over X on which E is a prime divisor [KM98, Lemma 2.30], we obtain

$$a(E, X, B_X) = 1 - \operatorname{mult}_E B_{X'} = e(1 - \operatorname{mult}_{E_Y} B_{Y'}) = ea(E_Y, Y, B). \qquad \square$$

2.4. **Int-amplified and polarized endomorphisms.** The following definitions are from [Men20, Section 1].

Definition 2.11. Let X be a projective variety and $f : X \to X$ a surjective endomorphism. We say that f is *int-amplified* if there exists an ample Cartier divisor H on X such that $f^*H - H$ is ample. For an integer $q > 1$, we say that f is *q-polarized* if there exists an ample Cartier divisor H on X such that $f^*H \sim qH$. We say that f is *polarized* if f is q-polarized for some integer $q > 1$.

Lemma 2.12. *Let X be a projective variety of dimension d. Then any int-amplified endomorphism of X is finite. Moreover, if g is a q-polarized endomorphism of X, then g is int-amplified, $\deg g = q^d$, and g^n is q^n-polarized for any positive integer n.*

Proof. Let f be an int-amplified endomorphism of X and H an ample Cartier divisor on X such that $f^*H - H$ is ample. Then $f^*H = (f^*H - H) + H$ is ample. Let C be a curve on X. If $f(C)$ is a point, then $f^*H|_C$ is the pullback of a Cartier divisor on a point, so $(f^*H \cdot C) = 0$, which is impossible. Thus f is quasi-finite. Since f is projective, f is finite [Har77, Exercise III.11.2].

Now let g be a q-polarized endomorphism of X and H an ample Cartier divisor on X such that $g^*H \sim qH$. Then $g^*H - H \sim (q-1)H$ is ample, so g is int-amplified. In particular, g is finite. By the projection formula [Ful98, Proposition 2.5(c)],

$$q^d(H^d) = ((g^*H)^d) = (\deg g)(H^d).$$

Since $(H^d) > 0$, $\deg g = q^d$. For any positive integer n, g^n is surjective and $(g^n)^*H = (g^*)^n H \sim q^n H$, since the pullback of Cartier divisors preserves linear equivalence. Thus g^n is q^n-polarized. $\square$

Theorem 2.13 ([MZ18, Proposition 1.1 and Lemma 3.4])**.** *Let X be a projective variety, $f : X \to X$ a surjective endomorphism, and L a big $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X such that $f^*L \equiv qL$ for some rational number $q > 1$. Then q is an integer and f is q-polarized.*

2.5. **Nef envelopes and pseudo-lc varieties.**

Definition 2.14 ([BdFF12, Sections 1.2, 1.3, 1.5 and 3.1])**.** Let X be a normal variety. An *$\mathbb{R}$-$\boldsymbol{b}$-divisor* $\mathbf{D}$ over X assigns a real number $\operatorname{mult}_E \mathbf{D}$ to any prime divisor E over X, such that for any normal variety X' with a projective birational morphism $X' \to X$, only finitely many prime divisors E on X' satisfy $\operatorname{mult}_E \mathbf{D} \neq 0$. The *trace* of $\mathbf{D}$ on such X' is $\mathbf{D}_{X'} := \sum_E (\operatorname{mult}_E \mathbf{D})E$, where E runs over the prime divisors on X'. For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on such X', $\overline{D}$ denotes the $\boldsymbol{b}$-divisor with $\operatorname{mult}_E \overline{D} := \operatorname{mult}_E D$ for any prime divisor E over X'. Let ω be a non-zero rational differential form of top degree on X such that $K_X = \operatorname{div}(\omega)$. The *canonical $\boldsymbol{b}$-divisor* $\mathbf{K}_X$ is the $\boldsymbol{b}$-divisor whose trace on any such X' is $\operatorname{div}_{X'}(\omega)$. The $\boldsymbol{b}$-divisor $\mathbf{E}_X$ is defined by $\operatorname{mult}_E \mathbf{E}_X = 1$ if E is exceptional over X and $\operatorname{mult}_E \mathbf{E}_X = 0$ otherwise.

For a finite surjective morphism $f : X' \to X$ between normal varieties and a $\boldsymbol{b}$-divisor $\mathbf{D}$ over X, the *pullback* $f^*\mathbf{D}$ is the $\boldsymbol{b}$-divisor over X' with $\operatorname{mult}_{E'} f^*\mathbf{D} := e \operatorname{mult}_E \mathbf{D}$ for any prime divisor E' over X', where E is the prime divisor over X and e is the positive integer such that the restriction of $\operatorname{ord}_{E'}$ to $\mathbb{C}(X)$ is $e \operatorname{ord}_E$. For such f, $\mathbf{R}(f)$ denotes the $\boldsymbol{b}$-divisor over X' whose trace on X' is $R(f)$ and whose coefficient at any prime divisor exceptional over X' is 0.

We recall the nef envelope of [BdFF12], see also [Yos22, Definition 2.8].

Definition 2.15 ([BdFF12, Definition 2.3 and Proposition 2.1])**.** Let X be a normal variety. For a non-zero coherent fractional ideal sheaf $\mathfrak{a}$ on X and a prime divisor E over X, let $\operatorname{mult}_E \mathfrak{a}$ be the minimum of ord_E on the stalk of $\mathfrak{a}$ at the generic point of $\operatorname{center}_X E$. We denote by $Z(\mathfrak{a})$ the $\boldsymbol{b}$-divisor over X with $\operatorname{mult}_E Z(\mathfrak{a}) := -\operatorname{mult}_E \mathfrak{a}$ for any prime divisor E over X. For a $\mathbb{Q}$-divisor D on X, the *nef envelope* of D is

$$\operatorname{Env}_X(D) := \lim_{m\to+\infty} \frac{1}{m} Z(\mathcal{O}_X(mD)),$$

where the limit is taken coefficientwise over the positive integers m such that mD is integral.

Lemma 2.16. *Let $f : X' \to X$ be a finite surjective morphism between normal varieties, and D and L $\mathbb{Q}$-divisors on X. Then the following hold.*

(1) $\operatorname{Env}_X(D+L) \geq \operatorname{Env}_X(D) + \operatorname{Env}_X(L)$, $\operatorname{Env}_X(cD) = c\operatorname{Env}_X(D)$ *for any rational number* $c > 0$, *and* $\operatorname{Env}_X(D) \leq \operatorname{Env}_X(L)$ *if* $D \leq L$.

(2) The trace of $\operatorname{Env}_X(D)$ *on X is D, and* $\operatorname{Env}_X(D) = \overline{D}$ *if D is $\mathbb{Q}$-Cartier.*

(3) $\operatorname{Env}_{X'}(f^*D) = f^*\operatorname{Env}_X(D)$.

(4) $\mathbf{K}_{X'} + \mathbf{E}_{X'} = f^*(\mathbf{K}_X + \mathbf{E}_X) + \mathbf{R}(f)$, *where $\mathbf{K}_{X'}$ is defined by $f^*\omega$, that is, $\mathbf{K}_{X'}$ has trace $f^*K_X + R(f)$ on X' (cf.* [BdFF12, Lemma 3.3]*). If $X' = X$, the $\boldsymbol{b}$-divisor defined by $f^*\omega$ differs from $\mathbf{K}_X$ in general.*

Proof. (1) The superadditivity and the homogeneity are [BdFF12, Proposition 2.6]. Assume that $D \leq L$. Then $\mathcal{O}_X(mD) \subset \mathcal{O}_X(mL)$ for any positive integer m such that mD and mL are integral, so $Z(\mathcal{O}_X(mD)) \leq Z(\mathcal{O}_X(mL))$ by [BdFF12, Lemma 1.5]. Taking the limit over the positive integers m such that mD and mL are integral, we obtain $\operatorname{Env}_X(D) \leq \operatorname{Env}_X(L)$.

(2) The first statement is [BdFF12, Proposition 2.8]. Assume that D is $\mathbb{Q}$-Cartier, and choose a positive integer m such that mD is Cartier. For any integer $k \geq 1$, $\mathcal{O}_X(kmD)$ is invertible and $Z(\mathcal{O}_X(kmD)) = km\overline{D}$. Thus, taking the limit over the integers km, we obtain $\operatorname{Env}_X(D) = \overline{D}$.

(3) This is [BdFF12, Proposition 2.19].

(4) Let E' be a prime divisor over X', E the prime divisor over X, and e the positive integer such that the restriction of $\operatorname{ord}_{E'}$ to $\mathbb{C}(X)$ is $e\operatorname{ord}_E$. Let Y be a normal variety with a projective birational morphism $Y \to X$ such that E is a prime divisor on Y, Y' the normalization of Y in $\mathbb{C}(X')$, and $g : Y' \to Y$ the induced finite surjective morphism. Since X' is the normalization of X in $\mathbb{C}(X')$, the universal property of the normalization gives a projective birational morphism $Y' \to X'$. Since g is finite and $g(\operatorname{center}_{Y'} E') = \operatorname{center}_Y E = E$, E' is a prime divisor on Y', and the ramification index of g along E' is e. The traces of $\mathbf{K}_{X'}$ and $\mathbf{K}_X$ on Y' and Y are $K_{Y'} = \operatorname{div}_{Y'}(f^*\omega)$ and $K_Y = \operatorname{div}_Y(\omega)$, so Definition-Proposition 2.10(1) for g implies that $\operatorname{mult}_{E'} \mathbf{K}_{X'} = e\operatorname{mult}_E \mathbf{K}_X + e - 1$, that is,

$$\operatorname{mult}_{E'} \mathbf{K}_{X'} + 1 = e(\operatorname{mult}_E \mathbf{K}_X + 1)$$

(cf. [BdFF12, Lemma 3.3]). Since f is finite and $f(\operatorname{center}_{X'} E') = \operatorname{center}_X E$, E' is exceptional over X' if and only if E is exceptional over X.

If E' is exceptional over X', then $\operatorname{mult}_{E'} \mathbf{E}_{X'} = \operatorname{mult}_E \mathbf{E}_X = 1$ and $\operatorname{mult}_{E'} \mathbf{R}(f) = 0$, so

$$\operatorname{mult}_{E'}(\mathbf{K}_{X'} + \mathbf{E}_{X'}) = e(\operatorname{mult}_E \mathbf{K}_X + 1) = \operatorname{mult}_{E'} f^*(\mathbf{K}_X + \mathbf{E}_X) + \operatorname{mult}_{E'} \mathbf{R}(f).$$

Otherwise, E' is a prime divisor on X', E is a prime divisor on X, $\operatorname{mult}_{E'} \mathbf{E}_{X'} = \operatorname{mult}_E \mathbf{E}_X = 0$, and e is the ramification index of f along E', so

$$\operatorname{mult}_{E'} \mathbf{K}_{X'} - e\operatorname{mult}_E \mathbf{K}_X = e - 1 = \operatorname{mult}_{E'} R(f) = \operatorname{mult}_{E'} \mathbf{R}(f).$$

Since E' is arbitrary, (4) holds. □

Definition 2.17 ([Has20, Definitions 4.1 and 4.2])**.** Let X be a normal variety and $\Delta \geq 0$ a $\mathbb{Q}$-divisor on X whose coefficients are at most one. We do not require that $K_X + \Delta$ is $\mathbb{R}$-Cartier, and write $\langle X, \Delta\rangle$ for such data. For a prime divisor E over X, let $\alpha(E, X, \Delta)$ be the supremum of $a(E, U, \Delta|_U + G)$, where U runs over the affine open subsets of X such that $U \cap \operatorname{center}_X E \neq \emptyset$, and G runs over the $\mathbb{R}$-divisors $G \geq 0$ on U such that $K_U + \Delta|_U + G$ is $\mathbb{R}$-Cartier. We say that

$\langle X, \Delta\rangle$ is *pseudo-lc* if $\alpha(E, X, \Delta) \geq 0$ for any prime divisor E over X. We say that X is *pseudo-lc* if $\langle X, 0\rangle$ is pseudo-lc.

In [Has20], $\alpha(E, X, \Delta)$ is defined with discrepancies instead of log discrepancies, so the number defined in [Has20] is one less than $\alpha(E, X, \Delta)$, and the pseudo-lc condition of [Has20] is $\alpha(E, X, \Delta) - 1 \geq -1$ for any prime divisor E over X. Pseudo-lc is weaker than the existence of a boundary G on X such that $(X, \Delta + G)$ is lc, because the boundary G in Definition 2.17 may change with E and U, and only the supremum is bounded from below. It is exactly the hypothesis of Theorem 2.18, and we obtain it for varieties with int-amplified endomorphisms in Theorem 4.1.

Theorem 2.18 ([Has20, Theorem 4.17 and Definition 2.3])**.** *Let X be a normal quasi-projective variety and $\Delta \geq 0$ a $\mathbb{Q}$-divisor on X whose coefficients are at most one. Assume that $\langle X, \Delta\rangle$ is pseudo-lc. Then there exists a small projective birational morphism $\pi : X' \to X$ from a normal variety X' such that $K_{X'} + \pi_*^{-1}\Delta$ is $\mathbb{R}$-Cartier and π-ample and $(X', \pi_*^{-1}\Delta)$ is lc.*

2.6. **Small birational morphisms.**

Lemma 2.19. *Let $\pi : X' \to X$ be a small birational morphism between normal projective varieties. Then sending B to $\pi_*^{-1}B$ is a bijection from the set of $\mathbb{Q}$-divisors $B \geq 0$ on X such that $K_X + B \sim_{\mathbb{Q}} 0$ onto the set of $\mathbb{Q}$-divisors $B' \geq 0$ on X' such that $K_{X'} + B' \sim_{\mathbb{Q}} 0$, and $K_{X'} + \pi_*^{-1}B = \pi^*(K_X + B)$ for any such B. In particular, (X, B) is lc if and only if $(X', \pi_*^{-1}B)$ is lc, and X is of Calabi–Yau type if and only if X' is of Calabi–Yau type. Moreover, $\pi_*^{-1}D = \pi^*D$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X, and $K_{X'} + \pi_*^{-1}B = \pi^*(K_X + B)$ for any $\mathbb{Q}$-divisor B on X such that $K_X + B$ is $\mathbb{Q}$-Cartier.*

Proof. Since X is normal and π is a small birational morphism, π is an isomorphism over the generic point of any prime divisor on X, and π_*^{-1} is a bijection from the set of prime divisors on X onto the set of prime divisors on X', with inverse π_*. Thus π_*^{-1} and π_* are mutually inverse bijections between the $\mathbb{Q}$-divisors on X and the $\mathbb{Q}$-divisors on X', they preserve effectivity, and $\operatorname{div}_{X'}(\varphi) = \pi_*^{-1}\operatorname{div}_X(\varphi)$ for any non-zero rational function φ on X. Moreover, $\pi_*^{-1}K_X = K_{X'}$.

Let $B \geq 0$ be a $\mathbb{Q}$-divisor on X such that $K_X + B \sim_{\mathbb{Q}} 0$. Choose a positive integer m and a non-zero rational function φ on X such that $m(K_X + B) = \operatorname{div}_X(\varphi)$. Then

$$m(K_{X'} + \pi_*^{-1}B) = \pi_*^{-1}\operatorname{div}_X(\varphi) = \operatorname{div}_{X'}(\varphi) = \pi^*\operatorname{div}_X(\varphi) = m\pi^*(K_X + B).$$

Thus $K_{X'} + \pi_*^{-1}B \sim_{\mathbb{Q}} 0$ and $K_{X'} + \pi_*^{-1}B = \pi^*(K_X + B)$. Conversely, let $B' \geq 0$ be a $\mathbb{Q}$-divisor on X' such that $K_{X'} + B' \sim_{\mathbb{Q}} 0$, and choose a positive integer m and a non-zero rational function φ such that $m(K_{X'} + B') = \operatorname{div}_{X'}(\varphi)$. Then

$$m(K_X + \pi_*B') = \pi_*\operatorname{div}_{X'}(\varphi) = \operatorname{div}_X(\varphi),$$

so $K_X + \pi_*B' \sim_{\mathbb{Q}} 0$, $\pi_*B' \geq 0$, and $B' = \pi_*^{-1}\pi_*B'$. This proves the bijection and the equality.

Let B be as in the bijection. Then $K_X + B$ and $K_{X'} + \pi_*^{-1}B$ are $\mathbb{Q}$-Cartier, since they have principal positive multiples. Since $K_{X'} + \pi_*^{-1}B = \pi^*(K_X + B)$ and $\pi_*\pi_*^{-1}B = B$, we have $a(E, X', \pi_*^{-1}B) = a(E, X, B)$ for any prime divisor E over X by [KM98, Lemma 2.30]. Thus (X, B) is lc if and only if $(X', \pi_*^{-1}B)$ is lc. By Definition 2.3 and the bijection, X is of Calabi–Yau type if and only if X' is of Calabi–Yau type.

Let D be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. The $\mathbb{Q}$-divisor $\pi^*D - \pi_*^{-1}D$ has zero pushforward to X, so it is supported on prime divisors contracted by π, and there exist no such prime divisors since π is small. Thus $\pi^*D = \pi_*^{-1}D$. Applying the last equality to $D = K_X + B$ and using $\pi_*^{-1}K_X = K_{X'}$, we obtain $K_{X'} + \pi_*^{-1}B = \pi^*(K_X + B)$. □

Lemma 2.20. *Let $\pi : X' \to X$ be a small birational morphism between normal varieties and D a Weil divisor on X. Then $\pi_*\mathcal{O}_{X'}(m\pi_*^{-1}D) = \mathcal{O}_X(mD)$ for any integer $m \geq 0$. If moreover $\pi_*^{-1}D$ is $\mathbb{Q}$-Cartier and π-ample, then the $\mathcal{O}_X$-algebra $\bigoplus_{m\geq 0}\mathcal{O}_X(mD)$ is finitely generated and $X' \cong \operatorname{Proj}_X \bigoplus_{m\geq 0}\mathcal{O}_X(mD)$ over X.*

Proof. Let U be an open subset of X and $m \geq 0$ an integer. The sections of $\pi_*\mathcal{O}_{X'}(m\pi_*^{-1}D)$ over U are 0 and the non-zero rational functions φ on X such that $\operatorname{div}(\varphi) + m\pi_*^{-1}D \geq 0$ on $\pi^{-1}(U)$, and the sections of $\mathcal{O}_X(mD)$ over U are 0 and the non-zero rational functions φ such that $\operatorname{div}(\varphi) + mD \geq 0$ on U. Since X is normal and π is small, π_* is a bijection from the set of prime divisors on $\pi^{-1}(U)$ onto the set of prime divisors on U, π is an isomorphism at the generic point of any prime divisor on $\pi^{-1}(U)$, and the coefficients of $m\pi_*^{-1}D$ and mD at corresponding prime divisors coincide. Thus φ satisfies the condition on $\pi^{-1}(U)$ if and only if φ satisfies the condition on U, and $\pi_*\mathcal{O}_{X'}(m\pi_*^{-1}D) = \mathcal{O}_X(mD)$.

If moreover $\pi_*^{-1}D$ is $\mathbb{Q}$-Cartier and π-ample, then the remaining statements are [Urb17, Lemma 2.5], see also [KM98, Lemma 6.2]. □

2.7. **Inversion of adjunction.** We refer to [FH23, Definition 1.3] and [Kol13, Definition 4.2] for the definition of the different.

Theorem 2.21 ([Kaw07, Theorem], [FH23, Corollary 1.4], [Kol13, Proposition 4.5])**.** *Let X be a normal quasi-projective variety, $\Delta \geq 0$ a $\mathbb{Q}$-divisor on X, and F a prime divisor on X. Assume that $K_X + \Delta$ is $\mathbb{Q}$-Cartier and $\operatorname{mult}_F \Delta = 1$. Let $F^\nu \to F$ be the normalization and Δ_{F^ν} the different of (X, Δ) on F^ν, so that*

$$K_{F^\nu} + \Delta_{F^\nu} = (K_X + \Delta)|_{F^\nu}.$$

Then $\Delta_{F^\nu} \geq 0$, and (X, Δ) is lc on a neighborhood of F if and only if (F^ν, Δ_{F^ν}) is lc.

3. The class group of an int-amplified variety

In this section, we prove Construction-Proposition 3.1. It turns the dynamics of an int-amplified endomorphism h of degree r into an exact $\mathbb{Q}$-linear relation in $\operatorname{Cl}(X)_\mathbb{Q}$ between K_X and the $\mathbb{Q}$-divisors $r^{-j}(h^j)_*R(h)$, with correction coefficients that tend to zero. Construction-Proposition 3.1 is the class relation behind the lc anti-canonical series (Definition 7.1) in Section 5, and it also bounds the dimension of the spaces of classes used in Section 6. We need a $\mathbb{Q}$-linear relation and not only a numerical one, because in Section 12 we take products and spans of sections attached to corrected boundaries (Definition 7.1), and any two corrected boundaries must be $\mathbb{Q}$-linearly equivalent.

We expand $-K_X$ along $r^{-1}h_*$, not along h^*. Since $K_X \sim h^*K_X + R(h)$ and $r^{-1}h_*h^*$ is the identity,

$$-K_X \sim_\mathbb{Q} \sum_{j=1}^{N} r^{-j}(h^j)_*R(h) + r^{-N}(h^N)_*(-K_X)$$

with $r^{-j}(h^j)_*R(h) \geq 0$, and Construction-Proposition 3.1 with $D = -K_X$ and $\Theta = R(h)$ turns the remainder into a combination of the first ℓ terms with coefficients tending to 0, as $r^{-1}h_* = (h^*)^{-1}$ has an annihilating polynomial with all roots in $\{|\lambda| < 1\}$. Along h^*,

$$-K_X \sim (h^N)^*(-K_X) - \sum_{j=0}^{N-1} (h^j)^*R(h)$$

with $-(h^j)^*R(h) \leq 0$, which does not converge, since h^* is expanding, with eigenvalues 2 and 3 in Example 3.2.

Construction-Proposition 3.1. Let X be a normal projective variety of dimension $d \geq 1$ and $h : X \to X$ an int-amplified endomorphism of degree r. Then there exists a monic polynomial $p(t) \in \mathbb{Q}[t]$ of degree $\ell \geq 1$ such that $p(0) \neq 0$, any complex root λ of $p(t)$ satisfies $|\lambda| < 1$, and $p(r^{-1}h_*) = 0$ on $\operatorname{Cl}(X)_\mathbb{Q}$. In particular, h^* is the inverse of $r^{-1}h_*$ on $\operatorname{Cl}(X)_\mathbb{Q}$. Moreover, there exist rational numbers $a_{j,N}$ for $1 \leq j \leq \ell$ and $N \geq \ell$, depending only on h, such that

$\lim_{N\to+\infty} a_{j,N} = 0$ for any j and

$$D \sim_{\mathbb{Q}} \sum_{j=1}^{N} r^{-j}(h^j)_*\Theta + \sum_{j=1}^{\ell} a_{j,N} r^{-j}(h^j)_*\Theta$$

for any integer $N \geq \ell$ and any $\mathbb{Q}$-divisors D and Θ on X such that $h^*D - D \sim_{\mathbb{Q}} \Theta$. In particular, for $D_j := r^{-j}(h^j)_*R(h)$, we have $K_X + \sum_{j=1}^{N} D_j + \sum_{j=1}^{\ell} a_{j,N} D_j \sim_{\mathbb{Q}} 0$ for any $N \geq \ell$.

By [Men20, Theorem 1.5], $-K_X$ is weakly numerically equivalent [MZ18, Definition 2.2] to an effective $\mathbb{Q}$-divisor, and if X is $\mathbb{Q}$-Gorenstein, then $-K_X$ is $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor by [MZ22, Theorem 1.5]. Construction-Proposition 3.1 is an exact $\mathbb{Q}$-linear relation in $\mathrm{Cl}(X)_{\mathbb{Q}}$, and X is not assumed to be $\mathbb{Q}$-Gorenstein.

Example 3.2. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, and let h and the prime divisors P_i, P_i' $(i = 0, 1)$ be as in Remark 1.7. Then $r = 6$, and by the formula of Remark 1.7 with $n = 1$,

$$R(h) = P_0 + P_1 + 2(P_0' + P_1').$$

Since h maps P_i onto P_i with degree 3 and P_i' onto P_i' with degree 2, we have $h_*P_i = 3P_i$ and $h_*P_i' = 2P_i'$. Thus

$$D_j = 2^{-j}(P_0 + P_1) + 2 \cdot 3^{-j}(P_0' + P_1')$$

for any $j \geq 1$. The $\mathbb{Q}$-vector space $\mathrm{Cl}(X)_{\mathbb{Q}}$ is spanned by $[P_0]$ and $[P_0']$, and

$$r^{-1}h_*[P_0] = \tfrac{1}{2}[P_0], \qquad r^{-1}h_*[P_0'] = \tfrac{1}{3}[P_0'].$$

Thus we may take $p(t) = (t - \frac{1}{2})(t - \frac{1}{3})$ and $\ell = 2$. Moreover,

$$\sum_{j=1}^{+\infty} D_j = P_0 + P_1 + P_0' + P_1' \sim -K_X.$$

We first show that the pushforward under a finite morphism preserves the $\mathbb{Q}$-Cartier property and nefness of $\mathbb{Q}$-divisors.

Lemma 3.3. *Let $f : X \to Y$ be a finite surjective morphism between normal varieties and D a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Then the following hold.*

(1) f_*D is $\mathbb{Q}$-Cartier, and $f_*D \geq 0$ if $D \geq 0$.

(2) For any prime divisor E over Y,

$$\mathrm{mult}_E f_*D = \sum_i [\mathbb{C}(E_i) : \mathbb{C}(E)] \,\mathrm{mult}_{E_i} D,$$

where E_i runs over the prime divisors over X such that the restriction of ord_{E_i} to $\mathbb{C}(Y)$ is a positive multiple of ord_E.

(3) If X and Y are projective and D is nef, then f_*D is nef. In particular, f_* induces a linear map $N^1(X)_{\mathbb{R}} \to N^1(Y)_{\mathbb{R}}$ that maps nef classes to nef classes.

Proof. For $\phi \in \mathbb{C}(X)^*$, the norm $\mathrm{N}_{\mathbb{C}(X)/\mathbb{C}(Y)}(\phi) \in \mathbb{C}(Y)^*$ is the determinant of the multiplication by ϕ on $\mathbb{C}(X)$, regarded as a $\mathbb{C}(Y)$-linear map. By [Ful98, Proposition 1.4(b)],

$$f_* \,\mathrm{div}(\phi) = \mathrm{div}\big(\mathrm{N}_{\mathbb{C}(X)/\mathbb{C}(Y)}(\phi)\big), \tag{3.1}$$

and (3.1) also holds for any finite surjective morphism between normal varieties whose function fields are $\mathbb{C}(X)$ and $\mathbb{C}(Y)$.

Fix a positive integer l such that lD is Cartier. For any point $y \in Y$, there exist an affine open neighborhood U of y and $\phi \in \mathbb{C}(X)^*$ such that ϕ is a local equation of lD on $f^{-1}(U)$, since for an affine open neighborhood U of y, $f^{-1}(U)$ is affine, the localization at y of the coordinate ring of $f^{-1}(U)$ is a semilocal ring, and an invertible module over a semilocal ring is free [Bou89, Chapter II, §5, no. 3, Proposition 5]. By (3.1), $\mathrm{N}_{\mathbb{C}(X)/\mathbb{C}(Y)}(\phi)$ is a local equation of lf_*D on U.

(1) Since the neighborhoods U of the points $y \in Y$ cover Y, lf_*D is Cartier, so f_*D is $\mathbb{Q}$-Cartier. If $D \geq 0$, then $f_*D \geq 0$ by the definition of the pushforward of cycles.

(2) Let $Y' \to Y$ be a projective birational morphism from a normal variety Y' such that E is a prime divisor on Y', and let $f' : X' \to Y'$ be the normalization of Y' in $\mathbb{C}(X)$. Let $\mathcal{O}'$ be the integral closure of $\mathcal{O}_{Y',E}$ in $\mathbb{C}(X)$. The valuation rings of $\mathbb{C}(X)$ dominating $\mathcal{O}_{Y',E}$ are the localizations of $\mathcal{O}'$ at the maximal ideals of $\mathcal{O}'$. Thus the E_i in (2) are the prime divisors on X' such that $f'(E_i) = E$, and $f'_*E_i = [\mathbb{C}(E_i) : \mathbb{C}(E)]E$.

Let y be the generic point of $\operatorname{center}_Y E$, let U be an affine open neighborhood of y, and let $\phi \in \mathbb{C}(X)^*$ be a local equation of lD on $f^{-1}(U)$. Since $\mathrm{N}_{\mathbb{C}(X)/\mathbb{C}(Y)}(\phi)$ is a local equation of lf_*D near y, Definition 2.6 implies that $\operatorname{mult}_E f_*D = l^{-1} \operatorname{ord}_E(\mathrm{N}_{\mathbb{C}(X)/\mathbb{C}(Y)}(\phi))$. Since the restriction of ord_{E_i} to $\mathbb{C}(Y)$ is a positive multiple of ord_E, f maps $\operatorname{center}_X E_i$ onto $\operatorname{center}_Y E$, so the generic point of $\operatorname{center}_X E_i$ belongs to $f^{-1}(U)$. Thus $\operatorname{mult}_{E_i} D = l^{-1} \operatorname{ord}_{E_i}(\phi)$. Applying (3.1) to f' and comparing the coefficients of E, we obtain

$$\operatorname{ord}_E\big(\mathrm{N}_{\mathbb{C}(X)/\mathbb{C}(Y)}(\phi)\big) = \sum_i [\mathbb{C}(E_i) : \mathbb{C}(E)] \operatorname{ord}_{E_i}(\phi).$$

Dividing by l proves (2).

(3) Let L be a finite Galois extension of $\mathbb{C}(Y)$ containing $\mathbb{C}(X)$, $G := \operatorname{Gal}(L/\mathbb{C}(Y))$, Z the normalization of X in L, $g : Z \to X$ the induced finite surjective morphism, and $q := f \circ g$. Since X is the normalization of Y in $\mathbb{C}(X)$, Z is also the normalization of Y in L, so G acts on Z by automorphisms over Y. With the pullback q^* of Definition-Proposition 2.10, for any prime divisor S on Z,

$$q^*q_*S = \sum_{\sigma \in G} \sigma^* S. \tag{3.2}$$

This holds since G acts transitively on the prime divisors on Z over $q(S)$, and the stabilizer of S has order $e \cdot [\mathbb{C}(S) : \mathbb{C}(q(S))]$, where e is the ramification index of q along S. By linearity, (3.2) holds for any $\mathbb{Q}$-divisor on Z.

For a Cartier divisor on a normal variety, the pullback of Definition-Proposition 2.10 under a finite surjective morphism is the usual pullback, since both are determined by the orders of local equations at codimension one points. By Definition-Proposition 2.10(2), $q_*g^*D = f_*g_*g^*D = (\deg g) f_*D$. Thus (3.2) applied to g^*D implies

$$(\deg g) q^*(f_*D) = \sum_{\sigma \in G} \sigma^* g^* D, \tag{3.3}$$

where f_*D is $\mathbb{Q}$-Cartier by (1). Both sides of (3.3) are linear in D, so (3.3) holds for any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X, and f_*D is $\mathbb{R}$-Cartier by (1).

Assume that D is nef. Since σ is an automorphism and nefness is preserved under pullback [Laz04, Example 1.4.4(i)], the right-hand side of (3.3) is nef. Thus $q^*(f_*D)$ is nef. Since nefness descends along the surjective morphism q [Laz04, Example 1.4.4(ii)], f_*D is nef. If D is numerically trivial, then D and $-D$ are nef, so f_*D and $-f_*D$ are nef, and f_*D is numerically trivial. Thus f_* induces a linear map $N^1(X)_{\mathbb{R}} \to N^1(Y)_{\mathbb{R}}$ that maps nef classes to nef classes. □

The next lemma is where the int-amplified hypothesis enters quantitatively: for an ample Cartier divisor H on X with $h^*H - H$ ample, $c^{-k}H - r^{-k}(h^k)_*H$ is nef for a rational number $c > 1$, and hence $v(r^{-k}(h^k)_*D)$ tends to 0 at any divisorial valuation v for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} aH$ for a rational number $a > 0$. We use it three times: in the proof of Construction-Proposition 3.1, in Theorem 4.1, and for the lc property of the partial sums in Construction-Proposition 5.2.

Lemma 3.4. *Let X be a normal projective variety of dimension $d \geq 1$, $h : X \to X$ an int-amplified endomorphism of degree r, and H an ample Cartier divisor on X such that $h^*H - H$ is ample. Then there exists a rational number $c > 1$ such that $r^{-k}(h^k)_*H$ and $c^{-k}H - r^{-k}(h^k)_*H$ are nef for any integer $k \geq 0$. Moreover, for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ on X such that*

$D \sim_{\mathbb{Q}} aH$ for some rational number $a > 0$, and for any divisorial valuation v of $\mathbb{C}(X)$, we have $\lim_{k\to+\infty} v(r^{-k}(h^k)_*D) = 0$.

Proof. By Lemma 2.12, h is finite, so we may use Lemma 3.3 for h^k for any $k \geq 1$, and $(h^k)_* = (h_*)^k$ on divisors. Since $h^*H - H$ is ample and the ample cone is open [Laz04, Theorem 1.4.23(ii)], there exists a rational number $c > 1$ such that $h^*H - cH = (h^*H - H) - (c-1)H$ is ample. Since $h_*h^*H = rH$ by Definition-Proposition 2.10(2), Lemma 3.3(3) applied to $h^*H - cH$ shows that

$$r^{-1}h_*(h^*H - cH) = H - cr^{-1}h_*H$$

is nef. Thus $c^{-1}H - r^{-1}h_*H$ is nef.

We prove the first statement by induction on k. For $k = 0$, H and 0 are nef. Assume that $r^{-k}(h^k)_*H$ and $c^{-k}H - r^{-k}(h^k)_*H$ are nef. Then $r^{-(k+1)}(h^{k+1})_*H = r^{-1}h_*(r^{-k}(h^k)_*H)$ is nef by Lemma 3.3(3), and

$$c^{-(k+1)}H - r^{-(k+1)}(h^{k+1})_*H = r^{-1}h_*\big(c^{-k}H - r^{-k}(h^k)_*H\big) + c^{-k}\big(c^{-1}H - r^{-1}h_*H\big)$$

is nef by Lemma 3.3(3).

We prove the second statement. By Definition 2.1, $v = t\operatorname{ord}_E$ for a real number $t > 0$ and a prime divisor E over X. Thus $v(D') = t\operatorname{mult}_E D'$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D' on X (Definition 2.6). By Lemma 3.3(1), $r^{-k}(h^k)_*D \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, and $r^{-k}(h^k)_*D \sim_{\mathbb{Q}} ar^{-k}(h^k)_*H$ by Definition-Proposition 2.10(2). Let $g : W \to X$ be a birational morphism from a smooth projective variety W such that E is a prime divisor on W, and let A be an ample Cartier divisor on W. Since $g^*(r^{-k}(h^k)_*D) - \operatorname{mult}_E(r^{-k}(h^k)_*D)E \geq 0$, and since $g^*(c^{-k}H - r^{-k}(h^k)_*H)$ is nef [Laz04, Example 1.4.4(i)] and nef classes have non-negative intersection with A^{d-1} [Laz04, Example 1.4.16],

$$\begin{aligned} 0 \leq \operatorname{mult}_E\big(r^{-k}(h^k)_*D\big)(E \cdot A^{d-1}) &\leq \big(g^*r^{-k}(h^k)_*D \cdot A^{d-1}\big) \\ &= a\big(g^*r^{-k}(h^k)_*H \cdot A^{d-1}\big) \leq ac^{-k}(g^*H \cdot A^{d-1}). \end{aligned}$$

Since $(E \cdot A^{d-1}) > 0$ does not depend on k and $c > 1$, we obtain $\lim_{k\to+\infty} \operatorname{mult}_E(r^{-k}(h^k)_*D) = 0$, and hence $\lim_{k\to+\infty} v(r^{-k}(h^k)_*D) = 0$. □

Proof of Construction-Proposition 3.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix the ample divisor H and the constant c, and we show that $r^{-1}h_* \circ h^* = \mathrm{id}$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$.

By Lemma 2.12, h is finite. By Definition-Proposition 2.10(2), h^* and h_* preserve $\mathbb{Q}$-linear equivalence, so they induce endomorphisms of $\mathrm{Cl}(X)_{\mathbb{Q}}$, and $h_*h^* = r \cdot \mathrm{id}$. Thus

$$r^{-1}h_* \circ h^* = \mathrm{id} \quad \text{on } \mathrm{Cl}(X)_{\mathbb{Q}}.$$

In particular, h^* is injective on $\mathrm{Cl}(X)_{\mathbb{Q}}$. Since $(h^k)_* = (h_*)^k$, we have $(r^{-1}h_*)^k = r^{-k}(h^k)_*$ for any $k \geq 0$. By Definition 2.11, there exists an ample Cartier divisor H on X such that $h^*H - H$ is ample. By Lemma 3.4, there exists a rational number $c > 1$ such that $c^{-k}H - r^{-k}(h^k)_*H$ is nef for any integer $k \geq 0$. We fix such a number c.

Step 2. In this step, we construct a subspace $\mathrm{Cl}^0(X)_{\mathbb{Q}}$ of $\mathrm{Cl}(X)_{\mathbb{Q}}$ coming from the Picard variety of a resolution, and a polynomial $p_1(t)$ that annihilates $r^{-1}h_*$ on $\mathrm{Cl}^0(X)_{\mathbb{Q}}$.

Let $g : Y \to X$ be a resolution with Y smooth projective, and let F_i be the g-exceptional prime divisors. Since Y is smooth, $\mathrm{Cl}(Y)_{\mathbb{Q}} = \mathrm{Pic}(Y)_{\mathbb{Q}} := \mathrm{Pic}(Y) \otimes_{\mathbb{Z}} \mathbb{Q}$. Let $\mathrm{Pic}^0(Y)_{\mathbb{Q}} := \mathrm{Pic}^0(Y) \otimes_{\mathbb{Z}} \mathbb{Q} \subset \mathrm{Pic}(Y)_{\mathbb{Q}}$ and

$$\mathrm{Cl}^0(X)_{\mathbb{Q}} := g_*\,\mathrm{Pic}^0(Y)_{\mathbb{Q}} \subset \mathrm{Cl}(X)_{\mathbb{Q}}.$$

The pushforward $g_* : \mathrm{Cl}(Y)_{\mathbb{Q}} \to \mathrm{Cl}(X)_{\mathbb{Q}}$ is surjective with kernel spanned by the classes $[F_i]$. Since X is normal, g is an isomorphism over an open subset $X^\circ \subset X$ with $\operatorname{codim}(X\backslash X^\circ, X) \geq 2$. Surjectivity follows from [Har77, Proposition II.6.5(a)] for Y and the open subset $g^{-1}(X^\circ)$, together with [Har77, Proposition II.6.5(b)] for $X\backslash X^\circ$. The kernel is spanned by the $[F_i]$ by

[Har77, Proposition II.6.5(c)] for each F_i in turn and [Har77, Proposition II.6.5(b)] for the union of the components of $Y\backslash g^{-1}(X^\circ)$ of codimension at least 2. Any element of $\mathrm{Pic}^0(Y)$ is numerically trivial, and a positive power of any numerically trivial line bundle on Y belongs to $\mathrm{Pic}^0(Y)$ by [Kle05, Theorem 9.6.3]. Thus the kernel of $\mathrm{Pic}(Y)_\mathbb{Q} \to N^1(Y)_\mathbb{Q}$ is $\mathrm{Pic}^0(Y)_\mathbb{Q}$, and g_* induces an isomorphism

$$N^1(Y)_\mathbb{Q}/\mathrm{Span}_\mathbb{Q}\{[F_i]\} \cong \mathrm{Cl}(X)_\mathbb{Q}/\mathrm{Cl}^0(X)_\mathbb{Q}. \tag{3.4}$$

In particular, $\dim_\mathbb{Q} \mathrm{Cl}(X)_\mathbb{Q}/\mathrm{Cl}^0(X)_\mathbb{Q} < +\infty$.

Resolving the indeterminacy of the rational map $g^{-1} \circ h \circ g : Y \dashrightarrow Y$, we obtain a smooth projective variety W, a birational morphism $g_1 : W \to Y$, and a generically finite surjective morphism $g_2 : W \to Y$ such that $g \circ g_2 = h \circ g \circ g_1$. By [BL04, Theorem 4.9.4], g_1 induces an isomorphism $\mathrm{Alb}(W) \to \mathrm{Alb}(Y)$, so $g_1^* : \mathrm{Pic}^0(Y) \to \mathrm{Pic}^0(W)$ is an isomorphism by the duality between Albanese and Picard varieties [BL04, Section 11.11]. In this proof, Φ denotes the group endomorphism $(g_1^*)^{-1} \circ g_2^*$ of $\mathrm{Pic}^0(Y)$, and also the induced endomorphism of $\mathrm{Pic}^0(Y)_\mathbb{Q}$. We show that

$$h^*[g_*M] = g_*\Phi[M] \quad \text{in } \mathrm{Cl}(X)_\mathbb{Q} \text{ for any } \mathbb{Q}\text{-divisor } M \text{ on } Y \text{ with } [M] \in \mathrm{Pic}^0(Y)_\mathbb{Q}. \tag{3.5}$$

Let M be such a $\mathbb{Q}$-divisor, and M' a $\mathbb{Q}$-divisor on Y with $[M'] = \Phi[M]$. Then $g_2^*M \sim_\mathbb{Q} g_1^*M'$. Since proper pushforward preserves $\mathbb{Q}$-linear equivalence [Ful98, Proposition 1.4(b)], $(g \circ g_1)_* g_2^* M \sim_\mathbb{Q} g_* M'$. Moreover, $(g \circ g_1)_* g_2^* M = h^* g_* M$, since for any prime divisor P on X, $g \circ g_1$ is an isomorphism over the generic point of P, g is an isomorphism over the generic point of the prime divisor $h(P)$, and $g_2 = g^{-1} \circ h \circ (g \circ g_1)$ near the generic point of the strict transform of P on W. Thus the coefficient of the strict transform of P in g_2^*M is the coefficient of P in h^*g_*M, and the $(g \circ g_1)$-exceptional components of g_2^*M have zero pushforward. This proves (3.5).

Let $\alpha : Y \to \mathrm{Alb}(Y)$ be an Albanese morphism. If $\dim \mathrm{Alb}(Y) = 0$, then $\mathrm{Pic}^0(Y) = 0$ [BL04, Section 11.11], so $\mathrm{Cl}^0(X)_\mathbb{Q} = 0$, and we let $p_1(t) := 1$. Assume that $\dim \mathrm{Alb}(Y) > 0$. Since any rational map from Y to an abelian variety is a morphism [BL04, Theorem 4.9.4], any rational map from X to an abelian variety factors uniquely through $\alpha \circ g^{-1} : X \dashrightarrow \mathrm{Alb}(Y)$. Thus $\alpha \circ g^{-1}$ is the birational Albanese map of X [Men20, Definition 7.1]. By [Men20, Theorem 1.8(2)], there exists an int-amplified endomorphism γ of $\mathrm{Alb}(Y)$ such that $\alpha \circ g^{-1} \circ h = \gamma \circ \alpha \circ g^{-1}$ as rational maps. Composing with $g \circ g_1$ and using $g \circ g_2 = h \circ g \circ g_1$, we obtain $\alpha \circ g_2 = \gamma \circ \alpha \circ g_1$.

Write $\gamma = t_x \circ \psi$, where ψ is a group endomorphism of $\mathrm{Alb}(Y)$ and t_x is the translation by a point $x \in \mathrm{Alb}(Y)$. By Definition 2.11, there exists an ample Cartier divisor J on $\mathrm{Alb}(Y)$ such that $\gamma^*J - J$ is ample. Translations act trivially on $N^1(\mathrm{Alb}(Y))$ and on $\mathrm{Pic}^0(\mathrm{Alb}(Y))$ [BL04, Section 2.3]. Thus $\psi^*J - J \equiv \gamma^*J - J$, so $\psi^*J - J$ is ample [Laz04, Proposition 1.3.13]. For any $N' \in \mathrm{Pic}^0(\mathrm{Alb}(Y))$,

$$g_2^*\alpha^*N' = g_1^*\alpha^*\gamma^*N' = g_1^*\alpha^*\psi^*N',$$

so $\Phi \circ \alpha^* = \alpha^* \circ \psi^*$ on $\mathrm{Pic}^0(\mathrm{Alb}(Y))$. By [BL04, Section 11.11], $\alpha^* : \mathrm{Pic}^0(\mathrm{Alb}(Y)) \to \mathrm{Pic}^0(Y)$ is an isomorphism, and ψ^* on $\mathrm{Pic}^0(\mathrm{Alb}(Y))$ is the dual of ψ.

By [BL04, Theorem 2.2.3 and Proposition 4.5.2], J corresponds to a positive definite Hermitian form $\langle \cdot, \cdot \rangle_J$ on the tangent space $T_0 \mathrm{Alb}(Y)$, and by [BL04, Section 2.3], $\psi^*J - J$ corresponds to the Hermitian form $\langle d\psi(\cdot), d\psi(\cdot) \rangle_J - \langle \cdot, \cdot \rangle_J$, where $d\psi$ is the differential of ψ at 0. Since $\psi^*J - J$ is ample, $\langle d\psi(\cdot), d\psi(\cdot) \rangle_J - \langle \cdot, \cdot \rangle_J$ is positive definite. Thus for any eigenvector w of $d\psi$ with eigenvalue λ,

$$(|\lambda|^2 - 1)\langle w, w \rangle_J = \langle d\psi(w), d\psi(w) \rangle_J - \langle w, w \rangle_J > 0,$$

so $|\lambda| > 1$. Let $\chi(t) \in \mathbb{Z}[t]$ be the characteristic polynomial of the action of ψ on $H_1(\mathrm{Alb}(Y), \mathbb{Z})$. By [BL04, Section 1.2], the complex roots of $\chi(t)$ are the eigenvalues of $d\psi$ and their complex conjugates. Thus any complex root λ of $\chi(t)$ satisfies $|\lambda| > 1$, and in particular $\chi(0) \neq 0$. Since the action of $\mathrm{End}(\mathrm{Alb}(Y))$ on $H_1(\mathrm{Alb}(Y), \mathbb{Z})$ is faithful [BL04, Section 1.2], the Cayley–Hamilton theorem implies $\chi(\psi) = 0$ in $\mathrm{End}(\mathrm{Alb}(Y))$. Taking duals, $\chi(\psi^*) = 0$ on $\mathrm{Pic}^0(\mathrm{Alb}(Y))$, hence $\chi(\Phi) = 0$ on $\mathrm{Pic}^0(Y)$.

Since $\chi(\Phi)=0$ and $\chi(0)\neq 0$, Φ is bijective on $\mathrm{Pic}^0(Y)_{\mathbb{Q}}$. By (3.5),

$$h^*\,\mathrm{Cl}^0(X)_{\mathbb{Q}} = g_*\Phi\,\mathrm{Pic}^0(Y)_{\mathbb{Q}} = \mathrm{Cl}^0(X)_{\mathbb{Q}},$$

and $\chi(h^*)g_* = g_*\chi(\Phi) = 0$ on $\mathrm{Pic}^0(Y)_{\mathbb{Q}}$, so $\chi(h^*)=0$ on $\mathrm{Cl}^0(X)_{\mathbb{Q}}$. Since h^* is injective (**Step 1**), h^* is bijective on $\mathrm{Cl}^0(X)_{\mathbb{Q}}$. For any $z\in\mathrm{Cl}^0(X)_{\mathbb{Q}}$, write $z=h^*z'$ with $z'\in\mathrm{Cl}^0(X)_{\mathbb{Q}}$. Then $r^{-1}h_*z=z'$ by **Step 1**. Thus $r^{-1}h_*$ maps $\mathrm{Cl}^0(X)_{\mathbb{Q}}$ into itself and is the inverse of h^* on $\mathrm{Cl}^0(X)_{\mathbb{Q}}$. Let $n_1:=\deg\chi(t)$ and

$$p_1(t) := \chi(0)^{-1}t^{n_1}\chi(t^{-1})\in\mathbb{Q}[t].$$

Then $p_1(t)$ is monic, $p_1(0)=\chi(0)^{-1}\neq 0$, any complex root of $p_1(t)$ is λ^{-1} for a complex root λ of $\chi(t)$, hence has modulus less than 1, and

$$p_1(r^{-1}h_*) = \chi(0)^{-1}(r^{-1}h_*)^{n_1}\chi(h^*) = 0 \quad \text{on } \mathrm{Cl}^0(X)_{\mathbb{Q}}.$$

Step 3. In this step, we construct a polynomial $p_2(t)$ that annihilates the endomorphism of $\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}}$ induced by $r^{-1}h_*$, and we show that any complex root of $p_2(t)$ has modulus less than 1.

By **Step 2**, h^* and $r^{-1}h_*$ map $\mathrm{Cl}^0(X)_{\mathbb{Q}}$ into itself, so they induce endomorphisms of $\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}}$, and the composite $r^{-1}h_*\circ h^*$ induces the identity by **Step 1**. Since $\dim_{\mathbb{Q}}\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}}<+\infty$ (**Step 2**), the endomorphism induced by $r^{-1}h_*$ is bijective. Let $p_2(t)\in\mathbb{Q}[t]$ be the characteristic polynomial of the endomorphism of $\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}}$ induced by $r^{-1}h_*$. Then $p_2(t)$ is monic, $p_2(0)\neq 0$, and $p_2(r^{-1}h_*)$ maps $\mathrm{Cl}(X)_{\mathbb{Q}}$ into $\mathrm{Cl}^0(X)_{\mathbb{Q}}$ by the Cayley–Hamilton theorem.

Recall that H and $c>1$ are fixed in **Step 1**. Let $\Gamma\geq 0$ be a $\mathbb{Q}$-divisor on X, and fix a positive integer m such that $m\Gamma$ is a divisor. By [Har77, Theorems II.7.6 and II.5.17], there exist a positive integer b and a non-zero global section u of $\mathcal{O}_X(bH-m\Gamma)$. Regarding u as a rational function, $\mathrm{div}(u)+bH-m\Gamma\geq 0$. Let $L:=m^{-1}(\mathrm{div}(u)+bH)$. Then L is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor such that $L\geq\Gamma\geq 0$ and $L\sim_{\mathbb{Q}}\frac{b}{m}H$. For any integer $k\geq 0$, $r^{-k}(h^k)_*L\geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor by Lemma 3.3(1),

$$r^{-k}(h^k)_*L\geq r^{-k}(h^k)_*\Gamma, \qquad r^{-k}(h^k)_*L\sim_{\mathbb{Q}}\tfrac{b}{m}r^{-k}(h^k)_*H.$$

Let $\Gamma_k:=g_*^{-1}(r^{-k}(h^k)_*\Gamma)$ be the strict transform on Y. Since Γ_k has no g-exceptional component and $g^*(r^{-k}(h^k)_*L)\geq 0$, we have $0\leq\Gamma_k\leq g^*(r^{-k}(h^k)_*L)$. Fix an ample Cartier divisor A on Y. Since $g^*(c^{-k}H-r^{-k}(h^k)_*H)$ is nef [Laz04, Example 1.4.4(i)], [Laz04, Example 1.4.16] implies

$$\begin{aligned}0\leq(\Gamma_k\cdot A^{d-1}) &\leq \left(g^*r^{-k}(h^k)_*L\cdot A^{d-1}\right)\\ &= \tfrac{b}{m}\left(g^*r^{-k}(h^k)_*H\cdot A^{d-1}\right)\leq\tfrac{b}{m}c^{-k}(g^*H\cdot A^{d-1}).\end{aligned}\tag{3.6}$$

Next, we construct a norm $\|\cdot\|$ on $N^1(Y)_{\mathbb{R}}$ and a real number $\kappa>0$ such that $\|[\Gamma']\|\leq\kappa(\Gamma'\cdot A^{d-1})$ for any $\mathbb{Q}$-divisor $\Gamma'\geq 0$ on Y. If $d=1$, then $N^1(Y)_{\mathbb{R}}=\mathbb{R}$ by the degree, and we take the absolute value of the degree and $\kappa=1$. Assume that $d\geq 2$. Fix Cartier divisors $M_1,\dots,M_s$ on Y whose classes form a basis of $N^1(Y)_{\mathbb{R}}$, and a rational number $\epsilon>0$ such that $A+\epsilon M_i$ and $A-\epsilon M_i$ are ample for any i. For a $\mathbb{Q}$-divisor $\Gamma'\geq 0$ on Y, $(\Gamma'\cdot A^{d-2}\cdot(A\pm\epsilon M_i))\geq 0$, so

$$|(\Gamma'\cdot A^{d-2}\cdot M_i)|\leq\epsilon^{-1}(\Gamma'\cdot A^{d-1}).$$

By the Hodge index theorem [Voi02, Theorem 6.32], the bilinear form $(U,U')\to(U\cdot U'\cdot A^{d-2})$ on $N^1(Y)_{\mathbb{R}}$ is non-degenerate. Thus $\|U\|:=\max_i|(U\cdot M_i\cdot A^{d-2})|$ is a norm on $N^1(Y)_{\mathbb{R}}$, and we take $\kappa=\epsilon^{-1}$. By (3.6),

$$\|[\Gamma_k]\|\leq\kappa\tfrac{b}{m}c^{-k}(g^*H\cdot A^{d-1}) \quad \text{for any } k\geq 0.$$

Since $g_*\Gamma_k=r^{-k}(h^k)_*\Gamma$, the isomorphism (3.4) maps the class of $[\Gamma_k]$ to the class of $(r^{-1}h_*)^k[\Gamma]$. Fix a norm on $(\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}})\otimes_{\mathbb{Q}}\mathbb{C}$. The linear map $N^1(Y)_{\mathbb{R}}\to(\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}})\otimes_{\mathbb{Q}}\mathbb{R}$ induced by (3.4) is continuous, so there exists a real number C_Γ such that the class of $(r^{-1}h_*)^k[\Gamma]$ has norm at most $C_\Gamma c^{-k}$ for any $k\geq 0$. Since any $\mathbb{Q}$-divisor on X is a difference $\Gamma_1-\Gamma_2$ of two

$\mathbb{Q}$-divisors $\Gamma_1, \Gamma_2 \geq 0$, for any $z \in (\mathrm{Cl}(X)_{\mathbb{Q}}/\mathrm{Cl}^0(X)_{\mathbb{Q}}) \otimes_{\mathbb{Q}} \mathbb{C}$ there exists a real number C_z such that the image of z under the k-th power of the endomorphism induced by $r^{-1}h_*$ has norm at most $C_z c^{-k}$ for any $k \geq 0$. If $z \neq 0$ is an eigenvector with eigenvalue λ, then $|\lambda|^k \|z\| \leq C_z c^{-k}$ for any $k \geq 0$, so $|\lambda| \leq c^{-1} < 1$. Thus any complex root λ of $p_2(t)$ satisfies $|\lambda| < 1$.

Finally, we show that $\deg p_2(t) \geq 1$. By (3.4), it suffices to show that $[g^*H] \notin \mathrm{Span}_{\mathbb{Q}}\{[F_i]\}$ in $N^1(Y)_{\mathbb{Q}}$, since $g_* g^* H = H$. If $d = 1$, then g is an isomorphism, so the set of F_i is empty, and $\deg H > 0$. If $d \geq 2$, then $(F_i \cdot (g^*H)^{d-1}) = 0$ for any i by the projection formula, since $\dim g(F_i) \leq d-2$, whereas $(g^*H \cdot (g^*H)^{d-1}) = (H^d) > 0$.

Step 4. In this step, we construct $p(t)$ and show that h^* is the inverse of $r^{-1}h_*$.

Let $p(t) := p_1(t)p_2(t)$ and $\ell := \deg p(t)$. By **Step 3**, $p_2(r^{-1}h_*)$ maps $\mathrm{Cl}(X)_{\mathbb{Q}}$ into $\mathrm{Cl}^0(X)_{\mathbb{Q}}$, and by **Step 2**, $p_1(r^{-1}h_*) = 0$ on $\mathrm{Cl}^0(X)_{\mathbb{Q}}$. Thus

$$p(r^{-1}h_*) = p_1(r^{-1}h_*) \circ p_2(r^{-1}h_*) = 0 \quad \text{on } \mathrm{Cl}(X)_{\mathbb{Q}}.$$

Moreover, $p(t) \in \mathbb{Q}[t]$ is monic, $p(0) = p_1(0)p_2(0) \neq 0$, any complex root λ of $p(t)$ satisfies $|\lambda| < 1$, and $\ell \geq \deg p_2(t) \geq 1$. Since $p(0) \neq 0$, the polynomials t and $p(t)$ are coprime, so there exists $e_2(t) \in \mathbb{Q}[t]$ such that $te_2(t) \equiv 1 \bmod p(t)$. Thus

$$r^{-1}h_* \circ e_2(r^{-1}h_*) = e_2(r^{-1}h_*) \circ r^{-1}h_* = \mathrm{id} \quad \text{on } \mathrm{Cl}(X)_{\mathbb{Q}}.$$

Since $r^{-1}h_* \circ h^* = \mathrm{id}$ by **Step 1**, $h^* = e_2(r^{-1}h_*) \circ r^{-1}h_* \circ h^* = e_2(r^{-1}h_*)$, which is the inverse of $r^{-1}h_*$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$.

Step 5. In this step, we construct the rational numbers $a_{j,N}$ and prove the relation for D and Θ.

Recall that $p(t)$ is the polynomial of degree ℓ constructed in **Step 4**. Since any complex root of $p(t)$ has modulus less than 1, $p(1) \neq 0$. Thus $1 - t$ is a unit in $\mathbb{Q}[t]/(p(t))$, and there exists a unique polynomial $e_1(t) \in \mathbb{Q}[t]$ of degree less than ℓ such that $(1-t)e_1(t) \equiv 1 \bmod p(t)$. For any integer $N \geq 0$, let $s_N(t) \in \mathbb{Q}[t]$ be the unique polynomial of degree less than ℓ such that

$$s_N(t) \equiv t^N e_1(t) \bmod p(t).$$

Since the multiplication by t on the ℓ-dimensional $\mathbb{Q}$-vector space $\mathbb{Q}[t]/(p(t))$ has characteristic polynomial $p(t)$, any complex eigenvalue λ of the multiplication by t satisfies $|\lambda| < 1$. In the Jordan normal form, the entries of the N-th power of a Jordan block with eigenvalue λ are of the form $\binom{N}{i}\lambda^{N-i}$ with $0 \leq i < \ell$, which tend to 0 as $N \to +\infty$. Thus the N-th power of the multiplication by t tends to 0, and the coefficients of $s_N(t)$ tend to 0 as $N \to +\infty$. For any integer $N \geq 1$, let

$$Q_N(t) := \sum_{i=0}^{N-1} t^i + s_N(t).$$

Then

$$(1-t)Q_N(t) = 1 - t^N + (1-t)s_N(t) \equiv 1 - t^N + t^N(1-t)e_1(t) \equiv 1 \bmod p(t).$$

Since $p(r^{-1}h_*) = 0$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$ (**Step 4**), we obtain

$$Q_N(r^{-1}h_*) \circ (\mathrm{id} - r^{-1}h_*) = \mathrm{id} \quad \text{on } \mathrm{Cl}(X)_{\mathbb{Q}}. \tag{3.7}$$

The argument below works for any integer $N \geq 1$. We state it for $N \geq \ell$, which is the range of Definition 7.1(2) with $m_0 = \ell$ (see the proof of Theorem 1.3). For any integers $1 \leq j \leq \ell$ and $N \geq \ell$, let $a_{j,N} \in \mathbb{Q}$ be the coefficient of t^{j-1} in $s_N(t)$, so that

$$s_N(t) = \sum_{j=1}^{\ell} a_{j,N} t^{j-1}.$$

The $a_{j,N}$ depend only on $p(t)$, hence only on h, and $\lim_{N \to +\infty} a_{j,N} = 0$ for any j. Let $N \geq \ell$ be an integer, and let D and Θ be $\mathbb{Q}$-divisors on X such that $h^*D - D \sim_{\mathbb{Q}} \Theta$. Applying $r^{-1}h_*$ and

using $r^{-1}h_* \circ h^* = \mathrm{id}$ (**Step 1**), we obtain $[D] - r^{-1}h_*[D] = r^{-1}h_*[\Theta]$ in $\mathrm{Cl}(X)_{\mathbb{Q}}$. By (3.7),

$$[D] = Q_N(r^{-1}h_*)r^{-1}h_*[\Theta] = \sum_{j=1}^{N}(r^{-1}h_*)^j[\Theta] + \sum_{j=1}^{\ell} a_{j,N}(r^{-1}h_*)^j[\Theta].$$

Since $(r^{-1}h_*)^j = r^{-j}(h^j)_*$ (**Step 1**), we obtain

$$D \sim_{\mathbb{Q}} \sum_{j=1}^{N} r^{-j}(h^j)_*\Theta + \sum_{j=1}^{\ell} a_{j,N}r^{-j}(h^j)_*\Theta.$$

Step 6. In this step, we prove the relation for K_X.

By the Hurwitz formula, Definition-Proposition 2.10(1), $K_X \sim h^*K_X + R(h)$. Thus $h^*(-K_X) - (-K_X) \sim R(h)$. By **Step 5** applied to $D = -K_X$ and $\Theta = R(h)$,

$$-K_X \sim_{\mathbb{Q}} \sum_{j=1}^{N} D_j + \sum_{j=1}^{\ell} a_{j,N}D_j$$

for any integer $N \geq \ell$, where $D_j = r^{-j}(h^j)_*R(h)$. Thus $K_X + \sum_{j=1}^{N} D_j + \sum_{j=1}^{\ell} a_{j,N}D_j \sim_{\mathbb{Q}} 0$. □

4. The canonical model of an int-amplified variety

In this section, we construct the canonical model of a normal projective variety admitting an int-amplified endomorphism (Construction-Proposition 4.2). All small models in the rest of the paper are built over the canonical model. We then show that the lc centers of the canonical model are not contained in the exceptional locus of the structure morphism (Theorem 4.4), nor in the support of the strict transform of $R(h^j)$ for any int-amplified endomorphism h and any positive integer j (Proposition 4.5). Section 5 and Section 6 use Theorem 4.4 and Proposition 4.5 to make finitely many divisors $\mathbb{Q}$-Cartier.

4.1. Pseudo-lc singularities and the canonical model.

In this subsection, we prove that a normal projective variety admitting an int-amplified endomorphism is pseudo-lc, and we construct the canonical model.

The finite generation [Has20, Theorem 4.19] and the small lc modification (Theorem 2.18) depend on the pseudo-lc condition, which is a condition on $\mathrm{Env}_X(-K_X)$. A normal projective variety X admitting a non-invertible polarized endomorphism is valuative lc [Yos22, Theorem 1.4], that is, $\mathbf{K}_X + \mathbf{E}_X - \mathrm{Env}_X(K_X) \geq 0$ [Yos22, Definitions 4.1 and 4.3], and a normal projective variety admitting an int-amplified endomorphism is also valuative lc [Yos22, Corollary 7.5]. The pseudo-lc condition is stronger, by the superadditivity in Lemma 2.16(1) applied to K_X and $-K_X$ and by $\mathrm{Env}_X(0) = 0$ (Lemma 2.16(2)), and valuative lc does not imply pseudo-lc in general [Yos22, Remarks 4.2 and 4.5].

The proof compares $\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X)$ with its pullbacks by the iterates h^k of an int-amplified endomorphism h of X and uses Lemma 3.4.

Theorem 4.1. *Let X be a normal projective variety admitting an int-amplified endomorphism. Then*

$$\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X) \geq 0.$$

Equivalently, X is pseudo-lc.

Proof. Let $h : X \to X$ be an int-amplified endomorphism of degree r and H an ample Cartier divisor on X such that $h^*H - H$ is ample (Definition 2.11). By Lemma 2.12, h is finite, so h^k is a finite surjective morphism of degree r^k for any integer $k \geq 0$. If $\dim X = 0$, then there exists no prime divisor over X, so we may assume that $\dim X \geq 1$. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a Cartier divisor $D \geq 0$ on X such that $D \sim lbH$ for some positive integers l and b, and

$$\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X) \geq -\overline{D}.$$

By [Har77, Theorems II.7.6 and II.5.17], there exists an integer $a > 0$ such that $\mathcal{O}_X(-K_X + aH)$ is globally generated. Let s be a non-zero global section of $\mathcal{O}_X(-K_X + aH)$, viewed as a rational function on X, and let $B := \mathrm{div}(s) - K_X + aH$. Then $B \geq 0$ and $K_X + B = \mathrm{div}(s) + aH$ is Cartier. By Lemma 2.16(1)(2),

$$\mathrm{Env}_X(-K_X) \geq \mathrm{Env}_X(-K_X - B) = -\overline{K_X + B}.$$

Let $g : Y \to X$ be a log resolution of $(X, \mathrm{Supp}\, B)$ and E_Y the sum of the g-exceptional prime divisors. Since Y is smooth and E_Y is an snc divisor with coefficients one, (Y, E_Y) is lc [KM98, Corollary 2.31(3)], so $\mathbf{K}_X + \mathbf{E}_X \geq \overline{K_Y + E_Y}$. Let $L := K_Y + E_Y - g^*(K_X + B)$. Then L is Cartier and

$$\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X) \geq \overline{K_Y + E_Y} - \overline{g^*(K_X + B)} = \overline{L}.$$

Let $S_1, \ldots, S_p$ be the prime divisors on Y such that $\mathrm{mult}_{S_i} L < 0$. Since g is projective and birational, $V := g(S_1) \cup \cdots \cup g(S_p)$ is a proper closed subset of X. Let $\mathcal{I}_V$ be the ideal sheaf of V. By [Har77, Theorems II.7.6 and II.5.17], there exists an integer $b > 0$ such that $\mathcal{I}_V \otimes \mathcal{O}_X(bH)$ has a non-zero global section, so there exists a Cartier divisor $D_0 \geq 0$ such that $D_0 \sim bH$ and $V \subset \mathrm{Supp}\, D_0$. Then $S_i \subset \mathrm{Supp}\, g^*D_0$, so $\mathrm{mult}_{S_i} g^*D_0 > 0$ for any i. Let $l > 0$ be an integer such that

$$l \,\mathrm{mult}_{S_i} g^*D_0 \geq -\mathrm{mult}_{S_i} L$$

for any i, and let $D := lD_0$. Then $D \sim lbH$, and $L + g^*D \geq 0$. Thus $\overline{L} \geq -\overline{D}$, and **Step 1** follows.

Step 2. In this step, we show that for any integer $k \geq 0$,

$$\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X) \leq (h^k)^*(\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X)).$$

For a rational function $\varphi \neq 0$ on X and a $\mathbb{Q}$-divisor D' on X, $\mathcal{O}_X(m(D' + \mathrm{div}\,\varphi)) = \varphi^{-m}\mathcal{O}_X(mD')$ for any positive integer m such that mD' is integral, so

$$\mathrm{Env}_X(D' + \mathrm{div}\,\varphi) = \mathrm{Env}_X(D') + \overline{\mathrm{div}\,\varphi}.$$

Replacing K_X by $K_X + \mathrm{div}\,\varphi$ adds $\overline{\mathrm{div}\,\varphi}$ to $\mathbf{K}_X$, so $\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X)$ does not depend on the choice of K_X. By Definition-Proposition 2.10(1), $(h^k)^*K_X + R(h^k)$ is a canonical divisor of X, and we use it to compute the left-hand side of the inequality of **Step 2**. By Lemma 2.16(4), with $\mathbf{K}_X$ on the left-hand side defined by $(h^k)^*K_X + R(h^k)$,

$$\mathbf{K}_X + \mathbf{E}_X = (h^k)^*(\mathbf{K}_X + \mathbf{E}_X) + \mathbf{R}(h^k).$$

Since $R(h^k) \geq 0$, Lemma 2.16(1)(2) implies that $\mathrm{Env}_X(R(h^k)) \geq \mathrm{Env}_X(0) = 0$ and that the trace of $\mathrm{Env}_X(R(h^k))$ on X is $R(h^k)$. Thus $\mathrm{Env}_X(R(h^k)) \geq \mathbf{R}(h^k)$. By Lemma 2.16(1)(3),

$$(h^k)^* \,\mathrm{Env}_X(-K_X) = \mathrm{Env}_X(-(h^k)^*K_X) \geq \mathrm{Env}_X(-(h^k)^*K_X - R(h^k)) + \mathrm{Env}_X(R(h^k)).$$

Therefore,

$$\begin{aligned}
&\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-(h^k)^*K_X - R(h^k)) \\
&\quad = (h^k)^*(\mathbf{K}_X + \mathbf{E}_X) + \mathbf{R}(h^k) + \mathrm{Env}_X(-(h^k)^*K_X - R(h^k)) \\
&\quad \leq (h^k)^*(\mathbf{K}_X + \mathbf{E}_X) + \mathrm{Env}_X(R(h^k)) + \mathrm{Env}_X(-(h^k)^*K_X - R(h^k)) \\
&\quad \leq (h^k)^*(\mathbf{K}_X + \mathbf{E}_X + \mathrm{Env}_X(-K_X)).
\end{aligned}$$

By the first paragraph of this step, the left-hand side of the last display, which is computed with the canonical divisor $(h^k)^*K_X + R(h^k)$, is equal to the left-hand side of the inequality of **Step 2**, and **Step 2** follows.

Step 3. In this step, we show that for any integer $k \geq 0$ and any prime divisor E over X,

$$r^k \operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \geq -\operatorname{mult}_E(h^k)_*D,$$

where $D \geq 0$ is the Cartier divisor of **Step 1**.

By Lemma 3.3(1), $(h^k)_*D \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. Let $E_1, \dots, E_q$ be the prime divisors over X such that the restriction of ord_{E_i} to the subfield $(h^k)^*\mathbb{C}(X)$ of $\mathbb{C}(X)$ is $e_i \operatorname{ord}_E$ for some positive integer e_i. By **Step 2** and Definition 2.14,

$$\operatorname{mult}_{E_i}(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \leq e_i \operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)),$$

and by **Step 1**,

$$\operatorname{mult}_{E_i}(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \geq -\operatorname{mult}_{E_i} D.$$

Thus

$$e_i \operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \geq -\operatorname{mult}_{E_i} D \tag{4.1}$$

for any i. Since $\sum_i e_i[\mathbb{C}(E_i) : \mathbb{C}(E)] = r^k$ [BdFF12, Section 1.5, Proposition 1.15], multiplying (4.1) by $[\mathbb{C}(E_i) : \mathbb{C}(E)]$ and taking the sum over i, we obtain

$$r^k \operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \geq -\sum_i [\mathbb{C}(E_i) : \mathbb{C}(E)] \operatorname{mult}_{E_i} D = -\operatorname{mult}_E(h^k)_*D,$$

where the last equality follows from Lemma 3.3(2).

Step 4. In this step, we show that $\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X) \geq 0$.

Recall that $D \geq 0$ is a Cartier divisor on X such that $D \sim lbH$ (**Step 1**). Let E be a prime divisor over X. By Lemma 3.4, applied to D and the divisorial valuation ord_E,

$$\lim_{k \to +\infty} r^{-k} \operatorname{mult}_E(h^k)_*D = 0.$$

By **Step 3**,

$$\operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \geq -r^{-k} \operatorname{mult}_E(h^k)_*D$$

for any integer $k \geq 0$. Letting $k \to +\infty$, we obtain $\operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) \geq 0$. Since E is an arbitrary prime divisor over X, **Step 4** follows.

Step 5. In this step, we show that $\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X) \geq 0$ if and only if X is pseudo-lc.

Let F be a prime divisor over X. If F is exceptional over X, then by [Has20, Proposition 4.7 and Theorem 4.9],

$$\alpha(F, X, 0) = \operatorname{mult}_F(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)).$$

If F is a prime divisor on X, then $\alpha(F, X, 0) = 1$ by [Has20, Lemma 4.3(ii)], and $\operatorname{mult}_F(\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X)) = 0$ by Lemma 2.16(2). Hence $\mathbf{K}_X + \mathbf{E}_X + \operatorname{Env}_X(-K_X) \geq 0$ if and only if $\alpha(F, X, 0) \geq 0$ for any prime divisor F over X. By Definition 2.17, **Step 5** follows, and Theorem 4.1 follows from **Steps 4 and 5**. □

Construction-Proposition 4.2. Let X be a normal projective variety admitting an int-amplified endomorphism. Then the $\mathcal{O}_X$-algebra $\bigoplus_{m \geq 0} \mathcal{O}_X(mK_X)$ is finitely generated. Moreover, $X' := \operatorname{Proj}_X \bigoplus_{m \geq 0} \mathcal{O}_X(mK_X)$ is a normal projective variety, the structure morphism $\pi : X' \to X$ is a small birational morphism, $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample, $(X', 0)$ is lc, and $\pi_*\mathcal{O}_{X'}(mK_{X'}) = \mathcal{O}_X(mK_X)$ for any integer $m \geq 0$. We call X' the *canonical model* of X.

Proof. By Theorem 4.1, X is pseudo-lc. Thus the finite generation is the first part of [Has20, Theorem 4.19], and the small lc modification is Theorem 2.18 with $\Delta = 0$. The following argument proves all the statements, including the finite generation, from Theorem 2.18 and Lemma 2.20.

By Theorem 2.18 with $\Delta = 0$, there exists a small birational morphism $\pi_Y : Y \to X$ from a normal variety Y such that K_Y is $\mathbb{R}$-Cartier and π_Y-ample and $(Y, 0)$ is lc. Since π_Y is small, $K_Y = (\pi_Y)_*^{-1}K_X$. By Theorem 2.18 and Lemma 2.5, K_Y is $\mathbb{Q}$-Cartier. Since K_Y is π_Y-ample as an $\mathbb{R}$-divisor, K_Y is positive on $\overline{\mathrm{NE}}(Y/X)\backslash\{0\}$. Let $m_0 > 0$ be an integer such

that m_0K_Y is Cartier. Since π_Y is projective by Theorem 2.18, the relative Kleiman criterion [KM98, Theorem 1.44] implies that m_0K_Y is a π_Y-ample Cartier divisor. By Lemma 2.20 with $D = K_X$, the $\mathcal{O}_X$-algebra $\bigoplus_{m\geq 0}\mathcal{O}_X(mK_X)$ is finitely generated, $(\pi_Y)_*\mathcal{O}_Y(mK_Y) = \mathcal{O}_X(mK_X)$ for any integer $m \geq 0$, and $Y \cong X'$ over X. Under the isomorphism $Y \cong X'$, π_Y is the structure morphism π. Thus X' is a normal variety, π is a small birational morphism, $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample, $(X', 0)$ is lc, and $\pi_*\mathcal{O}_{X'}(mK_{X'}) = \mathcal{O}_X(mK_X)$ for any integer $m \geq 0$. Since π is projective and X is projective, X' is projective. □

4.2. **Lc centers of the canonical model.** In this subsection, we show that the lc centers of the canonical model $\pi : X' \to X$ of Construction-Proposition 4.2 are not contained in $\mathrm{Exc}(\pi)$ (Theorem 4.4). The pair $(X', 0)$ is lc, but if an lc center V of $(X', 0)$ is contained in $\mathrm{Exc}(\pi)$, then Proposition 4.5 does not apply, since the ramification formula on X cannot be used at the generic point of V. Section 5 uses Proposition 4.5 to verify the hypothesis of Lemma 5.1, which makes finitely many divisors on the canonical model $\mathbb{Q}$-Cartier.

The difficulty in the proof of Theorem 4.4 is uniformity. For each positive integer j, we construct from [Has20, Theorem 4.8] a $\mathbb{Q}$-divisor $\Delta_j \geq 0$ on X such that $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier and the strict transform of Δ_j on X' has order at most $1/j$ at a fixed lc place of $(X', 0)$. The classes of the Δ_j and the Cartier indices of their strict transforms on X' depend on j, but on a curve C contracted by π each strict transform has degree $-(K_{X'} \cdot C) < 0$, independent of j. Lemma 4.3 turns a fixed negative degree on a fixed curve into a uniform positive lower bound for the order at a fixed point, which contradicts the order at most $1/j$ of the j-th strict transform at an lc place.

Lemma 4.3. *Let X be a normal projective variety of dimension at least two, $C \subset X$ an integral projective curve, $x \in C$ a closed point, v a divisorial valuation of $\mathbb{C}(X)$, and $a > 0$ a real number. Assume that $\mathrm{center}_X v = \{x\}$. Then there exists a real number $\epsilon > 0$ such that $v(D) \geq \epsilon$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ on X with $(D \cdot C) \leq -a$.*

Proof. We first construct a smooth curve Γ over C, a line bundle A_Γ on Γ, and a constant c, which depend only on X, C, x, and v.

By [KM98, Theorem 0.2] applied to X and the ideal sheaf $\mathcal{I}_C$ of C, there exist a smooth variety Y and a projective birational morphism $\tau : Y \to X$ such that $\tau^{-1}\mathcal{I}_C \cdot \mathcal{O}_Y = \mathcal{O}_Y(-F)$ for a divisor $F \geq 0$ on Y with snc support. Since τ is surjective and $\mathrm{Supp}\, F = \tau^{-1}(C)$, some irreducible component P of $\mathrm{Supp}\, F$ satisfies $\tau(P) = C$, and P is smooth. If $\dim P = 1$, then let $\Gamma := P$. If $\dim P \geq 2$, then let Γ be the intersection of P with $\dim P - 1$ general members of a very ample linear system $|H_Y|$ on Y. By [Tan24, Theorem 5.11, Remark 5.12(a)], Γ is a smooth irreducible curve. We show $\tau(\Gamma) = C$. Let $H_C \geq 0$ be an ample Cartier divisor on C. Since $\tau(P) = C$, $(\tau|_P)^*H_C \geq 0$ is a non-zero Cartier divisor on P, so

$$\deg\left((\tau|_\Gamma)^*H_C\right) = \left((\tau|_P)^*H_C \cdot H_Y^{\dim P-1}\right)_P > 0,$$

and $\tau(\Gamma)$ is not a point. Thus $\tau|_\Gamma : \Gamma \to C$ is a finite surjective morphism in both cases. Let δ be the degree of $\tau|_\Gamma$, and fix a closed point $w \in \Gamma$ with $\tau(w) = x$.

Let $N^* := \mathcal{I}_\Gamma/\mathcal{I}_\Gamma^2$ be the conormal bundle of Γ in Y, and N the dual of N^*. Since Γ and Y are smooth, N^* is a vector bundle of rank $\dim X - 1 \geq 1$ on Γ, and $\mathcal{I}_\Gamma^l/\mathcal{I}_\Gamma^{l+1} \cong \mathrm{Sym}^l N^*$ for any integer $l \geq 0$ [Har77, Theorem II.8.17 and Theorem II.8.21A]. Let A_Γ be a line bundle on Γ such that $b := \deg A_\Gamma > 0$ and $N \otimes A_\Gamma$ is globally generated, and let $\mathcal{O}_\Gamma^{\oplus n} \to N \otimes A_\Gamma$ be a surjection for some positive integer n. Dualizing and tensoring with A_Γ, we obtain an injection $N^* \to A_\Gamma^{\oplus n}$ with locally free cokernel. Hence for any integer $k \geq 0$, the induced morphism

$$\mathrm{Sym}^k N^* \to \mathrm{Sym}^k\left(A_\Gamma^{\oplus n}\right) \cong \left(A_\Gamma^{\otimes k}\right)^{\oplus\binom{n+k-1}{k}}$$

is injective.

Let u be the divisorial valuation of $\mathbb{C}(X) = \mathbb{C}(Y)$ given by the exceptional divisor of the blow-up of Y at w. Then $u(\phi) \geq 1$ for any ϕ in the maximal ideal of $\mathcal{O}_{Y,w}$. Since $\tau(w) = x$

and $\operatorname{center}_X v = \{x\}$, both u and v are divisorial valuations centered at the maximal ideal of $R := \mathcal{O}_{X,x}$. Since R is the local ring of a normal variety at x, the completion of R is a normal local domain [Gro65, Scholie 7.8.3]. In particular, R is analytically irreducible. By Izumi's theorem [Izu85, Main Theorem (3.4)], in the algebraic form due to Rees (see [RS14, Theorem 1.3]), there exists a real number $c > 0$ such that

$$u(\phi) \leq cv(\phi) \quad \text{for any } \phi \in R\backslash\{0\}. \tag{4.2}$$

Now let $D \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X with $(D \cdot C) \leq -a$, and m a positive integer such that mD is Cartier. Let s_D be the global section of $\mathcal{O}_Y(\tau^*(mD))$ whose divisor of zeros is $\tau^*(mD)$, and η the generic point of Γ. Let k be the largest integer such that $s_D \in \mathcal{I}_\Gamma^k \cdot \mathcal{O}_Y(\tau^*(mD))$ at η. Since $s_D \neq 0$, $k < +\infty$ by the Krull intersection theorem in the Noetherian local ring $\mathcal{O}_{Y,\eta}$ [Mat86, Theorem 8.10].

We show that $s_D \in \mathcal{I}_\Gamma^k \cdot \mathcal{O}_Y(\tau^*(mD))$ on a neighborhood of Γ. Let $l < k$ be a non-negative integer, and assume that $s_D \in \mathcal{I}_\Gamma^l \cdot \mathcal{O}_Y(\tau^*(mD))$ on a neighborhood of Γ. The image of s_D in

$$\left(\mathcal{I}_\Gamma^l/\mathcal{I}_\Gamma^{l+1}\right) \otimes \mathcal{O}_Y(\tau^*(mD)) \cong \operatorname{Sym}^l N^* \otimes \mathcal{O}_Y(\tau^*(mD))|_\Gamma$$

is a section of a locally free sheaf on the integral curve Γ that vanishes at η, so the image vanishes. Thus $s_D \in \mathcal{I}_\Gamma^{l+1} \cdot \mathcal{O}_Y(\tau^*(mD))$ on a neighborhood of Γ. Induction on l starting from $l = 0$ proves the claim.

By the maximality of k, the image σ of s_D in $\operatorname{Sym}^k N^* \otimes \mathcal{O}_Y(\tau^*(mD))|_\Gamma$ is a non-zero global section. Applying the injection above to σ, we obtain a non-zero global section of a direct sum of copies of the line bundle $A_\Gamma^{\otimes k} \otimes \mathcal{O}_Y(\tau^*(mD))|_\Gamma$ on Γ, hence, by projecting to a summand, a non-zero global section of $A_\Gamma^{\otimes k} \otimes \mathcal{O}_Y(\tau^*(mD))|_\Gamma$. By the projection formula, $\deg\left(\mathcal{O}_Y(\tau^*(mD))|_\Gamma\right) = m\delta(D \cdot C)$. Hence

$$0 \leq kb + m\delta(D \cdot C) \leq kb - m\delta a,$$

that is, $k \geq m\delta a/b$.

Let $\phi_D \in R$ be a local equation of mD at x. Near w, $\tau^*\phi_D$ is a local equation of $\tau^*(mD)$, so in a local trivialization of $\mathcal{O}_Y(\tau^*(mD))$ at w, $\tau^*\phi_D$ is the product of s_D and a unit. Since $s_D \in \mathcal{I}_\Gamma^k \cdot \mathcal{O}_Y(\tau^*(mD))$ near w and $\mathcal{I}_\Gamma$ is contained in the maximal ideal of $\mathcal{O}_{Y,w}$, we have $u(\phi_D) \geq k$. By (4.2),

$$mv(D) = v(\phi_D) \geq \frac{u(\phi_D)}{c} \geq \frac{k}{c} \geq \frac{m\delta a}{bc}.$$

Let $\epsilon := \delta a/(bc)$. Since δ, b, and c do not depend on D, $v(D) \geq \epsilon$ for any D as in the lemma. □

Theorem 4.4. *Let X be a normal projective pseudo-lc variety and $\pi : X' \to X$ a small birational morphism from a normal projective variety X'. Assume that $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample, and $(X', 0)$ is lc. Then no lc center of $(X', 0)$ is contained in $\operatorname{Exc}(\pi)$. In particular, π is an isomorphism on a neighborhood of the generic point of any lc center of $(X', 0)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct $\mathbb{Q}$-divisors $\Delta_j \geq 0$ on X such that $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier and the strict transform of Δ_j on X' has order at most $1/j$ at a fixed lc place of $(X', 0)$.

Let S be an lc center of $(X', 0)$. Let E be an lc place of $(X', 0)$ with $\operatorname{center}_{X'} E = S$. By [KM98, Lemma 2.45], E is a prime divisor on some normal variety W_0 with a projective birational morphism $W_0 \to X'$. Taking a log resolution of $(W_0, \operatorname{Exc}(W_0 \to X'))$ [KM98, Theorem 0.2], we obtain a log resolution $\rho : W \to X'$ of $(X', 0)$ on which the strict transform of E, still denoted E, is a prime divisor. Since $a(E, X', 0) = 0$, E is exceptional over X'. Thus E is a component of the snc divisor $\operatorname{Exc}(\rho)$, and E is smooth.

Fix a positive integer j. By [Has20, Theorem 4.8] applied to $\langle X, 0\rangle$, the morphism $\pi \circ \rho : W \to X$, and $1/(2j)$, there exists an $\mathbb{R}$-divisor $G = \sum_{i=1}^n g_i G_i \geq 0$ on X, where the G_i are distinct

prime divisors and $g_i > 0$ for any i, such that $K_X + G$ is $\mathbb{R}$-Cartier and

$$\alpha(E, X, 0) - a(E, X, G) \leq \frac{1}{2j}.$$

We replace G with a $\mathbb{Q}$-divisor whose support is $\operatorname{Supp} G$. Let $U \subset \mathbb{R}^{1+n}$ be the set of $(c_0, c_1, \ldots, c_n)$ such that $c_0 K_X + \sum_{i=1}^n c_i G_i$ is $\mathbb{R}$-Cartier. Let $\lambda : \mathbb{Q}^{1+n} \to \operatorname{Div}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$ be the linear map $(c_0, \ldots, c_n) \to c_0 K_X + \sum_i c_i G_i$, where $\operatorname{Div}(X)$ is the group of Weil divisors on X, and let $\mathcal{C} \subset \operatorname{Div}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$ be the $\mathbb{Q}$-linear subspace of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors, that is, the image of $\operatorname{CDiv}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$, where $\operatorname{CDiv}(X)$ is the group of Cartier divisors on X. Then U is the kernel of the composition of $\lambda \otimes_{\mathbb{Q}} \mathbb{R}$ with the quotient map to $\big((\operatorname{Div}(X) \otimes_{\mathbb{Z}} \mathbb{Q})/\mathcal{C}\big) \otimes_{\mathbb{Q}} \mathbb{R}$. Since $\mathbb{R}$ is flat over $\mathbb{Q}$, taking kernels commutes with $\otimes_{\mathbb{Q}}\mathbb{R}$, so U is defined by linear equations with rational coefficients. The affine subspace $U_1 := U \cap \{c_0 = 1\}$ contains the real point $(1, g_1, \ldots, g_n)$ and is defined by linear equations with rational coefficients, so the rational points of U_1 are dense in U_1. For $(1, d_1, \ldots, d_n) \in U_1$, the divisor $K_X + \sum_i d_i G_i$ is $\mathbb{R}$-Cartier, and the function

$$(1, d_1, \ldots, d_n) \to a\Big(E, X, \sum_{i=1}^n d_i G_i\Big)$$

is affine on U_1, since the pullback of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors is linear. Hence there exists a rational point $(1, d_1, \ldots, d_n) \in U_1$ such that $d_i > 0$ for any i and, letting $\Delta_j := \sum_{i=1}^n d_i G_i$,

$$a(E, X, \Delta_j) \geq a(E, X, G) - \frac{1}{2j}.$$

Then $\Delta_j \geq 0$ is a $\mathbb{Q}$-divisor with $\operatorname{Supp} \Delta_j = \operatorname{Supp} G$, and $K_X + \Delta_j$ is $\mathbb{R}$-Cartier, hence $\mathbb{Q}$-Cartier by Lemma 2.5. Since X is pseudo-lc, $\alpha(E, X, 0) \geq 0$, so

$$a(E, X, \Delta_j) \geq \alpha(E, X, 0) - \frac{1}{j} \geq -\frac{1}{j}. \tag{4.3}$$

Let $B'_j := \pi_*^{-1} \Delta_j$. Then

$$K_{X'} + B'_j = \pi^*(K_X + \Delta_j). \tag{4.4}$$

Equality (4.4) holds by Lemma 2.19, since $K_X + \Delta_j$ is $\mathbb{Q}$-Cartier. In particular, $B'_j \geq 0$ is $\mathbb{Q}$-Cartier. Let $\beta_j := \operatorname{mult}_E B'_j$. By (4.4) and [KM98, Lemma 2.30], $a(E, X', B'_j) = a(E, X, \Delta_j)$. Since B'_j is $\mathbb{Q}$-Cartier and $a(E, X', 0) = 0$, (4.3) implies

$$0 \leq \beta_j = a(E, X', 0) - a(E, X', B'_j) = -a(E, X, \Delta_j) \leq \frac{1}{j}, \tag{4.5}$$

where $\beta_j \geq 0$ because $\rho^* B'_j \geq 0$. Let

$$J_j := \rho^* B'_j - \beta_j E.$$

Then $J_j \geq 0$ is a $\mathbb{Q}$-divisor on the smooth variety W and $E \not\subset \operatorname{Supp} J_j$, so $J_j|_E \geq 0$. By (4.4), $\rho^* B'_j = (\pi \circ \rho)^*(K_X + \Delta_j) - \rho^* K_{X'}$, hence

$$J_j|_E \sim_{\mathbb{Q}} \big((\pi \circ \rho)^*(K_X + \Delta_j)\big)|_E - (\rho^* K_{X'})|_E - \beta_j E|_E. \tag{4.6}$$

Step 2. In this step, we show that $\dim \pi(S) = \dim S$.

Let $T := \pi(S)$, $s := \dim S$, $t := \dim T$, $e := \dim E = \dim X' - 1$, and $\varphi := (\pi \circ \rho)|_E : E \to T$. Then φ is surjective. Suppose that $s > t$. Let A be a very ample divisor on E, H a very ample divisor on X, and $H_T := H|_T$. Since $\varphi^* H_T$ and A are nef,

$$(L \cdot \varphi^* H_T^t \cdot A^{e-t-1}) \geq 0 \quad \text{for any } \mathbb{Q}\text{-Cartier } \mathbb{Q}\text{-divisor } L \geq 0 \text{ on } E. \tag{4.7}$$

For any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L_T on T, we have $(\varphi^* L_T \cdot \varphi^* H_T^t \cdot A^{e-t-1}) = 0$, because a positive multiple of L_T is linearly equivalent to the difference of two very ample divisors on

T, and $t+1$ general very ample divisors on T have empty intersection since $\dim T = t$. Since $\big((\pi\circ\rho)^*(K_X+\Delta_j)\big)|_E \sim_{\mathbb{Q}} \varphi^*\big((K_X+\Delta_j)|_T\big)$, we obtain

$$\Big(\big((\pi\circ\rho)^*(K_X+\Delta_j)\big)|_E\cdot\varphi^*H_T^t\cdot A^{e-t-1}\Big)=0. \tag{4.8}$$

We claim that

$$\kappa := \big((\rho^*K_{X'})|_E\cdot\varphi^*H_T^t\cdot A^{e-t-1}\big)>0. \tag{4.9}$$

If $t=0$, then let $E_T := E$. If $t>0$, then let $H_1,\dots,H_t$ be general members of $|H|$ and $E_T := \varphi^*H_1\cap\cdots\cap\varphi^*H_t$. Since $|\varphi^*H|$ is base point free on the smooth variety E, [Tan24, Theorem 5.11] applied t times shows that E_T is smooth of pure dimension $e-t$, and $E_T\neq\emptyset$ because $T\cap H_1\cap\cdots\cap H_t\neq\emptyset$. The set $T\cap H_1\cap\cdots\cap H_t$ is finite and $E_T=\varphi^{-1}(T\cap H_1\cap\cdots\cap H_t)$ as sets. In both cases, $\varphi^*H_T^t$ is represented by the cycle $\sum_F[F]$, where F runs over the irreducible components of E_T, so

$$\kappa=\sum_F\big((\rho^*K_{X'})|_F\cdot A^{e-t-1}\big).$$

Any component F is contained in $\varphi^{-1}(y)$ for some $y\in T$, so $\rho(F)\subset\pi^{-1}(y)$. Since $K_{X'}$ is π-ample, $K_{X'}|_{\rho(F)}$ is ample, so $(\rho^*K_{X'})|_F$ is nef and $\big((\rho^*K_{X'})|_F\cdot A^{e-t-1}\big)\geq 0$. Fix a point $y\in\varphi(E_T)$. Since $\rho(E)=S$, we have $\rho(\varphi^{-1}(y))=S\cap\pi^{-1}(y)$, and $\dim V\geq s-t\geq 1$ for any irreducible component V of $S\cap\pi^{-1}(y)$ by [Har77, Exercise II.3.22(b)(d)]. Since $\varphi^{-1}(y)$ is a union of components of E_T, there exists a component F with $\dim\rho(F)\geq 1$. Let l be a positive integer such that $lK_{X'}|_{\rho(F)}$ is very ample, and H' a hyperplane section of $\rho(F)$ in the embedding defined by $lK_{X'}|_{\rho(F)}$. Then $(\rho|_F)^*H'\geq 0$ is a non-zero Cartier divisor on F, and since $\dim F=e-t$ and A is very ample,

$$l\big((\rho^*K_{X'})|_F\cdot A^{e-t-1}\big)=\big((\rho|_F)^*H'\cdot A^{e-t-1}\big)>0.$$

This proves (4.9).

Let $\gamma := (E|_E\cdot\varphi^*H_T^t\cdot A^{e-t-1})$, a real number independent of j. By (4.7) applied to $L=J_j|_E$, (4.6), (4.8), and (4.9),

$$0\leq(J_j|_E\cdot\varphi^*H_T^t\cdot A^{e-t-1})=-\kappa-\beta_j\gamma.$$

By (4.5), $|\beta_j\gamma|\leq|\gamma|/j$, so $-\kappa-\beta_j\gamma<0$ for any $j>|\gamma|/\kappa$, a contradiction. Thus $\dim\pi(S)=\dim S$.

Step 3. In this step, we show that the fibers of π through points of $\mathrm{Exc}(\pi)$ are connected and positive-dimensional, and we give a dimension estimate: $\dim X'-\dim S\geq 3$ if $S\subset\mathrm{Exc}(\pi)$.

Let $X^\circ\subset X$ be the set of points y such that $\pi^{-1}(y)$ is finite. The set of points $z\in X'$ such that $\pi^{-1}(\pi(z))$ is positive-dimensional at z is closed by [Har77, Exercise II.3.22(d)], and $X\backslash X^\circ$ is the image of the set under the proper morphism π. Thus X° is open. Since X' is projective, π is projective [Har77, Exercise II.4.9], so $\pi^{-1}(X^\circ)\to X^\circ$ is projective with finite fibers, hence finite [Har77, Exercise III.11.2]. Since $\pi_*\mathcal{O}_{\pi^{-1}(X^\circ)}$ is a finite $\mathcal{O}_{X^\circ}$-algebra contained in the function field and X° is normal, $\pi_*\mathcal{O}_{\pi^{-1}(X^\circ)}=\mathcal{O}_{X^\circ}$, so $\pi^{-1}(X^\circ)\to X^\circ$ is an isomorphism. For $y\in X\backslash X^\circ$, the fiber $\pi^{-1}(y)$ is connected [Har77, Corollary III.11.4] and positive-dimensional, so no point of $\pi^{-1}(y)$ is isolated in $\pi^{-1}(y)$, and π is not a local isomorphism at any point of $\pi^{-1}(y)$. Thus

$$\mathrm{Exc}(\pi)=\pi^{-1}(X\backslash X^\circ), \tag{4.10}$$

and for any $z\in\mathrm{Exc}(\pi)$, the fiber $\pi^{-1}(\pi(z))$ is connected and positive-dimensional. Moreover, $\dim\mathrm{Exc}(\pi)\leq\dim X'-2$, which we now show. Since X is normal, X° contains the generic point of any prime divisor on X [Har77, Theorem I.6.2A and Theorem II.4.7], so $\dim(X\backslash X^\circ)\leq\dim X-2$. If a prime divisor P on X' were contained in $\mathrm{Exc}(\pi)$, then $\dim\pi(P)\leq\dim X-2<\dim P$ by (4.10), and P would be contracted by π, which is impossible because π is small.

Now suppose that $S\subset\mathrm{Exc}(\pi)$. Recall that $T=\pi(S)$ and $t=\dim T$, and $\dim S=t$ by **Step 2**. Moreover, $\pi^{-1}(T)\subset\mathrm{Exc}(\pi)$ by (4.10). Any fiber of $\pi^{-1}(T)\to T$ is positive-dimensional. The irreducible components of $\pi^{-1}(T)$ that do not dominate T have proper closed images in T,

so for a general point $y \in T$, some irreducible component Q of $\pi^{-1}(T)$ dominating T satisfies $\dim(Q \cap \pi^{-1}(y)) \geq 1$. Hence $\dim Q \geq t+1$ by [Har77, Exercise II.3.22(e)], and

$$t+1 \leq \dim \pi^{-1}(T) \leq \dim \mathrm{Exc}(\pi) \leq \dim X' - 2.$$

Thus $\dim X' - t \geq 3$. Below we only use $\dim X' - t \geq 2$.

Step 4. In this step, we assume that $S \subset \mathrm{Exc}(\pi)$, and we construct a normal projective subvariety $Z \subset X'$ with $\dim Z \geq 2$, a closed point $x \in S \cap Z$, and a divisorial valuation v of $\mathbb{C}(Z)$ with $\mathrm{center}_Z v = \{x\}$, such that $Z \not\subset \mathrm{Supp}\, B'_j$ and $v(B'_j|_Z) \leq 1/j$ for any j. Recall that $t = \dim S = \dim T$ (**Step 2**).

Case $t = 0$. Then $S = \{x\}$ for a closed point x. Let $Z := X'$ and $v := \mathrm{ord}_E$. Then $\mathrm{center}_{X'} v = S = \{x\}$, $\dim Z \geq 2$ by **Step 3**, and $v(B'_j|_Z) = \mathrm{mult}_E B'_j = \beta_j \leq 1/j$ by (4.5).

Case $t > 0$. Recall from **Step 1** that E is a smooth prime divisor on W with $\rho(E) = S$, $\beta_j = \mathrm{mult}_E B'_j$, and $J_j = \rho^* B'_j - \beta_j E \geq 0$ with $E \not\subset \mathrm{Supp}\, J_j$.

We first define proper closed subsets $\Sigma_j \subsetneq S$ for $j \geq 0$. Let Σ_0 be the set of points $z \in S$ such that $S \cap \pi^{-1}(\pi(z))$ is positive-dimensional at z. Then Σ_0 is closed by [Har77, Exercise II.3.22(d)], and $\Sigma_0 \neq S$ because $\dim S = \dim T$. For $j \geq 1$, by [Har77, Exercise II.3.22(b)(e)] applied to $\rho|_E : E \to S$ and to the components of $\mathrm{Supp}(J_j|_E)$ dominating S, there exists a dense open subset $S_j^\circ \subset S$ with the following property: for any $z \in S_j^\circ$, $\dim V = e - t$ for any irreducible component V of $(\rho|_E)^{-1}(z)$, and $\dim V' = e - t - 1$ for any irreducible component V' of the fiber over z of a component of $\mathrm{Supp}(J_j|_E)$ dominating S. The components of $\mathrm{Supp}(J_j|_E)$ that do not dominate S have proper closed images in S. Let $\Sigma_j \subsetneq S$ be the union of $S \backslash S_j^\circ$ and the images in S of the components of $\mathrm{Supp}(J_j|_E)$ that do not dominate S. Then for any $z \in S \backslash \Sigma_j$, no irreducible component of $(\rho|_E)^{-1}(z)$ is contained in $\mathrm{Supp}\, J_j$. For any $j \geq 0$, $\dim \pi(\Sigma_j) \leq \dim \Sigma_j < t$.

We choose anew a very ample complete linear system $|H|$ on X. The members $H_1, \dots, H_t$ of $|H|$ below are unrelated to those in **Step 2**. Let $U_\rho \subset X'$ be a dense open subset over which ρ is an isomorphism. For $(H_1, \dots, H_t) \in |H|^t$, let

$$Z := \pi^* H_1 \cap \cdots \cap \pi^* H_t \subset X', \qquad W' := (\pi \circ \rho)^* H_1 \cap \cdots \cap (\pi \circ \rho)^* H_t \subset W.$$

We choose $(H_1, \dots, H_t)$ such that the following conditions hold.

(1) Z is normal and irreducible, W' is smooth and irreducible, and $E \cap W'$ is smooth of pure dimension $e - t$.
(2) $Z \not\subset X' \backslash U_\rho$.

We require in addition that the following countably many conditions hold.

(3) $H_1 \cap \cdots \cap H_t \cap \pi(\Sigma_j) = \emptyset$ for any $j \geq 0$.
(4) $Z \not\subset \mathrm{Supp}\, B'_j$ for any $j \geq 1$.

Each of the conditions (1)–(4) holds on a dense open subset of $|H|^t$. For (1), the restrictions of $|\pi^* H|$ and $|(\pi \circ \rho)^* H|$ to X', to W, to E, and to the intersections with the first k members are base point free. The image of X' or W cut by $k < t$ members under the morphism defined by the restricted system is $X \cap H_1 \cap \cdots \cap H_k$, whose dimension is $\dim X' - k \geq 2$ by **Step 3**. Thus [Tan24, Theorem 5.11, Remark 5.12(a)], applied t times, implies (1). For (2) and (4), a general member of a base point free linear system does not contain a given closed point, so cutting by general members of $|\pi^* H|$ lowers the dimension of each irreducible component of $X' \backslash U_\rho$ and of $\mathrm{Supp}\, B'_j$ by one at each cut. Hence $\dim\big((X' \backslash U_\rho) \cap Z\big) < \dim Z$ and $\dim(\mathrm{Supp}\, B'_j \cap Z) < \dim Z$. For (3), t general members of $|H|$ have no common point on $\pi(\Sigma_j)$ because $\dim \pi(\Sigma_j) < t$. Since $\mathbb{C}$ is uncountable, the irreducible variety $|H|^t$ is not a countable union of proper closed subsets, so there exists $(H_1, \dots, H_t)$ satisfying (1)–(4).

By **Step 3**, $\dim Z = \dim X' - t \geq 2$. Since $W' = \rho^{-1}(Z)$ and ρ is an isomorphism over U_ρ, (1) and (2) imply that $\rho' := \rho|_{W'} : W' \to Z$ is a projective birational morphism. The set $T \cap H_1 \cap \cdots \cap H_t$ is non-empty because $\dim T = t$, so $S \cap Z = (\pi|_S)^{-1}(T \cap H_1 \cap \cdots \cap H_t)$ is non-empty, and $S \cap Z$ is finite by (3) for $j = 0$. Since $E \cap W' = (\rho|_E)^{-1}(S \cap Z)$ as sets, any

irreducible component of $E \cap W'$ is mapped by ρ to a point of $S \cap Z$. Let E' be an irreducible component of $E \cap W'$ and $x \in S \cap Z$ the point with $\rho(E') = \{x\}$. By (1), E' is a prime divisor on the smooth variety W', so $v := \mathrm{ord}_{E'}$ is a divisorial valuation of $\mathbb{C}(Z) = \mathbb{C}(W')$ with $\mathrm{center}_Z v = \{x\}$.

Since $E' \subset (\rho|_E)^{-1}(x) \subset E \cap W'$, E' is an irreducible component of $(\rho|_E)^{-1}(x)$. By (3), $\pi(x) \notin \pi(\Sigma_j)$, so $x \notin \Sigma_j$ and $E' \not\subset \operatorname{Supp} J_j$ for any $j \geq 1$. In particular, $W' \not\subset \operatorname{Supp} J_j$ and $W' \not\subset E$, so $W' \not\subset \operatorname{Supp} \rho^* B'_j$. By (4), $B'_j|_Z \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Z, and

$$\rho'^*(B'_j|_Z) = (\rho^* B'_j)|_{W'} = \beta_j E|_{W'} + J_j|_{W'}.$$

Since $E \cap W'$ is smooth, $\mathrm{mult}_{E'}(E|_{W'}) = 1$, and $\mathrm{mult}_{E'}(J_j|_{W'}) = 0$ since $E' \not\subset \operatorname{Supp} J_j$. Thus $v(B'_j|_Z) = \beta_j \leq 1/j$ by (4.5).

In both cases, Z is a normal projective variety with $\dim Z \geq 2$, $x \in S \cap Z$, $\mathrm{center}_Z v = \{x\}$, and

$$v(B'_j|_Z) \leq \frac{1}{j} \quad \text{for any positive integer } j. \tag{4.11}$$

Step 5. In this step, we derive a contradiction from $S \subset \mathrm{Exc}(\pi)$, and we prove the last statement of the theorem.

Assume that $S \subset \mathrm{Exc}(\pi)$, and let Z, x, and v be as constructed in **Step 4**. Since $x \in S \subset \mathrm{Exc}(\pi)$, the fiber $\pi^{-1}(\pi(x))$ is connected and positive-dimensional by **Step 3**, so there exists an integral projective curve $C \subset \pi^{-1}(\pi(x))$ with $x \in C$. If $t > 0$, then $\pi(x) \in H_1 \cap \cdots \cap H_t$ because $x \in Z$, so $\pi^{-1}(\pi(x)) \subset Z$ and $C \subset Z$. If $t = 0$, then $C \subset X' = Z$. Let $a_C := (K_{X'} \cdot C)$. Since $K_{X'}$ is π-ample and $\pi(C)$ is a point, $a_C > 0$. By (4.4) and the projection formula, for any j,

$$(B'_j|_Z \cdot C) = (B'_j \cdot C) = \big((\pi^*(K_X + \Delta_j) - K_{X'}) \cdot C\big) = -(K_{X'} \cdot C) = -a_C.$$

By Lemma 4.3 applied to Z, C, x, and v, with $a = a_C$, there exists a real number $\epsilon > 0$ such that $v(B'_j|_Z) \geq \epsilon$ for any positive integer j. This contradicts (4.11) for $j > 1/\epsilon$. Thus $S \not\subset \mathrm{Exc}(\pi)$.

Since S is an arbitrary lc center of $(X', 0)$, no lc center of $(X', 0)$ is contained in $\mathrm{Exc}(\pi)$. By (4.10), $\mathrm{Exc}(\pi)$ is closed, so the generic point of S belongs to the open subset $\pi^{-1}(X^\circ)$, and $\pi^{-1}(X^\circ) \to X^\circ$ is an isomorphism by **Step 3**. □

4.3. **Images of lc centers.** In this subsection, we consider a finite surjective endomorphism f of X and a small birational morphism $\pi : X' \to X$ such that $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample, $(X', 0)$ is lc, and no lc center of $(X', 0)$ is contained in $\mathrm{Exc}(\pi)$. We show that the set $\mathcal{T}$ of images on X of the lc centers of $(X', 0)$ is finite, that $f(T) \in \mathcal{T}$ and $f^{-1}(T) \in \mathcal{T}$ for any $T \in \mathcal{T}$, and that for any integer $j \geq 1$, no lc center of $(X', 0)$ is contained in $\operatorname{Supp} \pi_*^{-1} R(f^j)$ (Proposition 4.5). Sections 5 and 6 use Proposition 4.5 for the canonical model of Construction-Proposition 4.2 and an int-amplified endomorphism (any polarized endomorphism is int-amplified by Lemma 2.12).

For pairs with an int-amplified endomorphism, the analogous statement follows from the fact that an extension of an lc place along a finite morphism is again an lc place (cf. [Yos26, Proposition 3.4]), and the argument there uses that the log canonical divisor is $\mathbb{Q}$-Cartier. Since K_X need not be $\mathbb{Q}$-Cartier here, we use instead the $\boldsymbol{b}$-divisor $\mathbf{K}_X + \mathbf{E}_X - \mathrm{Env}_X(K_X)$ of Definitions 2.14 and 2.15, which computes the log discrepancies of $(X', 0)$ at prime divisors exceptional over X and does not increase under pullback by a finite surjective morphism (cf. [Yos22, Proposition 4.11]). For the canonical model of Construction-Proposition 4.2, no lc center of $(X', 0)$ is contained in $\mathrm{Exc}(\pi)$ by Theorems 4.1 and 4.4.

Proposition 4.5. *Let X be a normal projective variety, $f : X \to X$ a finite surjective endomorphism, and $\pi : X' \to X$ a small birational morphism from a normal projective variety X'. Assume that $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample, $(X', 0)$ is lc, and no lc center of $(X', 0)$ is contained in* $\mathrm{Exc}(\pi)$. *Let $\mathcal{T}$ be the set of the images $\pi(S)$ of the lc centers S of $(X', 0)$. Then $\mathcal{T}$ is finite, and for any $T \in \mathcal{T}$, $f(T) \in \mathcal{T}$ and the reduced closed subset $f^{-1}(T)$ is a member of $\mathcal{T}$. Moreover, for any integer $j \geq 1$, no lc center of $(X', 0)$ is contained in* $\operatorname{Supp} \pi_*^{-1} R(f^j)$.

Proof. The proof is long, so we divide it into several steps. Let $f^* : \mathbb{C}(X) \to \mathbb{C}(X)$ be the homomorphism of function fields induced by f. For prime divisors F and E over X and a positive integer e, we write $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ if the restriction of ord_F to $\mathbb{C}(X)$ via f^* is $e \operatorname{ord}_E$.

Step 1. In this step, we show that for any lc place E of $(X', 0)$ and any prime divisor F over X such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ for some positive integer e, F is an lc place of $(X', 0)$.

Since π is small, π induces a bijection between the prime divisors on X' and the prime divisors on X. Thus a prime divisor over X is exceptional over X if and only if it is exceptional over X', and

$$\pi_* \mathcal{O}_{X'}(mK_{X'}) = \mathcal{O}_X(mK_X)$$

for any integer m. Since $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample, for any sufficiently divisible positive integer m, $mK_{X'}$ is Cartier and $\mathcal{O}_{X'}(mK_{X'})$ is π-generated. For such m, $\mathcal{O}_X(mK_X) \cdot \mathcal{O}_{X'} = \mathcal{O}_{X'}(mK_{X'})$, which is locally generated by a rational function ψ with $\operatorname{div} \psi = -mK_{X'}$. Thus $\operatorname{mult}_G \mathcal{O}_X(mK_X) = -m \operatorname{mult}_G \overline{K_{X'}}$ for any prime divisor G over X, that is,

$$\frac{1}{m} Z(\mathcal{O}_X(mK_X)) = \overline{K_{X'}}.$$

The limit in Definition 2.15 exists [BdFF12, Proposition 2.1], and a subsequence of a convergent sequence converges to the limit of the sequence, so we may compute it along sufficiently divisible m. Thus

$$\operatorname{Env}_X(K_X) = \overline{K_{X'}}. \tag{4.12}$$

Let E be a prime divisor exceptional over X. By (4.12) and Definition 2.1,

$$\operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X)) = \operatorname{mult}_E \mathbf{K}_X + 1 - \operatorname{mult}_E \overline{K_{X'}} = a(E, X', 0). \tag{4.13}$$

At any prime divisor on X, the coefficient of $\mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X)$ is 0 by Lemma 2.16(2).

For any $\varphi \in \mathbb{C}(X)^*$, any $\mathbb{Q}$-divisor D on X, and any positive integer m such that mD is integral,

$$\mathcal{O}_X(m(D + \operatorname{div} \varphi)) = \varphi^{-m} \mathcal{O}_X(mD).$$

Thus

$$\operatorname{Env}_X(D + \operatorname{div} \varphi) = \operatorname{Env}_X(D) + \overline{\operatorname{div} \varphi}.$$

Replacing K_X with $K_X + \operatorname{div} \varphi$ replaces $\mathbf{K}_X$ with $\mathbf{K}_X + \overline{\operatorname{div} \varphi}$. Therefore, the $\boldsymbol{b}$-divisor $\mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X)$ only depends on the linear equivalence class of K_X. By Definition-Proposition 2.10(1), $R(f) \geq 0$ and $f^* K_X + R(f) = K_X + \operatorname{div} \varphi$ for some $\varphi \in \mathbb{C}(X)^*$. Let $\mathbf{K} := \mathbf{K}_X + \overline{\operatorname{div} \varphi}$, the canonical $\boldsymbol{b}$-divisor whose trace on X is $f^* K_X + R(f)$. Then

$$\mathbf{K} + \mathbf{E}_X - \operatorname{Env}_X(f^* K_X + R(f)) = \mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X),$$

and Lemma 2.16(4) applied to f reads $\mathbf{K} + \mathbf{E}_X = f^*(\mathbf{K}_X + \mathbf{E}_X) + \mathbf{R}(f)$. By Lemma 2.16(1)(3), we obtain

$$\begin{aligned} \mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X) &= f^*(\mathbf{K}_X + \mathbf{E}_X) + \mathbf{R}(f) - \operatorname{Env}_X(f^* K_X + R(f)) \\ &\leq f^*(\mathbf{K}_X + \mathbf{E}_X) + \mathbf{R}(f) - \operatorname{Env}_X(f^* K_X) - \operatorname{Env}_X(R(f)) \\ &= f^*(\mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X)) - (\operatorname{Env}_X(R(f)) - \mathbf{R}(f)). \end{aligned}$$

Since $R(f) \geq 0$, Lemma 2.16(1)(2) implies that $\operatorname{Env}_X(R(f)) \geq \operatorname{Env}_X(0) = 0$ and that the trace of $\operatorname{Env}_X(R(f))$ on X is $R(f)$. Since $\mathbf{R}(f)$ has trace $R(f)$ on X and coefficient 0 at any prime divisor exceptional over X, we have $\operatorname{Env}_X(R(f)) - \mathbf{R}(f) \geq 0$. Thus, by the definition of f^* in Definition 2.14, for any prime divisors F and E over X such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ for some positive integer e,

$$\operatorname{mult}_F(\mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X)) \leq e \operatorname{mult}_E(\mathbf{K}_X + \mathbf{E}_X - \operatorname{Env}_X(K_X)) \tag{4.14}$$

(cf. [Yos22, Proposition 4.11]).

Now let E be an lc place of $(X', 0)$, and let F be a prime divisor over X such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ for some positive integer e. Since $a(G, X', 0) = 1$ for any prime divisor G on X', E is

exceptional over X', hence over X. Suppose that F is a prime divisor on X. Then $f(F)$ is a prime divisor on X and $\operatorname{ord}_F \circ f^* = e_F \operatorname{ord}_{f(F)}$, where e_F is the ramification index of f along F. Thus E is the prime divisor $f(F)$ on X, a contradiction. Hence F is exceptional over X. Since $(X', 0)$ is lc, (4.13) and (4.14) imply that

$$0 \leq a(F, X', 0) \leq ea(E, X', 0) = 0.$$

Thus F is an lc place of $(X', 0)$.

Step 2. In this step, we show that for any prime divisor E over X, the irreducible components of $f^{-1}(\operatorname{center}_X E)$ are exactly the subvarieties $\operatorname{center}_X F$, where F runs over the prime divisors over X such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ for some positive integer e, and that f maps any of them onto $\operatorname{center}_X E$.

Let F be a prime divisor over X such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ for some positive integer e. Since f is proper, $f(\operatorname{center}_X F) = \operatorname{center}_X E$. Since f is finite, $\dim \operatorname{center}_X F = \dim \operatorname{center}_X E$, and $\dim T_0 \leq \dim \operatorname{center}_X E$ for any irreducible component T_0 of $f^{-1}(\operatorname{center}_X E)$. Thus $\operatorname{center}_X F$ is an irreducible component of $f^{-1}(\operatorname{center}_X E)$. Conversely, let u be a real valuation of $\mathbb{C}(X)$ such that $u \circ f^* = \operatorname{ord}_E$. Since f is finite, u has a center on X, $u(\mathbb{C}(X)^*) = \frac{1}{e}\mathbb{Z}$ for a positive integer e, and the residue field of u has transcendence degree $\dim X - 1$, so $eu = \operatorname{ord}_F$ for a prime divisor F over X [KM98, Lemma 2.45], and $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$. Any irreducible component of $f^{-1}(\operatorname{center}_X E)$ is the center on X of such a valuation u [Yos26, Lemma 3.2].

Step 3. In this step, we show that $\mathcal{T}$ is finite, that no member of $\mathcal{T}$ is contained in $N := \pi(\operatorname{Exc}(\pi))$, and that for any $T \in \mathcal{T}$, $f(T) \in \mathcal{T}$ and $f^{-1}(T) \in \mathcal{T}$.

If $\mu : W \to X'$ is a log resolution of $(X', 0)$ and $K_W + B_W = \mu^* K_{X'}$, then any lc center of $(X', 0)$ is the image of a stratum of the components of B_W with coefficient one (cf. [KM98, Corollary 2.31]). Thus $(X', 0)$ has only finitely many lc centers, and $\mathcal{T}$ is finite. Since π is a local isomorphism at any point of $X' \setminus \operatorname{Exc}(\pi)$ and the fibers of π are connected [Har77, Corollary III.11.4], we have $\pi^{-1}(N) = \operatorname{Exc}(\pi)$, and π is an isomorphism over $X \setminus N$. For any lc center S of $(X', 0)$, $S \not\subset \operatorname{Exc}(\pi) = \pi^{-1}(N)$, so $\pi(S) \not\subset N$. Thus no member of $\mathcal{T}$ is contained in N.

Let $T \in \mathcal{T}$. Then $T = \pi(S)$ for an lc center S of $(X', 0)$, and $S = \operatorname{center}_{X'} E$ for an lc place E of $(X', 0)$, so $\operatorname{center}_X E = T$. Let T'' be an irreducible component of $f^{-1}(T)$. By **Step 2**, $f(T'') = T$ and $T'' = \operatorname{center}_X F$ for a prime divisor F over X such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ for some positive integer e. By **Step 1**, F is an lc place of $(X', 0)$, so $T'' = \pi(\operatorname{center}_{X'} F) \in \mathcal{T}$.

For any $T \in \mathcal{T}$, choose an irreducible component $g(T)$ of $f^{-1}(T)$. Then $g(T) \in \mathcal{T}$ and $f(g(T)) = T$, so $g : \mathcal{T} \to \mathcal{T}$ is injective. Since $\mathcal{T}$ is finite, g is bijective. Thus any $T' \in \mathcal{T}$ is $g(T)$ for some $T \in \mathcal{T}$, and $f(T') = T \in \mathcal{T}$. Therefore, f induces a map $\mathcal{T} \to \mathcal{T}$, which is the inverse of g. For any $T \in \mathcal{T}$ and any irreducible component T'' of $f^{-1}(T)$, we have $T'' \in \mathcal{T}$ and $f(T'') = T = f(g(T))$, so $T'' = g(T)$. Thus the reduced closed subset $f^{-1}(T)$ is $g(T) \in \mathcal{T}$. In particular, f is a bijection on $\mathcal{T}$, that is, f permutes $\mathcal{T}$.

Step 4. In this step, we show that no member of $\mathcal{T}$ is contained in $\operatorname{Supp} R(f^j)$ for any integer $j \geq 1$, and conclude the proof.

Recall that $N = \pi(\operatorname{Exc}(\pi))$ and that π is an isomorphism over $X \setminus N$. In particular, K_X is $\mathbb{Q}$-Cartier on $X \setminus N$. Since f is finite, $U := X \setminus (N \cup f(N))$ is open in X. Since $f^{-1}(U) \subset X \setminus N$, the restriction $f^{-1}(U) \to U$ of f is a finite surjective morphism between normal varieties such that $K_{f^{-1}(U)}$ and K_U are $\mathbb{Q}$-Cartier. In particular, the restriction of $R(f)$ to $f^{-1}(U)$ is $\mathbb{Q}$-Cartier.

Fix $T' \in \mathcal{T}$ and let $T := f(T')$. By **Step 3**, $T \in \mathcal{T}$ and $f^{-1}(T) = T'$, so $\dim T' = \dim T$. By **Step 3**, $T \not\subset N$. Suppose that $T \subset f(N)$. Then $f(N \cap T') = T$, so $\dim(N \cap T') \geq \dim T = \dim T'$. Thus $T' \subset N$, which contradicts **Step 3**. Hence $T \not\subset N \cup f(N)$, so the generic point of T is in U. Since $f(T') = T$, the generic point of T' is in $f^{-1}(U)$.

Let E be an lc place of $(X', 0)$ with $\operatorname{center}_X E = T$. Since $f^{-1}(T) = T'$, **Step 2** implies that there exist a prime divisor F over X and a positive integer e such that $\operatorname{ord}_F \circ f^* = e \operatorname{ord}_E$ and $\operatorname{center}_X F = T'$. By **Step 1**, F is an lc place of $(X', 0)$. Since π is an isomorphism over $X \setminus N$,

which contains U and $f^{-1}(U)$, and log discrepancies can be computed over any open subset intersecting the center, we have $a(E,U,0)=a(E,X',0)=0$ and $a(F,f^{-1}(U),0)=a(F,X',0)=0$. By Definition-Proposition 2.10(4) applied to $f^{-1}(U)\to U$ with $B=0$,

$$a(F,f^{-1}(U),0)+\operatorname{mult}_F R(f)=a(F,f^{-1}(U),-R(f))=ea(E,U,0), \tag{4.15}$$

where the first equality holds because $R(f)$ is $\mathbb{Q}$-Cartier on $f^{-1}(U)$. Thus $\operatorname{mult}_F R(f)=0$. Suppose that $T'\subset \operatorname{Supp} R(f)$. Since $R(f)\geq 0$, a local equation at the generic point of T' of a Cartier multiple of $R(f)$ belongs to the ideal of $T'=\operatorname{center}_X F$, on which ord_F is positive. Thus $\operatorname{mult}_F R(f)>0$, a contradiction. Since $T'\in\mathcal{T}$ is arbitrary, no member of $\mathcal{T}$ is contained in $\operatorname{Supp} R(f)$.

Let $j\geq 1$ be an integer. By Definition-Proposition 2.10(3) and induction on j,

$$R(f^j)=\sum_{i=0}^{j-1}(f^i)^*R(f),$$

where $f^0:=\mathrm{id}_X$. By the definition of the pullback in Definition-Proposition 2.10,

$$\operatorname{Supp}(f^i)^*R(f)\subset (f^i)^{-1}(\operatorname{Supp} R(f)).$$

Suppose that $T'\subset \operatorname{Supp} R(f^j)$ for some $T'\in\mathcal{T}$. Since $R(f)\geq 0$ and T' is irreducible, $T'\subset\operatorname{Supp}(f^i)^*R(f)$ for some $0\leq i\leq j-1$, so $f^i(T')\subset\operatorname{Supp} R(f)$. By **Step 3**, $f^i(T')\in\mathcal{T}$, a contradiction. Thus no member of $\mathcal{T}$ is contained in $\operatorname{Supp} R(f^j)$.

Finally, let S be an lc center of $(X',0)$ and $T:=\pi(S)$. Suppose that $S\subset\operatorname{Supp}\pi_*^{-1}R(f^j)$. Since π is an isomorphism over $X\backslash N$,

$$\operatorname{Supp}\pi_*^{-1}R(f^j)\backslash\operatorname{Exc}(\pi)=\pi^{-1}(\operatorname{Supp} R(f^j)\backslash N).$$

Since $S\not\subset\operatorname{Exc}(\pi)$, $T\backslash N$ is a dense open subset of T contained in $\operatorname{Supp} R(f^j)$, so $T\subset\operatorname{Supp} R(f^j)$. This contradicts $T\in\mathcal{T}$. Thus no lc center of $(X',0)$ is contained in $\operatorname{Supp}\pi_*^{-1}R(f^j)$. □

5. A small model for the pushforwards of the ramification divisor

The goal of this section is to prove Construction-Proposition 5.2. For an int-amplified endomorphism h of a normal projective variety X, we construct a small birational morphism $\pi:X'\to X$, factoring through the canonical model of X, such that the strict transforms D'_j of the $\mathbb{Q}$-divisors D_j of Construction-Proposition 3.1 are $\mathbb{Q}$-Cartier, and the pair formed by X' and any partial sum of the D'_j is lc. In the terminology of Definition 7.1, $D'_\bullet$ is an lc anti-canonical series on $(X',0)$. From Section 7 on, we work with $D'_\bullet$ and no longer use h.

Outside klt type, a small $\mathbb{Q}$-factorial modification need not exist. For example, the affine cone over an elliptic curve in a projectively normal embedding is lc, not of klt type, and not $\mathbb{Q}$-factorial at the vertex [Har77, Exercise II.6.3], while a small proper birational morphism onto a normal surface is an isomorphism, so the cone has no small $\mathbb{Q}$-factorial modification. In our situation, we only need finitely many divisors, namely $D_1,\dots,D_\ell$, to become $\mathbb{Q}$-Cartier. Lemma 5.1 treats each of them by one application of Theorem 2.18, under the condition that its support contains no lc center. In the proof of Construction-Proposition 5.2, we check the condition for the strict transforms of the D_j on the canonical model of X using Proposition 4.5.

Theorem 2.18 for $\langle X,\epsilon D_j\rangle$ only makes the sum of the canonical divisor and the strict transform of ϵD_j $\mathbb{Q}$-Cartier, while we need the canonical divisor and the strict transforms of $D_1,\dots,D_\ell$ to be $\mathbb{Q}$-Cartier separately on one model. Thus we first pass to the canonical model of X, on which the canonical divisor is $\mathbb{Q}$-Cartier and the pair with zero boundary is lc, and then make $D_1,\dots,D_\ell$ $\mathbb{Q}$-Cartier one at a time by Lemma 5.1.

Lemma 5.1. *Let X be a normal projective variety and $D_1,\dots,D_k\geq 0$ $\mathbb{Q}$-divisors on X. Assume that K_X is $\mathbb{Q}$-Cartier, $(X,0)$ is lc, and no lc center of $(X,0)$ is contained in $\operatorname{Supp} D_i$ for any i. Then there exists a small birational morphism $\pi:X'\to X$ from a normal projective variety X' such that $K_{X'}=\pi^*K_X$, $(X',0)$ is lc, and $\pi_*^{-1}D_i$ is $\mathbb{Q}$-Cartier for any i. Moreover, for any ample*

Cartier divisor A on X, there exist rational numbers $c_1, \dots, c_k \geq 0$ such that $\pi^*A + \sum_{i=1}^{k} c_i \pi_*^{-1} D_i$ is ample.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we treat one divisor. Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X such that no lc center of $(X, 0)$ is contained in $\operatorname{Supp} D$. We show that there exists a small birational morphism $\mu : Y \to X$ from a normal projective variety Y such that $K_Y = \mu^* K_X$, $(Y, 0)$ is lc, and $\mu_*^{-1} D$ is $\mathbb{Q}$-Cartier and μ-ample.

If $D = 0$, we take μ to be the identity of X. Assume that $D \neq 0$. The support of D and the conclusion on $\mu_*^{-1} D$ do not change when D is replaced with a positive multiple. Possibly replacing D with a positive multiple, we may assume that D has integer coefficients. Then $\mathcal{O}_X(-D)$ is a coherent ideal sheaf of $\mathcal{O}_X$. Let H be an ample Cartier divisor on X. Since H is ample, there exists a positive integer b such that $\mathcal{O}_X(-D) \otimes \mathcal{O}_X(bH)$ is globally generated.

If $f : W \to X$ is a log resolution of $(X, 0)$ and $K_W + B_W = f^* K_X$, then any lc center of $(X, 0)$ is the image of a stratum of the components of B_W with coefficient one (cf. [KM98, Corollary 2.31]). Thus $(X, 0)$ has only finitely many lc centers. For any lc center T of $(X, 0)$, we have $T \not\subset \operatorname{Supp} D$, so $\mathcal{O}_X(-D) \otimes \mathcal{O}_X(bH)$ coincides with $\mathcal{O}_X(bH)$ at the generic point of T. Since $\mathcal{O}_X(-D) \otimes \mathcal{O}_X(bH)$ is globally generated, the sections in

$$H^0(X, \mathcal{O}_X(-D) \otimes \mathcal{O}_X(bH)) \subset H^0(X, \mathcal{O}_X(bH))$$

vanishing at the generic point of a given lc center form a proper linear subspace. Thus there exists $s \in H^0(X, \mathcal{O}_X(-D) \otimes \mathcal{O}_X(bH))$ that does not vanish at the generic point of any lc center of $(X, 0)$. Let C be the zero divisor of s as a section of $\mathcal{O}_X(bH)$. Then $C \geq 0$ is a Cartier divisor, $C \sim bH$, $C \geq D$, and no lc center of $(X, 0)$ is contained in $\operatorname{Supp} C$.

We show that $(X, \epsilon C)$ is lc for some rational number $\epsilon > 0$. Let $f : W \to X$ be a log resolution of (X, C), and write $K_W + B_W = f^* K_X$. Since $(X, 0)$ is lc, the coefficients of B_W are at most one. Let P be a prime divisor on W such that $\operatorname{mult}_P f^* C > 0$. Then $f(P) \subset \operatorname{Supp} C$, so $f(P)$ is not an lc center of $(X, 0)$, and

$$a(P, X, 0) = 1 - \operatorname{mult}_P B_W > 0.$$

Choose a rational number $\epsilon > 0$ such that $\epsilon \operatorname{mult}_P f^* C \leq a(P, X, 0)$ for any of the finitely many prime divisors P on W with $\operatorname{mult}_P f^* C > 0$. Then

$$K_W + B_W + \epsilon f^* C = f^*(K_X + \epsilon C),$$

the coefficients of $B_W + \epsilon f^* C$ are at most one, and $\operatorname{Supp}(B_W + f^* C)$ is simple normal crossing. Thus $(X, \epsilon C)$ is lc [KM98, Lemma 2.30, Corollary 2.31(3)]. In particular, the coefficients of ϵC, and hence those of $\epsilon D \leq \epsilon C$, are at most one.

We show that $\langle X, \epsilon D \rangle$ is pseudo-lc. Let E be a prime divisor over X, and let U be an affine open subset of X such that $U \cap \operatorname{center}_X E \neq \emptyset$. Then $\epsilon(C - D)|_U \geq 0$, and

$$K_U + \epsilon D|_U + \epsilon(C - D)|_U = (K_X + \epsilon C)|_U$$

is $\mathbb{Q}$-Cartier. Thus

$$\alpha(E, X, \epsilon D) \geq a(E, U, \epsilon C|_U) = a(E, X, \epsilon C) \geq 0.$$

By Definition 2.17, $\langle X, \epsilon D \rangle$ is pseudo-lc.

By Theorem 2.18, there exists a small birational morphism $\mu : Y \to X$ from a normal variety Y such that $K_Y + \epsilon \mu_*^{-1} D$ is $\mathbb{R}$-Cartier and μ-ample, and $(Y, \epsilon \mu_*^{-1} D)$ is lc. By Lemma 2.5, $K_Y + \epsilon \mu_*^{-1} D$ is $\mathbb{Q}$-Cartier. Since μ is projective by Theorem 2.18 and X is projective, Y is projective. Since K_X is $\mathbb{Q}$-Cartier, Lemma 2.19 applied to μ implies $K_Y = \mu^* K_X$. Thus K_Y is $\mathbb{Q}$-Cartier, and $(Y, 0)$ is lc since $a(E, Y, 0) = a(E, X, 0)$ for any prime divisor E over X [KM98, Lemma 2.30]. Moreover,

$$\mu_*^{-1} D = \epsilon^{-1}\big((K_Y + \epsilon \mu_*^{-1} D) - \mu^* K_X\big)$$

is $\mathbb{Q}$-Cartier and μ-ample.

Step 2. In this step, we prove the lemma by induction on k. For $k = 0$, we take π to be the identity of X, and the moreover part holds since A is ample.

Let $k \geq 1$. By the induction hypothesis applied to $D_1, \dots, D_{k-1}$, there exists a small birational morphism $\pi_1 : X_1 \to X$ from a normal projective variety X_1 such that $K_{X_1} = \pi_1^* K_X$, $(X_1, 0)$ is lc, $\pi_{1*}^{-1} D_i$ is $\mathbb{Q}$-Cartier for any $i \leq k-1$, and for any ample Cartier divisor A on X there exist rational numbers $c_1, \dots, c_{k-1} \geq 0$ such that $\pi_1^* A + \sum_{i=1}^{k-1} c_i \pi_{1*}^{-1} D_i$ is ample.

We show that no lc center of $(X_1, 0)$ is contained in $\operatorname{Supp} \pi_{1*}^{-1} D_k$. Let S be an lc center of $(X_1, 0)$. Then $S = \operatorname{center}_{X_1} E$ for some prime divisor E over X_1 with $a(E, X_1, 0) = 0$. Since $K_{X_1} = \pi_1^* K_X$, we have $a(E, X, 0) = a(E, X_1, 0) = 0$ [KM98, Lemma 2.30], so $\pi_1(S) = \operatorname{center}_X E$ is an lc center of $(X, 0)$. If $S \subset \operatorname{Supp} \pi_{1*}^{-1} D_k$, then $\pi_1(S) \subset \pi_1(\operatorname{Supp} \pi_{1*}^{-1} D_k) = \operatorname{Supp} D_k$, which contradicts the assumption.

By **Step 1** applied to X_1 and $\pi_{1*}^{-1} D_k$, there exists a small birational morphism $\mu : X' \to X_1$ from a normal projective variety X' such that $K_{X'} = \mu^* K_{X_1}$, $(X', 0)$ is lc, and $\pi_*^{-1} D_k = \mu_*^{-1} \pi_{1*}^{-1} D_k$ is $\mathbb{Q}$-Cartier and μ-ample, where $\pi := \pi_1 \circ \mu$. The morphism π is small birational, since a prime divisor on X' contracted by π is contracted by μ or is mapped by μ onto a prime divisor contracted by π_1. We have $K_{X'} = \mu^* \pi_1^* K_X = \pi^* K_X$. For any $i \leq k-1$, $\pi_{1*}^{-1} D_i$ is $\mathbb{Q}$-Cartier, so Lemma 2.19 applied to μ implies that

$$\pi_*^{-1} D_i = \mu^* \pi_{1*}^{-1} D_i$$

is $\mathbb{Q}$-Cartier for any $i \leq k-1$.

Let A be an ample Cartier divisor on X. By the induction hypothesis, there exist rational numbers $c_1, \dots, c_{k-1} \geq 0$ such that

$$A_1 := \pi_1^* A + \sum_{i=1}^{k-1} c_i \pi_{1*}^{-1} D_i$$

is ample. Choose a positive integer m such that mA_1 and $m\pi_*^{-1} D_k$ are Cartier. Since $m\pi_*^{-1} D_k$ is μ-ample, there exists a positive integer b such that $m\pi_*^{-1} D_k + b\mu^*(mA_1)$ is ample [KM98, Proposition 1.45]. Let $c_k := b^{-1}$. Since $\mu^* \pi_1^* A = \pi^* A$ and $\mu^* \pi_{1*}^{-1} D_i = \pi_*^{-1} D_i$ for any $i \leq k-1$,

$$\pi^* A + \sum_{i=1}^{k} c_i \pi_*^{-1} D_i = \mu^* A_1 + b^{-1} \pi_*^{-1} D_k = (mb)^{-1} \big(b\mu^*(mA_1) + m\pi_*^{-1} D_k \big)$$

is ample. □

The partial sums of the D_j need not be anti-canonical, so their log discrepancies cannot be compared through h directly. In **Steps 3 and 4** of the proof of Construction-Proposition 5.2, we bound each partial sum above by $\mathbb{Q}$-divisors $\Delta_k \geq 0$ with $K_X + \Delta_k \sim_{\mathbb{Q}} 0$ and $\Delta_{k+1} = r^{-1} h_*(R(h) + \Delta_k)$. For such divisors and a prime divisor E over X, $a(E, X, \Delta_{k+1})$ is the weighted average, with weights $e_i[\mathbb{C}(E_i) : \mathbb{C}(E)]/r$, of the log discrepancies $a(e_i^{-1} \operatorname{ord}_{E_i}, X, \Delta_k) = e_i^{-1} a(E_i, X, \Delta_k)$ at the extensions $e_i^{-1} \operatorname{ord}_{E_i}$ of ord_E, where E_i runs over the prime divisors over X with $\operatorname{ord}_{E_i} \circ h^* = e_i \operatorname{ord}_E$ (cf. [Yos26, Lemma 3.3, (3.6)] for pairs), and Lemma 3.4 shows that the error term $\operatorname{mult}_E(r^{-k}(h^k)_* L)$, for the Cartier divisor $L \geq 0$ of **Step 3**, tends to zero.

Construction-Proposition 5.2. Let X be a normal projective variety of dimension $d \geq 1$, $h : X \to X$ an int-amplified endomorphism of degree r, $D_j := r^{-j}(h^j)_* R(h)$ for $j \geq 1$, and ℓ and $a_{j,N}$ as in Construction-Proposition 3.1. Then there exists a small birational morphism $\pi : X' \to X$ from a normal projective variety X' such that the following hold.

(1) $K_{X'}$ is $\mathbb{Q}$-Cartier and $(X', 0)$ is lc.
(2) For any $j \geq 1$, $D'_j := \pi_*^{-1} D_j$ is $\mathbb{Q}$-Cartier, and $(X', \sum_{j=1}^{N} D'_j)$ is lc for any $N \geq 1$.
(3) $K_{X'} + \sum_{j=1}^{N} D'_j + \sum_{j=1}^{\ell} a_{j,N} D'_j \sim_{\mathbb{Q}} 0$ for any $N \geq \ell$.

If $R(h) \neq 0$, then for any $N \geq \ell$ the boundary in (3) is not bounded above by any partial sum of the D'_j. Let H' be an ample Cartier divisor on X' and $M \geq 1$. Since $R(h) \geq 0$ and $R(h) \neq 0$,

we have $D'_j \geq 0$, $D'_j \neq 0$, and hence $(D'_j \cdot H'^{d-1}) > 0$ for any j. By (3) and $\lim_{N' \to +\infty} a_{j,N'} = 0$ for any $j \leq \ell$ (Construction-Proposition 3.1),

$$(-K_{X'} \cdot H'^{d-1}) = \lim_{N' \to +\infty} \sum_{j=1}^{N'} (D'_j \cdot H'^{d-1}) > \sum_{j=1}^{M} (D'_j \cdot H'^{d-1}),$$

while the intersection number of the boundary in (3) with H'^{d-1} is $(-K_{X'} \cdot H'^{d-1})$, since the boundary is $\mathbb{Q}$-linearly equivalent to $-K_{X'}$. The lc property of such boundaries is the subject of Theorem 12.1.

Proof of Construction-Proposition 5.2. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that no lc center of the canonical model of X is contained in the support of the strict transform of any D_j, and that the class of any D_j in $\mathrm{Cl}(X)_{\mathbb{Q}}$ is a linear combination of the classes of $D_1, \ldots, D_\ell$.

Let X_0 be the canonical model of X, that is, the variety denoted by X' in Construction-Proposition 4.2, and let $\pi_0 : X_0 \to X$ be the structure morphism. By Construction-Proposition 4.2, π_0 is a small birational morphism, X_0 is a normal projective variety, K_{X_0} is $\mathbb{Q}$-Cartier and π_0-ample, and $(X_0, 0)$ is lc. By Theorem 4.1, X is pseudo-lc. Thus, by Theorem 4.4, no lc center of $(X_0, 0)$ is contained in $\mathrm{Exc}(\pi_0)$, and π_0 is an isomorphism on a neighborhood of the generic point of any lc center of $(X_0, 0)$. By Lemma 2.12, h is finite, so h^j is a finite surjective endomorphism of X for any $j \geq 1$. Let $\mathcal{T}$ be the set of the images $\pi_0(S)$ of the lc centers S of $(X_0, 0)$. By Proposition 4.5 applied to π_0 and h^j for $j \geq 1$, $\mathcal{T}$ is finite, the reduced closed subset $(h^j)^{-1}(T)$ is a member of $\mathcal{T}$ for any $T \in \mathcal{T}$ and any $j \geq 1$, and no lc center of $(X_0, 0)$ is contained in $\operatorname{Supp} \pi_{0*}^{-1} R(h)$.

Fix $j \geq 1$. We have $R(h) \geq 0$ by Definition-Proposition 2.10(1), and the pushforward of a prime divisor P on X under h^j is a positive multiple of $h^j(P)$. Thus $D_j \geq 0$ and

$$\operatorname{Supp} D_j = h^j(\operatorname{Supp} R(h)).$$

Suppose that $T \subset h^j(\operatorname{Supp} R(h))$ for some $T \in \mathcal{T}$, and let $T' := (h^j)^{-1}(T)$. Then $T' \in \mathcal{T}$, so T' is irreducible, and $\dim T' = \dim T$ since h^j is finite and surjective. Since

$$h^j\big(\operatorname{Supp} R(h) \cap T'\big) = h^j(\operatorname{Supp} R(h)) \cap T = T,$$

some irreducible component V of $\operatorname{Supp} R(h) \cap T'$ satisfies $h^j(V) = T$. Since h^j is finite, $\dim V = \dim T = \dim T'$. Thus $V = T'$, and $T' \subset \operatorname{Supp} R(h)$. Let S' be an lc center of $(X_0, 0)$ such that $\pi_0(S') = T'$. Since π_0 is an isomorphism on a neighborhood of the generic point η of S' and $\pi_0(\eta) \in \operatorname{Supp} R(h)$, we have $\eta \in \operatorname{Supp} \pi_{0*}^{-1} R(h)$. Thus $S' \subset \operatorname{Supp} \pi_{0*}^{-1} R(h)$, a contradiction. Therefore no member of $\mathcal{T}$ is contained in $\operatorname{Supp} D_j$. If an lc center S of $(X_0, 0)$ is contained in $\operatorname{Supp} \pi_{0*}^{-1} D_j$, then

$$\pi_0(S) \subset \pi_0(\operatorname{Supp} \pi_{0*}^{-1} D_j) = \operatorname{Supp} D_j$$

with $\pi_0(S) \in \mathcal{T}$, which is impossible. Thus no lc center of $(X_0, 0)$ is contained in $\operatorname{Supp} \pi_{0*}^{-1} D_j$.

Since $(h^{j+1})_* = h_*(h^j)_*$, we have $D_{j+1} = r^{-1} h_* D_j$, so $[D_j] = (r^{-1} h_*)^{j-1} [D_1]$ in $\mathrm{Cl}(X)_{\mathbb{Q}}$ for any $j \geq 1$. By Construction-Proposition 3.1, there exists a monic polynomial $p(t) = t^\ell + \sum_{i=0}^{\ell-1} p_i t^i \in \mathbb{Q}[t]$ such that $p(r^{-1} h_*) = 0$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$. Applying $(r^{-1} h_*)^{j-\ell-1}$ to $p(r^{-1} h_*)[D_1] = 0$, we obtain

$$[D_j] = -\sum_{i=0}^{\ell-1} p_i [D_{j-\ell+i}]$$

for any $j > \ell$. By induction on j, the class $[D_j]$ belongs to the $\mathbb{Q}$-span of $[D_1], \ldots, [D_\ell]$ in $\mathrm{Cl}(X)_{\mathbb{Q}}$ for any $j \geq 1$.

Step 2. In this step, we construct π and prove (1), (3), and the first half of (2).

By **Step 1**, Lemma 5.1 applied to X_0 and $\pi_{0*}^{-1} D_1, \ldots, \pi_{0*}^{-1} D_\ell$ implies that there exists a small birational morphism $g : X' \to X_0$ from a normal projective variety X' such that $K_{X'} = g^* K_{X_0}$

and $(X',0)$ is lc, and such that, for any $1\le j\le \ell$, $g_*^{-1}\pi_{0*}^{-1}D_j$ is $\mathbb{Q}$-Cartier. Let $\pi:=\pi_0\circ g$. Then π is a small birational morphism, and $K_{X'}=g^*K_{X_0}$ is $\mathbb{Q}$-Cartier. This proves (1). Moreover, $D'_j=\pi_*^{-1}D_j=g_*^{-1}\pi_{0*}^{-1}D_j$ is $\mathbb{Q}$-Cartier for any $1\le j\le \ell$.

Since π is small, X and X' contain isomorphic open subsets whose complements have codimension at least two, so π_*^{-1} induces an isomorphism $\mathrm{Cl}(X)_{\mathbb{Q}}\to \mathrm{Cl}(X')_{\mathbb{Q}}$ preserving $\mathbb{Q}$-linear equivalence (cf. [Har77, Proposition II.6.5(b)]). Let $j\ge 1$. By **Step 1**, there exist rational numbers $b_1,\dots,b_\ell$ such that $D_j\sim_{\mathbb{Q}}\sum_{i=1}^{\ell}b_iD_i$. Thus $D'_j\sim_{\mathbb{Q}}\sum_{i=1}^{\ell}b_iD'_i$, and D'_j is $\mathbb{Q}$-Cartier. Let $N\ge\ell$. By Construction-Proposition 3.1,

$$K_X+\sum_{j=1}^{N}D_j+\sum_{j=1}^{\ell}a_{j,N}D_j\sim_{\mathbb{Q}}0.$$

Since $K_{X'}=\pi_*^{-1}K_X$, applying π_*^{-1} to both sides implies (3).

Step 3. In this step, we construct $\mathbb{Q}$-divisors $\Delta_k\ge 0$ on X with $K_X+\Delta_k\sim_{\mathbb{Q}}0$ for $k\ge 0$ and bound their log discrepancies from below.

By Construction-Proposition 3.1, $\lim_{N\to+\infty}a_{j,N}=0$ for any $j\le\ell$. Thus there exists an integer $N_*\ge\ell$ such that $1+a_{j,N_*}>0$ for any $j\le\ell$. Let

$$\Delta_0:=\sum_{j=1}^{N_*}D_j+\sum_{j=1}^{\ell}a_{j,N_*}D_j=\sum_{j=1}^{\ell}(1+a_{j,N_*})D_j+\sum_{j=\ell+1}^{N_*}D_j.$$

Since $D_j\ge 0$ for any j, $\Delta_0\ge 0$, and $K_X+\Delta_0\sim_{\mathbb{Q}}0$ by Construction-Proposition 3.1. For any integer $k\ge 0$, let $D_{\le k}:=\sum_{j=1}^{k}D_j$, where $D_{\le 0}=0$, and let

$$\Delta_k:=D_{\le k}+r^{-k}(h^k)_*\Delta_0=D_{\le k}+\sum_{j=1}^{N_*}D_{j+k}+\sum_{j=1}^{\ell}a_{j,N_*}D_{j+k}.$$

Then $\Delta_k\ge 0$, and since $r^{-1}h_*R(h)=D_1$ and $r^{-1}h_*D_j=D_{j+1}$,

$$\Delta_{k+1}=r^{-1}h_*\big(R(h)+\Delta_k\big).$$

We show that $K_X+\Delta_k\sim_{\mathbb{Q}}0$ for any $k\ge 0$ by induction on k. Assume that $K_X+\Delta_k\sim_{\mathbb{Q}}0$. By Definition-Proposition 2.10(1), $K_X\sim h^*K_X+R(h)$, so

$$R(h)+\Delta_k\sim_{\mathbb{Q}}-h^*K_X+(K_X+\Delta_k)\sim_{\mathbb{Q}}-h^*K_X.$$

By Definition-Proposition 2.10(2), $h_*(R(h)+\Delta_k)\sim_{\mathbb{Q}}-h_*h^*K_X=-rK_X$, that is,

$$K_X+\Delta_{k+1}\sim_{\mathbb{Q}}0.$$

In particular, $K_X+\Delta_k$ is $\mathbb{Q}$-Cartier for any $k\ge 0$.

Fix a prime divisor E over X. Let E_i run over the prime divisors over X such that $\mathrm{ord}_{E_i}\circ h^*=e_i\,\mathrm{ord}_E$ on $\mathbb{C}(X)$ for some positive integer e_i. We show that

$$\sum_i e_i[\mathbb{C}(E_i):\mathbb{C}(E)]=r. \tag{5.1}$$

Let $\varphi\in\mathbb{C}(X)$ be a rational function such that $\mathrm{ord}_E\varphi=1$. The divisor $h^*\operatorname{div}\varphi=\operatorname{div}(h^*\varphi)$ is principal, and $h_*h^*\operatorname{div}\varphi=r\operatorname{div}\varphi$ by Definition-Proposition 2.10(2). By Lemma 3.3(2),

$$r=\mathrm{mult}_E\big(h_*h^*\operatorname{div}\varphi\big)=\sum_i[\mathbb{C}(E_i):\mathbb{C}(E)]\,\mathrm{ord}_{E_i}(h^*\varphi)=\sum_i e_i[\mathbb{C}(E_i):\mathbb{C}(E)].$$

Let $k\ge 0$. We show that

$$a(E,X,\Delta_{k+1})=\frac{1}{r}\sum_i[\mathbb{C}(E_i):\mathbb{C}(E)]a(E_i,X,\Delta_k). \tag{5.2}$$

By Definition-Proposition 2.10(4) applied to the pair (X, Δ_{k+1}), the sub-pair $(X, h^*\Delta_{k+1} - R(h))$ satisfies

$$a(E_i, X, h^*\Delta_{k+1} - R(h)) = e_i a(E, X, \Delta_{k+1})$$

for any i. Since $K_X \sim h^*K_X + R(h)$ by Definition-Proposition 2.10(1), the $\mathbb{Q}$-divisor

$$\Delta_k - h^*\Delta_{k+1} + R(h) \sim (K_X + \Delta_k) - h^*(K_X + \Delta_{k+1})$$

is $\mathbb{Q}$-Cartier, and by Definition-Proposition 2.10(2),

$$h_*\big(\Delta_k - h^*\Delta_{k+1} + R(h)\big) = h_*(R(h) + \Delta_k) - r\Delta_{k+1} = 0.$$

Thus, for any i,

$$a(E_i, X, \Delta_k) = e_i a(E, X, \Delta_{k+1}) - \operatorname{mult}_{E_i}\big(\Delta_k - h^*\Delta_{k+1} + R(h)\big).$$

Multiplying by $[\mathbb{C}(E_i) : \mathbb{C}(E)]$ and summing over i, Lemma 3.3(2) and (5.1) imply

$$\begin{aligned}\sum_i [\mathbb{C}(E_i) : \mathbb{C}(E)] a(E_i, X, \Delta_k) &= r a(E, X, \Delta_{k+1}) - \operatorname{mult}_E h_*\big(\Delta_k - h^*\Delta_{k+1} + R(h)\big) \\ &= r a(E, X, \Delta_{k+1}).\end{aligned}$$

This proves (5.2).

Since h is int-amplified, there exists an ample Cartier divisor H on X such that $h^*H - H$ is ample. We show that there exist a positive integer m and a Cartier divisor $L \geq 0$ with $L \sim mH$ such that $(X, \Delta_0 - L)$ is sub-lc. Let $f : W \to X$ be a log resolution of (X, Δ_0), and write $K_W + B_W = f^*(K_X + \Delta_0)$. Let $V \subset X$ be the union of the images of the finitely many prime divisors P on W such that $\operatorname{mult}_P B_W > 1$. Since $V \neq X$ and H is ample, there exists a positive integer m such that $\mathcal{I}_V \otimes \mathcal{O}_X(mH)$ is globally generated and non-zero, where $\mathcal{I}_V$ is the ideal sheaf of V. The divisor of a non-zero section of $\mathcal{I}_V \otimes \mathcal{O}_X(mH)$ is a Cartier divisor $L \geq 0$ with $L \sim mH$ and $V \subset \operatorname{Supp} L$. Then $\operatorname{mult}_P f^*L > 0$ for any prime divisor P on W such that $\operatorname{mult}_P B_W > 1$. Possibly replacing m and L with positive multiples, we may assume that $\operatorname{mult}_P(B_W - f^*L) \leq 1$ for any such P. Let G_W be the reduced divisor with $\operatorname{Supp} G_W = \operatorname{Supp} B_W$. Then $B_W - f^*L \leq G_W$, and (W, G_W) is lc [KM98, Corollary 2.31(3)]. Since W is smooth, $G_W - (B_W - f^*L) \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, so for any prime divisor F over X,

$$a(F, X, \Delta_0 - L) = a(F, W, B_W - f^*L) \geq a(F, W, G_W) \geq 0,$$

where the equality follows from $K_W + B_W - f^*L = f^*(K_X + \Delta_0 - L)$ [KM98, Lemma 2.30]. Thus $(X, \Delta_0 - L)$ is sub-lc.

By Lemma 3.3(1), $r^{-k}(h^k)_*L \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor for any $k \geq 0$. We show that

$$a(E, X, \Delta_k) \geq -\operatorname{mult}_E\big(r^{-k}(h^k)_*L\big) \tag{5.3}$$

for any $k \geq 0$ and any prime divisor E over X, by induction on k. For $k = 0$, $a(E, X, \Delta_0) + \operatorname{mult}_E L = a(E, X, \Delta_0 - L) \geq 0$. Assume that (5.3) holds for k and any prime divisor over X. Let E_i run over the prime divisors over X such that $\operatorname{ord}_{E_i} \circ h^* = e_i \operatorname{ord}_E$ on $\mathbb{C}(X)$ for some positive integer e_i. Then (5.2) and Lemma 3.3(2) imply

$$a(E, X, \Delta_{k+1}) \geq -\frac{1}{r}\sum_i [\mathbb{C}(E_i) : \mathbb{C}(E)] \operatorname{mult}_{E_i}\big(r^{-k}(h^k)_*L\big) = -\operatorname{mult}_E\big(r^{-k-1}(h^{k+1})_*L\big).$$

This proves (5.3).

Step 4. In this step, we prove that $(X', D'_{\leq N})$ is lc for any $N \geq 1$, where $D'_{\leq N} := \sum_{j=1}^N D'_j$. This finishes the proof of (2).

Fix $N \geq 1$ and a prime divisor E over X', which is also a prime divisor over X since π is birational, and let $k \geq N$. Recall from **Step 3** that $\Delta_k = D_{\leq k} + \sum_{j=1}^{N_*} D_{j+k} + \sum_{j=1}^{\ell} a_{j,N_*} D_{j+k}$.

Thus

$$\pi_*^{-1}\Delta_k - D'_{\le N} = \sum_{j=N+1}^{k} D'_j + \sum_{j=1}^{\ell}(1+a_{j,N_*})D'_{j+k} + \sum_{j=\ell+1}^{N_*} D'_{j+k} \ge 0,$$

since $1 + a_{j,N_*} > 0$, and this $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier by (2). Since $\Delta_k \ge 0$ and $K_X + \Delta_k \sim_{\mathbb{Q}} 0$, Lemma 2.19 implies $K_{X'} + \pi_*^{-1}\Delta_k = \pi^*(K_X + \Delta_k)$, so $a(E, X', \pi_*^{-1}\Delta_k) = a(E, X, \Delta_k)$ [KM98, Lemma 2.30]. By (5.3),

$$a(E, X', D'_{\le N}) \ge a(E, X', \pi_*^{-1}\Delta_k) = a(E, X, \Delta_k) \ge -\operatorname{mult}_E\big(r^{-k}(h^k)_*L\big).$$

Since $L \ge 0$ is Cartier and $L \sim mH$, Lemma 3.4 applied to L and $v = \operatorname{ord}_E$ implies

$$\lim_{k\to+\infty} \operatorname{mult}_E\big(r^{-k}(h^k)_*L\big) = 0.$$

The left-hand side $a(E, X', D'_{\le N})$ does not depend on k, so $a(E, X', D'_{\le N}) \ge 0$. Since E is an arbitrary prime divisor over X' and $K_{X'} + D'_{\le N}$ is $\mathbb{Q}$-Cartier by (1) and (2), $(X', D'_{\le N})$ is lc. □

6. Polarized endomorphisms

In this section, we prove Construction-Proposition 6.2 and Lemma 6.4, from which Section 13 deduces Theorem 1.5. For a q-polarized endomorphism f, we construct an equivariant small birational morphism $\pi : X' \to X$ for an iterate of f such that $K_{X'}$ is $\mathbb{Q}$-Cartier and $(X', 0)$ is lc (Construction-Proposition 6.2). In Subsection 6.3, we first prove the conclusion of Theorem 1.5 when K_X is $\mathbb{Q}$-Cartier (Proposition 6.3), and then deduce it whenever an iterate of f lifts to a higher model of X with $\mathbb{Q}$-Cartier canonical divisor (Lemma 6.4). This section depends on Sections 3–5, but not on Sections 7–12.

6.1. Lifting to small models.

In this subsection, we prove a criterion for lifting a finite surjective endomorphism along a small birational morphism. Lemma 6.1 reduces the lifting problem to relative ampleness: if the strict transforms of D and f^*D are $\mathbb{Q}$-Cartier and relatively ample, then the description of the small birational morphism as a relative Proj (Lemma 2.20) implies the existence of a regular lift.

Lemma 6.1. *Let X be a normal projective variety, $f : X \to X$ a finite surjective endomorphism, and $\pi : X' \to X$ a small birational morphism from a normal projective variety X'. Assume that there exists a $\mathbb{Q}$-divisor D on X such that $\pi_*^{-1}D$ and $\pi_*^{-1}f^*D$ are $\mathbb{Q}$-Cartier and π-ample. Then there exists a finite surjective endomorphism $g : X' \to X'$ such that $\pi \circ g = f \circ \pi$.*

Proof. Let n be a positive integer such that nD is a Weil divisor. Since $\pi_*^{-1}(nD) = n\pi_*^{-1}D$, $f^*(nD) = nf^*D$, and $\mathbb{Q}$-Cartierness and π-ampleness are preserved under multiplication by positive integers, possibly replacing D with nD, we may assume that D is a Weil divisor. Then f^*D is also a Weil divisor.

Let $U \subset X$ be a non-empty open subset over which π is an isomorphism. Over $f^{-1}(U)$, the second projection of the fiber product $X' \times_X X$ of π and f is an isomorphism. Thus $X' \times_X X$ has a unique irreducible component dominating the second factor X. Let X'' be the normalization of the unique irreducible component of $X' \times_X X$ dominating the second factor, with the reduced structure, and let $p : X'' \to X'$ and $\tau : X'' \to X$ be the morphisms induced by the first and the second projections respectively. Then the diagram

$$\begin{array}{ccc} X'' & \xrightarrow{p} & X' \\ {\scriptstyle\tau}\downarrow & & \downarrow{\scriptstyle\pi} \\ X & \xrightarrow{f} & X \end{array}$$

commutes, that is, $\pi \circ p = f \circ \tau$. The first projection $X' \times_X X \to X'$ is the base change of the finite morphism f, and normalization is finite [Har77, Exercise II.3.8]. Thus p is finite. Since $\dim X'' = \dim X'$ and X' is irreducible, p is surjective, and X'' is projective. Since X is normal

and the component of $X' \times_X X$ dominating the second factor is isomorphic to $f^{-1}(U)$ over $f^{-1}(U)$, τ is an isomorphism over $f^{-1}(U)$. In particular, τ is birational.

We show that τ is small. Let E be a prime divisor on X''. Since p is finite and surjective, $p(E)$ is a prime divisor on X'. Since π is small, $\pi(p(E))$ is a prime divisor on X. Since f is finite and $f(\tau(E)) = \pi(p(E))$, we have $\dim \tau(E) = \dim X - 1$. Thus τ contracts no prime divisor, that is, τ is small.

Next, we show that

$$\tau_*^{-1} f^* D = p^* \pi_*^{-1} D, \tag{6.1}$$

where p^* is the pullback of Definition-Proposition 2.10. Let E be a prime divisor on X'', and let $P := \tau(E)$ and $S := \pi(p(E))$. Then P and S are prime divisors on X and $f(P) = S$. Since τ and π are proper birational morphisms onto the normal variety X, they are isomorphisms over neighborhoods of the generic points of P and S respectively [Har77, Theorem I.6.2A and Theorem II.4.7]. We regard τ and π as equalities of function fields $\mathbb{C}(X'') = \mathbb{C}(X)$ and $\mathbb{C}(X') = \mathbb{C}(X)$. Then $\mathrm{ord}_E = \mathrm{ord}_P$, $\mathrm{ord}_{p(E)} = \mathrm{ord}_S$, and the field homomorphism $p^* : \mathbb{C}(X') \to \mathbb{C}(X'')$ becomes $f^* : \mathbb{C}(X) \to \mathbb{C}(X)$, since $\pi \circ p = f \circ \tau$. Thus the ramification index of p along E equals the ramification index e_P of f along P. Since τ is small, E is the strict transform of P, so the coefficients of E in both sides of (6.1) are equal to e_P times the coefficient of S in D. This proves (6.1).

Let l be a positive integer such that $l\pi_*^{-1}D$ is Cartier. For a prime divisor E on X'', let ϕ be a local equation of $l\pi_*^{-1}D$ near the generic point of $p(E)$. Then $\mathrm{ord}_E(\phi \circ p)$ is the ramification index of p along E times $\mathrm{ord}_{p(E)}(\phi)$. Thus $p^*\pi_*^{-1}D$ is the pullback of the π-ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\pi_*^{-1}D$ as a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. Since relative ampleness is preserved under base change, restriction to closed subschemes, and pullback by finite morphisms [Gro61, Proposition 4.6.13], the pullback of $l\pi_*^{-1}D$ to $X' \times_X X$ is relatively ample over the second factor X, the restriction of the pullback to the irreducible component dominating the second factor is relatively ample over X, and the pullback $lp^*\pi_*^{-1}D$ of $l\pi_*^{-1}D$ to X'' is τ-ample. By (6.1), $\tau_*^{-1}f^*D$ is $\mathbb{Q}$-Cartier and τ-ample.

Now Lemma 2.20, applied to τ and the Weil divisor f^*D, implies that the $\mathcal{O}_X$-algebra $\bigoplus_{m\geq 0} \mathcal{O}_X(mf^*D)$ is finitely generated and

$$X'' \cong \mathrm{Proj}_X \bigoplus_{m\geq 0} \mathcal{O}_X(mf^*D)$$

over X. Lemma 2.20, applied to π and f^*D, implies that $X' \cong \mathrm{Proj}_X \bigoplus_{m\geq 0} \mathcal{O}_X(mf^*D)$ over X. Thus there exists an isomorphism $\iota : X' \to X''$ such that $\tau \circ \iota = \pi$. Let $g := p \circ \iota$. Then g is finite and surjective, and

$$\pi \circ g = \pi \circ p \circ \iota = f \circ \tau \circ \iota = f \circ \pi. \qquad \square$$

6.2. **A polarized small model.** In this subsection, we construct, for a polarized endomorphism f, a small birational morphism $\pi : X' \to X$ such that $K_{X'}$ is $\mathbb{Q}$-Cartier, $(X', 0)$ is lc, and an iterate of f lifts to X'.

Lemma 6.1 does not apply directly to the canonical model of X, since the relative ampleness of the strict transform of the pullback of K_X by an iterate of f is only obtained in Corollary 1.6. Instead, in **Steps 2 and 3** of the proof of Construction-Proposition 6.2, we make the strict transforms on X' of all $\mathbb{Q}$-divisors on X whose classes belong to W simultaneously $\mathbb{Q}$-Cartier, where W is a $\mathbb{Q}$-linear subspace of $\mathrm{Cl}(X)_{\mathbb{Q}}$ of finite dimension containing the classes of K_X, of an ample Cartier divisor H with $f^*H \sim qH$, and of the ramification divisors of all iterates of f. In **Steps 4 and 5**, we find the iterate by a compactness argument, which uses that the integral powers of $q^{-1}\varphi$ are uniformly bounded, where φ is the linear map induced by f^* on the span V_W of the numerical classes of the strict transforms. They are uniformly bounded since $q^{-1}\varphi$ maps a closed convex cone $\mathcal{C}_W$ containing no line onto itself and fixes the class of π^*H, an interior point of $\mathcal{C}_W$. For the int-amplified endomorphism of Remark 1.7, the pullback has eigenvalues 2 and 3

on $N^1(\mathbb{P}^1\times\mathbb{P}^1)_{\mathbb{R}}$, so for any real number $c>0$, the integral powers of c^{-1} times the pullback are not uniformly bounded, and the compactness argument does not apply.

Construction-Proposition 6.2. Let X be a normal projective variety and $f: X\to X$ a q-polarized endomorphism. Then there exist a small birational morphism $\pi: X'\to X$ from a normal projective variety X' such that $K_{X'}$ is $\mathbb{Q}$-Cartier and $(X',0)$ is lc, a positive integer a, and a q^a-polarized endomorphism $g: X'\to X'$ such that $\pi\circ g = f^a\circ\pi$.

Proof. If $\dim X=0$, then X is a point, and $X':=X$, $\pi:=\mathrm{id}_X$, $a:=1$, and $g:=f$ satisfy the conclusion. From now on, we assume that $d:=\dim X\geq 1$. Let H be an ample Cartier divisor on X such that $f^*H\sim qH$. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a $\mathbb{Q}$-linear subspace $W\subset \mathrm{Cl}(X)_{\mathbb{Q}}$ with $\dim W<+\infty$ such that $f^*W=W$ and $q^{-d}f_*W=W$.

By Lemma 2.12, f is a finite int-amplified endomorphism of degree q^d. By Construction-Proposition 3.1 applied to f, there exists a monic polynomial $t^\ell+\sum_{i=0}^{\ell-1}b_it^i\in\mathbb{Q}[t]$ of degree $\ell\geq 1$ such that $b_0\neq 0$ and

$$(q^{-d}f_*)^\ell+\sum_{i=0}^{\ell-1}b_i(q^{-d}f_*)^i=0$$

on $\mathrm{Cl}(X)_{\mathbb{Q}}$, and f^* is the inverse of $q^{-d}f_*$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$. Composing with $(f^*)^\ell$, we obtain

$$1+\sum_{i=0}^{\ell-1}b_i(f^*)^{\ell-i}=0 \tag{6.2}$$

on $\mathrm{Cl}(X)_{\mathbb{Q}}$. Since $b_0\neq 0$, (6.2) expresses $(f^*)^\ell$ as a $\mathbb{Q}$-linear combination of $1, f^*, \dots, (f^*)^{\ell-1}$.

Let $\kappa:=[K_X]\in\mathrm{Cl}(X)_{\mathbb{Q}}$, and let W be the $\mathbb{Q}$-linear span of $\kappa, f^*\kappa, \dots, (f^*)^{\ell-1}\kappa$ and $[H]$. By (6.2) and $f^*[H]=q[H]$, $f^*W\subset W$. Since f^* is injective on $\mathrm{Cl}(X)_{\mathbb{Q}}$ and $\dim W<+\infty$, $f^*W=W$, and hence $q^{-d}f_*W=W$. In particular, $(f^*)^j\kappa\in W$ for any integer $j\geq 0$.

Let $j\geq 2$ be an integer and E a prime divisor on X, and let e_E be the ramification index of f along E. For any $\phi\in\mathbb{C}(X)^*$,

$$\mathrm{ord}_E(\phi\circ f^j)=e_E\,\mathrm{ord}_{f(E)}(\phi\circ f^{j-1}).$$

Thus the ramification index of f^j along E is e_E times the ramification index of f^{j-1} along $f(E)$, and by induction on j and the definition of the pullback in Definition-Proposition 2.10, $(f^j)^*=(f^*)^j$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$ for any integer $j\geq 1$. By Definition-Proposition 2.10(1) applied to f^j,

$$[R(f^j)]=\kappa-(f^*)^j\kappa \quad \text{for any integer } j\geq 1. \tag{6.3}$$

By (6.3), W is the $\mathbb{Q}$-linear span of κ, $[H]$, and $[R(f^i)]$ for $1\leq i\leq \ell$.

Step 2. In this step, we construct $\pi: X'\to X$.

Since f is int-amplified, X is pseudo-lc (Theorem 4.1). Let $\pi_0: X_0\to X$ be the canonical model of X (Construction-Proposition 4.2). Then K_{X_0} is $\mathbb{Q}$-Cartier and π_0-ample, $(X_0,0)$ is lc, and no lc center of $(X_0,0)$ is contained in $\mathrm{Exc}(\pi_0)$ (Theorem 4.4). By Proposition 4.5 applied to f and π_0, no lc center of $(X_0,0)$ is contained in $\mathrm{Supp}\big(\pi_{0*}^{-1}R(f^i)\big)$ for any integer $i\geq 1$. For $1\leq i\leq \ell$, let

$$D_i:=\pi_{0*}^{-1}R(f^i)\geq 0.$$

By Lemma 5.1 applied to X_0 and $D_1,\dots,D_\ell$, there exists a small birational morphism $\mu: X'\to X_0$ from a normal projective variety X' such that $K_{X'}=\mu^*K_{X_0}$, $(X',0)$ is lc, and $\mu_*^{-1}D_i$ is $\mathbb{Q}$-Cartier for any i. Let $\pi:=\pi_0\circ\mu: X'\to X$. Then π is a small birational morphism, $K_{X'}$ is $\mathbb{Q}$-Cartier, and $(X',0)$ is lc.

Step 3. In this step, we show that $\pi_*^{-1}D$ is $\mathbb{Q}$-Cartier for any $\mathbb{Q}$-divisor D on X with $[D]\in W$, and we construct an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor A' on X' such that $[\pi_*A']\in W$.

Since π is small, π_*^{-1} induces an isomorphism $\mathrm{Cl}(X)_{\mathbb{Q}} \to \mathrm{Cl}(X')_{\mathbb{Q}}$ with inverse π_* [Har77, Proposition II.6.5(b)]. Since $\mathbb{Q}$-Cartierness is preserved under $\mathbb{Q}$-linear equivalence and $\mathbb{Q}$-linear combinations, by **Step 1** it suffices to show that $\pi_*^{-1}K_X$, $\pi_*^{-1}H$, and $\pi_*^{-1}R(f^i)$ for $1 \leq i \leq \ell$ are $\mathbb{Q}$-Cartier. Since π is small, $\pi_*^{-1}K_X = K_{X'}$ and $\pi_*^{-1}H = \pi^*H$. By **Step 2**, $K_{X'}$ is $\mathbb{Q}$-Cartier, and $\pi_*^{-1}R(f^i) = \mu_*^{-1}D_i$ is $\mathbb{Q}$-Cartier for $1 \leq i \leq \ell$.

Since K_{X_0} is $\mathbb{Q}$-Cartier and π_0-ample, there exists a positive integer b such that $K_{X_0} + b\pi_0^*H$ is ample [Laz04, Proposition 1.7.10]. Let n be a positive integer such that $A := n(K_{X_0} + b\pi_0^*H)$ is Cartier. By Lemma 5.1, there exist rational numbers $c_1, \ldots, c_\ell \geq 0$ such that

$$A' := \mu^*A + \sum_{i=1}^{\ell} c_i \mu_*^{-1} D_i$$

is ample. Since μ and π_0 are small,

$$\pi_* A' = nK_X + nbH + \sum_{i=1}^{\ell} c_i R(f^i),$$

and hence $[\pi_* A'] \in W$ by **Step 1**.

Step 4. In this step, we construct a real vector space $V_W \subset N^1(X')_{\mathbb{R}}$, a linear automorphism φ of V_W induced by f^*, and a closed convex cone $\mathcal{C}_W \subset V_W$ such that $\varphi(\mathcal{C}_W) = \mathcal{C}_W$, $\mathcal{C}_W$ contains no line, and $[\pi^*H]$ is an interior point of $\mathcal{C}_W$ in V_W.

In **Steps 4 and 5**, for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D' on X', $[D']$ denotes the numerical class of D' in $N^1(X')_{\mathbb{R}}$. By **Step 3**, the classes $[\pi_*^{-1}D]$, where D runs over the $\mathbb{Q}$-divisors on X with $[D] \in W$, form a $\mathbb{Q}$-linear subspace of $N^1(X')_{\mathbb{Q}}$. Let V_W be the $\mathbb{R}$-linear span of the classes $[\pi_*^{-1}D]$ in $N^1(X')_{\mathbb{R}} = N^1(X')_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}$.

We claim that, for any $\mathbb{Q}$-divisor D on X such that $[D] \in W$ and $[\pi_*^{-1}D] = 0$, we have $[\pi_*^{-1}f^*D] = 0$ and $[\pi_*^{-1}f_*D] = 0$. Let X'' be a resolution of the closure of the graph of the rational map $\pi^{-1} \circ f \circ \pi : X' \dashrightarrow X'$, and let $p : X'' \to X'$ and $h : X'' \to X'$ be the morphisms induced by the first and the second projections respectively. Then p is birational, and

$$\pi \circ h = f \circ \pi \circ p, \tag{6.4}$$

since both sides coincide on a non-empty open subset of X''. Since $f \circ \pi \circ p$ is surjective and $\dim X'' = \dim X'$, h is surjective and generically finite. Let E be a prime divisor on X''. By (6.4), $f(\pi(p(E))) = \pi(h(E))$. Since f is finite and π is small, $\dim p(E) = d-1$ if and only if $\dim \pi(p(E)) = d-1$, if and only if $\dim \pi(h(E)) = d-1$, if and only if $\dim h(E) = d-1$. Thus E is p-exceptional if and only if E is h-exceptional.

Let $D' := \pi_*^{-1}D$. We show that

$$\pi_*^{-1}f^*D = p_*h^*D' \quad \text{and} \quad \pi_*^{-1}f_*D = h_*p^*D'. \tag{6.5}$$

Let E be a prime divisor on X'' that is not p-exceptional, and let $P := \pi(p(E))$ and $S := \pi(h(E))$. Then P and S are prime divisors on X and $f(P) = S$. Since p and π are isomorphisms over neighborhoods of the generic points of $p(E)$ and P [Har77, Theorem I.6.2A and Theorem II.4.7], we may regard p and π as equalities of function fields, and then $\mathrm{ord}_E = \mathrm{ord}_P$, $\mathrm{ord}_{h(E)} = \mathrm{ord}_S$, the field homomorphism h^* becomes f^* by (6.4), and $h|_E : E \to h(E)$ is birationally equivalent to $f|_P : P \to S$. Thus the coefficient of E in h^*D' is e_P times the coefficient of S in D, where e_P is the ramification index of f along P, and $\deg(h|_E) = \deg(f|_P)$. Since $\pi \circ p$ induces a bijection between the prime divisors on X'' that are not p-exceptional and the prime divisors on X, and since p_* and h_* vanish on p-exceptional prime divisors, comparing coefficients with the definitions of f^* and f_* (Definition-Proposition 2.10) implies (6.5).

Now assume that $[D'] = 0$. Then h^*D' and p^*D' are numerically trivial. For any Cartier divisors $H_1, \ldots, H_{d-1}$ on X', by (6.5) and the projection formula,

$$(\pi_*^{-1}f^*D \cdot H_1 \cdots H_{d-1}) = (h^*D' \cdot p^*H_1 \cdots p^*H_{d-1}) = 0$$

and

$$(\pi_*^{-1}f_*D\cdot H_1\cdots H_{d-1})=(p^*D'\cdot h^*H_1\cdots h^*H_{d-1})=0.$$

Thus $\pi_*^{-1}f^*D$ and $\pi_*^{-1}f_*D$ are weakly numerically trivial [MZ18, Definition 2.2]. Since $f^*[D]$ and $q^{-d}f_*[D]$ belong to W (**Step 1**), $\pi_*^{-1}f^*D$ and $\pi_*^{-1}f_*D$ are $\mathbb{Q}$-Cartier by **Step 3**. Since a weakly numerically trivial $\mathbb{R}$-Cartier $\mathbb{R}$-divisor is numerically trivial [MZ18, Lemma 2.3], $[\pi_*^{-1}f^*D]=0$ and $[\pi_*^{-1}f_*D]=0$. This proves the claim.

By the claim and **Step 1**, there exist unique $\mathbb{R}$-linear maps $\varphi,\psi:V_W\to V_W$ such that

$$\varphi([\pi_*^{-1}D])=[\pi_*^{-1}f^*D]\quad\text{and}\quad\psi([\pi_*^{-1}D])=[\pi_*^{-1}q^{-d}f_*D]$$

for any $\mathbb{Q}$-divisor D on X with $[D]\in W$. Since f^* and $q^{-d}f_*$ are inverse to each other on W, $\psi\circ\varphi=\varphi\circ\psi=\mathrm{id}_{V_W}$, that is, φ is invertible and $\psi=\varphi^{-1}$. Since $f^*H\sim qH$, $\varphi([\pi^*H])=q[\pi^*H]$.

Let $\mathcal{C}_W\subset V_W$ be the closed convex cone generated by the classes $[\pi_*^{-1}D]$, where D runs over the $\mathbb{Q}$-divisors $D\ge 0$ on X with $[D]\in W$. If $D\ge 0$, then $f^*D\ge 0$ and $q^{-d}f_*D\ge 0$, and both classes belong to W. Thus $\varphi(\mathcal{C}_W)\subset\mathcal{C}_W$ and $\varphi^{-1}(\mathcal{C}_W)\subset\mathcal{C}_W$, that is, $\varphi(\mathcal{C}_W)=\mathcal{C}_W$. Since $\mathcal{C}_W$ is contained in the pseudo-effective cone of X' in $N^1(X')_{\mathbb{R}}$, which contains no line (see the sentence after [MZ18, Definition 2.4]), $\mathcal{C}_W$ contains no line.

It remains to show that $[\pi^*H]$ is an interior point of $\mathcal{C}_W$ in V_W. Since π^*H is nef and $((\pi^*H)^d)=(H^d)>0$, π^*H is big [Laz04, Theorem 2.2.16]. Let $L_1,\dots,L_k$ be $\mathbb{Q}$-divisors on X with $[L_i]\in W$ such that $[\pi_*^{-1}L_1],\dots,[\pi_*^{-1}L_k]$ form a basis of V_W. Since the big cone is open [Laz04, Theorem 2.2.26] and bigness depends only on the numerical class [Laz04, Corollary 2.2.8], there exists a rational number $\delta>0$ such that $\pi^*H+\delta\pi_*^{-1}L_i$ and $\pi^*H-\delta\pi_*^{-1}L_i$ are big for any i. Since a big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor is $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-divisor [Laz04, Corollary 2.2.7], there exist $\mathbb{Q}$-divisors $L_i^+\ge 0$ and $L_i^-\ge 0$ on X' such that

$$L_i^+\sim_{\mathbb{Q}}\pi^*H+\delta\pi_*^{-1}L_i\quad\text{and}\quad L_i^-\sim_{\mathbb{Q}}\pi^*H-\delta\pi_*^{-1}L_i.$$

Then $\pi_*L_i^{\pm}\ge 0$, $[\pi_*L_i^{\pm}]=[H]\pm\delta[L_i]\in W$, and $\pi_*^{-1}\pi_*L_i^{\pm}=L_i^{\pm}$ since π is small. Thus

$$[\pi^*H]\pm\delta[\pi_*^{-1}L_i]\in\mathcal{C}_W$$

for any i. The convex hull of the $2k$ classes $[\pi^*H]\pm\delta[\pi_*^{-1}L_i]$ contains a neighborhood of $[\pi^*H]$ in V_W, and hence $[\pi^*H]$ is an interior point of $\mathcal{C}_W$ in V_W.

Step 5. In this step, we find a and construct g.

Let

$$\mathcal{B}:=\{z\in V_W\mid[\pi^*H]+z\in\mathcal{C}_W\text{ and }[\pi^*H]-z\in\mathcal{C}_W\}.$$

Then $\mathcal{B}$ is closed, convex, and symmetric, and $\mathcal{B}$ contains a neighborhood of 0 in V_W by **Step 4**. We show that $\mathcal{B}$ is bounded. Suppose otherwise. Then there exist $z_k\in\mathcal{B}$ with $\|z_k\|\to+\infty$ for some norm $\|\cdot\|$ on V_W, and possibly passing to a subsequence, $z_k/\|z_k\|$ converges to a vector $z\ne 0$. Since $([\pi^*H]\pm z_k)/\|z_k\|\in\mathcal{C}_W$ and $\mathcal{C}_W$ is closed, $z\in\mathcal{C}_W$ and $-z\in\mathcal{C}_W$, which contradicts the fact that $\mathcal{C}_W$ contains no line. Thus $\mathcal{B}$ is the unit ball of a norm $\|\cdot\|_{\mathcal{B}}$ on V_W.

By **Step 4**, $q^{-1}\varphi([\pi^*H])=[\pi^*H]$ and $(q^{-1}\varphi)^j(\mathcal{C}_W)=\mathcal{C}_W$ for any integer j. Thus $(q^{-1}\varphi)^j(\mathcal{B})=\mathcal{B}$, that is, $(q^{-1}\varphi)^j$ is an isometry of $(V_W,\|\cdot\|_{\mathcal{B}})$ for any integer j. The isometries of $(V_W,\|\cdot\|_{\mathcal{B}})$ form a compact subset of the space of linear endomorphisms of V_W with the operator norm, again denoted by $\|\cdot\|_{\mathcal{B}}$. Thus there exist integers $1\le j_1<j_2<\cdots$ such that $(q^{-1}\varphi)^{j_k}$ converges as $k\to+\infty$, and

$$\begin{aligned}\|(q^{-1}\varphi)^{j_{k+1}-j_k}-\mathrm{id}_{V_W}\|_{\mathcal{B}}&=\|(q^{-1}\varphi)^{-j_k}\big((q^{-1}\varphi)^{j_{k+1}}-(q^{-1}\varphi)^{j_k}\big)\|_{\mathcal{B}}\\&=\|(q^{-1}\varphi)^{j_{k+1}}-(q^{-1}\varphi)^{j_k}\|_{\mathcal{B}}\to 0\end{aligned}$$

as $k\to+\infty$ (cf. [MZ18, Proposition 2.9]).

Recall that A' is the ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X' constructed in **Step 3**, with $[\pi_*A']\in W$. Since π is small, $\pi_*^{-1}\pi_*A'=A'$, and hence $[A']\in V_W$. Since the ample cone of X' is open in $N^1(X')_{\mathbb{R}}$ [Laz04, Theorem 1.4.23(ii)], there exists a positive integer a of the form $j_{k+1}-j_k$ such

that $(q^{-1}\varphi)^a([A'])$ belongs to the ample cone of X', and hence so does $\varphi^a([A']) = q^a(q^{-1}\varphi)^a([A'])$. Let $D := \pi_* A'$. Since $(f^a)^* = (f^*)^a$ on $\mathrm{Cl}(X)_{\mathbb{Q}}$ (**Step 1**),

$$\varphi^a([A']) = [\pi_*^{-1}(f^a)^* D].$$

By **Step 1**, $[(f^a)^* D] \in W$, so $\pi_*^{-1}(f^a)^* D$ is $\mathbb{Q}$-Cartier by **Step 3**. Since ampleness depends only on the numerical class [Laz04, Proposition 1.3.13], $\pi_*^{-1}(f^a)^* D$ is ample. Thus $\pi_*^{-1} D = A'$ and $\pi_*^{-1}(f^a)^* D$ are $\mathbb{Q}$-Cartier and π-ample. Since f^a is a finite surjective endomorphism of X (Lemma 2.12), Lemma 6.1 applied to f^a, π, and D implies that there exists a finite surjective endomorphism $g : X' \to X'$ such that $\pi \circ g = f^a \circ \pi$.

Finally, $\pi^* H$ is a big Cartier divisor on X' by **Step 4**, and

$$g^* \pi^* H = \pi^* (f^a)^* H \sim q^a \pi^* H.$$

By Theorem 2.13 applied to g and $\pi^* H$, g is q^a-polarized. □

6.3. **Normalized ramification divisors.** In this subsection, we prove the conclusion of Theorem 1.5 when K_X is $\mathbb{Q}$-Cartier (Proposition 6.3), and deduce from it Lemma 6.4.

When K_X is $\mathbb{Q}$-Cartier, the lc property in Proposition 6.3 follows from [Yos26, Theorem 8.1], and what remains is the $\mathbb{Q}$-linear triviality. The idea is that an iterate of f fixes any contractible extremal ray, the cone theorem for lc pairs then implies nefness, and the polarization implies that the intersection number with powers of the polarization class is zero.

Proposition 6.3. *Let X be a normal projective variety of dimension $d \geq 1$ such that K_X is $\mathbb{Q}$-Cartier, and $f : X \to X$ a q-polarized endomorphism. Then there exists a positive integer s such that for any positive integer m, the pair $(X, \frac{1}{q^{ms}-1} R(f^{ms}))$ is lc and $K_X + \frac{1}{q^{ms}-1} R(f^{ms}) \sim_{\mathbb{Q}} 0$. In particular, $(f^{ms})^* K_X \sim_{\mathbb{Q}} q^{ms} K_X$ for any positive integer m.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix a polarization and express the normalized ramification divisors of the iterates of f in terms of K_X.

Let H be an ample Cartier divisor on X such that $f^* H \sim qH$. Then $(f^k)^* H \sim q^k H$ for any positive integer k, so f^k is q^k-polarized. By Lemma 2.12, f is finite and int-amplified, and $\deg f^k = (q^k)^d$ for any positive integer k. By Definition-Proposition 2.10(1), $R(f^k) \geq 0$ and $K_X \sim (f^k)^* K_X + R(f^k)$. Since K_X is $\mathbb{Q}$-Cartier, we obtain

$$(q^k - 1)\left(K_X + \frac{1}{q^k - 1} R(f^k)\right) \sim_{\mathbb{Q}} q^k K_X - (f^k)^* K_X \tag{6.6}$$

for any positive integer k. In particular, $K_X + \frac{1}{q^k-1} R(f^k)$ is $\mathbb{Q}$-Cartier.

Step 2. In this step, we find a positive integer t such that f^t fixes any contractible extremal ray of $\overline{\mathrm{NE}}(X)$, and we compute the action of $(f^k)^*$ on such rays when $t \mid k$.

We first show that f_* is invertible on $N_1(X)_{\mathbb{R}}$. Since f is finite and surjective, any curve C on X is the image of a curve C' on X, and $f_*[C'] = \deg(C' \to C)[C]$. Thus f_* is surjective on $N_1(X)_{\mathbb{R}}$. Since $\dim N_1(X)_{\mathbb{R}} < +\infty$, f_* is bijective.

Since f is int-amplified, by [MZ20, Definition 4.1, Theorem 4.5], the set $\mathcal{S}$ of rays $(f_*)^i(R)$, where $i \in \mathbb{Z}$ and R runs over the contractible extremal rays of $\overline{\mathrm{NE}}(X)$, is finite. The map f_* sends $\mathcal{S}$ into $\mathcal{S}$ and is injective, hence f_* permutes $\mathcal{S}$. Thus there exists a positive integer t such that $(f_*)^t(R) = R$ for any $R \in \mathcal{S}$. If $\mathcal{S} = \emptyset$, we let $t := 1$.

Let k be a positive multiple of t, and let R be a contractible extremal ray of $\overline{\mathrm{NE}}(X)$. By [MZ20, Definition 4.1], R is generated by the class γ of a curve on X. Since $(f^k)_* R = R$ and $(f^k)_* \gamma \neq 0$, we have $(f^k)_* \gamma = \lambda \gamma$ for some real number $\lambda > 0$. By the projection formula [Ful98, Proposition 2.5(c)],

$$\lambda(H \cdot \gamma) = (H \cdot (f^k)_* \gamma) = ((f^k)^* H \cdot \gamma) = q^k (H \cdot \gamma).$$

Since H is ample, $(H \cdot \gamma) > 0$, so $\lambda = q^k$. By the projection formula again, for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on X, we have

$$((q^k L - (f^k)^* L) \cdot \gamma) = q^k (L \cdot \gamma) - (L \cdot (f^k)_* \gamma) = 0. \tag{6.7}$$

Step 3. In this step, we define the integer s and the boundaries B_m, and we show that (X, B_m) is lc and $K_X + B_m$ is nef.

By Definition-Proposition 2.10(1), $R(f) \geq 0$, and K_X is $\mathbb{Q}$-Cartier, so $(X, 0, f)$ is a q-polarized dynamical triple in the sense of [Yos26, Definition 3.1]. By [Yos26, Theorem 8.1], there exists a positive integer r such that the pair $(X, \frac{1}{q^{jr}-1} R(f^{jr}))$ is lc for any positive integer j. We let $s := \mathrm{lcm}(r, t)$, where t is the integer in **Step 2**. For any positive integer m, we let

$$B_m := \frac{1}{q^{ms} - 1} R(f^{ms}) \quad \text{and} \quad M_m := K_X + B_m.$$

Since $r \mid ms$, (X, B_m) is lc. By (6.6) with $k = ms$,

$$(q^{ms} - 1) M_m \sim_{\mathbb{Q}} q^{ms} K_X - (f^{ms})^* K_X. \tag{6.8}$$

Since $t \mid ms$, (6.7) with $k = ms$ and $L = K_X$ implies that $(M_m \cdot \gamma) = 0$ for the class γ of any curve generating a contractible extremal ray of $\overline{\mathrm{NE}}(X)$.

Suppose that M_m is not nef. By the cone and contraction theorems for lc pairs [Fuj11, Theorem 1.1(4), Theorems 16.4 and 16.6] applied to (X, B_m), there exists an extremal ray R of $\overline{\mathrm{NE}}(X)$ generated by the class γ of a curve such that $(M_m \cdot \gamma) < 0$, together with a projective surjective morphism $\varphi_R : X \to Y$ with connected fibers which contracts exactly the curves whose classes are in R. By [Fuj11, Theorem 1.1(4)(ii) and (i)], φ_R satisfies conditions (1) and (2) of [MZ20, Definition 4.1]. Let D be a $\mathbb{Q}$-Cartier divisor on X and l a positive integer such that lD is Cartier. If $(D \cdot \gamma) = 0$, then $\mathcal{O}_X(lD) \cong \varphi_R^* L_Y$ for a line bundle L_Y on Y [Fuj11, Theorem 1.1(4)(iii)], so $D \equiv \varphi_R^* D_Y$ for a $\mathbb{Q}$-Cartier divisor D_Y on Y with $\mathcal{O}_Y(lD_Y) \cong L_Y$. Conversely, if $D \equiv \varphi_R^* D_Y$, then $(D \cdot \gamma) = 0$ by the projection formula, since φ_R contracts γ. Thus φ_R also satisfies condition (3) of [MZ20, Definition 4.1], and R is contractible in the sense of [MZ20, Definition 4.1]. Thus $(M_m \cdot \gamma) = 0$, a contradiction. Therefore, M_m is nef.

Step 4. In this step, we show that $M_m \equiv 0$, and we conclude the proof.

Recall that H is the ample Cartier divisor in **Step 1** with $(f^{ms})^* H \sim q^{ms} H$, and that $\deg f^{ms} = (q^{ms})^d$. By the projection formula [Ful98, Proposition 2.5(c)],

$$(q^{ms})^{d-1}((f^{ms})^* K_X \cdot H^{d-1}) = ((f^{ms})^* K_X \cdot ((f^{ms})^* H)^{d-1}) = (q^{ms})^d (K_X \cdot H^{d-1}).$$

Thus $((f^{ms})^* K_X \cdot H^{d-1}) = q^{ms}(K_X \cdot H^{d-1})$, and by (6.8),

$$(M_m \cdot H^{d-1}) = 0.$$

If $d = 1$, then $M_m \equiv 0$. Suppose that $d \geq 2$. Since H is ample, there exists a positive integer N such that $NH - M_m$ is ample. Since M_m, $NH - M_m$, and H are nef, [Laz04, Example 1.4.16] implies

$$0 \leq (M_m \cdot (NH - M_m) \cdot H^{d-2}) = N(M_m \cdot H^{d-1}) - (M_m^2 \cdot H^{d-2}) = -(M_m^2 \cdot H^{d-2})$$

and

$$0 \leq (M_m^2 \cdot H^{d-2}).$$

Thus $(M_m^2 \cdot H^{d-2}) = 0$. Since $(M_m \cdot H^{d-1}) = 0$, by [MZ18, Lemma 2.3], $M_m \equiv 0$.

Since (X, B_m) is a projective lc pair, B_m is a $\mathbb{Q}$-divisor, and $M_m = K_X + B_m \equiv 0$, by [Gon13, Theorem 1.2], $K_X + B_m \sim_{\mathbb{Q}} 0$. By (6.8), $(f^{ms})^* K_X \sim_{\mathbb{Q}} q^{ms} K_X$. Since m is an arbitrary positive integer, Proposition 6.3 follows. □

Lemma 6.4. *Let X be a normal projective variety of dimension $d \geq 1$, $f : X \to X$ a q-polarized endomorphism, and $h : X' \to X$ a projective birational morphism from a normal projective variety X' such that $K_{X'}$ is $\mathbb{Q}$-Cartier. Assume that there exist a positive integer a and a surjective*

endomorphism $g : X' \to X'$ such that $h \circ g = f^a \circ h$. Then there exists a positive integer s such that for any positive integer m, the pair $(X, \frac{1}{q^{ms}-1}R(f^{ms}))$ is lc and $K_X + \frac{1}{q^{ms}-1}R(f^{ms}) \sim_{\mathbb{Q}} 0$.

Proof. Let H be an ample Cartier divisor on X such that $f^*H \sim qH$. Since h is birational, $((h^*H)^d) = (H^d) > 0$, so the nef Cartier divisor h^*H is big [Laz04, Theorem 2.2.16]. Since $h \circ g = f^a \circ h$,

$$g^*h^*H = h^*(f^a)^*H \sim q^a h^*H.$$

By Theorem 2.13, g is q^a-polarized. In particular, g is finite (Lemma 2.12). Since $\dim X' = d$ and $K_{X'}$ is $\mathbb{Q}$-Cartier, by Proposition 6.3 applied to g, there exists a positive integer b such that for any positive integer m, the pair (X', B'_m) is lc and $K_{X'} + B'_m \sim_{\mathbb{Q}} 0$, where

$$B'_m := \frac{1}{q^{amb}-1}R(g^{mb}).$$

We let $s := ab$. For any positive integer m, we let $B_m := \frac{1}{q^{ms}-1}R(f^{ms})$.

Fix a positive integer m. We show that $h_*R(g^{mb}) = R(f^{amb})$. We have $h \circ g^{mb} = f^{amb} \circ h$. Let P be a prime divisor on X and $P' := h_*^{-1}P$. Since $h(g^{mb}(P')) = f^{amb}(P)$ and f is finite, $f^{amb}(P)$ is a prime divisor on X, so $g^{mb}(P')$ is not h-exceptional. Since h is an isomorphism at the generic points of P' and $g^{mb}(P')$, we have

$$\mathrm{ord}_{P'} \circ h^* = \mathrm{ord}_P \quad \text{and} \quad \mathrm{ord}_{g^{mb}(P')} \circ h^* = \mathrm{ord}_{f^{amb}(P)}$$

on $\mathbb{C}(X)$. Since $(g^{mb})^* \circ h^* = h^* \circ (f^{amb})^*$ on $\mathbb{C}(X)$, the ramification index of g^{mb} along P' is equal to the ramification index of f^{amb} along P. Thus the coefficient of P' in $R(g^{mb})$ is equal to the coefficient of P in $R(f^{amb})$. Since P is an arbitrary prime divisor on X and h_* sends any h-exceptional divisor to zero, $h_*R(g^{mb}) = R(f^{amb})$. Since $s = ab$, we have $q^{amb} = q^{ms}$ and $R(f^{amb}) = R(f^{ms})$, so

$$h_*B'_m = B_m \quad \text{and} \quad h_*(K_{X'} + B'_m) = K_X + B_m.$$

Since $K_{X'} + B'_m \sim_{\mathbb{Q}} 0$, there exist a positive integer N and a rational function $u \in \mathbb{C}(X')^* = \mathbb{C}(X)^*$ such that

$$N(K_{X'} + B'_m) = \mathrm{div}_{X'}(u).$$

Since the pushforward of a principal divisor under a proper birational morphism is principal [Ful98, Proposition 1.4(b)],

$$N(K_X + B_m) = h_* \mathrm{div}_{X'}(u) = \mathrm{div}_X(u).$$

Thus $K_X + B_m$ is $\mathbb{Q}$-Cartier and $K_X + B_m \sim_{\mathbb{Q}} 0$. Moreover,

$$h^*(K_X + B_m) = \frac{1}{N}\mathrm{div}_{X'}(u) = K_{X'} + B'_m.$$

Since $h_*B'_m = B_m$, by [KM98, Lemma 2.30], $a(E, X, B_m) = a(E, X', B'_m)$ for any prime divisor E over X. Since (X', B'_m) is lc and $B_m \geq 0$, (X, B_m) is lc. □

7. Lc anti-canonical series

In this section, we introduce lc anti-canonical series and prove Lemma 7.2. Sections 7–12 deal with arbitrary projective lc pairs, no endomorphism appears there, and they prove Theorem 12.1. In particular, the residue models of Construction-Proposition 11.4, used in the induction on the dimension, carry no endomorphism. Section 13 deduces Theorem 1.3 from Theorem 12.1 and Construction-Proposition 5.2.

Definition 7.1. Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor. An *lc anti-canonical series* on (X, B) is a sequence $D_\bullet = (D_j)_{j\geq 1}$ of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors $D_j \geq 0$ on X satisfying the following conditions.

(1) $(X, B + D_{\leq N})$ is lc for any $N \geq 1$, where $D_{\leq N} := \sum_{j=1}^{N} D_j$.

(2) There exist an integer $m_0 \geq 1$ and rational numbers $a_{j,N}$ for $1 \leq j \leq m_0$ and $N \geq m_0$, such that $\lim_{N\to+\infty} a_{j,N} = 0$ for any j and

$$K_X + B + D_{\leq N} + \sum_{j=1}^{m_0} a_{j,N} D_j \sim_{\mathbb{Q}} 0$$

for any $N \geq m_0$.

A *correction family* for $D_\bullet$ is a family of rational numbers $c_{j,N}$ for $1 \leq j \leq m$ and $N \geq N_0$, for some integers $N_0 \geq m \geq 1$, such that $\lim_{N\to+\infty} c_{j,N} = 0$ for any j and $K_X + B + G_N \sim_{\mathbb{Q}} 0$ for any $N \geq N_0$, where

$$G_N := D_{\leq N} + \sum_{j=1}^{m} c_{j,N} D_j.$$

We call G_N the *corrected boundaries* of $(c_{j,N})$. A *moment row* for $D_\bullet$ is a sequence $x_\bullet = (x_j)_{j\geq 1}$ of vectors $x_j \in \mathbb{Q}^n$, for some integer $n \geq 1$, such that $\sum_{j=1}^{N} x_j$ converges to a vector $x_\infty \in \mathbb{Q}^n$ as $N \to +\infty$. We say that $(c_{j,N})$ *satisfies* $x_\bullet$ if $\sum_{j=1}^{N} x_j + \sum_{j=1}^{m} c_{j,N} x_j = x_\infty$ for any $N \geq N_0$. A quasi-monomial valuation v of $\mathbb{C}(X)$ is *stationary* for $(X, B; D_\bullet)$ if $\sum_{j\geq 1} v(D_j) = a(v, X, B)$.

Condition (1) of Definition 7.1 implies that, at any quasi-monomial valuation of $\mathbb{C}(X)$, the sum of the values of the D_j is at most the log discrepancy with respect to (X, B). Roughly speaking, the corrected boundaries of a correction family can therefore fail to be lc only near the centers of the valuations at which equality holds, that is, of the stationary valuations. Lemma 12.3 is the precise statement. Condition (2) implies that the classes of the partial sums $D_{\leq N}$ converge to $-[K_X + B]$ in the finite-dimensional space spanned by $[D_1], \ldots, [D_{m_0}]$. Lemma 7.2 uses the convergence of the classes of the $D_{\leq N}$ to provide corrections tending to 0, so that for large N a corrected boundary has non-negative coefficients and differs from an lc partial sum only by small multiples of finitely many D_j, which is what Lemma 12.3 needs. If the $a_{j,N}$ were only bounded, the corrections could destroy the lc property also away from the centers of stationary valuations.

At a stationary valuation v, the vanishing of the log discrepancy of v with respect to $(X, B+G_N)$ is a rational linear equation on the coefficients $c_{j,N}$ (see Lemma 8.5), and the condition that a correction family satisfies a moment row is also a rational linear equation on the coefficients $c_{j,N}$. Lemma 7.2 shows that one correction family can satisfy finitely many moment rows simultaneously, because the classes of all D_j in $\mathrm{Cl}(X)_{\mathbb{Q}}$ belong to the $\mathbb{Q}$-span of the classes of $D_1, \ldots, D_{m_0}$.

Lemma 7.2. *Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B), and $x_\bullet^{(1)}, \ldots, x_\bullet^{(p)}$ moment rows for $D_\bullet$. Then there exists a correction family $(c_{j,N})$ for $D_\bullet$ satisfying $x_\bullet^{(1)}, \ldots, x_\bullet^{(p)}$. Moreover, for any correction family $(c_{j,N})$ for $D_\bullet$ and any sufficiently large N, we have $1 + c_{j,N} > 0$ for any j and $G_N \geq 0$.*

Proof. Let m_0 and $a_{j,N}$ be as in Definition 7.1(2), and let $\Lambda \subset \mathrm{Cl}(X)_{\mathbb{Q}}$ be the $\mathbb{Q}$-span of $[D_1], \ldots, [D_{m_0}]$. Subtracting the relation in Definition 7.1(2) for N from the relation for $N+1$ (with $N \geq m_0$), we obtain

$$D_{N+1} + \sum_{j=1}^{m_0} (a_{j,N+1} - a_{j,N}) D_j \sim_{\mathbb{Q}} 0.$$

Thus $[D_{N+1}] \in \Lambda$ for any $N \geq m_0$, hence $[D_j] \in \Lambda$ for any $j \geq 1$. Moreover, Definition 7.1(2) for $N = m_0$ implies that $[K_X + B] \in \Lambda$, and Definition 7.1(2) implies that

$$\sum_{j=1}^{N} [D_j] = -[K_X + B] - \sum_{j=1}^{m_0} a_{j,N} [D_j] \tag{7.1}$$

for any $N \geq m_0$.

Let $\mathcal{V}$ be the direct sum of Λ and of the $\mathbb{Q}$-vector spaces in which $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ take values. Then $\dim \mathcal{V} < +\infty$. Let $\mathcal{V}_{\mathbb{R}} := \mathcal{V} \otimes_{\mathbb{Q}} \mathbb{R}$ with the Euclidean topology. For $1 \leq k \leq p$, let $x_\infty^{(k)}$ be the limit of $\sum_{j=1}^N x_j^{(k)}$ as $N \to +\infty$. Let

$$y_j := \big([D_j], x_j^{(1)}, \dots, x_j^{(p)}\big) \in \mathcal{V} \quad \text{and} \quad y_\infty := \big(-[K_X + B], x_\infty^{(1)}, \dots, x_\infty^{(p)}\big) \in \mathcal{V}.$$

Since $\lim_{N\to+\infty} a_{j,N} = 0$ for any j, (7.1) implies that

$$\lim_{N\to+\infty} \sum_{j=1}^N y_j = y_\infty$$

in $\mathcal{V}_{\mathbb{R}}$.

Let $\mathcal{V}' \subset \mathcal{V}$ be the $\mathbb{Q}$-span of y_j for $j \geq 1$, $\phi : \mathcal{V} \to \mathcal{V}/\mathcal{V}'$ the quotient map, and $\phi_{\mathbb{R}} := \phi \otimes_{\mathbb{Q}} \mathbb{R}$ the induced $\mathbb{R}$-linear map on $\mathcal{V}_{\mathbb{R}}$. Since $\phi_{\mathbb{R}}(\sum_{j=1}^N y_j) = 0$ for any N and $\phi_{\mathbb{R}}$ is continuous, we have $\phi(y_\infty) = \phi_{\mathbb{R}}(y_\infty) = 0$. Thus $y_\infty \in \mathcal{V}'$.

Since $\dim \mathcal{V}' < +\infty$, there exists an integer $m \geq 1$ such that $y_1, \dots, y_m$ span $\mathcal{V}'$. Let $\psi : \mathbb{Q}^m \to \mathcal{V}'$ be the $\mathbb{Q}$-linear map sending $(t_1, \dots, t_m)$ to $\sum_{j=1}^m t_j y_j$. Since ψ is surjective, there exists a $\mathbb{Q}$-linear map $\iota : \mathcal{V}' \to \mathbb{Q}^m$ such that $\psi \circ \iota = \mathrm{id}_{\mathcal{V}'}$. For any $N \geq m$, we have $y_\infty - \sum_{j=1}^N y_j \in \mathcal{V}'$, and we define

$$(c_{1,N}, \dots, c_{m,N}) := \iota\Big(y_\infty - \sum_{j=1}^N y_j\Big) \in \mathbb{Q}^m.$$

Since ι extends to a continuous $\mathbb{R}$-linear map $\mathcal{V}' \otimes_{\mathbb{Q}} \mathbb{R} \to \mathbb{R}^m$ and $\lim_{N\to+\infty}(y_\infty - \sum_{j=1}^N y_j) = 0$, we have $\lim_{N\to+\infty} c_{j,N} = 0$ for any j. Since $\psi \circ \iota = \mathrm{id}_{\mathcal{V}'}$, we have

$$\sum_{j=1}^N y_j + \sum_{j=1}^m c_{j,N} y_j = y_\infty$$

for any $N \geq m$. Taking components in Λ, we obtain $[K_X + B + G_N] = 0$, where $G_N = D_{\leq N} + \sum_{j=1}^m c_{j,N} D_j$, that is, $K_X + B + G_N \sim_{\mathbb{Q}} 0$. Taking components in the space of $x_\bullet^{(k)}$, we obtain

$$\sum_{j=1}^N x_j^{(k)} + \sum_{j=1}^m c_{j,N} x_j^{(k)} = x_\infty^{(k)}.$$

Thus, with $N_0 := m$, $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$.

Now let $(c_{j,N})$ be any correction family for $D_\bullet$, with $1 \leq j \leq m$ and $N \geq N_0$. Since $\lim_{N\to+\infty} c_{j,N} = 0$ for each of the finitely many j, there exists an integer $N_1 \geq N_0$ such that $|c_{j,N}| < 1$ for any $1 \leq j \leq m$ and any $N \geq N_1$. Then $1 + c_{j,N} > 0$ for any j and any $N \geq N_1$. Since $N \geq N_0 \geq m$ and $D_j \geq 0$ for any j, we have

$$G_N = \sum_{j=1}^m (1 + c_{j,N}) D_j + \sum_{j=m+1}^N D_j \geq 0$$

for any $N \geq N_1$. □

Definition 7.3. Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B), and $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ moment rows for $D_\bullet$, where $p \geq 0$. A *corrected system* for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ is a linear subspace $W \subset H^0(X, \mathcal{O}_X(-e(K_X + B)))$, where e is a positive integer such that $-e(K_X + B)$ is Cartier, spanned by finitely many products $s_1 \cdots s_k$ with the following properties.

(1) $e = e_1 + \cdots + e_k$ for positive integers e_i such that $-e_i(K_X + B)$ is Cartier.

(2) For any i, $s_i \in H^0(X, \mathcal{O}_X(-e_i(K_X+B)))$ and $\operatorname{div}(s_i) = e_i G^{(i)}$, where $G^{(i)}$ is the corrected boundary at an index N_i of a correction family $(c^{(i)}_{j,N})$ for $D_\bullet$ satisfying $x^{(1)}_\bullet, \dots, x^{(p)}_\bullet$, and $1 + c^{(i)}_{j,N_i} \geq 0$ for any j. The correction families $(c^{(i)}_{j,N})$ may depend on i.

For $s \in W \backslash \{0\}$, let $G_s := \frac{1}{e}\operatorname{div}(s)$. We say that a property holds for a *general member* G of W if it holds for $G = G_s$ for any s in a non-empty Zariski open subset of W.

We use spans of products of sections instead of single corrected boundaries for two reasons. At any valuation, a general member of a linear system attains the minimum of the values of the divisors of the generators, so that corrected systems which are lc near different centers can be merged (Lemma 12.4). Moreover, the induction on dimension lifts systems of sections, not single sections, from the residue model (Construction-Proposition 11.4(4)). The condition on the numbers $1 + c^{(i)}_{j,N_i}$ ensures that the corrected boundaries on X corresponding to the lifted sections have non-negative coefficients. By Lemma 7.2, the condition holds at any sufficiently large index, and equality is allowed for the reindexing in Construction-Proposition 10.1(5).

Lemma 7.4. *Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B), $x^{(1)}_\bullet, \dots, x^{(p)}_\bullet$ moment rows for $D_\bullet$, and $W \subset H^0(X, \mathcal{O}_X(-e(K_X + B)))$ a corrected system for $D_\bullet$ satisfying $x^{(1)}_\bullet, \dots, x^{(p)}_\bullet$, spanned by products $\sigma_1, \dots, \sigma_n$ as in Definition 7.3. Let k be a positive integer and $W' \subset H^0(X, \mathcal{O}_X(-ke(K_X + B)))$ the linear subspace spanned by $\sigma_1^k, \dots, \sigma_n^k$. Then W' is a corrected system for $D_\bullet$ satisfying $x^{(1)}_\bullet, \dots, x^{(p)}_\bullet$, and $\operatorname{Nlc}(X, B + G) = \operatorname{Nlc}(X, B + G')$ for general members G of W and G' of W'.*

Proof. Write $\sigma_i = s_1 \cdots s_l$ as in Definition 7.3, with $s_t \in H^0(X, \mathcal{O}_X(-e_t(K_X+B)))$ and $\operatorname{div}(s_t) = e_t G^{(t)}$ for $1 \leq t \leq l$. Then σ_i^k is the product of the kl sections obtained by repeating each s_t exactly k times, and ke is the sum of the corresponding kl integers e_t. Since $-ke(K_X + B)$ is Cartier, W' is a corrected system for $D_\bullet$ satisfying $x^{(1)}_\bullet, \dots, x^{(p)}_\bullet$.

Let $\mathcal{L} := \mathcal{O}_X(-e(K_X + B))$, and let $\mathfrak{b}, \mathfrak{b}' \subset \mathcal{O}_X$ be the base ideals of $W \subset H^0(X, \mathcal{L})$ and $W' \subset H^0(X, \mathcal{L}^{\otimes k})$ respectively. Let $f : Y \to X$ be a log resolution of (X, B) such that $\mathfrak{b} \cdot \mathcal{O}_Y = \mathcal{O}_Y(-F)$ for a divisor $F \geq 0$ on Y and $\operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B \cup \operatorname{Supp} F$ is a simple normal crossing divisor, and write $K_Y + B_Y := f^*(K_X + B)$. Near any point of Y, after trivializing $f^*\mathcal{L}$, we may write $f^*\sigma_i = g u_i$ for $1 \leq i \leq n$, where g is a local equation of F and $u_1, \dots, u_n$ are regular functions without common zero. Then $f^*\sigma_i^k = g^k u_i^k$, and $u_1^k, \dots, u_n^k$ have no common zero. Thus $\mathfrak{b}' \cdot \mathcal{O}_Y = \mathcal{O}_Y(-kF)$.

Hence the images of f^*W in $H^0(Y, f^*\mathcal{L} \otimes \mathcal{O}_Y(-F))$ and of f^*W' in $H^0(Y, f^*\mathcal{L}^{\otimes k} \otimes \mathcal{O}_Y(-kF))$ define base point free linear systems $\mathfrak{d}$ and $\mathfrak{d}'$ on Y. For any $s \in W \backslash \{0\}$ and $s' \in W' \backslash \{0\}$, we have

$$f^* \operatorname{div}(s) = F + M_s \quad \text{and} \quad f^* \operatorname{div}(s') = kF + M'_{s'}$$

for members $M_s \in \mathfrak{d}$ and $M'_{s'} \in \mathfrak{d}'$, and M_s and $M'_{s'}$ are general members of $\mathfrak{d}$ and $\mathfrak{d}'$ for general s and s'. By Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], applied to Y and to the intersections of components of $\operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B \cup \operatorname{Supp} F$, on which the restrictions of $\mathfrak{d}$ and $\mathfrak{d}'$ have no base points, for general s and s', M_s and $M'_{s'}$ are smooth, have no common component with $\operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B \cup \operatorname{Supp} F$, and $\operatorname{Supp} M_s \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B \cup \operatorname{Supp} F$ and $\operatorname{Supp} M'_{s'} \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B \cup \operatorname{Supp} F$ are simple normal crossing divisors. Here a general member contains no given prime divisor, since the members through a given point form a hyperplane. For such s and s', let $G := G_s = \frac{1}{e}\operatorname{div}(s)$ and $G' := \frac{1}{ke}\operatorname{div}(s')$, and let (Γ, Δ) be either $(G, \frac{1}{e}M_s)$ or $(G', \frac{1}{ke}M'_{s'})$. Then

$$K_Y + B_Y + \frac{1}{e}F + \Delta = f^*(K_X + B + \Gamma),$$

and the components of Δ have coefficient $\frac{1}{e} \leq 1$ or $\frac{1}{ke} \leq 1$ in $B_Y + \frac{1}{e}F + \Delta$. By [KM98, Lemma 2.30], $a(E, X, B + \Gamma) = a(E, Y, B_Y + \frac{1}{e}F + \Delta)$ for any prime divisor E over X. Since

$\operatorname{center}_X E = f(\operatorname{center}_Y E)$, we have

$$\operatorname{Nlc}(X, B+\Gamma) = f\Big(\operatorname{Nlc}\big(Y, B_Y + \tfrac{1}{e}F + \Delta\big)\Big).$$

Let Σ be the union of the components of $B_Y + \frac{1}{e}F$ with coefficient > 1. By [KM98, Corollary 2.31(3)], $(Y\backslash\Sigma, (B_Y + \frac{1}{e}F + \Delta)|_{Y\backslash\Sigma})$ is sub-lc, and any component P of Σ satisfies $a(P, Y, B_Y + \frac{1}{e}F + \Delta) < 0$. Thus $\operatorname{Nlc}(Y, B_Y + \frac{1}{e}F + \Delta) = \Sigma$, and $\operatorname{Nlc}(X, B+\Gamma) = f(\Sigma)$. Since Σ depends only on W, we have $\operatorname{Nlc}(X, B+G) = \operatorname{Nlc}(X, B+G')$. □

8. Leading divisors

In this section, we introduce adapted carriers, leading exponents, and leading divisors (Construction-Proposition 8.1). For a stationary valuation v, the value of v does not show at which points of $\operatorname{center}_X v$ the lc condition fails. The leading divisors are $\mathbb{Q}$-divisors on a stratum S of a log resolution at whose generic point v is monomial, and they move the lc problem near v to S. We show that the leading divisors of the partial sums $D_{\le N}$ define sub-lc sub-pairs on S (Lemma 8.6), and Proposition 9.1 in Section 9 goes in the other direction, from sub-lc on S to lc near the center of v.

8.1. **Adapted carriers.** In this subsection, we construct adapted carriers, leading exponents, and leading divisors, and the valuations w_t of Construction-Proposition 8.3.

Let X' be a smooth variety and S a smooth closed subvariety of X'. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D' on X' whose support may contain S, we denote by $D'|_S$ the class of $\mathcal{O}_{X'}(lD')|_S$ in $\operatorname{Pic}(S)\otimes_{\mathbb{Z}}\mathbb{Q}$ divided by l, where l is a positive integer such that lD' is Cartier. If no component of $\operatorname{Supp} D'$ contains S, then $D'|_S$ also denotes the restriction of D' to S as a $\mathbb{Q}$-divisor, whose class is the class above.

In Construction-Proposition 8.1, v is monomial at the generic point of a stratum S of a log resolution X' of (X, B), with weights $\alpha_1, \dots, \alpha_r$ linearly independent over $\mathbb{Q}$. Thus the local equation of a Cartier divisor $D' \ge 0$ on X' at the generic point of S has a unique monomial of lowest weight, whose coefficient extends to a section of a line bundle on S, and the divisor of the section gives the leading divisor (Construction-Proposition 8.1(4)). The set Σ and the morphism μ are used in Section 10.

Construction-Proposition 8.1. Let (X, B) be a sub-pair such that X is projective of dimension d and B is a $\mathbb{Q}$-divisor, v a quasi-monomial valuation of $\mathbb{C}(X)$ of rational rank $r \ge 1$, Σ a finite set of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X, and $\mu : \widetilde{X} \to X$ a projective birational morphism from a normal variety $\widetilde{X}$. Then there exist a projective log resolution $h : X' \to X$ of (X, B) that factors through μ, a reduced simple normal crossing divisor E on X', components $E_1, \dots, E_r$ of E, an irreducible component S of $E_1 \cap \cdots \cap E_r$, and $\alpha = (\alpha_1, \dots, \alpha_r) \in \mathbb{R}^r_{>0}$ satisfying (1) and (2) below. We call (h, S) an *adapted carrier* of v for (X, B) and Σ. Moreover, any adapted carrier of v for (X, B) and Σ satisfies (3)–(7) below.

(1) E contains $\operatorname{Exc}(h)$, $\operatorname{Supp} B'$, and $\operatorname{Supp} h^*D$ for any $D \in \Sigma$, where $K_{X'} + B' := h^*(K_X + B)$.
(2) $\alpha_1, \dots, \alpha_r$ are linearly independent over $\mathbb{Q}$, and v is the monomial valuation with weights α in $E_1, \dots, E_r$ at the generic point of S.
(3) S is a smooth projective variety of dimension $d - r$, $h(S) = \operatorname{center}_X v$, and $k_v = \mathbb{C}(S)$.
(4) For any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \ge 0$ on X', there exist a unique vector $b(D') = (b(D')_1, \dots, b(D')_r) \in \mathbb{Q}^r_{\ge 0}$, called the *leading exponent* of D', and a unique $\mathbb{Q}$-divisor $\operatorname{lead}_S(D') \ge 0$ on S, called the *leading divisor* of D' along S, with the following property: if l is a positive integer such that lD' is Cartier, $y \in S$ is a closed point, $z_1, \dots, z_r$ are local equations of $E_1, \dots, E_r$ near y, and ϕ is a local equation of lD' near y, then $\phi \equiv \psi z^{lb(D')}$ modulo the ideal generated by the monomials z^β with $\sum_i \alpha_i\beta_i > l\sum_i \alpha_i b(D')_i$, for a regular function ψ such that $\operatorname{div}(\psi|_S) = l \operatorname{lead}_S(D')$ near y.

(5) For any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors $D', D'' \geq 0$ on X' and any positive integer l such that lD' is Cartier, we have $v(D') = \sum_i \alpha_i b(D')_i$, $b(D' + D'') = b(D') + b(D'')$, $\operatorname{lead}_S(D' + D'') = \operatorname{lead}_S(D') + \operatorname{lead}_S(D'')$, and

$$\mathcal{O}_S(l \operatorname{lead}_S(D')) \cong \mathcal{O}_{X'}\Big(lD' - \sum_{i=1}^{r} lb(D')_i E_i\Big)\Big|_S.$$

(6) The $\mathbb{Q}$-divisor $B_S := \big(B' - \sum_{i=1}^{r}(1 - a(E_i, X, B))E_i\big)|_S$ satisfies

$$K_S + B_S \sim_{\mathbb{Q}} (K_{X'} + B')|_S + \sum_{i=1}^{r} a(E_i, X, B)E_i|_S.$$

(7) For any $D \in \Sigma$ with $D \geq 0$, we have $b(h^*D)_i = \operatorname{mult}_{E_i} h^*D$ for any i, and $\operatorname{lead}_S(h^*D) = (h^*D - \sum_i (\operatorname{mult}_{E_i} h^*D)E_i)|_S$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct an adapted carrier of v for (X, B) and Σ.

By Definition 2.7, there exist a projective birational morphism $h_0 : X_0 \to X$ from a smooth variety X_0 and a reduced simple normal crossing divisor on X_0 such that v is a monomial valuation in some components of that divisor at the generic point of an irreducible component of their intersection. Since X_0 is regular, by [JM12, Lemma 3.6(ii)], there exists a log smooth pair (Y, D_Y) over X_0 which is good for v in the sense of [JM12, Definition 3.5]: v is a monomial valuation in some components of D_Y at the generic point of an irreducible component of their intersection, and the values of v on the components of D_Y on which v is positive are linearly independent over $\mathbb{Q}$. By Chow's lemma [Har77, Exercise II.4.10] and [KM98, Theorem 0.2], there exists a projective log resolution $h : X' \to X$ of $(X, B + \sum_{D\in\Sigma} \operatorname{Supp} D)$ which factors through μ and through $Y \to X$, such that the reduced divisor E whose support is the union of the preimage of $\operatorname{Supp} D_Y$, $\operatorname{Exc}(h)$, $\operatorname{Supp} B'$, and $\operatorname{Supp} h^*D$ for $D \in \Sigma$ is a simple normal crossing divisor, where $K_{X'} + B' := h^*(K_X + B)$. Since the preimage of $\operatorname{Supp} D_Y$ is contained in E, (X', E) is good for v by [JM12, Lemma 3.6(i)]. Thus there exist distinct components $E_1, \dots, E_s$ of E, an irreducible component S of $E_1 \cap \cdots \cap E_s$, and $\alpha \in \mathbb{R}^s_{>0}$ with $\alpha_1, \dots, \alpha_s$ linearly independent over $\mathbb{Q}$, such that v is the monomial valuation with weights α in $E_1, \dots, E_s$ at the generic point η_S of S.

By Definition 2.7, v takes values in $\sum_i \mathbb{N}\alpha_i$ on $\mathcal{O}_{X',\eta_S}\backslash\{0\}$, and $v(z_i) = \alpha_i$ for a local equation z_i of E_i at η_S. Since $\mathbb{C}(X)$ is the fraction field of $\mathcal{O}_{X',\eta_S}$, we have $v(\mathbb{C}(X)^*) = \sum_i \mathbb{Z}\alpha_i$. Hence s is the rational rank of v, that is, $s = r$. Thus h, E, $E_1, \dots, E_r$, S, and α satisfy (1) and (2).

Step 2. In this step, we fix an adapted carrier (h, S) of v for (X, B) and Σ, with E, $E_1, \dots, E_r$, and α satisfying (1) and (2), and prove (3).

Since E is a simple normal crossing divisor, $E_1 \cap \cdots \cap E_r$ is smooth of pure dimension $d - r$, and the irreducible components of $E_1 \cap \cdots \cap E_r$ are pairwise disjoint. Since X and h are projective, S is a smooth projective variety of dimension $d - r$. If a component of E other than $E_1, \dots, E_r$ contained S, then $r + 1$ components of E would contain the generic point η_S of S, which is a point of codimension r. Since E is a simple normal crossing divisor, this is not possible. Thus no component of E other than $E_1, \dots, E_r$ contains S. In particular, by (1), B_S in (6) and $(h^*D - \sum_i (\operatorname{mult}_{E_i} h^*D)E_i)|_S$ in (7) are restrictions of $\mathbb{Q}$-divisors whose supports do not contain S.

Since $\alpha \in \mathbb{R}^r_{>0}$, $v \geq 0$ on $\mathcal{O}_{X',\eta_S}$ and $v > 0$ on the maximal ideal of $\mathcal{O}_{X',\eta_S}$, so η_S is the center of v on X'. Since h induces a local homomorphism $\mathcal{O}_{X,h(\eta_S)} \to \mathcal{O}_{X',\eta_S}$, $h(\eta_S)$ is the center of v on X. Since h is proper, $h(S)$ is the closure of $h(\eta_S)$, that is, $h(S) = \operatorname{center}_X v$.

We show that $k_v = \mathbb{C}(S)$. Since $\mathcal{O}_{X',\eta_S}$ is contained in the valuation ring of v and the maximal ideal of $\mathcal{O}_{X',\eta_S}$ is contained in the maximal ideal of the valuation ring, there exists an embedding of fields $\mathbb{C}(S) \to k_v$. Any non-zero element of k_v is the image of f/g for some $f, g \in \mathcal{O}_{X',\eta_S}\backslash\{0\}$ with $v(f) = v(g)$. Write $\widehat{\mathcal{O}}_{X',\eta_S} = \kappa[[z_1, \dots, z_r]]$ for a coefficient field κ as in Definition 2.7. Since $\alpha_1, \dots, \alpha_r$ are linearly independent over $\mathbb{Q}$, distinct exponents $\beta \in \mathbb{N}^r$ have distinct weights

$\sum_i \alpha_i\beta_i$. By (2), the monomials of lowest weight in f and in g are unique, and since $v(f) = v(g)$, they are $c_f z^\beta$ and $c_g z^\beta$ for some $\beta \in \mathbb{N}^r$ and some $c_f, c_g \in \kappa\backslash\{0\}$. Let $\phi \in \mathcal{O}_{X',\eta_S}$ be an element whose image in the residue field $\mathbb{C}(S)$ is the image of c_f/c_g. Then $\phi - c_f/c_g$ belongs to the ideal $(z_1, \dots, z_r)$ of $\kappa[[z_1, \dots, z_r]]$, so any monomial in $f - \phi g$ has weight $> v(g)$. Hence $v(f - \phi g) > v(g)$, that is, $v(f/g - \phi) > 0$. Thus the image of f/g in k_v is the image of ϕ, which belongs to $\mathbb{C}(S)$. Therefore, $k_v = \mathbb{C}(S)$.

Step 3. In this step, we construct leading exponents and leading divisors and prove (4).

For a real number $\lambda \geq 0$, let

$$I_{\geq\lambda} := \sum_{\beta\in\mathbb{N}^r,\ \sum_i \alpha_i\beta_i\geq\lambda} \mathcal{O}_{X'}\Big(-\sum_{i=1}^r \beta_i E_i\Big),$$

and let $I_{>\lambda}$ be the sum of the ideals $\mathcal{O}_{X'}(-\sum_{i=1}^r \beta_i E_i)$ over $\beta \in \mathbb{N}^r$ with $\sum_i \alpha_i\beta_i > \lambda$. Since $\mathcal{O}_{X'}$ is Noetherian, $I_{\geq\lambda}$ and $I_{>\lambda}$ are coherent ideal sheaves. If $y \in S$ is a closed point and $z_1, \dots, z_r$ are local equations of $E_1, \dots, E_r$ near y, then near y, $I_{\geq\lambda}$ (resp. $I_{>\lambda}$) is generated by the monomials z^β with $\sum_i \alpha_i\beta_i \geq \lambda$ (resp. $\sum_i \alpha_i\beta_i > \lambda$). Since $\alpha \in \mathbb{R}^r_{>0}$, for any λ, only finitely many $\beta \in \mathbb{N}^r$ satisfy $\sum_i \alpha_i\beta_i \leq \lambda$. We have $I_{\geq\lambda}I_{\geq\lambda'} \subset I_{\geq\lambda+\lambda'}$ and $I_{\geq\lambda}I_{>\lambda'} \subset I_{>\lambda+\lambda'}$ for any λ, λ'. Let U be the complement in X' of the irreducible components of $E_1 \cap \cdots \cap E_r$ other than S.

Let $\beta \in \mathbb{N}^r$ and $\lambda := \sum_i \alpha_i\beta_i$. We show that the inclusion $\mathcal{O}_{X'}(-\sum_i \beta_i E_i) \subset I_{\geq\lambda}$ induces an isomorphism

$$\mathcal{O}_{X'}\Big(-\sum_{i=1}^r \beta_i E_i\Big)\Big|_S \cong (I_{\geq\lambda}/I_{>\lambda})|_U. \tag{8.1}$$

Since $\alpha_1, \dots, \alpha_r$ are linearly independent over $\mathbb{Q}$, β is the only element of $\mathbb{N}^r$ of weight λ. Thus near any closed point $y \in S$, we have $I_{\geq\lambda} = (z^\beta) + I_{>\lambda}$ and $z_i z^\beta \in I_{>\lambda}$ for any i. Since $S = V(z_1, \dots, z_r)$ near y, the inclusion induces a surjection $\mathcal{O}_{X'}(-\sum_i \beta_i E_i)|_S \to (I_{\geq\lambda}/I_{>\lambda})|_U$. Let $z_1, \dots, z_r, x_1, \dots, x_{d-r}$ be a regular system of parameters of $\mathcal{O}_{X',y}$. An element of $\widehat{\mathcal{O}}_{X',y} = \mathbb{C}[[z_1, \dots, z_r, x_1, \dots, x_{d-r}]]$ belongs to $I_{>\lambda}\widehat{\mathcal{O}}_{X',y}$ if and only if any monomial $z^{\beta'}x^\tau$ with non-zero coefficient in the expansion of the element satisfies $\sum_i \alpha_i\beta'_i > \lambda$. Hence if $gz^\beta \in I_{>\lambda}$ for some $g \in \mathcal{O}_{X',y}$, then $g \in (z_1, \dots, z_r)$. Thus the surjection is injective, which proves (8.1).

Let $D' \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X' and l a positive integer such that lD' is Cartier. Let ϕ_0 be a generator of $\mathcal{O}_{X'}(-lD')$ at η_S. Write $\widehat{\mathcal{O}}_{X',\eta_S} = \kappa[[z_1, \dots, z_r]]$ as in **Step 2**. The expansion of ϕ_0 has a unique monomial cz^u of lowest weight, where $c \in \kappa\backslash\{0\}$ and $u \in \mathbb{N}^r$. Let $\lambda := \sum_i \alpha_i u_i$. We show that $\mathcal{O}_{X'}(-lD') \subset I_{\geq\lambda}$ on U. Let $\lambda_1 < \cdots < \lambda_k$ be the weights of the elements $\beta \in \mathbb{N}^r$ with $\sum_i \alpha_i\beta_i < \lambda$, and $\lambda_{k+1} := \lambda$. Then $\lambda_1 = 0$, $I_{\geq\lambda_1} = \mathcal{O}_{X'}$, and $I_{>\lambda_j} = I_{\geq\lambda_{j+1}}$ for any $j \leq k$. Suppose that $\mathcal{O}_{X'}(-lD') \subset I_{\geq\lambda_j}$ on U for some $j \leq k$. By (8.1), the composite map $\mathcal{O}_{X'}(-lD')|_U \to (I_{\geq\lambda_j}/I_{>\lambda_j})|_U$ is a homomorphism between line bundles on S. Any monomial in the expansion of ϕ_0 has weight $\geq \lambda > \lambda_j$, so $\phi_0 \in I_{>\lambda_j}\widehat{\mathcal{O}}_{X',\eta_S}$. Since $\widehat{\mathcal{O}}_{X',\eta_S}$ is faithfully flat over $\mathcal{O}_{X',\eta_S}$ [Mat86, Theorem 8.14], ϕ_0 belongs to the stalk of $I_{>\lambda_j}$ at η_S. Thus the homomorphism vanishes at η_S. Since S is integral, the homomorphism vanishes on S. Hence $\mathcal{O}_{X'}(-lD') \subset I_{>\lambda_j} = I_{\geq\lambda_{j+1}}$ on U. By induction on j, we have $\mathcal{O}_{X'}(-lD') \subset I_{\geq\lambda}$ on U. By (8.1), the composite map

$$\mathcal{O}_{X'}(-lD')|_U \to (I_{\geq\lambda}/I_{>\lambda})|_U \cong \mathcal{O}_{X'}\Big(-\sum_{i=1}^r u_i E_i\Big)\Big|_S$$

is a section σ_l of $\mathcal{O}_{X'}(lD' - \sum_i u_i E_i)|_S$, and σ_l does not vanish at η_S because $c \neq 0$. We define

$$b(D') := \frac{1}{l}u \quad \text{and} \quad \mathrm{lead}_S(D') := \frac{1}{l}\operatorname{div}(\sigma_l) \geq 0.$$

If l_0 is a positive integer, then $\phi_0^{l_0}$ generates $\mathcal{O}_{X'}(-l_0 lD')$ at η_S, the lowest monomial of $\phi_0^{l_0}$ is $c^{l_0}z^{l_0 u}$, and $\sigma_{l_0 l} = \sigma_l^{\otimes l_0}$ since $I_{\geq\lambda}^{l_0} \subset I_{\geq l_0\lambda}$ and $I_{\geq(l_0-1)\lambda}I_{>\lambda} \subset I_{>l_0\lambda}$. Thus $b(D')$ and $\operatorname{lead}_S(D')$ do not depend on l.

Let l, y, $z_1, \dots, z_r$, and ϕ be as in (4). Since $\phi \in I_{\geq\lambda} = (z^u) + I_{>\lambda}$ near y, we have $\phi = \psi z^u + f$ for a regular function ψ near y and some $f \in I_{>\lambda}$, and $I_{>\lambda}$ is generated near y by the monomials z^β with $\sum_i \alpha_i\beta_i > \lambda = l\sum_i \alpha_i b(D')_i$. With respect to the local generators ϕ of $\mathcal{O}_{X'}(-lD')$ and z^u of $\mathcal{O}_{X'}(-\sum_i u_i E_i)$, the section σ_l is $\psi|_S$. Hence $\operatorname{div}(\psi|_S) = l\operatorname{lead}_S(D')$ near y. The ideals $I_{\geq\lambda}$ and $I_{>\lambda}$ are defined without local equations, and replacing ϕ by $\epsilon\phi$ for a unit ϵ replaces ψ by $\epsilon\psi$, so the property in (4) holds for any choice of $z_1, \dots, z_r$ and ϕ.

We prove the uniqueness. Suppose that $b' \in \mathbb{Q}_{\geq 0}^r$ and a $\mathbb{Q}$-divisor $L' \geq 0$ on S satisfy the property in (4) in place of $b(D')$ and $\operatorname{lead}_S(D')$. Let l, y, $z_1, \dots, z_r$, and ϕ be as in (4), and let ψ' be a regular function such that $\phi \equiv \psi' z^{lb'}$ modulo the ideal generated by the monomials z^β with $\sum_i \alpha_i\beta_i > l\sum_i \alpha_i b'_i$, and $\operatorname{div}(\psi'|_S) = lL'$ near y. Since $\psi'|_S \neq 0$, the image of ψ' in $\mathbb{C}(S)$ is non-zero. Thus the monomial of lowest weight in the expansion of ϕ in $\kappa[[z_1, \dots, z_r]]$ has exponent lb'. Hence $lb' = u$, that is, $b' = b(D')$. Then $(\psi - \psi')z^u \in I_{>\lambda}$ near y, so $\psi|_S = \psi'|_S$ by (8.1). Thus $L' = \operatorname{lead}_S(D')$ near y. Since y is an arbitrary closed point of S, $L' = \operatorname{lead}_S(D')$. This proves (4).

Step 4. In this step, we prove (5).

Let D', D'', and l be as in (5), and ϕ_0, u, and σ_l as in **Step 3**. By (2), we have

$$v(D') = \frac{1}{l}v(\phi_0) = \frac{1}{l}\sum_{i=1}^r \alpha_i u_i = \sum_{i=1}^r \alpha_i b(D')_i.$$

Since σ_l is a non-zero section of $\mathcal{O}_{X'}(lD' - \sum_i lb(D')_i E_i)|_S$ with $\operatorname{div}(\sigma_l) = l\operatorname{lead}_S(D')$, we have

$$\mathcal{O}_S(l\operatorname{lead}_S(D')) \cong \mathcal{O}_{X'}\Big(lD' - \sum_{i=1}^r lb(D')_i E_i\Big)\Big|_S.$$

Let l_0 be a positive integer such that l_0D' and l_0D'' are Cartier, $y \in S$ a closed point, and $z_1, \dots, z_r$ local equations of $E_1, \dots, E_r$ near y. By (4), local equations ϕ' of l_0D' and ϕ'' of l_0D'' near y satisfy $\phi' \equiv \psi' z^{l_0 b(D')}$ and $\phi'' \equiv \psi'' z^{l_0 b(D'')}$ modulo the ideals generated by the monomials of weight $> l_0\sum_i \alpha_i b(D')_i$ and $> l_0\sum_i \alpha_i b(D'')_i$ respectively, with $\operatorname{div}(\psi'|_S) = l_0\operatorname{lead}_S(D')$ and $\operatorname{div}(\psi''|_S) = l_0\operatorname{lead}_S(D'')$ near y. Then $\phi'\phi''$ is a local equation of $l_0(D' + D'')$ near y, and

$$\phi'\phi'' \equiv \psi'\psi'' z^{l_0(b(D')+b(D''))}$$

modulo the ideal generated by the monomials z^β with $\sum_i \alpha_i\beta_i > l_0\sum_i \alpha_i(b(D')_i + b(D'')_i)$. Since $\operatorname{div}(\psi'\psi''|_S) = l_0(\operatorname{lead}_S(D') + \operatorname{lead}_S(D''))$ near y, the uniqueness in (4) implies that $b(D' + D'') = b(D') + b(D'')$ and $\operatorname{lead}_S(D' + D'') = \operatorname{lead}_S(D') + \operatorname{lead}_S(D'')$. This proves (5).

Step 5. In this step, we prove (6) and (7).

Applying the adjunction formula [Har77, Proposition II.8.20] r times, to the irreducible component of $E_1 \cap \cdots \cap E_k$ containing S for $k = 1, \dots, r$, we have

$$K_S \sim (K_{X'} + E_1 + \cdots + E_r)|_S.$$

Let $B'_0 := B' - \sum_i (1 - a(E_i, X, B))E_i$. Since $a(E_i, X, B) = 1 - \operatorname{mult}_{E_i} B'$, no E_i is a component of B'_0. By (1), $\operatorname{Supp} B'_0$ is contained in the union of the components of E other than $E_1, \dots, E_r$, so no component of B'_0 contains S by **Step 2**, and $B_S = B'_0|_S$. Since $B'_0 + \sum_i E_i = B' + \sum_i a(E_i, X, B)E_i$, we have

$$K_S + B_S \sim_{\mathbb{Q}} \Big(K_{X'} + B'_0 + \sum_{i=1}^r E_i\Big)\Big|_S = (K_{X'} + B')|_S + \sum_{i=1}^r a(E_i, X, B)E_i|_S.$$

This proves (6).

Let $D \in \Sigma$ with $D \geq 0$, l a positive integer such that lD is Cartier, and $\beta_i := \mathrm{mult}_{E_i} h^*D$. Then lh^*D is Cartier, and $\operatorname{Supp} h^*D \subset E$ by (1). Let $y \in S$ be a closed point. Let $T_1, \dots, T_s$ be the components of E other than $E_1, \dots, E_r$ which contain y, and $x_1, \dots, x_s$ local equations of $T_1, \dots, T_s$ near y. Then a local equation of lh^*D near y is $\phi = \epsilon z^{l\beta} \prod_k x_k^{lc_k}$, where $\beta = (\beta_1, \dots, \beta_r)$, $c_k := \mathrm{mult}_{T_k} h^*D$, and ϵ is a unit near y. Let $\psi := \epsilon \prod_k x_k^{lc_k}$. Then $\phi = \psi z^{l\beta}$, and $\operatorname{div}(\psi|_S) = (lh^*D - \sum_i l\beta_i E_i)|_S$ near y, since no T_k contains S. By the uniqueness in (4), $b(h^*D) = \beta$ and $\mathrm{lead}_S(h^*D) = (h^*D - \sum_i \beta_i E_i)|_S$ near y. Since y is an arbitrary closed point of S, this proves (7). □

In Subsection 8.2 and Sections 10 and 11, we compute the leading exponents and leading divisors of rational linear combinations of $\mathbb{Q}$-divisors on X' by the following lemma.

Lemma 8.2. *Let (h, S) be an adapted carrier of v for (X, B) and Σ as in Construction-Proposition 8.1, with $h : X' \to X$. Let $D'_1, \dots, D'_n \geq 0$ be $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X' and $q_1, \dots, q_n$ rational numbers such that $\sum_{k=1}^n q_k D'_k \geq 0$. Then*

$$b\Big(\sum_{k=1}^n q_k D'_k\Big) = \sum_{k=1}^n q_k b(D'_k) \quad \text{and} \quad \mathrm{lead}_S\Big(\sum_{k=1}^n q_k D'_k\Big) = \sum_{k=1}^n q_k \,\mathrm{lead}_S(D'_k).$$

Proof. By Construction-Proposition 8.1(5), b and lead_S are additive on $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors ≥ 0 on X'. Let $D' \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X'. Then $b(0) = b(0+0) = 2b(0)$, so $b(0) = 0$, $b(pD') = pb(D')$ for any positive integer p, and $p'b(\frac{p}{p'}D') = b(pD') = pb(D')$ for any positive integers p, p'. Thus $b(\lambda D') = \lambda b(D')$ for any rational number $\lambda \geq 0$. Replacing b with lead_S in the argument above, we have $\mathrm{lead}_S(\lambda D') = \lambda \,\mathrm{lead}_S(D')$. Applying the additivity to

$$\sum_{k=1}^n q_k D'_k + \sum_{q_k<0} (-q_k) D'_k = \sum_{q_k>0} q_k D'_k,$$

whose two sides are sums of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors ≥ 0, we obtain the equalities in Lemma 8.2. □

For a divisorial valuation u of $\mathbb{C}(S)$, the valuations w_t of Construction-Proposition 8.3 are monomial in the strict transforms of $E_1, \dots, E_r$ and a prime divisor over X' whose center on X' is contained in S, and as $t \to +\infty$, the difference $w_t(D') - tv(D')$ tends to $u(\mathrm{lead}_S(D'))$. Thus the valuations w_t turn the lc inequalities on X into inequalities on S.

Construction-Proposition 8.3. Let (X, B) be a projective sub-pair such that B is a $\mathbb{Q}$-divisor, v a quasi-monomial valuation of $\mathbb{C}(X)$, (h, S) an adapted carrier of v for (X, B) and a finite set Σ, with $h : X' \to X$, $E_1, \dots, E_r$, α, and B_S as in Construction-Proposition 8.1, and u a divisorial valuation of $\mathbb{C}(S)$. Then for any sufficiently large real number t, there exists a quasi-monomial valuation w_t of $\mathbb{C}(X)$ such that

$$a(w_t, X, B) = t \sum_{i=1}^r \alpha_i a(E_i, X, B) + a(u, S, B_S),$$

and such that for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \geq 0$ on X', we have $w_t(D') = tv(D') + e_{D'}(t)$ for a function $e_{D'}(t) \geq 0$ such that $\lim_{t\to+\infty} e_{D'}(t) = u(\mathrm{lead}_S(D'))$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a sequence of blow-ups of X' along smooth centers contained in S.

Let E be the reduced simple normal crossing divisor of Construction-Proposition 8.1, and R_u the valuation ring of u. Then R_u is a DVR whose residue field has transcendence degree $\dim S - 1$. Since S is projective, by [KM98, Lemma 2.45], a finite sequence of blow-ups of S, each along the center of u on the previous variety, yields a projective birational morphism $Y \to S$ such that $\mathcal{O}_{Y,\eta_Y} = R_u$, where η_Y is the generic point of $\mathrm{center}_Y u$. By [Har77, Theorem II.7.17], $Y \to S$ is the blow-up of S along a coherent sheaf of ideals $\mathcal{I}$. Let $\mathcal{I}_\Gamma$ be the ideal sheaf of $\Gamma := \mathrm{center}_S u$

with the reduced structure. By [Wło05, Theorem 2.0.1] applied to $\mathcal{I}\cdot\mathcal{I}_\Gamma$, there exists a sequence of blow-ups $S^\circ_{k+1}\to S^\circ_k$ along smooth closed subvarieties $Z^\circ_k\subset S^\circ_k$ for $0\le k\le m-1$, where $S^\circ_0:=S$, with the following properties: let $E^\circ_{(k)}$ be the reduced divisor on S°_k whose support is the inverse image of $Z^\circ_0\cup\cdots\cup Z^\circ_{k-1}$, then $E^\circ_{(m)}$ is a simple normal crossing divisor, Z°_k has simple normal crossings with $E^\circ_{(k)}$ for any k, and the pullback of $\mathcal{I}\cdot\mathcal{I}_\Gamma$ to S°_m is the ideal of a divisor supported on $E^\circ_{(m)}$. Since S°_m is smooth and a product of two non-zero ideals of a domain is invertible only if both of them are, the pullbacks of $\mathcal{I}$ and $\mathcal{I}_\Gamma$ to S°_m are invertible. Thus $S^\circ_m\to S$ factors through Y by [Har77, Proposition II.7.14].

Let η° be the generic point of $\operatorname{center}_{S^\circ_m}u$. Then $\mathcal{O}_{S^\circ_m,\eta^\circ}$ dominates $\mathcal{O}_{Y,\eta_Y}=R_u$ and is dominated by R_u, so $\mathcal{O}_{S^\circ_m,\eta^\circ}=R_u$. Thus the closure P of η° is a prime divisor on S°_m, and $u=c\operatorname{ord}_P$ for a real number $c>0$. Since P maps into Γ, P is contained in the zero locus of the pullback of $\mathcal{I}_\Gamma$, which is contained in $E^\circ_{(m)}$. Thus P is a component of $E^\circ_{(m)}$, so P is smooth, and the union of P and the exceptional divisor of $S^\circ_m\to S$ is a simple normal crossing divisor, since the exceptional divisor of $S^\circ_m\to S$ is a union of components of $E^\circ_{(m)}$. The connected components of each Z°_k are pairwise disjoint, and a blow-up along a smooth center of codimension one is an isomorphism. Replacing each blow-up along Z°_k with the sequence of blow-ups along the connected components of Z°_k, and removing the blow-ups along centers of codimension one, we obtain an integer $n\ge 0$ and a sequence of blow-ups $\rho_k: S_{k+1}\to S_k$ along smooth irreducible closed subvarieties $Z_k\subset S_k$ of codimension at least 2 for $0\le k\le n-1$, where $S_0:=S$, such that $S_n=S^\circ_m$, since the removed blow-ups are isomorphisms, and Z_k has simple normal crossings with the exceptional divisor of $S_k\to S$ for any k. Thus P is a smooth prime divisor on S_n, $u=c\operatorname{ord}_P$, and the union of P and the exceptional divisor of $S_n\to S$ is a simple normal crossing divisor. Let $Z_n:=P$, $S_{n+1}:=S_n$, and $\rho_n:=\mathrm{id}_{S_n}$.

Let $X'_0:=X'$ and $E_{i,0}:=E_i$. For $0\le k\le n$, we regard S_k as the strict transform of S on X'_k, let $\pi_k: X'_{k+1}\to X'_k$ be the blow-up along Z_k, $E_{i,k+1}$ the strict transform of $E_{i,k}$, and F_k the exceptional divisor of π_k. Then the strict transform of S_k on X'_{k+1} is S_{k+1}. Let Δ_k and $\Delta_{S,k}$ be the divisors supported on the exceptional loci of $X'_k\to X'$ and $S_k\to S$ respectively such that

$$K_{X'_k}+\sum_{i=1}^r E_{i,k}=(X'_k\to X')^*\Big(K_{X'}+\sum_{i=1}^r E_i\Big)+\Delta_k\quad\text{and}\quad K_{S_k}=(S_k\to S)^*K_S+\Delta_{S,k}.$$

By induction on k, we have the following for $0\le k\le n+1$.

(1) X'_k is smooth, and the sum of $E_{1,k},\dots,E_{r,k}$ and the reduced exceptional divisor of $X'_k\to X'$ is a simple normal crossing divisor.
(2) S_k is an irreducible component of $E_{1,k}\cap\cdots\cap E_{r,k}$, and no exceptional divisor of $X'_k\to X'$ contains S_k. If $k\le n$, then the restriction to S_k of the reduced exceptional divisor of $X'_k\to X'$ is the reduced exceptional divisor of $S_k\to S$. Moreover, $F_{k-1}|_{S_k}$ is the exceptional divisor of ρ_{k-1} if $1\le k\le n$, and $F_n|_{S_{n+1}}=P$.
(3) No component of Δ_k contains S_k, and $\Delta_k|_{S_k}=\Delta_{S,k}$.

Since E is a simple normal crossing divisor, (1)–(3) hold for $k=0$. Assume that (1) and (2) hold for some $k\le n$. By (2), the restriction to S_k of the reduced exceptional divisor of $X'_k\to X'$ is the reduced exceptional divisor of $S_k\to S$, and Z_k has simple normal crossings with the reduced exceptional divisor of $S_k\to S$. Thus near any closed point of Z_k, there exist local coordinates $z_1,\dots,z_r,x_1,\dots,x_{d-r}$ of X'_k such that $E_{i,k}=V(z_i)$ for any i, $S_k=V(z_1,\dots,z_r)$, each exceptional divisor of $X'_k\to X'$ through the point is $V(x_a)$ for some a, and $Z_k=V(z_1,\dots,z_r,x_1,\dots,x_s)$, where $s:=\operatorname{codim}_{S_k}Z_k$. On the standard affine charts of the blow-up of X'_k along Z_k, X'_{k+1} is smooth, and $E_{1,k+1},\dots,E_{r,k+1}$, F_k, and the strict transforms of the exceptional divisors of $X'_k\to X'$ are coordinate hyperplanes. Thus (1) holds for $k+1$. By [Har77, Corollary II.7.15], the strict transform S_{k+1} of S_k is the blow-up of S_k along $\mathcal{I}_{Z_k}\cdot\mathcal{O}_{S_k}$, that is, ρ_k. Since $S_{k+1}\subset E_{i,k+1}$ and $E_{i,k+1}$ does not meet the chart on which z_i generates the pullback of the ideal of Z_k, S_{k+1} is covered by the charts with coordinates $z_1/x_b,\dots,z_r/x_b$, x_b, x_a/x_b for $a\le s$ with $a\ne b$,

and x_a for $a > s$, where $1 \leq b \leq s$. On such a chart, $E_{i,k+1} = V(z_i/x_b)$, $F_k = V(x_b)$, and $S_{k+1} = V(z_1/x_b, \dots, z_r/x_b)$. Thus S_{k+1} is an irreducible component of $E_{1,k+1} \cap \cdots \cap E_{r,k+1}$, no exceptional divisor of $X'_{k+1} \to X'$ contains S_{k+1}, $F_k|_{S_{k+1}}$ is the exceptional divisor of ρ_k, and the strict transform of $V(x_a)$ restricts to the strict transform of $V(x_a|_{S_k})$ on S_{k+1}. Thus (2) holds for $k+1$. If $k = n$, then $Z_n = P$ is a divisor on S_n, so $s = 1$, $x_1|_{S_n}$ is a local equation of P, and $F_n|_{S_{n+1}} = V(x_1)|_{S_{n+1}} = P$.

For (3), since $Z_k \subset E_{i,k}$ for any i and Z_k has codimension $r + \operatorname{codim}_{S_k} Z_k$ in X'_k, we have $\pi_k^* E_{i,k} = E_{i,k+1} + F_k$ and $K_{X'_{k+1}} = \pi_k^* K_{X'_k} + (r + \operatorname{codim}_{S_k} Z_k - 1)F_k$. Thus

$$\Delta_{k+1} = \pi_k^* \Delta_k + (\operatorname{codim}_{S_k} Z_k - 1)F_k \quad \text{and} \quad \Delta_{S,k+1} = \rho_k^* \Delta_{S,k} + (\operatorname{codim}_{S_k} Z_k - 1)F_k|_{S_{k+1}}.$$

Since no component of Δ_k contains S_k, we have $(\pi_k^* \Delta_k)|_{S_{k+1}} = \rho_k^*(\Delta_k|_{S_k})$, and (3) follows by induction.

Let $X'' := X'_{n+1}$, $S'' := S_{n+1}$, $E''_i := E_{i,n+1}$, $F := F_n$, $\Delta'' := \Delta_{n+1}$, $\Delta_S := \Delta_{S,n+1}$, and let $\pi : X'' \to X'$ and $\rho : S'' \to S$ be the composite morphisms. Let η_P be the generic point of $P \subset S'' \subset X''$. By (1) and (2), η_P is the generic point of an irreducible component of $E''_1 \cap \cdots \cap E''_r \cap F$. By the choice of P, ρ is a log resolution of $(S, 0)$, and by (3) and Definition 2.1,

$$a(u, S, 0) = ca(P, S, 0) = c(1 + \operatorname{mult}_P \Delta_S).$$

Step 2. In this step, we define w_t and compute the values of w_t on functions whose restrictions to S'' are non-zero.

Let $z''_1, \dots, z''_r$ and z''_0 be local equations of $E''_1, \dots, E''_r$ and F at η_P. By (1) of **Step 1**, $z''_1, \dots, z''_r, z''_0$ form a regular system of parameters of $\mathcal{O}_{X'',\eta_P}$, and $\widehat{\mathcal{O}}_{X'',\eta_P} = \kappa[[z''_1, \dots, z''_r, z''_0]]$ for a coefficient field κ. By (2) of **Step 1**, $S'' = V(z''_1, \dots, z''_r)$ near η_P and $P = V(z''_0)$ in S'' near η_P. Thus if $g \in \mathcal{O}_{X'',\eta_P}$ has the expansion $g = \sum_{\beta,k} q_{\beta,k} z''^{\beta} (z''_0)^k$, then $\operatorname{ord}_P(g|_{S''}) = \min\{k \mid q_{0,k} \neq 0\}$.

Since E is a simple normal crossing divisor and S is a component of $E_1 \cap \cdots \cap E_r$, no component of E other than $E_1, \dots, E_r$ contains S. Let $B'_0 := B' - \sum_i (1 - a(E_i, X, B))E_i$, where $K_{X'} + B' = h^*(K_X + B)$. Then $\operatorname{Supp} B'_0 \subset E$, no E_i is a component of B'_0, and $B_S = B'_0|_S$. Let $\mathcal{T}$ be the finite set of prime divisors on X'' which contain η_P and are components of Δ'', of $\pi^* E_i - E''_i$ for some i, or of $\pi^* B'_0$. No element of $\mathcal{T}$ contains S'': the exceptional divisors of π do not contain S'' by (2) of **Step 1**, and the strict transforms of the components of B'_0 do not contain S'' since no component of B'_0 contains $S = \pi(S'')$. Let $p_i := \operatorname{ord}_P((\pi^* E_i - E''_i)|_{S''})$ for any i. Choose $t_0 > 0$ such that for any $t > t_0$ and any i,

$$\gamma_i(t) := t\alpha_i - cp_i > c \max\Big(\{p_1, \dots, p_r\} \cup \{\operatorname{mult}_P(T|_{S''}) \mid T \in \mathcal{T}\}\Big).$$

For $t > t_0$, let w_t be the monomial valuation with weights $\gamma_1(t), \dots, \gamma_r(t), c$ in $E''_1, \dots, E''_r, F$ at η_P. Since $h \circ \pi : X'' \to X$ is a projective birational morphism from a smooth variety, w_t is a quasi-monomial valuation of $\mathbb{C}(X)$ by (1) of **Step 1** and Definition 2.7. The center of w_t on X'' is η_P, so the center ξ of w_t on X' is $\pi(\eta_P) = \rho(\eta_P)$, which is the generic point of $\operatorname{center}_S u$.

We claim that if $t > t_0$, $g \in \mathcal{O}_{X'',\eta_P}$, $g|_{S''} \neq 0$, and $\min_i \gamma_i(t) > c \operatorname{ord}_P(g|_{S''})$, then

$$w_t(g) = c \operatorname{ord}_P(g|_{S''}) = u(g|_{S''}).$$

To prove the claim, let $e := \operatorname{ord}_P(g|_{S''})$ and write $g = \sum_{\beta,k} q_{\beta,k} z''^{\beta} (z''_0)^k$ in $\kappa[[z''_1, \dots, z''_r, z''_0]]$. By Definition 2.7, $w_t(g) = \min\{\sum_i \gamma_i(t)\beta_i + ck \mid q_{\beta,k} \neq 0\}$. The term with $\beta = 0$ and $k = e$ has weight ce, any other term with $\beta = 0$ has $k \geq e$, and any term with $\beta \neq 0$ has weight $\geq \min_i \gamma_i(t) > ce$. Thus $w_t(g) = ce$.

Let $t > t_0$. By the claim, $w_t(T) = c \operatorname{mult}_P(T|_{S''})$ for any $T \in \mathcal{T}$. Thus by (3) of **Step 1**,

$$w_t(\Delta'') = c \operatorname{mult}_P \Delta_S, \quad w_t(\pi^* E_i - E''_i) = cp_i,$$
$$w_t(B'_0) = w_t(\pi^* B'_0) = c \operatorname{mult}_P \rho^* B_S = u(B_S),$$

where we use $(\pi^*B'_0)|_{S''} = \rho^*(B'_0|_S) = \rho^*B_S$. In particular, $w_t(E_i) = w_t(E''_i) + w_t(\pi^*E_i - E''_i) = \gamma_i(t) + cp_i = t\alpha_i$ for any i.

Step 3. In this step, we compute $a(w_t, X, B)$ for $t > t_0$.

By (1) of **Step 1**, π is a log resolution of the sub-pair $(X', \sum_i E_i)$, and $K_{X''} + \sum_i E''_i - \Delta'' = \pi^*(K_{X'} + \sum_i E_i)$. Let $A_{X'}$ and $A_{X''}$ be the log discrepancy functions of Definition-Proposition 2.8. By Definition-Proposition 2.8 applied to $(X'', 0)$ with the identity of X'', $A_{X''}(w_t) = a(w_t, X'', 0) = \sum_i \gamma_i(t) + c$. By Definition-Proposition 2.8 applied to $(X', \sum_i E_i)$ with π and by **Step 2**,

$$a\Big(w_t, X', \sum_{i=1}^r E_i\Big) = A_{X''}(w_t) - \sum_{i=1}^r w_t(E''_i) + w_t(\Delta'') = c(1 + \mathrm{mult}_P \Delta_S) = a(u, S, 0),$$

where the last equality holds by **Step 1**. By Definition-Proposition 2.8 applied to (X, B) with h and to $(X', \sum_i E_i)$ with the identity of X', we have $a(w_t, X, B) = A_{X'}(w_t) - w_t(B')$ and $a(w_t, X', \sum_i E_i) = A_{X'}(w_t) - \sum_i w_t(E_i)$. Since $B' = \sum_i (1 - a(E_i, X, B))E_i + B'_0$, **Step 2** implies that

$$\begin{aligned} a(w_t, X, B) &= a\Big(w_t, X', \sum_{i=1}^r E_i\Big) + \sum_{i=1}^r a(E_i, X, B) w_t(E_i) - w_t(B'_0) \\ &= a(u, S, 0) + t\sum_{i=1}^r \alpha_i a(E_i, X, B) - u(B_S). \end{aligned}$$

Since B_S has simple normal crossing support, the identity of S is a log resolution of (S, B_S), and Definition-Proposition 2.8 applied to (S, B_S) and to $(S, 0)$ with the identity of S implies that $a(u, S, B_S) = a(u, S, 0) - u(B_S)$. Thus

$$a(w_t, X, B) = t\sum_{i=1}^r \alpha_i a(E_i, X, B) + a(u, S, B_S).$$

Step 4. In this step, we compute $w_t(D')$ for $t > t_0$.

Let $D' \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X', l a positive integer such that lD' is Cartier, and $\lambda := l\sum_i \alpha_i b(D')_i$. By Construction-Proposition 8.1(5), $v(D') = \lambda/l$. Recall that ξ is the generic point of $\mathrm{center}_S u$. Let y be a closed point of $\mathrm{center}_S u$, and $z_1, \ldots, z_r$ and ϕ local equations of $E_1, \ldots, E_r$ and lD' near y. By Construction-Proposition 8.1(4), in $\mathcal{O}_{X',\xi}$ we have

$$\phi = \psi z^{lb(D')} + f,$$

where ψ is regular near y with $\mathrm{div}(\psi|_S) = l\,\mathrm{lead}_S(D')$ near y, and $f = \sum_\beta g_\beta z^\beta$ for finitely many $\beta \in \mathbb{N}^r$ with $\sum_i \alpha_i\beta_i > \lambda$ and some $g_\beta \in \mathcal{O}_{X',\xi}$. Let

$$\lambda' := \min\Big\{ \sum_{i=1}^r \alpha_i\beta_i \;\Big|\; \beta \in \mathbb{N}^r,\ \sum_{i=1}^r \alpha_i\beta_i > \lambda \Big\},$$

which exists and satisfies $\lambda' > \lambda$ since $\alpha \in \mathbb{R}^r_{>0}$. Since $w_t(z_i) = w_t(E_i) = t\alpha_i$ by **Step 2**, we have $w_t(z^{lb(D')}) = t\lambda$. Since the center of w_t on X' is ξ, we have $w_t(\psi) \geq 0$ and $w_t(g_\beta) \geq 0$ for any β, so $w_t(f) \geq t\lambda'$. Thus $w_t(\phi) \geq t\lambda$, so

$$e_{D'}(t) := w_t(D') - tv(D') = \frac{1}{l}\big(w_t(\phi) - t\lambda\big) \geq 0.$$

Since $\psi|_S \neq 0$ and $(\pi^*\psi)|_{S''} = \rho^*(\psi|_S)$, the claim in **Step 2** implies that $w_t(\psi) = u(\psi|_S) = lu(\mathrm{lead}_S(D'))$ for any sufficiently large t. If moreover $t(\lambda' - \lambda) > lu(\mathrm{lead}_S(D'))$, then $w_t(f) > t\lambda + w_t(\psi) = w_t(\psi z^{lb(D')})$, so $w_t(\phi) = t\lambda + lu(\mathrm{lead}_S(D'))$. Hence $e_{D'}(t) = u(\mathrm{lead}_S(D'))$ for any sufficiently large t. In particular, $\lim_{t\to+\infty} e_{D'}(t) = u(\mathrm{lead}_S(D'))$. □

8.2. **Stationary valuations.** In this subsection, we study the leading exponents and the leading divisors of an lc anti-canonical series along an adapted carrier of a stationary valuation.

Set-up 8.4. Let (X, B), $D_\bullet$, v, and (h, S) be as follows.

(1) (X, B) is a projective lc pair of dimension d such that B is a $\mathbb{Q}$-divisor.
(2) $D_\bullet$ is an lc anti-canonical series on (X, B).
(3) v is a quasi-monomial valuation of $\mathbb{C}(X)$ of rational rank $r \geq 1$ that is stationary for $(X, B; D_\bullet)$.
(4) (h, S) is an adapted carrier of v for (X, B) and a finite set Σ of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X, with $h : X' \to X$, $E_1, \dots, E_r$, α, and B_S as in Construction-Proposition 8.1.

Let $V := \operatorname{center}_X v$. Let $\mathbf{a} := (a(E_1, X, B), \dots, a(E_r, X, B))$. Let $b_j := b(h^*D_j)$ for $j \geq 1$, and write $b_j = (b_{j,1}, \dots, b_{j,r})$. For $\gamma \in \mathbb{R}^r_{>0}$, let v_γ be the monomial valuation with weights γ in $E_1, \dots, E_r$ at the generic point of S. Let

$$\delta(\gamma) := \sum_{i=1}^{r} \gamma_i a(E_i, X, B) - \sum_{j \geq 1} v_\gamma(D_j)$$

for $\gamma \in \mathbb{R}^r_{>0}$.

The next lemma computes the log discrepancy of v after adding a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ to B through the leading exponent of h^*G. It also shows that $(b_j)_{j \geq 1}$ is a moment row for $D_\bullet$, so that Lemma 7.2 provides correction families satisfying it. The Lipschitz property of δ is used in Construction-Proposition 9.3 to choose integral weight vectors.

Lemma 8.5. *Notation and conditions as in Set-up 8.4. For $j \geq 1$ and $\gamma \in \mathbb{R}^r_{>0}$, let $f_j(\gamma) := b_j \cdot \gamma - v_\gamma(D_j)$. Then the following hold.*

(1) $\sum_{j \geq 1} b_j = \mathbf{a}$. In particular, $(b_j)_{j \geq 1}$ is a moment row for $D_\bullet$.
(2) For any $j \geq 1$, $f_j \geq 0$ on $\mathbb{R}^r_{>0}$, f_j vanishes on a neighborhood of α, and $f_j(t\gamma) = t f_j(\gamma)$ for any $\gamma \in \mathbb{R}^r_{>0}$ and any real number $t > 0$. Moreover, there exist a compact neighborhood U of α in $\mathbb{R}^r_{>0}$ and a real number $c > 0$ such that f_j is $cv(D_j)$-Lipschitz on U for any $j \geq 1$.
(3) $\delta = \sum_{j \geq 1} f_j$ on $\mathbb{R}^r_{>0}$. In particular, $\delta \geq 0$ on $\mathbb{R}^r_{>0}$, $\delta(\alpha) = 0$, $\delta(t\gamma) = t\delta(\gamma)$ for any $\gamma \in \mathbb{R}^r_{>0}$ and any real number $t > 0$, and δ is $ca(v, X, B)$-Lipschitz on U.
(4) For any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on X, we have $a(v, X, B+G) = \sum_{i=1}^r \alpha_i \big(a(E_i, X, B) - b(h^*G)_i\big)$. In particular, $a(v, X, B + G) = 0$ if and only if $b(h^*G) = \mathbf{a}$.

Proof. The proof is long, so we divide it into several steps. For $x \in \mathbb{R}^r$, we write $\|x\|$ for the Euclidean norm of x and $|x|_1 := \sum_i |x_i|$.

Step 1. In this step, we compute the log discrepancies of the valuations v_γ.

Let $\gamma \in \mathbb{R}^r_{>0}$. By Construction-Proposition 8.1(1), $E_1, \dots, E_r$ are components of a reduced simple normal crossing divisor on X' containing $\operatorname{Exc}(h) \cup \operatorname{Supp} B'$, where $K_{X'} + B' := h^*(K_X + B)$. Thus Definition-Proposition 2.8 implies

$$a(v_\gamma, X, B) = \sum_{i=1}^{r} \gamma_i a(E_i, X, B). \tag{8.2}$$

By Construction-Proposition 8.1(2), $v = v_\alpha$. Thus $a(v, X, B) = \sum_{i=1}^r \alpha_i a(E_i, X, B)$.

Step 2. In this step, we prove (2).

Fix $j \geq 1$, and let l be a positive integer such that lD_j is Cartier. Let $y \in S$ be a closed point, $z_1, \dots, z_r$ local equations of $E_1, \dots, E_r$ near y, and ϕ a local equation of $h^*(lD_j)$ near y. Let η_S be the generic point of S. Then $z_1, \dots, z_r$ is a regular system of parameters of $\mathcal{O}_{X', \eta_S}$. Fix a coefficient field κ of $\widehat{\mathcal{O}}_{X', \eta_S}$, so that $\widehat{\mathcal{O}}_{X', \eta_S} = \kappa[[z_1, \dots, z_r]]$, and write

$$\phi = \sum_{\beta \in \mathbb{N}^r} c_\beta z^\beta, \quad c_\beta \in \kappa.$$

Let $\Gamma \subset \mathbb{N}^r$ be the set of β such that $c_\beta \neq 0$. Then $v_\gamma(\phi) = \min_{\beta\in\Gamma} \gamma\cdot\beta$ for any $\gamma \in \mathbb{R}^r_{>0}$ [JM12, Proposition 3.1]. By Dickson's lemma [Dic13, Lemma A], the set $\Gamma_{\min}$ of the minimal elements of Γ for the coordinatewise order is finite, and any element of Γ is coordinatewise at least some element of $\Gamma_{\min}$. Since $v_\gamma(D_j) = \frac{1}{l} v_\gamma(\phi)$ and γ has positive coordinates, we have

$$v_\gamma(D_j) = \frac{1}{l} \min_{\beta\in\Gamma_{\min}} \gamma\cdot\beta \quad \text{for any } \gamma \in \mathbb{R}^r_{>0}. \tag{8.3}$$

By Construction-Proposition 8.1(4) applied to $D' = h^*D_j$, after shrinking the neighborhood of y, we have $\phi \equiv \psi z^{lb_j}$ modulo the ideal generated by the monomials z^β with $\alpha\cdot\beta > l\alpha\cdot b_j$, where ψ is a regular function such that $\operatorname{div}(\psi|_S) = l\operatorname{lead}_S(h^*D_j)$. In particular, $\psi|_S \neq 0$, so ψ is a unit of $\mathcal{O}_{X',\eta_S}$, and $\psi \equiv \psi_0$ modulo $(z_1,\dots,z_r)$ in $\kappa[[z_1,\dots,z_r]]$ for some $\psi_0 \in \kappa\backslash\{0\}$. Thus $c_{lb_j} = \psi_0 \neq 0$, and $\alpha\cdot\beta > \alpha\cdot(lb_j)$ for any $\beta \in \Gamma\backslash\{lb_j\}$. In particular, $lb_j \in \Gamma_{\min}$. For $\beta \in \Gamma_{\min}$, let $\theta_\beta(\gamma) := \gamma\cdot(b_j - \frac{1}{l}\beta)$. By (8.3),

$$f_j(\gamma) = \max_{\beta\in\Gamma_{\min}} \theta_\beta(\gamma) \quad \text{for any } \gamma \in \mathbb{R}^r_{>0}.$$

Since $\theta_{lb_j} = 0$, $f_j \geq 0$ on $\mathbb{R}^r_{>0}$. For any $\beta \in \Gamma_{\min}\backslash\{lb_j\}$, $\theta_\beta(\alpha) < 0$. Since $\Gamma_{\min}$ is finite, there exists a neighborhood of α on which $\theta_\beta < 0$ for any $\beta \in \Gamma_{\min}\backslash\{lb_j\}$. Thus f_j vanishes on a neighborhood of α. By Definition 2.7, $v_{t\gamma} = tv_\gamma$ for any $\gamma \in \mathbb{R}^r_{>0}$ and any real number $t > 0$, so $f_j(t\gamma) = tf_j(\gamma)$.

Let $m_\alpha := \min_i \alpha_i$, let

$$U := \{\gamma \in \mathbb{R}^r \mid |\gamma_i - \alpha_i| \leq \tfrac{1}{2}\alpha_i \text{ for any } i\},$$

and let $c := 4/m_\alpha$. Then U is a compact neighborhood of α in $\mathbb{R}^r_{>0}$, and U and c do not depend on j. By Construction-Proposition 8.1(5), $\alpha\cdot b_j = v(h^*D_j) = v(D_j)$. Let Γ_U be the set of $\beta \in \Gamma_{\min}$ such that $\theta_\beta(\gamma) \geq 0$ for some $\gamma \in U$. Since $f_j \geq \theta_{lb_j} = 0$, we have $f_j = \max_{\beta\in\Gamma_U} \theta_\beta$ on U. Let $\beta \in \Gamma_U$ and $\gamma \in U$ such that $\theta_\beta(\gamma) \geq 0$. Since $\frac{1}{2}m_\alpha \leq \gamma_i \leq \frac{3}{2}\alpha_i$ for any i, we have

$$\frac{m_\alpha}{2l}|\beta|_1 \leq \frac{1}{l}\gamma\cdot\beta \leq \gamma\cdot b_j \leq \frac{3}{2}\alpha\cdot b_j = \frac{3}{2}v(D_j).$$

Moreover, $m_\alpha|b_j|_1 \leq \alpha\cdot b_j = v(D_j)$. Thus the gradient $b_j - \frac{1}{l}\beta$ of θ_β satisfies

$$\left\|b_j - \frac{1}{l}\beta\right\| \leq |b_j|_1 + \frac{1}{l}|\beta|_1 \leq \frac{4}{m_\alpha}v(D_j) = cv(D_j).$$

Since Γ_U is finite and the maximum of finitely many $cv(D_j)$-Lipschitz functions is $cv(D_j)$-Lipschitz, f_j is $cv(D_j)$-Lipschitz on U.

Step 3. In this step, we prove (1) and (3).

By **Step 1** and since v is stationary,

$$\sum_{j\geq 1} v(D_j) = a(v,X,B) = \sum_{i=1}^r \alpha_i a(E_i,X,B) < +\infty. \tag{8.4}$$

Since $b_{j,i} \geq 0$ and $\alpha_i b_{j,i} \leq \alpha\cdot b_j = v(D_j)$ for any i and j, the series $\sum_j b_j$ converges to a vector $b_\infty \in \mathbb{R}^r_{\geq 0}$. Let $\gamma \in \mathbb{R}^r_{>0}$. Since $0 \leq f_j(\gamma) \leq b_j\cdot\gamma$ by **Step 2**, the series $F(\gamma) := \sum_j f_j(\gamma)$ converges, and

$$\sum_{j\geq 1} v_\gamma(D_j) = \sum_{j\geq 1} \big(b_j\cdot\gamma - f_j(\gamma)\big) = b_\infty\cdot\gamma - F(\gamma).$$

Thus

$$\delta(\gamma) = (\mathbf{a} - b_\infty)\cdot\gamma + F(\gamma). \tag{8.5}$$

We show that $\delta(\gamma) \geq 0$. Let $N \geq 1$. By Definition 7.1(1), $(X, B + D_{\leq N})$ is lc. By Definition-Proposition 2.8 and (8.2),

$$0 \leq a(v_\gamma, X, B + D_{\leq N}) = \sum_{i=1}^{r} \gamma_i a(E_i, X, B) - \sum_{j=1}^{N} v_\gamma(D_j).$$

Letting $N \to +\infty$, we obtain $\delta(\gamma) \geq 0$. By (8.4) and $v = v_\alpha$, we have $\delta(\alpha) = 0$.

Since $f_j(\alpha) = 0$ for any j, $F(\alpha) = 0$. Thus (8.5) at $\gamma = \alpha$ implies $(\mathbf{a} - b_\infty) \cdot \alpha = 0$. Let $\nu \in \mathbb{R}^r$, and let $t_1 > 0$ be such that $\alpha + t\nu \in U$ for any $0 < t \leq t_1$. For any j, $f_j(\alpha + t\nu) = 0$ for t sufficiently small, and

$$0 \leq \frac{1}{t} f_j(\alpha + t\nu) = \frac{1}{t}\big(f_j(\alpha + t\nu) - f_j(\alpha)\big) \leq c v(D_j)\|\nu\|$$

for $0 < t \leq t_1$ by **Step 2**. By (8.4) and the dominated convergence theorem, $\frac{1}{t}F(\alpha + t\nu) \to 0$ as $t \to 0^+$. By (8.5) and $\delta \geq 0$,

$$0 \leq \frac{1}{t}\delta(\alpha + t\nu) = (\mathbf{a} - b_\infty) \cdot \nu + \frac{1}{t} F(\alpha + t\nu)$$

for $0 < t \leq t_1$. Letting $t \to 0^+$, we obtain $(\mathbf{a} - b_\infty) \cdot \nu \geq 0$. Since ν is arbitrary, $b_\infty = \mathbf{a}$. Since $b_j \in \mathbb{Q}^r$ by Construction-Proposition 8.1(4) and $\mathbf{a} \in \mathbb{Q}^r$ because B is a $\mathbb{Q}$-divisor, $(b_j)_{j \geq 1}$ is a moment row for $D_\bullet$.

Thus $\delta = F = \sum_j f_j$ on $\mathbb{R}^r_{>0}$ by (8.5). By **Step 2**, $\delta(t\gamma) = t\delta(\gamma)$ for any $\gamma \in \mathbb{R}^r_{>0}$ and any real number $t > 0$. By **Step 2** and (8.4), δ is $ca(v, X, B)$-Lipschitz on U.

Step 4. In this step, we prove (4).

Let $G \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Then $h^*G \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on the smooth variety X', and $v(G) = v(h^*G)$. By Definition-Proposition 2.8, **Step 1**, and Construction-Proposition 8.1(5),

$$\begin{aligned} a(v, X, B + G) &= a(v, X, B) - v(h^*G) = \sum_{i=1}^{r} \alpha_i a(E_i, X, B) - \sum_{i=1}^{r} \alpha_i b(h^*G)_i \\ &= \sum_{i=1}^{r} \alpha_i \big(a(E_i, X, B) - b(h^*G)_i\big). \end{aligned}$$

Since B is a $\mathbb{Q}$-divisor, $a(E_i, X, B) \in \mathbb{Q}$, and $b(h^*G) \in \mathbb{Q}^r$. Since $\alpha_1, \ldots, \alpha_r$ are linearly independent over $\mathbb{Q}$, $a(v, X, B + G) = 0$ if and only if $b(h^*G) = \mathbf{a}$. □

The next lemma describes the leading divisors on S. Part (1) is a condition on S that follows from the lc property of the pairs $(X, B + D_{\leq N})$ near the center of v. Part (2) shows that, for the corrected boundaries G_N of a correction family satisfying the moment row $(b_j)_{j \geq 1}$, the pair $(S, B_S + \mathrm{lead}_S(h^*G_N))$ has $\mathbb{Q}$-linearly trivial log canonical divisor, as $(X, B + G_N)$ does.

Lemma 8.6. *Notation and conditions as in Set-up 8.4. Then the following hold.*

1. *For any divisorial valuation u of $\mathbb{C}(S)$, $\sum_{j \geq 1} u(\mathrm{lead}_S(h^*D_j)) \leq a(u, S, B_S)$. In particular, $(S, B_S + \sum_{j=1}^{N} \mathrm{lead}_S(h^*D_j))$ is sub-lc for any $N \geq 1$.*
2. *Let $(c_{j,N})_{1 \leq j \leq m, N \geq N_0}$ be a correction family for $D_\bullet$ satisfying the moment row $(b_j)_{j \geq 1}$. Then for any $N \geq N_0$ such that $G_N \geq 0$, we have $a(v, X, B + G_N) = 0$ and $K_S + B_S + \mathrm{lead}_S(h^*G_N) \sim_{\mathbb{Q}} 0$.*

Proof. (1) Let u be a divisorial valuation of $\mathbb{C}(S)$. By Construction-Proposition 8.3, there exists a real number t_0 such that for any $t \geq t_0$, there exists a quasi-monomial valuation w_t of $\mathbb{C}(X)$ satisfying

$$a(w_t, X, B) = t \sum_{i=1}^{r} \alpha_i a(E_i, X, B) + a(u, S, B_S)$$

and

$$w_t(h^*D_j) = tv(h^*D_j) + e_{h^*D_j}(t) \quad \text{for any } j \geq 1,$$

where $e_{h^*D_j}(t) \geq 0$ and $\lim_{t\to+\infty} e_{h^*D_j}(t) = u(\mathrm{lead}_S(h^*D_j))$.

Fix $t \geq t_0$ and $N \geq 1$. By Definition 7.1(1), $(X, B + D_{\leq N})$ is lc. By Definition-Proposition 2.8, $a(w_t, X, B + D_{\leq N}) \geq 0$ and $a(w_t, X, B + D_{\leq N}) = a(w_t, X, B) - \sum_{j\leq N} w_t(D_j)$, and $w_t(D_j) = w_t(h^*D_j)$, $v(D_j) = v(h^*D_j)$. Thus

$$t\sum_{j=1}^{N} v(D_j) + \sum_{j=1}^{N} e_{h^*D_j}(t) \leq t\sum_{i=1}^{r} \alpha_i a(E_i, X, B) + a(u, S, B_S). \tag{8.6}$$

Keep t fixed and let $N \to +\infty$ in (8.6). Since v is stationary and $a(v, X, B) = \sum_i \alpha_i a(E_i, X, B)$ by Definition-Proposition 2.8 and Construction-Proposition 8.1(1)(2), we have $\sum_{j\leq N} v(D_j) \to \sum_i \alpha_i a(E_i, X, B)$. Since $e_{h^*D_j}(t) \geq 0$, we obtain

$$\sum_{j\geq 1} e_{h^*D_j}(t) \leq a(u, S, B_S) \quad \text{for any } t \geq t_0.$$

Let $L \geq 1$ be an integer. Then $\sum_{j=1}^{L} e_{h^*D_j}(t) \leq a(u, S, B_S)$ for any $t \geq t_0$. Letting $t \to +\infty$, we obtain $\sum_{j=1}^{L} u(\mathrm{lead}_S(h^*D_j)) \leq a(u, S, B_S)$. Letting $L \to +\infty$, we obtain the inequality in (1).

Let $N \geq 1$ and let P be a prime divisor over S. Since S is smooth, each $\mathrm{lead}_S(h^*D_j)$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. By Definition-Proposition 2.8, computed on a log resolution of $(S, \mathrm{Supp}\, B_S + \sum_{j\leq N} \mathrm{Supp}\, \mathrm{lead}_S(h^*D_j))$, and by the inequality in (1) for $u = \mathrm{ord}_P$,

$$a\Big(P, S, B_S + \sum_{j=1}^{N} \mathrm{lead}_S(h^*D_j)\Big) = a(P, S, B_S) - \sum_{j=1}^{N} \mathrm{mult}_P\, \mathrm{lead}_S(h^*D_j) \geq 0.$$

Thus $(S, B_S + \sum_{j=1}^{N} \mathrm{lead}_S(h^*D_j))$ is sub-lc.

(2) By Lemma 8.5(1), $(b_j)_{j\geq 1}$ is a moment row for $D_\bullet$ and $\sum_{j\geq 1} b_j = \mathbf{a}$. Thus, for any $N \geq N_0$,

$$\sum_{j=1}^{N} b_j + \sum_{j=1}^{m} c_{j,N} b_j = \mathbf{a}. \tag{8.7}$$

Let $N \geq N_0$ such that $G_N \geq 0$, and let $c_{j,N} := 0$ for $m < j \leq N$. Then $G_N = \sum_{j=1}^{N}(1+c_{j,N})D_j$, and Lemma 8.2 applied to $h^*G_N = \sum_{j=1}^{N}(1 + c_{j,N})h^*D_j$ implies

$$b(h^*G_N) = \sum_{j=1}^{N}(1 + c_{j,N})b_j \quad \text{and} \quad \mathrm{lead}_S(h^*G_N) = \sum_{j=1}^{N}(1 + c_{j,N})\, \mathrm{lead}_S(h^*D_j).$$

By (8.7), $b(h^*G_N) = \mathbf{a}$. Thus $a(v, X, B + G_N) = 0$ by Lemma 8.5(4).

Let l be a positive integer such that lh^*G_N is Cartier. By Construction-Proposition 8.1(5) and $b(h^*G_N) = \mathbf{a}$,

$$\mathcal{O}_S\big(l\, \mathrm{lead}_S(h^*G_N)\big) \cong \mathcal{O}_{X'}\Big(lh^*G_N - \sum_{i=1}^{r} la(E_i, X, B)E_i\Big)\Big|_S.$$

By Construction-Proposition 8.1(6),

$$K_S + B_S \sim_{\mathbb{Q}} (K_{X'} + B')|_S + \sum_{i=1}^{r} a(E_i, X, B)E_i|_S,$$

where $K_{X'} + B' := h^*(K_X + B)$. Combining the isomorphism and the $\mathbb{Q}$-linear equivalence above, we obtain

$$K_S + B_S + \mathrm{lead}_S(h^*G_N) \sim_{\mathbb{Q}} \big(h^*(K_X + B + G_N)\big)\big|_S \sim_{\mathbb{Q}} 0,$$

where the last relation holds because $K_X + B + G_N \sim_{\mathbb{Q}} 0$ by Definition 7.1. □

The next definition-lemma attaches to any section s of $\mathcal{O}_X(-el(K_X+B))$ with $a(v,X,B+\frac{1}{el}\operatorname{div}(s))\leq 0$ a section $\operatorname{lead}(s)$ of a line bundle on S, the image of h^*s in a graded piece of the monomial filtration defined by α. The map lead is linear and multiplicative, and Construction-Proposition 11.4(4) uses lead to lift systems of sections from S to X.

Definition-Lemma 8.7. Notation and conditions as in Set-up 8.4. Let e be a positive integer such that $-e(K_X+B)$ is Cartier and $e\mathbf{a}\in\mathbb{Z}^r$, and let $\mathcal{L}:=\mathcal{O}_X(-e(K_X+B))$ and

$$\mathcal{L}_S:=(h^*\mathcal{L})|_S\otimes\mathcal{O}_{X'}\Big(-\sum_{i=1}^r ea(E_i,X,B)E_i\Big)\Big|_S.$$

For any positive integer l, let $W_l\subset H^0(X,\mathcal{L}^{\otimes l})$ be the set of sections s such that $s=0$ or $a(v,X,B+\frac{1}{el}\operatorname{div}(s))\leq 0$, let $\lambda_l:=el\sum_{i=1}^r\alpha_i a(E_i,X,B)$, and let $I_{\geq\lambda_l}\supset I_{>\lambda_l}$ be the ideal sheaves on a neighborhood of S in X' that near each closed point of S are generated by the monomials z^β with $\sum_i\alpha_i\beta_i\geq\lambda_l$ and with $\sum_i\alpha_i\beta_i>\lambda_l$, respectively, where $z_1,\dots,z_r$ are local equations of $E_1,\dots,E_r$. Then W_l is a linear subspace, $I_{\geq\lambda_l}/I_{>\lambda_l}$ is an invertible $\mathcal{O}_S$-module isomorphic to $\mathcal{O}_{X'}(-\sum_{i=1}^r ela(E_i,X,B)E_i)|_S$, and h^*s is a section of $h^*\mathcal{L}^{\otimes l}\otimes I_{\geq\lambda_l}$ on a neighborhood of S for any $s\in W_l$. For $s\in W_l$, we define $\operatorname{lead}(s)\in H^0(S,\mathcal{L}_S^{\otimes l})$ to be the image of h^*s in $h^*\mathcal{L}^{\otimes l}\otimes(I_{\geq\lambda_l}/I_{>\lambda_l})\cong\mathcal{L}_S^{\otimes l}$. Then the following hold.

(1) $\operatorname{lead}: W_l\to H^0(S,\mathcal{L}_S^{\otimes l})$ is linear.
(2) $W_lW_{l'}\subset W_{l+l'}$, and $\operatorname{lead}(ss')=\operatorname{lead}(s)\operatorname{lead}(s')$ for any $s\in W_l$ and $s'\in W_{l'}$.
(3) For any $s\in W_l\backslash\{0\}$ and $G:=\frac{1}{el}\operatorname{div}(s)$, we have $\operatorname{lead}(s)\neq 0$ if and only if $a(v,X,B+G)=0$, in which case $\operatorname{div}(\operatorname{lead}(s))=el\operatorname{lead}_S(h^*G)$.

Proof. The proof is long, so we divide it into several steps. Let η_S be the generic point of S.

Step 1. In this step, we show that W_l is a linear subspace and that $W_lW_{l'}\subset W_{l+l'}$.

Let $s\in H^0(X,\mathcal{L}^{\otimes l})\backslash\{0\}$ and $G:=\frac{1}{el}\operatorname{div}(s)$. Then $G\geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. By Definition-Proposition 2.8 and Construction-Proposition 8.1(1)(2),

$$a(v,X,B+G)=a(v,X,B)-v(G)=\sum_{i=1}^r\alpha_i a(E_i,X,B)-\frac{1}{el}v(\operatorname{div}(s)).$$

Thus

$$\begin{aligned}&s\in W_l\text{ if and only if }v(\operatorname{div}(s))\geq\lambda_l,\\ &a(v,X,B+G)<0\text{ if and only if }v(\operatorname{div}(s))>\lambda_l.\end{aligned}\tag{8.8}$$

Let σ_l be a local generator of $\mathcal{L}^{\otimes l}$ at the generic point of V, and for $s\in H^0(X,\mathcal{L}^{\otimes l})\backslash\{0\}$, let φ_s be the rational function such that $s=\varphi_s\sigma_l$. Since V is the center of v on X, $v(\operatorname{div}(s))=v(\varphi_s)$. Let $s,s'\in W_l\backslash\{0\}$ and $\xi\in\mathbb{C}$ such that $s+\xi s'\neq 0$. Then $v(\varphi_s+\xi\varphi_{s'})\geq\min\{v(\varphi_s),v(\varphi_{s'})\}\geq\lambda_l$, so $s+\xi s'\in W_l$ by (8.8). Thus W_l is a linear subspace. Let $s\in W_l\backslash\{0\}$ and $s'\in W_{l'}\backslash\{0\}$. Since X is integral, $ss'\neq 0$, and $v(\operatorname{div}(ss'))=v(\operatorname{div}(s))+v(\operatorname{div}(s'))\geq\lambda_l+\lambda_{l'}=\lambda_{l+l'}$. Thus $ss'\in W_{l+l'}$ by (8.8).

Step 2. In this step, we show that the graded pieces of the monomial filtration defined by α are line bundles on S.

For a real number μ, let $I_{\geq\mu}\supset I_{>\mu}$ be the ideal sheaves on a neighborhood of S in X' defined by the monomials z^β with $\alpha\cdot\beta\geq\mu$ and with $\alpha\cdot\beta>\mu$, respectively, near each closed point of S. For $\mu=\lambda_l$, $I_{\geq\mu}\supset I_{>\mu}$ are the ideal sheaves $I_{\geq\lambda_l}\supset I_{>\lambda_l}$ of the statement. They do not depend on the choice of $z_1,\dots,z_r$, since replacing z_i by a unit multiple of z_i does not change the ideals generated by monomials, and by Dickson's lemma [Dic13, Lemma A], they are generated by finitely many monomials near any closed point of S. Since $E_1\cap\cdots\cap E_r$ is smooth, S is a connected component of $E_1\cap\cdots\cap E_r$. We may shrink the neighborhood of S so that $(z_1,\dots,z_r)$ is the ideal of S near any closed point of S.

Let $\beta \in \mathbb{N}^r$ and $\mu := \alpha \cdot \beta$. We show that

$$I_{\geq \mu}/I_{>\mu} \cong \mathcal{O}_{X'}\Big(-\sum_{i=1}^{r} \beta_i E_i\Big)\Big|_S, \tag{8.9}$$

where near any closed point of S, the isomorphism sends the image of z^β to the image of the local generator z^β of $\mathcal{O}_{X'}(-\sum_i \beta_i E_i)$. Let $y \in S$ be a closed point and $z_1, \dots, z_r$ local equations of $E_1, \dots, E_r$ near y. If $\beta' \in \mathbb{N}^r$ and $\alpha \cdot \beta' = \mu$, then $\alpha \cdot (\beta' - \beta) = 0$ and $\beta' - \beta \in \mathbb{Z}^r$, so $\beta' = \beta$ because $\alpha_1, \dots, \alpha_r$ are linearly independent over $\mathbb{Q}$. Thus $I_{\geq \mu} = z^\beta \mathcal{O}_{X'} + I_{>\mu}$ near y. Since $z_i z^\beta \in I_{>\mu}$ for any i, $I_{\geq\mu}/I_{>\mu}$ is generated by the image of z^β as an $\mathcal{O}_S$-module near y. Let g be a regular function near y such that $gz^\beta \in I_{>\mu}$. Since $v = v_\alpha$ and $v(z^{\beta'}) = \alpha \cdot \beta'$ for any $\beta' \in \mathbb{N}^r$, we have $v(gz^\beta) > \mu$, so $v(g) > 0$. Since η_S is the center of v on X', $g|_S$ vanishes at η_S, so $g|_S = 0$ near y. Thus $I_{\geq\mu}/I_{>\mu}$ is a free $\mathcal{O}_S$-module of rank one near y with basis the image of z^β. If z_i is replaced by $u_i z_i$ for units u_i, the image of z^β is multiplied by $\prod_i u_i^{\beta_i}|_S$, which is the transition function of $\mathcal{O}_{X'}(-\sum_i \beta_i E_i)|_S$. Thus the local isomorphisms glue to (8.9).

Since (X, B) is lc, $a(E_i, X, B) \geq 0$ for any i, so $el\mathbf{a} \in \mathbb{N}^r$ and $\lambda_l = \alpha \cdot (el\mathbf{a})$. By (8.9) for $\beta = el\mathbf{a}$, $I_{\geq\lambda_l}/I_{>\lambda_l}$ is an invertible $\mathcal{O}_S$-module isomorphic to $\mathcal{O}_{X'}(-\sum_{i=1}^r el a(E_i, X, B)E_i)|_S$. Hence $h^*\mathcal{L}^{\otimes l} \otimes (I_{\geq\lambda_l}/I_{>\lambda_l}) \cong \mathcal{L}_S^{\otimes l}$.

Step 3. In this step, we show that h^*s is a section of $h^*\mathcal{L}^{\otimes l} \otimes I_{\geq\lambda_l}$ on a neighborhood of S for any $s \in W_l$, and that $\mathrm{lead}(s) = 0$ if $s \neq 0$ and $v(\mathrm{div}(s)) > \lambda_l$.

Let $s \in W_l \backslash \{0\}$. Let σ' be a local generator of $h^*\mathcal{L}^{\otimes l}$ at η_S, and let $\phi \in \mathcal{O}_{X',\eta_S}$ be such that $h^*s = \phi\sigma'$ at η_S. Then $v(\phi) = v(\mathrm{div}(s)) \geq \lambda_l$ by (8.8). Fix a coefficient field κ of $\widehat{\mathcal{O}}_{X',\eta_S} = \kappa[[z_1, \dots, z_r]]$ and write $\phi = \sum_\beta c_\beta z^\beta$ with $c_\beta \in \kappa$. Then $v(\phi) = \min\{\alpha \cdot \beta \mid c_\beta \neq 0\}$ [JM12, Proposition 3.1]. Thus $\alpha \cdot \beta \geq \lambda_l$ for any β with $c_\beta \neq 0$, and $\alpha \cdot \beta > \lambda_l$ for any such β if $v(\mathrm{div}(s)) > \lambda_l$. Near η_S, $I_{\geq\lambda_l}$ (resp. $I_{>\lambda_l}$) is generated by finitely many monomials, and any monomial z^β with $\alpha \cdot \beta \geq \lambda_l$ (resp. $\alpha \cdot \beta > \lambda_l$) is a multiple of one of the generators. Thus $\phi \in I_{\geq\lambda_l}\widehat{\mathcal{O}}_{X',\eta_S}$, and $\phi \in I_{>\lambda_l}\widehat{\mathcal{O}}_{X',\eta_S}$ if $v(\mathrm{div}(s)) > \lambda_l$. Since $\widehat{\mathcal{O}}_{X',\eta_S}$ is faithfully flat over $\mathcal{O}_{X',\eta_S}$ [Mat86, Theorem 8.14], we have $J\widehat{\mathcal{O}}_{X',\eta_S} \cap \mathcal{O}_{X',\eta_S} = J$ for any ideal J of $\mathcal{O}_{X',\eta_S}$. Thus we obtain

$$\phi \in I_{\geq\lambda_l}\mathcal{O}_{X',\eta_S}, \quad \text{and} \quad \phi \in I_{>\lambda_l}\mathcal{O}_{X',\eta_S} \text{ if } v(\mathrm{div}(s)) > \lambda_l. \tag{8.10}$$

If $\lambda_l = 0$, then $I_{\geq\lambda_l} = \mathcal{O}_{X'}$, so h^*s is a section of $h^*\mathcal{L}^{\otimes l} \otimes I_{\geq\lambda_l}$. Assume that $\lambda_l > 0$. Let $\mu_0 < \mu_1 < \cdots < \mu_k$ be the elements of $\{\alpha \cdot \beta \mid \beta \in \mathbb{N}^r\}$ that are smaller than λ_l, which form a finite set since $\alpha_i > 0$ for any i. Then $\mu_0 = 0$, $I_{\geq\mu_0} = \mathcal{O}_{X'}$, $I_{>\mu_p} = I_{\geq\mu_{p+1}}$ for $0 \leq p < k$, and $I_{>\mu_k} = I_{\geq\lambda_l}$ because $\lambda_l = \alpha \cdot (el\mathbf{a})$ with $el\mathbf{a} \in \mathbb{N}^r$. Assume that h^*s is a section of $h^*\mathcal{L}^{\otimes l} \otimes I_{\geq\mu_p}$ on a neighborhood X_0' of S. By (8.9), the image of h^*s in $h^*\mathcal{L}^{\otimes l} \otimes (I_{\geq\mu_p}/I_{>\mu_p})$ is a section of a line bundle on S. By (8.10) and $I_{\geq\lambda_l} \subset I_{>\mu_p}$, the image vanishes at η_S. Since S is integral, the image is zero, so h^*s is a section of $h^*\mathcal{L}^{\otimes l} \otimes I_{>\mu_p}$ on X_0'. By induction on p, h^*s is a section of $h^*\mathcal{L}^{\otimes l} \otimes I_{\geq\lambda_l}$ on a neighborhood of S.

In both cases, by **Step 2**, $\mathrm{lead}(s)$ is well defined. If moreover $v(\mathrm{div}(s)) > \lambda_l$, then by (8.10), $\mathrm{lead}(s)$ vanishes at η_S. Since $\mathcal{L}_S^{\otimes l}$ is a line bundle on the integral variety S, $\mathrm{lead}(s) = 0$.

Step 4. In this step, we prove (1) and (2).

The pullback $h^* : H^0(X, \mathcal{L}^{\otimes l}) \to H^0(X', h^*\mathcal{L}^{\otimes l})$ is linear, and taking the image in $h^*\mathcal{L}^{\otimes l} \otimes (I_{\geq\lambda_l}/I_{>\lambda_l})$ is $\mathcal{O}_{X'}$-linear. Thus lead is linear on the linear subspace W_l, which proves (1).

By **Step 1**, $W_l W_{l'} \subset W_{l+l'}$. Since $\lambda_l + \lambda_{l'} = \lambda_{l+l'}$ and $z^\beta z^{\beta'} = z^{\beta+\beta'}$, we have

$$I_{\geq\lambda_l} I_{\geq\lambda_{l'}} \subset I_{\geq\lambda_{l+l'}} \quad \text{and} \quad I_{\geq\lambda_l} I_{>\lambda_{l'}} + I_{>\lambda_l} I_{\geq\lambda_{l'}} \subset I_{>\lambda_{l+l'}}.$$

Thus multiplication induces a homomorphism

$$(I_{\geq\lambda_l}/I_{>\lambda_l}) \otimes (I_{\geq\lambda_{l'}}/I_{>\lambda_{l'}}) \to I_{\geq\lambda_{l+l'}}/I_{>\lambda_{l+l'}},$$

which sends the image of $z^{el\mathbf{a}} \otimes z^{el'\mathbf{a}}$ to the image of $z^{e(l+l')\mathbf{a}}$. Under (8.9), the multiplication homomorphism is the natural isomorphism $\mathcal{L}_S^{\otimes l} \otimes \mathcal{L}_S^{\otimes l'} \cong \mathcal{L}_S^{\otimes (l+l')}$ after tensoring with $h^*\mathcal{L}^{\otimes l} \otimes h^*\mathcal{L}^{\otimes l'} \cong h^*\mathcal{L}^{\otimes (l+l')}$. Since $h^*(ss') = h^*s \cdot h^*s'$, we obtain $\operatorname{lead}(ss') = \operatorname{lead}(s)\operatorname{lead}(s')$ for any $s \in W_l$ and $s' \in W_{l'}$. This proves (2).

Step 5. In this step, we prove (3).

Let $s \in W_l \setminus \{0\}$ and $G := \frac{1}{el}\operatorname{div}(s)$. Then $a(v, X, B+G) \leq 0$. If $a(v, X, B+G) < 0$, then $v(\operatorname{div}(s)) > \lambda_l$ by (8.8), so $\operatorname{lead}(s) = 0$ by **Step 3**.

Assume that $a(v, X, B+G) = 0$. By Lemma 8.5(4), $b(h^*G) = \mathbf{a}$. Let $y \in S$ be a closed point, $z_1, \dots, z_r$ local equations of $E_1, \dots, E_r$ near y, σ' a local generator of $h^*\mathcal{L}^{\otimes l}$ near y, and ϕ the regular function near y such that $h^*s = \phi\sigma'$. Then ϕ is a local equation of $elh^*G = h^*\operatorname{div}(s)$ near y. Since $el\alpha \cdot b(h^*G) = el\alpha \cdot \mathbf{a} = \lambda_l$, Construction-Proposition 8.1(4), applied to $D' = h^*G$ with el in place of l, implies that

$$\phi \equiv \psi z^{el\mathbf{a}} \mod I_{>\lambda_l}$$

near y, where ψ is a regular function such that $\operatorname{div}(\psi|_S) = el\operatorname{lead}_S(h^*G)$ near y. By (8.9) for $\beta = el\mathbf{a}$, near y, $\operatorname{lead}(s)$ is $\psi|_S$ times the local generator of $\mathcal{L}_S^{\otimes l}$ defined by σ' and $z^{el\mathbf{a}}$. Since $\psi|_S \neq 0$, we have $\operatorname{lead}(s) \neq 0$, and $\operatorname{div}(\operatorname{lead}(s)) = \operatorname{div}(\psi|_S) = el\operatorname{lead}_S(h^*G)$ near y. Since y is an arbitrary closed point of S, $\operatorname{div}(\operatorname{lead}(s)) = el\operatorname{lead}_S(h^*G)$. This proves (3). □

9. Return from leading divisors

In this section, we prove that if $G \geq 0$ satisfies $a(v, X, B+G) = 0$ and $(S, B_S + \operatorname{lead}_S(h^*G))$ is sub-lc, then $(X, B+G)$ is lc on a neighborhood of the center V of v (Proposition 9.1). In the proof of Theorem 12.1, Construction-Proposition 11.4(4), Lemma 10.2, and induction on dimension show that $(S, B_S + \operatorname{lead}_S(h^*G))$ is sub-lc for the $\mathbb{Q}$-divisors G to which Proposition 9.1 is applied.

Proposition 9.1. *Notation and conditions as in Set-up 8.4. Let $G \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that $a(v, X, B+G) = 0$. If $(S, B_S + \operatorname{lead}_S(h^*G))$ is sub-lc, then $(X, B+G)$ is lc on a neighborhood of V.*

We prove Proposition 9.1 at the end of this section.

We first compute the different on the exceptional divisor $E_{\mathbf{e}}$ of the weighted blow-up of X' along S with primitive integral weights $\mathbf{e}$. The different is the pullback of $B_S + \operatorname{lead}_S(h^*G)$ plus the reduced relative toric boundary $D_{\mathbf{e}}$, whose components have coefficient one. The coefficients of $D_{\mathbf{e}}$ equal one because $a(v, X, B+G) = 0$: in the normal direction of E_i, the coefficients $1 - a(E_i, X, B)$ and $b(h^*G)_i$ sum to one.

Construction-Proposition 9.2. Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, v a quasi-monomial valuation of $\mathbb{C}(X)$ of rational rank r, and (h, S) an adapted carrier of v for (X, B) and a finite set Σ, with $h : X' \to X$, $E_1, \dots, E_r$, α, and B_S as in Construction-Proposition 8.1. Let $G \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that $a(v, X, B+G) = 0$, and let $\mathbf{e} \in \mathbb{Z}^r_{>0}$ be a primitive vector with the following property: if l is a positive integer such that lh^*G is Cartier, then $lb(h^*G)$ is the unique exponent of least $\mathbf{e}$-weight in the expansion of a local equation of lh^*G in $\widehat{\mathcal{O}}_{X',\eta} = \kappa[[z_1, \dots, z_r]]$ as in Definition 2.7, where η is the generic point of S and $z_1, \dots, z_r$ are local equations of $E_1, \dots, E_r$ at η. Let w be the monomial valuation with weights $\mathbf{e}$ in $E_1, \dots, E_r$ at η. If $r \geq 2$, let

$$g : X'_{\mathbf{e}} := \operatorname{Proj}_{X'} \bigoplus_{k \geq 0} I_k^{\mathbf{e}} \to X'$$

be the *weighted blow-up* of X' along S with weights $\mathbf{e}$, where $I_k^{\mathbf{e}} \subset \mathcal{O}_{X'}$ is the ideal sheaf that is $\mathcal{O}_{X'}$ away from S and near each closed point of S is generated by the monomials z^β with $\mathbf{e} \cdot \beta \geq k$, $z_1, \dots, z_r$ being local equations of $E_1, \dots, E_r$, $E_{\mathbf{e}}$ the exceptional divisor of g, $p : E_{\mathbf{e}} \to S$ the induced morphism, and $D_{\mathbf{e}} := \sum_{i=1}^r (g_*^{-1}E_i \cap E_{\mathbf{e}})_{\mathrm{red}}$ the reduced relative

toric boundary of p. If $r = 1$, let $X'_{\mathbf{e}} := X'$, $g := \mathrm{id}_{X'}$, $E_{\mathbf{e}} := S = E_1$, $p := \mathrm{id}_S$, and $D_{\mathbf{e}} := 0$. Write $K_{X'_{\mathbf{e}}} + \Gamma := (h \circ g)^*(K_X + B + G)$. Then $w = \mathrm{ord}_{E_{\mathbf{e}}}$, $a(w, X, B + G) = 0$, $\mathrm{mult}_{E_{\mathbf{e}}} \Gamma = 1$, and the different $\Delta_{E_{\mathbf{e}}}$ of $(X'_{\mathbf{e}}, \Gamma)$ on $E_{\mathbf{e}}$ in the sense of [Kol13, Definition 4.2], so that $K_{E_{\mathbf{e}}} + \Delta_{E_{\mathbf{e}}} = (K_{X'_{\mathbf{e}}} + \Gamma)|_{E_{\mathbf{e}}}$, satisfies

$$\Delta_{E_{\mathbf{e}}} = D_{\mathbf{e}} + p^*(B_S + \mathrm{lead}_S(h^*G)).$$

In particular, for any open subset $U \subset S$ and any $\mathbb{Q}$-divisor L on S such that $(U, (B_S + \mathrm{lead}_S(h^*G) + L)|_U)$ is sub-lc, the sub-pair $(p^{-1}(U), (\Delta_{E_{\mathbf{e}}} + p^*L)|_{p^{-1}(U)})$ is sub-lc.

Proof. The proof is long, so we divide it into several steps. Let $a_i := a(E_i, X, B)$ for $1 \le i \le r$, and let $\mathbf{a} := (a_1, \dots, a_r)$.

Step 1. In this step, we show that $b(h^*G) = \mathbf{a}$ and $a(w, X, B + G) = 0$.

By Construction-Proposition 8.1(1)(2), $E_1, \dots, E_r$ are components of a reduced simple normal crossing divisor containing $\mathrm{Exc}(h) \cup \mathrm{Supp}\, B'$, where $K_{X'} + B' := h^*(K_X + B)$, and v and w are the monomial valuations with weights α and $\mathbf{e}$ in $E_1, \dots, E_r$ at η. Thus Definition-Proposition 2.8 implies $a(v, X, B) = \alpha \cdot \mathbf{a}$ and $a(w, X, B) = \mathbf{e} \cdot \mathbf{a}$. By Construction-Proposition 8.1(5), $v(G) = v(h^*G) = \alpha \cdot b(h^*G)$. By Definition-Proposition 2.8,

$$0 = a(v, X, B + G) = \sum_{i=1}^{r} \alpha_i \big(a_i - b(h^*G)_i\big).$$

Since $a_i - b(h^*G)_i \in \mathbb{Q}$ for any i and $\alpha_1, \dots, \alpha_r$ are linearly independent over $\mathbb{Q}$, we have $b(h^*G) = \mathbf{a}$. Let l be a positive integer such that lh^*G is Cartier. By assumption, $l\mathbf{a}$ is the unique exponent of least $\mathbf{e}$-weight in a local equation of lh^*G near η, so $w(h^*G) = \mathbf{e} \cdot \mathbf{a}$ by Definition 2.7. By Definition-Proposition 2.8, $a(w, X, B + G) = \mathbf{e} \cdot \mathbf{a} - \mathbf{e} \cdot \mathbf{a} = 0$.

Step 2. In this step, we describe g near a closed point $y \in S$, and we show that $w = \mathrm{ord}_{E_{\mathbf{e}}}$ and $\mathrm{mult}_{E_{\mathbf{e}}} \Gamma = 1$.

Since $E_1 + \cdots + E_r$ is a simple normal crossing divisor and the components of $E_1 \cap \cdots \cap E_r$ are pairwise disjoint, there exist an affine open neighborhood Y of y in X' and regular functions $z_1, \dots, z_r$ on Y such that $E_c \cap Y = \mathrm{div}(z_c)$ for any c, the subscheme of Y defined by $z_1 = \cdots = z_r = 0$ is $S \cap Y$, and the morphism $z := (z_1, \dots, z_r) : Y \to \mathbb{A}^r$ is smooth. For any integer $k \ge 0$, let $I_k \subset \mathbb{C}[z_1, \dots, z_r]$ be the ideal generated by the monomials z^β with $\mathbf{e} \cdot \beta \ge k$, and let $\mathbb{A}^r_{\mathbf{e}} := \mathrm{Proj} \bigoplus_{k \ge 0} I_k t^k$. Then $I_k^{\mathbf{e}}|_Y = I_k \mathcal{O}_Y$. Since z is flat, $g^{-1}(Y) = Y \times_{\mathbb{A}^r} \mathbb{A}^r_{\mathbf{e}}$. If $r = 1$, then $\mathbf{e} = (1)$ and $\mathbb{A}^1_{\mathbf{e}} = \mathbb{A}^1$, in accordance with $g = \mathrm{id}_{X'}$.

For $1 \le i \le r$, let μ_{e_i} act on $\mathbb{A}^r$ with coordinates $u_1, \dots, u_r$ by $\zeta \cdot u_i = \zeta u_i$ and $\zeta \cdot u_c = \zeta^{-e_c} u_c$ for $c \ne i$, and let $\pi_i : \mathbb{A}^r \to \mathbb{A}^r$ be the μ_{e_i}-invariant morphism defined by

$$z_i = u_i^{e_i}, \qquad z_c = u_c u_i^{e_c} \ (c \ne i).$$

A direct calculation shows that $\mathbb{A}^r_{\mathbf{e}}$ is covered by the open subsets $W_i := D_+(z_i t^{e_i})$, $1 \le i \le r$, and that π_i induces an isomorphism $\mathbb{A}^r / \mu_{e_i} \cong W_i$ over $\mathbb{A}^r$. The elements $z^\beta t^{m e_i} / (z_i t^{e_i})^m$ of degree zero, with $\mathbf{e} \cdot \beta \ge m e_i$, pull back under π_i to the monomials $u_i^{\mathbf{e} \cdot \beta - m e_i} \prod_{c \ne i} u_c^{\beta_c}$, that is, to the monomials u^γ with $\gamma \in \mathbb{N}^r$ and $\gamma_i \equiv \sum_{c \ne i} e_c \gamma_c \pmod{e_i}$, which are exactly the μ_{e_i}-invariant monomials in $u_1, \dots, u_r$.

Let $\widetilde{Y}_i := Y \times_{\mathbb{A}^r} \mathbb{A}^r$ be the fiber product of z and π_i, let $\varpi_i : \widetilde{Y}_i \to Y$ be the first projection, and let μ_{e_i} act on $\widetilde{Y}_i$ through the second factor. Since z is smooth, $\widetilde{Y}_i$ is smooth. Since taking μ_{e_i}-invariants is exact and commutes with flat base change, the induced morphism $q_i : \widetilde{Y}_i \to Y \times_{\mathbb{A}^r} W_i \subset X'_{\mathbf{e}}$ is the quotient by μ_{e_i}, and $\varpi_i = g \circ q_i$. In particular, $X'_{\mathbf{e}}$ is normal. Let $\widetilde{E}_c := \mathrm{div}(u_c)$ on $\widetilde{Y}_i$ for $1 \le c \le r$. Then

$$\varpi_i^* E_i = e_i \widetilde{E}_i, \qquad \varpi_i^* E_c = \widetilde{E}_c + e_c \widetilde{E}_i \ (c \ne i), \qquad \varpi_i^* z^\beta = u_i^{\mathbf{e} \cdot \beta} \prod_{c \ne i} u_c^{\beta_c}. \tag{9.1}$$

Since $\pi_i^{-1}(0) = \{u_i = 0\}$, we have $q_i^{-1}(E_{\mathbf{e}}) = \widetilde{E}_i = (S \cap Y) \times \mathbb{A}^{r-1}$, where $\mathbb{A}^{r-1}$ has the coordinates u_c, $c \neq i$, and $p \circ q_i|_{\widetilde{E}_i}$ is the first projection $\tilde{p}_i : \widetilde{E}_i \to S \cap Y$. Moreover, $q_i^{-1}(g_*^{-1}E_c) = \widetilde{E}_c$ for $c \neq i$, and $g_*^{-1}E_i$ is disjoint from $Y \times_{\mathbb{A}^r} W_i$. Since taking invariants is exact, $q_i|_{\widetilde{E}_i}$ is the quotient by μ_{e_i}, so

$$E_{\mathbf{e}} \cap (Y \times_{\mathbb{A}^r} W_i) \cong (S \cap Y) \times (\mathbb{A}^{r-1}/\mu_{e_i}).$$

In particular, $E_{\mathbf{e}}$ is normal.

An element $\zeta \neq 1$ of μ_{e_i} fixes a point $(u_1, \dots, u_r)$ only if $u_i = 0$ and $u_c = 0$ for any $c \neq i$ with $\zeta^{e_c} \neq 1$. Since $\mathbf{e}$ is primitive, such an index c exists. Thus q_i is étale in codimension one. For $c \neq i$, let $\widetilde{D}_{i,c} := \widetilde{E}_c \cap \widetilde{E}_i = (S \cap Y) \times \{u_c = 0\}$ and $n_c := \gcd(e_{c'} : c' \neq c)$. The stabilizer of the generic point of $\widetilde{D}_{i,c}$ is μ_{n_c}, it acts faithfully on u_c, and $q_i|_{\widetilde{E}_i}$ is étale at the other points of codimension one of $\widetilde{E}_i$. By the Hurwitz formula,

$$K_{\widetilde{E}_i} = q_i^* K_{E_{\mathbf{e}}} + \sum_{c \neq i} (n_c - 1)\widetilde{D}_{i,c}, \qquad q_i^* D_{\mathbf{e}} = \sum_{c \neq i} n_c \widetilde{D}_{i,c}, \tag{9.2}$$

where q_i^* denotes the pullback along $q_i|_{\widetilde{E}_i}$.

Since z is flat and the $\mathbb{C}$-vector space I_k/I_{k+1} has the basis z^β, $\mathbf{e} \cdot \beta = k$, the $\mathcal{O}_{S \cap Y}$-module $I_k\mathcal{O}_Y/I_{k+1}\mathcal{O}_Y$ is free with the basis z^β, $\mathbf{e} \cdot \beta = k$. For $f \in I_k\mathcal{O}_Y$, let $f_\beta \in \mathcal{O}(S \cap Y)$ be the coefficients of the image of f in $I_k\mathcal{O}_Y/I_{k+1}\mathcal{O}_Y$. By (9.1), $\varpi_i^* f \in u_i^k \mathcal{O}_{\widetilde{Y}_i}$ and

$$(u_i^{-k}\varpi_i^* f)|_{\widetilde{E}_i} = \sum_{\mathbf{e} \cdot \beta = k} \tilde{p}_i^*(f_\beta) \prod_{c \neq i} u_c^{\beta_c}. \tag{9.3}$$

Distinct exponents β with $\mathbf{e} \cdot \beta = k$ have distinct vectors $(\beta_c)_{c \neq i}$, so the right-hand side of (9.3) is zero only if $f_\beta = 0$ for any β.

Let $f \in \mathcal{O}_{X',\eta}$ be non-zero. In $\widehat{\mathcal{O}}_{X',\eta} = \kappa[[z_1, \dots, z_r]]$, the ideal $I_k\widehat{\mathcal{O}}_{X',\eta}$ consists of the power series whose exponents have $\mathbf{e}$-weight at least k, and $I_k\mathcal{O}_{X',\eta} = I_k\widehat{\mathcal{O}}_{X',\eta} \cap \mathcal{O}_{X',\eta}$ by the faithful flatness of the completion [Mat86, Theorem 8.14]. Thus $k := w(f)$ is the largest integer such that $f \in I_k\mathcal{O}_{X',\eta}$, and the image of f in $I_k\mathcal{O}_{X',\eta}/I_{k+1}\mathcal{O}_{X',\eta}$ is non-zero. By (9.3), $\operatorname{ord}_{\widetilde{E}_i}(\varpi_i^* f) = k$. Since q_i is étale at the generic point of $\widetilde{E}_i$, $\operatorname{ord}_{E_{\mathbf{e}}}(f) = \operatorname{ord}_{\widetilde{E}_i}(\varpi_i^* f) = w(f)$. Since $\mathbb{C}(X)$ is the fraction field of $\mathcal{O}_{X',\eta}$, we have $w = \operatorname{ord}_{E_{\mathbf{e}}}$. By **Step 1** and Definition-Proposition 2.8, $a(E_{\mathbf{e}}, X, B + G) = a(w, X, B + G) = 0$, so $\operatorname{mult}_{E_{\mathbf{e}}} \Gamma = 1$.

Step 3. In this step, we show that the part of least $\mathbf{e}$-weight of a local equation of lh^*G near y is $\psi z^{l\mathbf{a}}$, where $\operatorname{div}(\psi|_S) = l \operatorname{lead}_S(h^*G)$ near y.

Let l be a positive integer such that lh^*G is Cartier, and let $k_0 := l\mathbf{e} \cdot \mathbf{a}$. Let $J \subset \mathbb{C}[z_1, \dots, z_r]$ be the ideal generated by the monomials z^β with $\alpha \cdot \beta > l\alpha \cdot \mathbf{a}$. By Construction-Proposition 8.1(4) and $b(h^*G) = \mathbf{a}$, lh^*G has a local equation ϕ near y such that $\phi - \psi z^{l\mathbf{a}} \in J\mathcal{O}_{X',y}$ for a regular function ψ with $\operatorname{div}(\psi|_S) = l \operatorname{lead}_S(h^*G)$ near y. Possibly replacing Y with a smaller affine neighborhood of y, we may assume that ϕ and ψ are regular on Y, ϕ is a local equation of lh^*G on Y, $\phi - \psi z^{l\mathbf{a}} \in J\mathcal{O}_Y$, and $\operatorname{div}(\psi|_{S \cap Y}) = l \operatorname{lead}_S(h^*G)|_{S \cap Y}$. We show that

$$\phi - \psi z^{l\mathbf{a}} \in I_{k_0+1}\mathcal{O}_Y. \tag{9.4}$$

Let $0 \leq k < k_0$ be an integer such that $\phi \in I_k\mathcal{O}_Y$. For $\mathbf{e} \cdot \beta = k$, the image of ϕ_β in $\kappa(\eta)$ is the coefficient of z^β in the power series expansion of ϕ in $\kappa[[z_1, \dots, z_r]]$. By assumption, ϕ has no exponent of $\mathbf{e}$-weight less than k_0, so ϕ_β vanishes at η. Since $S \cap Y$ is integral, $\phi_\beta = 0$. Thus $\phi \in I_{k+1}\mathcal{O}_Y$. By induction on k, $\phi \in I_{k_0}\mathcal{O}_Y$. For $\mathbf{e} \cdot \beta = k_0$ and $\beta \neq l\mathbf{a}$, the image of ϕ_β in $\kappa(\eta)$ is zero by the uniqueness of the exponent $l\mathbf{a}$, so $\phi_\beta = 0$ since $S \cap Y$ is integral. Since z is flat, $J\mathcal{O}_Y \cap I_{k_0}\mathcal{O}_Y = (J \cap I_{k_0})\mathcal{O}_Y$, and $J \cap I_{k_0}$ is generated by the monomials z^β with $\alpha \cdot \beta > l\alpha \cdot \mathbf{a}$ and $\mathbf{e} \cdot \beta \geq k_0$, none of which is $z^{l\mathbf{a}}$. Since $\phi - \psi z^{l\mathbf{a}} \in J\mathcal{O}_Y \cap I_{k_0}\mathcal{O}_Y$, the coefficient of $z^{l\mathbf{a}}$ in the image of $\phi - \psi z^{l\mathbf{a}}$ in $I_{k_0}\mathcal{O}_Y/I_{k_0+1}\mathcal{O}_Y$ is zero. Thus $\phi_{l\mathbf{a}} = \psi|_{S \cap Y}$, and (9.4) follows.

Step 4. In this step, we compute $\Delta_{E_{\mathbf{e}}}$ on $E_{\mathbf{e}} \cap (Y \times_{\mathbb{A}^r} W_i)$.

Let $B_0' := B' - \sum_{c=1}^{r}(1-a_c)E_c$. Then $B_S = B_0'|_S$ by Construction-Proposition 8.1(6). The components of B_0' are components of a simple normal crossing divisor containing $E_1, \dots, E_r$ and different from $E_1, \dots, E_r$, so none of them contains S, and $(\varpi_i^* B_0')|_{\widetilde{E}_i} = \tilde{p}_i^* B_S$. Since z is smooth and

$$\pi_i^*\Big(\frac{dz_1}{z_1} \wedge \cdots \wedge \frac{dz_r}{z_r}\Big) = e_i \frac{du_1}{u_1} \wedge \cdots \wedge \frac{du_r}{u_r},$$

we have

$$K_{\widetilde{Y}_i} + \sum_{c=1}^{r} \widetilde{E}_c = \varpi_i^*\Big(K_Y + \sum_{c=1}^{r} E_c\Big),$$

in accordance with the formula for the canonical divisor of a toric variety [CLS11, Theorem 8.2.3]. By (9.1) and (9.4), $\varpi_i^*\phi = u_i^{k_0}\tilde{\phi}$ for a regular function $\tilde{\phi}$ on $\widetilde{Y}_i$ with

$$\tilde{\phi}|_{\widetilde{E}_i} = \tilde{p}_i^*(\psi|_{S\cap Y}) \prod_{c\neq i} u_c^{la_c}.$$

Since q_i is étale in codimension one,

$$K_{\widetilde{Y}_i} + q_i^*\Gamma = q_i^*(K_{X'_{\mathbf{e}}} + \Gamma) = \varpi_i^*(K_{X'} + B' + h^*G).$$

Since $B' = \sum_c E_c - \sum_c a_c E_c + B_0'$, $h^*G = \frac{1}{l}\operatorname{div}(\phi)$, and $k_0/l = \mathbf{e}\cdot\mathbf{a}$, (9.1) implies

$$q_i^*\Gamma = \widetilde{E}_i + \sum_{c\neq i}(1-a_c)\widetilde{E}_c + \varpi_i^* B_0' + \frac{1}{l}\operatorname{div}(\tilde{\phi}).$$

The prime divisor $\widetilde{E}_i$ is not contained in $\operatorname{Supp}(q_i^*\Gamma - \widetilde{E}_i)$, since $\tilde{\phi}|_{\widetilde{E}_i} \neq 0$. Since $\widetilde{Y}_i$ and $\widetilde{E}_i$ are smooth, the adjunction formula [Har77, Proposition II.8.20] implies

$$\begin{aligned}(K_{\widetilde{Y}_i} + q_i^*\Gamma)|_{\widetilde{E}_i} &= K_{\widetilde{E}_i} + \sum_{c\neq i}(1-a_c)\widetilde{D}_{i,c} + \tilde{p}_i^* B_S + \sum_{c\neq i} a_c \widetilde{D}_{i,c} + \tilde{p}_i^* \operatorname{lead}_S(h^*G)\\ &= K_{\widetilde{E}_i} + \sum_{c\neq i}\widetilde{D}_{i,c} + \tilde{p}_i^*\big(B_S + \operatorname{lead}_S(h^*G)\big).\end{aligned}$$

Since q_i is finite and étale in codimension one and $q_i^{-1}(E_{\mathbf{e}}) = \widetilde{E}_i$, the left-hand side is $q_i^*(K_{E_{\mathbf{e}}} + \Delta_{E_{\mathbf{e}}})$ [Kol13, Definition 4.2, Proposition 4.5]. By (9.2),

$$q_i^*\Big(K_{E_{\mathbf{e}}} + D_{\mathbf{e}} + p^*\big(B_S + \operatorname{lead}_S(h^*G)\big)\Big) = K_{\widetilde{E}_i} + \sum_{c\neq i}\widetilde{D}_{i,c} + \tilde{p}_i^*\big(B_S + \operatorname{lead}_S(h^*G)\big).$$

Since $q_i|_{\widetilde{E}_i}$ is finite and surjective, q_i^* is injective on $\mathbb{Q}$-divisors. Thus

$$\Delta_{E_{\mathbf{e}}} = D_{\mathbf{e}} + p^*\big(B_S + \operatorname{lead}_S(h^*G)\big) \quad \text{on } E_{\mathbf{e}} \cap (Y \times_{\mathbb{A}^r} W_i).$$

Since the open subsets $E_{\mathbf{e}} \cap (Y \times_{\mathbb{A}^r} W_i)$ cover $E_{\mathbf{e}}$ when y runs over the closed points of S and $1 \le i \le r$, the formula for $\Delta_{E_{\mathbf{e}}}$ holds.

Step 5. In this step, we prove that $(p^{-1}(U), (\Delta_{E_{\mathbf{e}}} + p^*L)|_{p^{-1}(U)})$ is sub-lc.

Let $\Theta := (B_S + \operatorname{lead}_S(h^*G) + L)|_U$. By **Step 4**, $(\Delta_{E_{\mathbf{e}}} + p^*L)|_{p^{-1}(U)} = D_{\mathbf{e}}|_{p^{-1}(U)} + p^*\Theta$. Since the sub-lc property is local, it suffices to prove that the restriction of $(E_{\mathbf{e}}, D_{\mathbf{e}} + p^*\Theta)$ to $p^{-1}(U\cap Y)\cap(Y\times_{\mathbb{A}^r} W_i)$ is sub-lc for any closed point $y \in U$ and any i. By (9.2),

$$q_i^*\big(K_{E_{\mathbf{e}}} + D_{\mathbf{e}} + p^*\Theta\big) = K_{\widetilde{E}_i} + \sum_{c\neq i}\widetilde{D}_{i,c} + \tilde{p}_i^*\Theta$$

on $\tilde{p}_i^{-1}(U\cap Y) = (U\cap Y)\times\mathbb{A}^{r-1}$. By [KM98, Proposition 5.20], it suffices to show that $((U\cap Y)\times\mathbb{A}^{r-1}, \sum_{c\neq i}\widetilde{D}_{i,c} + \tilde{p}_i^*\Theta)$ is sub-lc. Let $f : U' \to U\cap Y$ be a log resolution of

$(U\cap Y,\Theta|_{U\cap Y})$, and write $K_{U'}+\Theta':=f^*(K_{U\cap Y}+\Theta|_{U\cap Y})$. Since (U,Θ) is sub-lc, the coefficients of Θ' are at most 1. Let $\mathrm{pr}_1 : U'\times\mathbb{A}^{r-1}\to U'$ be the first projection. Then

$$K_{U'\times\mathbb{A}^{r-1}}+\sum_{c\neq i}U'\times\{u_c=0\}+\mathrm{pr}_1^*\,\Theta'=(f\times\mathrm{id}_{\mathbb{A}^{r-1}})^*\Big(K_{(U\cap Y)\times\mathbb{A}^{r-1}}+\sum_{c\neq i}\widetilde{D}_{i,c}+\tilde{p}_i^*\Theta\Big),$$

and $\sum_{c\neq i}U'\times\{u_c=0\}+\mathrm{pr}_1^*\,\Theta'$ has simple normal crossing support and coefficients at most 1. By [KM98, Corollary 2.31(3)], $(U'\times\mathbb{A}^{r-1},\sum_{c\neq i}U'\times\{u_c=0\}+\mathrm{pr}_1^*\,\Theta')$ is sub-lc. By [KM98, Lemma 2.30], $((U\cap Y)\times\mathbb{A}^{r-1},\sum_{c\neq i}\widetilde{D}_{i,c}+\tilde{p}_i^*\Theta)$ is sub-lc. □

Next we pass from $X'_{\mathbf{e}}$ to a higher model of X on which the crepant boundary has non-negative coefficients, as Theorem 2.21 requires. Since α may be irrational, v need not be divisorial, so we extract, by [Mor20, Theorem 1], the divisorial valuation w attached to an integral vector $\mathbf{e}$ near the ray of α. [Mor20, Theorem 1] extracts divisorial valuations whose log discrepancy with respect to an lc pair is less than one. Since $a(w,X,B+D_{\leq N})$ decreases to $\delta(\mathbf{e})$ as $N\to+\infty$, we choose $\mathbf{e}$ with $\delta(\mathbf{e})<1$, by simultaneous approximation [Cas57] and Lemma 8.5.

Construction-Proposition 9.3. Notation and conditions as in Set-up 8.4. Let $G\geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X such that $a(v,X,B+G)=0$. Then there exist a primitive vector $\mathbf{e}\in\mathbb{Z}^r_{>0}$, a projective birational morphism $\rho : X_{\mathbf{e}}\to X$ from a normal variety $X_{\mathbf{e}}$, and a prime divisor F on $X_{\mathbf{e}}$, such that the following hold, where $w:=v_{\mathbf{e}}$ is the monomial valuation of Set-up 8.4 with weights $\mathbf{e}$ in $E_1,\dots,E_r$ at the generic point of S.

(1) If l is a positive integer such that lh^*G is Cartier, then $lb(h^*G)$ is the unique exponent of least $\mathbf{e}$-weight in the expansion of a local equation of lh^*G in $\widehat{\mathcal{O}}_{X',\eta}=\kappa[[z_1,\dots,z_r]]$ as in Definition 2.7, where η is the generic point of S and $z_1,\dots,z_r$ are local equations of $E_1,\dots,E_r$ at η.
(2) $w=\mathrm{ord}_F$.
(3) Either $\rho=\mathrm{id}_X$, or $\mathrm{Exc}(\rho)=F$ and F is $\mathbb{Q}$-Cartier.
(4) $\rho(F)=V$, and $\rho^{-1}(x)\subset F$ for any point $x\in V$.
(5) The $\mathbb{Q}$-divisor Q defined by $K_{X_{\mathbf{e}}}+Q=\rho^*(K_X+B+G)$ satisfies $Q\geq 0$ and $\mathrm{mult}_F\,Q=1$.
(6) The different Δ_{F^ν} on the normalization F^ν of F, defined by $K_{F^\nu}+\Delta_{F^\nu}=(K_{X_{\mathbf{e}}}+Q)|_{F^\nu}$, satisfies $\Delta_{F^\nu}\geq 0$, and (F^ν,Δ_{F^ν}) is crepant birational over X to the sub-pair $(E_{\mathbf{e}},\Delta_{E_{\mathbf{e}}})$ of Construction-Proposition 9.2 for G and $\mathbf{e}$.

Proof. The proof is long, so we divide it into several steps. Let η be the generic point of S.

Step 1. In this step, we construct $\mathbf{e}$ satisfying (1) and $\delta(\mathbf{e})<1$.

By Lemma 8.5(4), $b(h^*G)=\mathbf{a}$. Let l_0 be the smallest positive integer such that l_0h^*G is Cartier, and write a local equation of l_0h^*G near η as $\sum_\beta c_\beta z^\beta\in\kappa[[z_1,\dots,z_r]]=\widehat{\mathcal{O}}_{X',\eta}$ as in Definition 2.7. By Dickson's lemma [Dic13, Lemma A], the set P of minimal elements of $\{\beta\in\mathbb{N}^r\mid c_\beta\neq 0\}$ for the coordinatewise partial order is finite. By Construction-Proposition 8.1(4), the local equation is $\psi z^{l_0\mathbf{a}}$ modulo monomials of larger α-weight, where ψ is a unit at η since $\mathrm{div}(\psi|_S)$ is a divisor on S. Thus $l_0\mathbf{a}$ is the unique exponent of least α-weight, so $l_0\mathbf{a}\in P$ and α belongs to the open cone

$$\mathcal{C}:=\{\gamma\in\mathbb{R}^r_{>0}\mid\gamma\cdot l_0\mathbf{a}<\gamma\cdot\beta\text{ for any }\beta\in P\backslash\{l_0\mathbf{a}\}\}.$$

Let $\gamma\in\mathcal{C}$, and let $\beta\neq l_0\mathbf{a}$ be an exponent with $c_\beta\neq 0$. Then $\beta\geq\beta'$ coordinatewise for some $\beta'\in P$. If $\beta'\neq l_0\mathbf{a}$, then $\gamma\cdot\beta\geq\gamma\cdot\beta'>\gamma\cdot l_0\mathbf{a}$. If $\beta'=l_0\mathbf{a}$, then $\gamma\cdot\beta>\gamma\cdot l_0\mathbf{a}$ since $\gamma\in\mathbb{R}^r_{>0}$. Thus $l_0\mathbf{a}$ is the unique exponent of least γ-weight. If lh^*G is Cartier, then $l=ml_0$ for a positive integer m, and a local equation of lh^*G near η is a unit times the m-th power of the local equation above. The part of least γ-weight of a product of power series is the product of their parts of least γ-weight, and the part of least γ-weight of a unit is a non-zero constant. Thus $lb(h^*G)=l\mathbf{a}$ is the unique exponent of least γ-weight in a local equation of lh^*G near η. In particular, any primitive vector $\mathbf{e}\in\mathcal{C}\cap\mathbb{Z}^r$ satisfies (1).

By Lemma 8.5(3), $\delta \geq 0$ on $\mathbb{R}^r_{>0}$, $\delta(\alpha) = 0$, $\delta(t\gamma) = t\delta(\gamma)$ for any $\gamma \in \mathbb{R}^r_{>0}$ and any real number $t > 0$, and δ is $ca(v, X, B)$-Lipschitz with respect to the Euclidean norm $\|\cdot\|$ on U, for the compact neighborhood U of α and the constant c of Lemma 8.5(2). By Dirichlet's theorem on simultaneous approximation [Cas57, Chapter I, Theorem VI], for any integer $k \geq 1$, there exist an integer $1 \leq q_k \leq k^r$ and a vector $\mathbf{u}_k \in \mathbb{Z}^r$ such that $\|\mathbf{u}_k - q_k\alpha\| \leq c_r/k$, where $c_r := \sqrt{r}$. Then $\|\mathbf{u}_k/q_k - \alpha\| \leq c_r/k$. Since $\alpha \in \mathcal{C} \cap U$ and $\mathcal{C}$ is an open cone, for any $k \gg 1$, we have $\mathbf{u}_k \in \mathcal{C} \cap \mathbb{Z}^r_{>0}$ and $\mathbf{u}_k/q_k \in U$. Write $\mathbf{u}_k = d_k\mathbf{e}_k$, where d_k is a positive integer and $\mathbf{e}_k \in \mathbb{Z}^r_{>0}$ is primitive. Then $\mathbf{e}_k \in \mathcal{C}$ and

$$\begin{aligned} 0 \leq \delta(\mathbf{e}_k) = \frac{q_k}{d_k}\delta\Big(\frac{\mathbf{u}_k}{q_k}\Big) = \frac{q_k}{d_k}\Big(\delta\Big(\frac{\mathbf{u}_k}{q_k}\Big) - \delta(\alpha)\Big) &\leq \frac{q_k ca(v, X, B)}{d_k}\Big\|\frac{\mathbf{u}_k}{q_k} - \alpha\Big\| \\ = \frac{ca(v, X, B)}{d_k}\|\mathbf{u}_k - q_k\alpha\| &\leq \frac{c_r ca(v, X, B)}{k}. \end{aligned}$$

We fix $k > c_r ca(v, X, B)$ such that $\mathbf{u}_k \in \mathcal{C} \cap \mathbb{Z}^r_{>0}$ and $\mathbf{u}_k/q_k \in U$, and we let $\mathbf{e} := \mathbf{e}_k$. Then $\mathbf{e}$ satisfies (1) and $\delta(\mathbf{e}) < 1$.

Step 2. In this step, we construct ρ and F and prove (2) and (3).

Let $g : X'_{\mathbf{e}} \to X'$, $E_{\mathbf{e}}$, Γ, and $\Delta_{E_{\mathbf{e}}}$ be the objects of Construction-Proposition 9.2 for G and $\mathbf{e}$. By **Step 1**, the hypotheses of Construction-Proposition 9.2 hold for G and $\mathbf{e}$, so $w = \mathrm{ord}_{E_{\mathbf{e}}}$ and $a(w, X, B + G) = 0$. For any $N \geq 1$, by Definition-Proposition 2.8,

$$a(w, X, B + D_{\leq N}) = a(w, X, B) - \sum_{j=1}^{N} w(D_j) = \mathbf{e} \cdot \mathbf{a} - \sum_{j=1}^{N} w(D_j),$$

and since $D_j \geq 0$ for any j and $w = v_{\mathbf{e}}$, $a(w, X, B + D_{\leq N})$ decreases to $\delta(\mathbf{e}) < 1$ as $N \to +\infty$. Since $(X, B + D_{\leq N})$ is lc (Definition 7.1) and $w = \mathrm{ord}_{E_{\mathbf{e}}}$, we have $a(w, X, B + D_{\leq N}) \geq 0$. We fix N such that $a(w, X, B + D_{\leq N}) \in [0, 1)$.

Suppose that $E_{\mathbf{e}}$ is exceptional over X. By [Mor20, Theorem 1], applied to the lc pair $(X, B + D_{\leq N})$ and the set $\{w\}$, there exists a projective birational morphism $\rho : X_{\mathbf{e}} \to X$ from a normal variety $X_{\mathbf{e}}$ such that $\mathrm{Exc}(\rho)$ is a $\mathbb{Q}$-Cartier prime divisor F with $\mathrm{ord}_F = w$. Suppose that $E_{\mathbf{e}}$ is not exceptional over X. Then $F := (h \circ g)(E_{\mathbf{e}})$ is a prime divisor on X with $\mathrm{ord}_F = w$, and we let $X_{\mathbf{e}} := X$ and $\rho := \mathrm{id}_X$. In both cases, (2) and (3) hold.

Step 3. In this step, we prove (4) and (5).

By (2), since $p : E_{\mathbf{e}} \to S$ is surjective, and by Construction-Proposition 8.1(3),

$$\rho(F) = \mathrm{center}_X w = h(g(E_{\mathbf{e}})) = h(S) = V.$$

If $\rho = \mathrm{id}_X$, then $F = V$ and $\rho^{-1}(x) = \{x\} \subset F$ for any $x \in V$. Suppose that $\rho \neq \mathrm{id}_X$, and let $x \in V$ be a closed point. Then $\rho^{-1}(x) \cap F \neq \emptyset$. Suppose that $\rho^{-1}(x)$ contains a point $x' \notin F$. Since $\mathrm{Exc}(\rho) = F$, ρ is a local isomorphism at x', so x' is an isolated point of $\rho^{-1}(x)$. Since $\rho^{-1}(x)$ is connected [Har77, Corollary III.11.4], $\rho^{-1}(x) = \{x'\}$, which contradicts $\rho^{-1}(x) \cap F \neq \emptyset$. Thus $\rho^{-1}(x) \subset F$. Since $\rho^{-1}(V)$ is closed, $\rho^{-1}(V) \subset F$, and (4) holds.

By (2), **Step 2**, and Definition-Proposition 2.8, $a(F, X, B + G) = a(w, X, B + G) = 0$. If $\rho \neq \mathrm{id}_X$, then F is the only exceptional prime divisor of ρ, so $Q = \rho_*^{-1}(B + G) + F \geq 0$ and $\mathrm{mult}_F Q = 1$. If $\rho = \mathrm{id}_X$, then $Q = B + G \geq 0$ and $\mathrm{mult}_F Q = 1 - a(F, X, B + G) = 1$. Thus (5) holds.

Step 4. In this step, we prove (6).

The variety $X_{\mathbf{e}}$ is normal and projective, $Q \geq 0$, $K_{X_{\mathbf{e}}} + Q$ is $\mathbb{Q}$-Cartier, and $\mathrm{mult}_F Q = 1$. By Theorem 2.21, $\Delta_{F^\nu} \geq 0$.

Away from $E_{\mathbf{e}}$, $X'_{\mathbf{e}}$ is isomorphic to an open subset of the smooth variety X'. Near $E_{\mathbf{e}}$, $X'_{\mathbf{e}}$ is covered by open subsets each of which is the quotient of a smooth variety by μ_{e_i} for some i, namely the images of the morphisms q_i of **Step 2** of the proof of Construction-Proposition 9.2. Thus $X'_{\mathbf{e}}$ has only quotient singularities, so $X'_{\mathbf{e}}$ is $\mathbb{Q}$-factorial [KM98, Proposition 5.15]. Write $\Gamma = E_{\mathbf{e}} + \Gamma_+ - \Gamma_-$, where $\Gamma_+, \Gamma_- \geq 0$ have no common components and do not contain $E_{\mathbf{e}}$. Then

Γ_- and $K_{X'_{\mathbf{e}}} + E_{\mathbf{e}} + \Gamma_+$ are $\mathbb{Q}$-Cartier. Since $X'_{\mathbf{e}}$ is normal, it is smooth at the generic point of $E_{\mathbf{e}}$, where $E_{\mathbf{e}} + \Gamma_+$ equals $E_{\mathbf{e}}$, so $(X'_{\mathbf{e}}, E_{\mathbf{e}} + \Gamma_+)$ is lc at the generic point of $E_{\mathbf{e}}$. Define $\Delta^+_{E_{\mathbf{e}}}$ by $K_{E_{\mathbf{e}}} + \Delta^+_{E_{\mathbf{e}}} = (K_{X'_{\mathbf{e}}} + E_{\mathbf{e}} + \Gamma_+)|_{E_{\mathbf{e}}}$. Then

$$\Delta_{E_{\mathbf{e}}} = \Delta^+_{E_{\mathbf{e}}} - \Gamma_-|_{E_{\mathbf{e}}}.$$

By [KM98, Theorem 0.2], applied to the closure of the graph of the birational map $X_{\mathbf{e}} \dashrightarrow X'_{\mathbf{e}}$ over X and to the product of the ideal sheaf $\mathcal{I}$ of the exceptional locus of the projection from the graph closure to $X_{\mathbf{e}}$ and the ideal sheaves of the exceptional locus of the projection to $X'_{\mathbf{e}}$ and of the preimages of $\operatorname{Supp} Q$ and $\operatorname{Supp} \Gamma$, there exist a smooth variety Y and projective birational morphisms $f : Y \to X_{\mathbf{e}}$ and $f' : Y \to X'_{\mathbf{e}}$ with $\rho \circ f = h \circ g \circ f'$ such that

$$\operatorname{Exc}(f) \cup \operatorname{Exc}(f') \cup f^{-1}(\operatorname{Supp} Q) \cup f'^{-1}(\operatorname{Supp} \Gamma)$$

is a simple normal crossing divisor. Since $\operatorname{ord}_F = w = \operatorname{ord}_{E_{\mathbf{e}}}$, the strict transform T of F on Y is also the strict transform of $E_{\mathbf{e}}$. Since $\operatorname{mult}_F Q = 1$, $T \subset f^{-1}(\operatorname{Supp} Q)$, so T is a component of the simple normal crossing divisor above, and T is smooth.

Moreover, $\operatorname{Exc}(f)$ is a divisor. By [KM98, Corollary 2.63], $\operatorname{Exc}(f')$ is a divisor, since $X'_{\mathbf{e}}$ is $\mathbb{Q}$-factorial. By (3) and $\operatorname{ord}_F = \operatorname{ord}_{E_{\mathbf{e}}}$, any prime divisor on $X_{\mathbf{e}}$ is the birational transform of a prime divisor on $X'_{\mathbf{e}}$, so any component of $\operatorname{Exc}(f')$ is exceptional over $X_{\mathbf{e}}$, and $\operatorname{Exc}(f') \subset \operatorname{Exc}(f)$. The zero locus of the invertible ideal $\mathcal{I} \cdot \mathcal{O}_Y$ is the preimage of the exceptional locus of the projection from the graph closure to $X_{\mathbf{e}}$, so it is contained in $\operatorname{Exc}(f)$, and f is a local isomorphism at any point of Y outside the zero locus and outside $\operatorname{Exc}(f')$. Thus $\operatorname{Exc}(f)$ is the union of $\operatorname{Exc}(f')$ and the support of the Cartier divisor defined by $\mathcal{I} \cdot \mathcal{O}_Y$, so Y is a log resolution of $(X_{\mathbf{e}}, Q)$ and of $(X'_{\mathbf{e}}, E_{\mathbf{e}} + \Gamma_+)$. Write

$$K_Y + \Delta_Y := f^*(K_{X_{\mathbf{e}}} + Q) = (\rho \circ f)^*(K_X + B + G) = f'^*(K_{X'_{\mathbf{e}}} + \Gamma),$$
$$K_Y + \Delta^+_Y := f'^*(K_{X'_{\mathbf{e}}} + E_{\mathbf{e}} + \Gamma_+).$$

Then $\operatorname{mult}_T \Delta_Y = 1 - a(w, X, B + G) = 1$. Let $\Delta_T := (\Delta_Y - T)|_T$, and let $f_T : T \to F^\nu$ and $f'_T : T \to E_{\mathbf{e}}$ be the induced projective birational morphisms. By [FH23, Definition 1.3 and Corollary 1.4], applied to $(X_{\mathbf{e}}, Q)$, where $Q \geq 0$, and to $(X'_{\mathbf{e}}, E_{\mathbf{e}} + \Gamma_+)$, we have

$$K_T + \Delta_T = f_T^*(K_{F^\nu} + \Delta_{F^\nu}), \qquad K_T + (\Delta^+_Y - T)|_T = f'^*_T(K_{E_{\mathbf{e}}} + \Delta^+_{E_{\mathbf{e}}}).$$

Since $\Delta_Y = \Delta^+_Y - f'^*\Gamma_-$ and $T \not\subset \operatorname{Supp} f'^*\Gamma_-$, we have $(f'^*\Gamma_-)|_T = f'^*_T(\Gamma_-|_{E_{\mathbf{e}}})$. Subtracting it from the second equality, we obtain

$$K_T + \Delta_T = f_T^*(K_{F^\nu} + \Delta_{F^\nu}) = f'^*_T(K_{E_{\mathbf{e}}} + \Delta_{E_{\mathbf{e}}}).$$

Since $\rho \circ f = h \circ g \circ f'$, the compositions $T \to F^\nu \to X$ and $T \to E_{\mathbf{e}} \to X$ coincide. By Definition 2.2, (F^ν, Δ_{F^ν}) and $(E_{\mathbf{e}}, \Delta_{E_{\mathbf{e}}})$ are crepant birational over X. □

Proof of Proposition 9.1. Let $\mathbf{e}$, $\rho : X_{\mathbf{e}} \to X$, F, and Q be the objects of Construction-Proposition 9.3, and let $E_{\mathbf{e}}$ and $\Delta_{E_{\mathbf{e}}}$ be the objects of Construction-Proposition 9.2 for G and $\mathbf{e}$. By Construction-Proposition 9.3(1), the hypotheses of Construction-Proposition 9.2 hold for G and $\mathbf{e}$. By Construction-Proposition 9.2 with $U = S$ and $L = 0$, $(E_{\mathbf{e}}, \Delta_{E_{\mathbf{e}}})$ is sub-lc.

By Construction-Proposition 9.3(6), there exist a normal variety W and projective birational morphisms $g_1 : W \to F^\nu$ and $g_2 : W \to E_{\mathbf{e}}$ such that

$$K_W + \Delta_W := g_1^*(K_{F^\nu} + \Delta_{F^\nu}) = g_2^*(K_{E_{\mathbf{e}}} + \Delta_{E_{\mathbf{e}}}).$$

Then $g_{1*}\Delta_W = \Delta_{F^\nu}$ and $g_{2*}\Delta_W = \Delta_{E_{\mathbf{e}}}$. By [KM98, Lemma 2.30], applied to g_1 and to g_2, for any prime divisor P over W,

$$a(P, F^\nu, \Delta_{F^\nu}) = a(P, W, \Delta_W) = a(P, E_{\mathbf{e}}, \Delta_{E_{\mathbf{e}}}) \geq 0.$$

Since any prime divisor over F^ν is a prime divisor over W, (F^ν, Δ_{F^ν}) is sub-lc. Since $\Delta_{F^\nu} \geq 0$ (Construction-Proposition 9.3(6)), (F^ν, Δ_{F^ν}) is lc.

By Construction-Proposition 9.3(5), $Q \geq 0$ and $\mathrm{mult}_F Q = 1$, and $K_{X_{\mathbf{e}}} + Q = \rho^*(K_X + B + G)$ is $\mathbb{Q}$-Cartier. By Theorem 2.21, $(X_{\mathbf{e}}, Q)$ is lc on an open neighborhood $X^\circ_{\mathbf{e}}$ of F. Since ρ is proper, $\rho(X_{\mathbf{e}} \backslash X^\circ_{\mathbf{e}})$ is closed. By Construction-Proposition 9.3(4), $\rho^{-1}(V) \subset F \subset X^\circ_{\mathbf{e}}$, so V is disjoint from $\rho(X_{\mathbf{e}} \backslash X^\circ_{\mathbf{e}})$. Let $X^\circ := X \backslash \rho(X_{\mathbf{e}} \backslash X^\circ_{\mathbf{e}})$. Then X° is an open neighborhood of V and $\rho^{-1}(X^\circ) \subset X^\circ_{\mathbf{e}}$. By Construction-Proposition 9.3(3)(5), $\rho_* Q = B + G$. By [KM98, Lemma 2.30], for any prime divisor P over X whose center on X intersects X°, we have $a(P, X, B + G) = a(P, X_{\mathbf{e}}, Q) \geq 0$. Thus $(X, B + G)$ is lc on X°. □

10. A dlt model on which the center has codimension equal to the rational rank

In this section, we prove Construction-Proposition 10.1 and Lemma 10.2. Section 11 requires that the center of the stationary valuation v has codimension equal to the rational rank r of v. Construction-Proposition 10.1 replaces (X, B) by a $\mathbb{Q}$-factorial dlt pair (X', B') on which $\mathrm{center}_{X'} v$ has codimension r. In this section and in Section 12, X' and B' always denote the variety and the boundary of Construction-Proposition 10.1, and an adapted carrier (Construction-Proposition 8.1) of v for (X, B) is written $h : X'' \to X$, and one for (X', B') is written $h' : X'' \to X'$. When Set-up 8.4 is used there for (X, B) or for (X', B'), the carrier of Set-up 8.4 is $h : X'' \to X$ or $h' : X'' \to X'$, respectively. Corrected boundaries on X and on X' correspond to each other by Construction-Proposition 10.1(2)–(4), and corrected systems on X' are corrected systems on X by Construction-Proposition 10.1(5).

On an adapted carrier of v (Construction-Proposition 8.1) the center of v already has codimension r, but the crepant boundary may have negative coefficients. We add $tD_{\leq N_1}$ to make the coefficients near the generic point of the center non-negative, and remove the remaining negative coefficients by [OX12, Theorem 1.1] and an MMP over X that is an isomorphism near the generic point of the center.

Construction-Proposition 10.1. Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B) with m_0 as in Definition 7.1, and v a quasi-monomial valuation of $\mathbb{C}(X)$ of rational rank r that is stationary for $(X, B; D_\bullet)$. Then there exist an integer $N_1 \geq m_0$, a rational number $0 < t < 1$, a $\mathbb{Q}$-factorial normal projective variety X', and a projective birational morphism $\mu : X' \to X$ such that the $\mathbb{Q}$-divisor B' defined by

$$K_{X'} + B' = \mu^*(K_X + B + tD_{\leq N_1})$$

satisfies $B' \geq 0$, (X', B') is dlt, and $\mathrm{center}_{X'} v$ is of codimension r in X'. Moreover, let $\kappa_j := 1 - t$ for $j \leq N_1$, $\kappa_j := 1$ for $j > N_1$, and $D'_j := \kappa_j \mu^* D_j$ for $j \geq 1$. Then the following hold.

(1) $D'_\bullet = (D'_j)_{j \geq 1}$ is an lc anti-canonical series on (X', B'), and v is stationary for $(X', B'; D'_\bullet)$.
(2) For any integers $N \geq N_1$ and $m \geq 1$ and any rational numbers $c_1, \ldots, c_m$,

$$K_{X'} + B' + D'_{\leq N} + \sum_{j=1}^{m} c_j D'_j = \mu^*\Big(K_X + B + D_{\leq N} + \sum_{j=1}^{m} \kappa_j c_j D_j\Big).$$

(3) For any family of rational numbers $(c_{j,N})_{1 \leq j \leq m, N \geq N_0}$ with $N_0 \geq \max\{m, N_1\}$, $(c_{j,N})$ is a correction family for $D'_\bullet$ if and only if $(\kappa_j c_{j,N})$ is a correction family for $D_\bullet$.
(4) For any such family $(c_{j,N})$ and any moment row $x_\bullet$ for $D_\bullet$, $(c_{j,N})$ satisfies the moment row $(\kappa_j x_j)_{j \geq 1}$ for $D'_\bullet$, whose sum is $x_\infty - t\sum_{j=1}^{N_1} x_j$, if and only if $(\kappa_j c_{j,N})$ satisfies $x_\bullet$.
(5) For any moment rows $x^{(1)}_\bullet, \ldots, x^{(p)}_\bullet$ for $D_\bullet$ and any corrected system W' for $D'_\bullet$ satisfying $(\kappa_j x^{(1)}_j)_{j \geq 1}, \ldots, (\kappa_j x^{(p)}_j)_{j \geq 1}$ such that $-e_i(K_X + B)$ is Cartier for the positive integers e_i in Definition 7.3 defining W', the space W' of rational functions is a corrected system for $D_\bullet$ satisfying $x^{(1)}_\bullet, \ldots, x^{(p)}_\bullet$, and $\mu^* G_s = G'_s + t\mu^* D_{\leq N_1}$ for any $s \in W' \backslash \{0\}$, where G'_s and G_s are the $\mathbb{Q}$-divisors of Definition 7.3 on X' and X, respectively.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct a $\mathbb{Q}$-factorial projective toroidal modification $g_0 : Y_0 \to X''$ of an adapted carrier $h : X'' \to X$ of v, on which $\operatorname{center}_{Y_0} v$ is a stratum of codimension r cut out by prime divisors $F_1, \dots, F_r$ with $\operatorname{ord}_{F_a} = v_{\mathbf{e}_a}$ and $\delta(\mathbf{e}_a) < 1/4$.

Let $d := \dim X$. By Definition 2.7, $r \geq 1$. By Construction-Proposition 8.1, there exists an adapted carrier (h, S) of v for (X, B) and the empty set, with $h : X'' \to X$, a reduced simple normal crossing divisor $E = \sum_b E_b$ on X'' whose components include $E_1, \dots, E_r$, and weights $\alpha \in \mathbb{R}^r_{>0}$. Thus (X, B), $D_\bullet$, v, and (h, S) satisfy the conditions of Set-up 8.4, and we use the notation $\mathbf{a}$, b_j, v_γ, and δ of Set-up 8.4.

Let f_j be the functions of Lemma 8.5. By Lemma 8.5(2) and (3), $\delta = \sum_{j\geq 1} f_j$ on $\mathbb{R}^r_{>0}$, and for any $j \geq 1$, $f_j \geq 0$ on $\mathbb{R}^r_{>0}$, f_j vanishes on a neighborhood of α, and $f_j(\lambda\gamma) = \lambda f_j(\gamma)$ and $\delta(\lambda\gamma) = \lambda\delta(\gamma)$ for any $\gamma \in \mathbb{R}^r_{>0}$ and any real number $\lambda > 0$. Moreover, there exist a compact neighborhood Ω of α in $\mathbb{R}^r_{>0}$ and a real number $c > 0$ such that f_j is $cv(D_j)$-Lipschitz on Ω for any $j \geq 1$.

Let $Q \subset \mathbb{R}^r$ be a compact set. We show that

$$\lim_{\lambda\to+\infty} \sup_{y\in Q} \delta(\lambda\alpha + y) = 0. \tag{10.1}$$

Since Q is bounded and Ω is a neighborhood of α, there exists $\lambda_0 > 0$ such that $\alpha + y/\lambda \in \Omega$ for any $y \in Q$ and any $\lambda \geq \lambda_0$. In particular, $\lambda\alpha + y = \lambda(\alpha + y/\lambda) \in \mathbb{R}^r_{>0}$. Since $f_j(\alpha) = 0$, for such y and λ and any $j \geq 1$, we have

$$0 \leq f_j(\lambda\alpha + y) = \lambda\big(f_j(\alpha + y/\lambda) - f_j(\alpha)\big) \leq \lambda \cdot cv(D_j) \cdot \|y/\lambda\| = cv(D_j)\|y\|.$$

For each fixed j, f_j vanishes on a neighborhood of α, so there exists $\lambda_j \geq \lambda_0$ such that $f_j(\lambda\alpha + y) = 0$ for any $y \in Q$ and any $\lambda \geq \lambda_j$. Let $\epsilon > 0$. Since $\sum_{j\geq 1} v(D_j) = a(v, X, B) < +\infty$, there exists an integer $J \geq 1$ such that

$$c \cdot \max_{y\in Q} \|y\| \cdot \sum_{j>J} v(D_j) < \epsilon.$$

Thus $\delta(\lambda\alpha + y) = \sum_{j>J} f_j(\lambda\alpha + y) < \epsilon$ for any $y \in Q$ and any $\lambda \geq \max\{\lambda_1, \dots, \lambda_J\}$. This proves (10.1).

Now we choose a simplicial cone. Let $Q := [-2, 2]^r$. By (10.1), there exists $\lambda > 0$ such that $\lambda\alpha + y \in \mathbb{R}^r_{>0}$ and $\delta(\lambda\alpha + y) < 1/4$ for any $y \in Q$. Let $\Lambda := \mathbb{Z}^r \cap (\lambda\alpha + Q)$. For each i, the interval $[\lambda\alpha_i - 2, \lambda\alpha_i + 2]$ contains integers $k_i^- \leq \lambda\alpha_i - 1$ and $k_i^+ \geq \lambda\alpha_i + 1$. Thus the vertices of the box $\prod_{i=1}^r [k_i^-, k_i^+]$ belong to Λ, and the box contains $\lambda\alpha + [-1, 1]^r$. Hence $\lambda\alpha$ is an interior point of the convex hull of Λ, and α is an interior point of the cone $\sigma_\Lambda \subset \mathbb{R}^r$ generated by Λ. For $p \in \Lambda$, let $\bar{p}$ be the primitive vector of $\mathbb{Z}^r$ on the ray $\mathbb{R}_{>0}p$. Then $\bar{p} \in \mathbb{Z}^r_{>0}$, and since p is a positive integer multiple of $\bar{p}$ and δ is homogeneous of degree one, $0 \leq \delta(\bar{p}) \leq \delta(p) < \frac{1}{4}$. Triangulating the polytope $\sigma_\Lambda \cap \{\gamma_1 + \cdots + \gamma_r = 1\}$ by simplices whose vertices are vertices of the polytope, we obtain a subdivision of σ_Λ into simplicial cones whose rays are generated by vectors $\bar{p}$ with $p \in \Lambda$. Let τ be the cone of the subdivision such that α is in the relative interior of τ. If $\dim \tau < r$, then the linear span of τ is a proper linear subspace of $\mathbb{R}^r$ defined by linear equations with rational coefficients, and α satisfies a non-zero linear equation with rational coefficients. This contradicts the linear independence of $\alpha_1, \dots, \alpha_r$ over $\mathbb{Q}$ (Construction-Proposition 8.1(2)). Hence τ is a simplicial cone of dimension r, and α is an interior point of τ. Let $\mathbf{e}_1, \dots, \mathbf{e}_r \in \mathbb{Z}^r_{>0}$ be the primitive vectors on the rays of τ. Then

$$0 \leq \delta(\mathbf{e}_a) < \frac{1}{4} \quad \text{for } a = 1, \dots, r. \tag{10.2}$$

Next we realize τ on a toroidal modification of X''. We regard τ as a cone in the cone $\mathbb{R}^r_{\geq 0}$ of the conical polyhedral complex of the toroidal embedding $X''\backslash E \subset X''$ [KKMS73, Chapter II] corresponding to the stratum S, the coordinates corresponding to $E_1, \dots, E_r$. For each facet of τ, choose integers $n_1, \dots, n_r$ such that the linear form $\ell = \sum_{i=1}^r n_i\gamma_i$ vanishes on the facet and is non-negative on τ, and let $n_b := 0$ for the components E_b of E other than $E_1, \dots, E_r$.

Let $n_b^+ := \max\{n_b, 0\}$ and $n_b^- := \max\{-n_b, 0\}$. The normalized blow-up of the product, over all facets of τ, of the monomial ideals

$$\mathcal{O}_{X''}\Big(-\sum_b n_b^+ E_b\Big) + \mathcal{O}_{X''}\Big(-\sum_b n_b^- E_b\Big)$$

is a projective toroidal modification of X'' whose conical polyhedral complex is the subdivision of that of (X'', E) by the hyperplanes $\ell = 0$ [KKMS73, Chapter I, §2, Theorem 10; Chapter II, §2, Theorem 10*]. Since $\tau \subset \mathbb{R}^r_{\geq 0}$ is the set of $\gamma \in \mathbb{R}^r_{\geq 0}$ with $\ell(\gamma) \geq 0$ for the forms ℓ of all facets, τ is a cone of the subdivision.

We then make the complex simplicial without subdividing τ. We star-subdivide the cones that are not simplicial, in increasing order of dimension, at lattice rays in their relative interiors. Any face of a simplicial cone is simplicial, so the star subdivisions do not change τ and the faces of τ, and after finitely many star subdivisions all cones are simplicial. Each star subdivision is realized by the normalized blow-up of a monomial ideal [KKMS73, Chapter I, §2, Theorem 10; Chapter II, §2, Theorem 10*]. We obtain a projective toroidal modification $g_0 : Y_0 \to X''$ whose conical polyhedral complex is simplicial and contains τ [KKMS73, Chapter II, §2, Theorem 6*]. Let $E_{Y_0} := g_0^{-1}(E)$ with the reduced structure. Then g_0 is an isomorphism over $X''\backslash E$. Analytically locally, Y_0 is isomorphic to the product of a smooth variety and the affine toric variety of a simplicial cone [KKMS73, Chapter II, §2, Theorem 6*], and the affine toric variety of a simplicial cone of dimension k is the quotient of $\mathbb{C}^k$ by a finite abelian group [CLS11, Proposition 1.3.18]. Thus Y_0 has only quotient singularities, and Y_0 is $\mathbb{Q}$-factorial by [KM98, Proposition 5.15].

For $a = 1, \dots, r$, let F_a be the prime divisor of Y_0 corresponding to the ray $\mathbb{R}_{\geq 0}\mathbf{e}_a$. Then F_a is a component of E_{Y_0}, and $\operatorname{ord}_{F_a} = v_{\mathbf{e}_a}$ [KKMS73, Chapter II, §2, Theorem 6*]. Let S_0 be the closure of the stratum of Y_0 corresponding to τ. Then S_0 is an irreducible component of $F_1 \cap \cdots \cap F_r$, $\dim S_0 = d - r$, and the components of E_{Y_0} containing S_0 are exactly $F_1, \dots, F_r$. Since α is an interior point of τ, $\operatorname{center}_{Y_0} v = S_0$ [KKMS73, Chapter II, §2, Theorem 6*].

Step 2. In this step we choose N_1 and t, and find an open subset $U \subset Y_0$ containing the generic point of S_0 such that $\Gamma|_U \geq 0$, where Γ is the crepant boundary of $(X, B + tD_{\leq N_1})$ on Y_0.

Let $a \in \{1, \dots, r\}$. Since $\operatorname{ord}_{F_a} = v_{\mathbf{e}_a}$, Definition-Proposition 2.8 implies that

$$a(F_a, X, B + D_{\leq N}) = a(v_{\mathbf{e}_a}, X, B) - \sum_{j=1}^{N} v_{\mathbf{e}_a}(D_j) = \mathbf{e}_a \cdot \mathbf{a} - \sum_{j=1}^{N} v_{\mathbf{e}_a}(D_j)$$

for any $N \geq 1$. As $N \to +\infty$, the right-hand side decreases to $\delta(\mathbf{e}_a)$, which is less than $1/4$ by (10.2). Thus there exists an integer $N_1 \geq m_0$ such that $a(F_a, X, B + D_{\leq N_1}) < 1/2$ for $a = 1, \dots, r$. Choose a rational number $0 < t < 1$ such that $(1-t)v_{\mathbf{e}_a}(D_{\leq N_1}) < 1/4$ for $a = 1, \dots, r$. Then

$$a(F_a, X, B + tD_{\leq N_1}) = a(F_a, X, B + D_{\leq N_1}) + (1-t)v_{\mathbf{e}_a}(D_{\leq N_1}) < \frac{3}{4} \quad \text{for } a = 1, \dots, r. \tag{10.3}$$

By Definition 7.1(1), $(X, B + D_{\leq N_1})$ is lc. For any prime divisor P over X,

$$a(P, X, B + tD_{\leq N_1}) = a(P, X, B + D_{\leq N_1}) + (1-t)\operatorname{mult}_P D_{\leq N_1} \geq 0,$$

by Definition-Proposition 2.8, so $(X, B + tD_{\leq N_1})$ is lc.

Define the $\mathbb{Q}$-divisor Γ on Y_0 by

$$K_{Y_0} + \Gamma = (h \circ g_0)^*(K_X + B + tD_{\leq N_1}).$$

For any prime divisor P on Y_0, $\operatorname{mult}_P \Gamma = 1 - a(P, X, B + tD_{\leq N_1}) \leq 1$. If P is not exceptional over X, then $\operatorname{mult}_P \Gamma$ is the coefficient of $(h \circ g_0)(P)$ in $B + tD_{\leq N_1}$, which is non-negative. Thus any prime divisor P with $\operatorname{mult}_P \Gamma < 0$ is exceptional over X. Such a P is either g_0-exceptional or the strict transform of an h-exceptional prime divisor, which is a component of E by Construction-Proposition 8.1(1). Since g_0 is an isomorphism over $X''\backslash E$, in both cases P is a component of

E_{Y_0}. By (10.3), $\mathrm{mult}_{F_a}\Gamma > 1/4$ for $a = 1, \dots, r$. Since the components of E_{Y_0} containing S_0 are $F_1, \dots, F_r$, no prime divisor P with $\mathrm{mult}_P\Gamma < 0$ contains S_0. Let

$$U := Y_0 \setminus \bigcup_{\mathrm{mult}_P\Gamma<0} P.$$

Then U is an open subset of Y_0 containing the generic point of S_0, $\Gamma|_U \geq 0$, and $(U, \Gamma|_U)$ is lc, since log discrepancies with respect to (Y_0, Γ) are equal to those with respect to $(X, B + tD_{\leq N_1})$ [KM98, Lemma 2.30].

Step 3. In this step we construct a $\mathbb{Q}$-factorial dlt pair (Y, Δ_Y) over Y_0 and a $\mathbb{Q}$-divisor $F \geq 0$ on Y that is exceptional over X and disjoint from the preimage of U.

Let $\Gamma^+ := \sum_P \max\{\mathrm{mult}_P\Gamma, 0\}P$, where P runs over the prime divisors on Y_0. The coefficients of Γ^+ belong to $[0, 1]$, and $K_{Y_0} + \Gamma^+$ is $\mathbb{Q}$-Cartier since Y_0 is $\mathbb{Q}$-factorial. By [OX12, Theorem 1.1], there exists a log canonical model $g_1 : Y_1 \to Y_0$ over the pair (Y_0, Γ^+) in the sense of [OX12, Definition 2.1]. In particular, g_1 is projective and birational, and (Y_1, Δ_1) is lc, where Δ_1 is the sum of the strict transform of Γ^+ and the g_1-exceptional prime divisors. Since $\Gamma^+|_U = \Gamma|_U$ and $(U, \Gamma|_U)$ is lc, g_1 is an isomorphism over U [OX12, Theorem 1.1 and the paragraph after Definition 2.1].

By [Fuj11, Theorem 10.4] (see also [Bir12, Corollary 3.6]), there exists a dlt blow-up $g_2 : Y \to Y_1$ of (Y_1, Δ_1): g_2 is projective and birational, Y is $\mathbb{Q}$-factorial, the $\mathbb{Q}$-divisor Δ_Y defined as the sum of the strict transform of Δ_1 and the g_2-exceptional prime divisors satisfies

$$K_Y + \Delta_Y = g_2^*(K_{Y_1} + \Delta_1),$$

and (Y, Δ_Y) is dlt. Define Γ_Y by $K_Y + \Gamma_Y = (h \circ g_0 \circ g_1 \circ g_2)^*(K_X + B + tD_{\leq N_1})$, and let $F := \Delta_Y - \Gamma_Y$. We show that

$$F \geq 0, \quad F \text{ is exceptional over } X, \quad \text{and} \quad \mathrm{Supp}\, F \cap (g_1 \circ g_2)^{-1}(U) = \emptyset. \tag{10.4}$$

Let P be a prime divisor on Y. If P is not exceptional over Y_0, let P_0 be the image of P on Y_0. Then $\mathrm{mult}_P\Delta_Y = \mathrm{mult}_{P_0}\Gamma^+$ and $\mathrm{mult}_P\Gamma_Y = \mathrm{mult}_{P_0}\Gamma$, so $\mathrm{mult}_P F = \mathrm{mult}_{P_0}(\Gamma^+ - \Gamma) \geq 0$. If moreover P is not exceptional over X, then $\mathrm{mult}_{P_0}\Gamma \geq 0$ by **Step 2**, so $\mathrm{mult}_P F = 0$. If P is exceptional over Y_0, then P is either g_2-exceptional or the strict transform of a g_1-exceptional prime divisor. In both cases $\mathrm{mult}_P\Delta_Y = 1$, while $\mathrm{mult}_P\Gamma_Y = 1 - a(P, X, B + tD_{\leq N_1}) \leq 1$ since $(X, B + tD_{\leq N_1})$ is lc. Thus $\mathrm{mult}_P F \geq 0$. Finally, g_1 is an isomorphism over U and Δ_1 is the strict transform of $\Gamma^+|_U = \Gamma|_U$ over U, so

$$K_Y + \Delta_Y = g_2^*(K_{Y_1} + \Delta_1) = (g_1 \circ g_2)^*(K_{Y_0} + \Gamma) = K_Y + \Gamma_Y$$

over U. Hence F is zero on $(g_1 \circ g_2)^{-1}(U)$. This proves (10.4).

Step 4. In this step we run an MMP over X starting with (Y, Δ_Y), construct X' and μ, and show that $\mathrm{center}_{X'} v$ has codimension r.

By the definition of F,

$$K_Y + \Delta_Y = (h \circ g_0 \circ g_1 \circ g_2)^*(K_X + B + tD_{\leq N_1}) + F,$$

so $K_Y + \Delta_Y \sim_{\mathbb{Q}} F$ over X. Since $h \circ g_0 \circ g_1 \circ g_2$ is birational and $F \geq 0$ is exceptional over X, F is very exceptional over X in the sense of [Bir12, Definition 3.1]. We run a $(K_Y + \Delta_Y)$-MMP over X with scaling of an ample divisor over X:

$$Y = Y_2 \dashrightarrow Y_3 \dashrightarrow \cdots \dashrightarrow Y_l.$$

The contraction of each extremal ray exists by [Fuj11, Theorem 1.1(4)], and each flip exists by [Bir12, Corollary 1.2]. Let Δ_{Y_i} and F_{Y_i} be the birational transforms of Δ_Y and F on Y_i. Then Y_i is $\mathbb{Q}$-factorial and (Y_i, Δ_{Y_i}) is dlt for any i [KM98, Propositions 3.36, 3.37, Corollary 3.44]. By [Bir12, Theorem 3.4], the MMP terminates, and $F_{Y_l} = 0$. Let $\pi_i : Y_i \to X$ be the induced

morphisms. Since $Y \dashrightarrow Y_i$ does not extract any divisor, pushing forward the equality $K_Y+\Delta_Y = (h\circ g_0\circ g_1\circ g_2)^*(K_X+B+tD_{\le N_1})+F$ implies that

$$K_{Y_i}+\Delta_{Y_i} = \pi_i^*(K_X+B+tD_{\le N_1})+F_{Y_i} \tag{10.5}$$

for any i. Let $X' := Y_l$, $\mu := \pi_l$, and $B' := \Delta_{X'}$. Since $F_{X'}=0$, (10.5) implies that $K_{X'}+B' = \mu^*(K_X+B+tD_{\le N_1})$. Thus X' is a $\mathbb{Q}$-factorial normal projective variety, μ is a projective birational morphism, $B'\ge 0$, and (X',B') is dlt.

We show that the birational map $Y \dashrightarrow X'$ restricts to an open immersion $\iota : Y\setminus \operatorname{Supp} F \to X'$ over X. Let $\varphi_i : Y_i \to Z_i$ be the contraction over X of the extremal ray R_i at the i-th step. For any curve C with $[C]\in R_i$, $\pi_i(C)$ is a point, so (10.5) implies that

$$(F_{Y_i}\cdot C) = \big((K_{Y_i}+\Delta_{Y_i})\cdot C\big) < 0.$$

Since $F_{Y_i}\ge 0$, $C\subset \operatorname{Supp} F_{Y_i}$. Let $z\in Z_i$ and $y\in \varphi_i^{-1}(z)\setminus \operatorname{Supp} F_{Y_i}$. Any curve in $\varphi_i^{-1}(z)$ has class in R_i, hence is contained in $\operatorname{Supp} F_{Y_i}$, so y is an isolated point of $\varphi_i^{-1}(z)$. Since φ_i is a projective birational morphism onto the normal variety Z_i, $\varphi_i^{-1}(z)$ is connected [Har77, Corollary III.11.4], so $\varphi_i^{-1}(z)=\{y\}$. Let $V_i := Z_i\setminus \varphi_i(\operatorname{Supp} F_{Y_i})$. Then $\varphi_i^{-1}(V_i) = Y_i\setminus \operatorname{Supp} F_{Y_i}$, and φ_i has finite fibers over V_i. Since φ_i is proper, φ_i is finite over V_i. Since φ_i is birational and Z_i is normal, $\varphi_{i*}\mathcal{O}_{Y_i}=\mathcal{O}_{Z_i}$, so φ_i is an isomorphism over V_i. If the i-th step is a divisorial contraction, then $Y_{i+1}=Z_i$ and $\operatorname{Supp} F_{Y_{i+1}} \subset \varphi_i(\operatorname{Supp} F_{Y_i})$. If the i-th step is a flip $\varphi_i^+ : Y_{i+1}\to Z_i$, then over V_i the φ_i^+-ample divisor $K_{Y_{i+1}}+\Delta_{Y_{i+1}}$ is the pullback of the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\varphi_{i*}(K_{Y_i}+\Delta_{Y_i})|_{V_i}$, so φ_i^+ contracts no curve over V_i and is an isomorphism over V_i. Moreover, $F_{Y_{i+1}}$ and F_{Y_i} coincide under the isomorphism $\varphi_i^{-1}(V_i)\cong(\varphi_i^+)^{-1}(V_i)$, so $(\varphi_i^+)^{-1}(V_i)$ is disjoint from $\operatorname{Supp} F_{Y_{i+1}}$. In both cases, $Y_i \dashrightarrow Y_{i+1}$ restricts to an open immersion of $Y_i\setminus\operatorname{Supp} F_{Y_i}$ into $Y_{i+1}\setminus\operatorname{Supp} F_{Y_{i+1}}$ over X. By induction on i, $Y\dashrightarrow X'$ restricts to an open immersion $\iota : Y\setminus\operatorname{Supp} F\to X'$ over X.

Let ξ be the generic point of $\operatorname{center}_Y v$. Since $g_1\circ g_2$ is proper, $(g_1\circ g_2)(\operatorname{center}_Y v) = \operatorname{center}_{Y_0} v = S_0$. Hence ξ is mapped to the generic point of S_0, which belongs to U by **Step 2**, and $\xi\in Y\setminus\operatorname{Supp} F$ by (10.4). Moreover, $\dim \operatorname{center}_Y v \ge \dim S_0 = d-r$. By Abhyankar's inequality [Abh56, Theorem 1], $r\le \dim \mathcal{O}_{Y,\xi} = d-\dim\operatorname{center}_Y v$. Thus $\dim\mathcal{O}_{Y,\xi}=r$. Since ι is an open immersion over X, $\mathcal{O}_{X',\iota(\xi)}=\mathcal{O}_{Y,\xi}$ as subrings of $\mathbb{C}(X)$. Hence $\iota(\xi)$ is the generic point of $\operatorname{center}_{X'} v$, and $\operatorname{center}_{X'} v$ is of codimension $\dim\mathcal{O}_{X',\iota(\xi)}=r$ in X'.

Step 5. In this step we prove (1)–(4).

We first prove (2). Let $N\ge N_1$ and $m\ge 1$ be integers, and $c_1,\dots,c_m$ rational numbers. Since $\kappa_j = 1-t$ for $j\le N_1$ and $\kappa_j=1$ for $j>N_1$, we have $D_{\le N}-tD_{\le N_1} = \sum_{j=1}^N \kappa_j D_j$. Thus

$$\begin{aligned}\mu^*\Big(K_X+B+D_{\le N}+\sum_{j=1}^m \kappa_j c_j D_j\Big) &= \mu^*(K_X+B+tD_{\le N_1}) + \sum_{j=1}^N \kappa_j\mu^*D_j + \sum_{j=1}^m c_j\kappa_j\mu^*D_j\\ &= K_{X'}+B'+D'_{\le N}+\sum_{j=1}^m c_jD'_j.\end{aligned}$$

This proves (2).

Next we prove (1). Each $D'_j\ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. If $N\ge N_1$, then (2) with $c_1=0$ implies that $K_{X'}+B'+D'_{\le N} = \mu^*(K_X+B+D_{\le N})$. Since $(X,B+D_{\le N})$ is lc and $B'+D'_{\le N}\ge 0$, $(X',B'+D'_{\le N})$ is lc [KM98, Lemma 2.30]. If $N<N_1$, then for any prime divisor P over X',

$$a(P,X',B'+D'_{\le N}) = a(P,X',B'+D'_{\le N_1}) + \operatorname{mult}_P\Big(\sum_{N<j\le N_1} D'_j\Big)\ge 0$$

by Definition-Proposition 2.8, so $(X',B'+D'_{\le N})$ is lc. For $N\ge N_1$, let $a'_{j,N} := a_{j,N}/(1-t)$ for $j\le m_0$ and $a'_{j,N}:=0$ for $m_0<j\le N_1$. Since $N_1\ge m_0$, $\kappa_j a'_{j,N} = a_{j,N}$ for $j\le m_0$, and (2)

implies that

$$K_{X'}+B'+D'_{\le N}+\sum_{j=1}^{N_1}a'_{j,N}D'_j=\mu^*\Big(K_X+B+D_{\le N}+\sum_{j=1}^{m_0}a_{j,N}D_j\Big)\sim_{\mathbb{Q}}0$$

for any $N\ge N_1$. Moreover, $\lim_{N\to+\infty}a'_{j,N}=0$ for any j. Thus $D'_\bullet$ is an lc anti-canonical series on (X',B'), with N_1 in place of m_0 in Definition 7.1. For stationarity, note that $a(v,X,B+D)=a(v,X,B)-v(D)$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X, by Definition-Proposition 2.8. Computing on a log resolution of X that factors through μ, on which the crepant boundaries of (X',B') and $(X,B+tD_{\le N_1})$ coincide, we also have $a(v,X',B')=a(v,X,B+tD_{\le N_1})$. Since $v(\mu^*D_j)=v(D_j)$,

$$\sum_{j\ge1}v(D'_j)=\sum_{j\ge1}v(D_j)-t\sum_{j=1}^{N_1}v(D_j)=a(v,X,B)-tv(D_{\le N_1})=a(v,X,B+tD_{\le N_1})=a(v,X',B').$$

Thus v is stationary for $(X',B';D'_\bullet)$. This proves (1).

We prove (3) and (4). Let $(c_{j,N})_{1\le j\le m,N\ge N_0}$ be a family of rational numbers with $N_0\ge\max\{m,N_1\}$. For $N\ge N_0$, let

$$G'_N:=D'_{\le N}+\sum_{j=1}^{m}c_{j,N}D'_j\quad\text{and}\quad G_N:=D_{\le N}+\sum_{j=1}^{m}\kappa_jc_{j,N}D_j.$$

Since $N\ge N_1$, (2) and $K_{X'}+B'=\mu^*(K_X+B)+t\mu^*D_{\le N_1}$ imply that

$$\begin{aligned}K_{X'}+B'+G'_N=\mu^*(K_X+B+G_N),\quad \mu^*G_N=G'_N+t\mu^*D_{\le N_1},\\ \text{and}\quad G_N=\mu_*G'_N+tD_{\le N_1}.\end{aligned}\tag{10.6}$$

Since μ^* and μ_* preserve $\mathbb{Q}$-linear equivalence and $\mu_*\mu^*$ is the identity, $K_{X'}+B'+G'_N\sim_{\mathbb{Q}}0$ if and only if $K_X+B+G_N\sim_{\mathbb{Q}}0$. Since $\kappa_j>0$, $\lim_{N\to+\infty}c_{j,N}=0$ if and only if $\lim_{N\to+\infty}\kappa_jc_{j,N}=0$. This proves (3). Let $x_\bullet$ be a moment row for $D_\bullet$. For $N\ge N_1$,

$$\sum_{j=1}^{N}\kappa_jx_j=\sum_{j=1}^{N}x_j-t\sum_{j=1}^{N_1}x_j,$$

which converges to $x_\infty-t\sum_{j=1}^{N_1}x_j$. Since t is rational, $(\kappa_jx_j)_{j\ge1}$ is a moment row for $D'_\bullet$. For $N\ge N_0$,

$$\sum_{j=1}^{N}\kappa_jx_j+\sum_{j=1}^{m}c_{j,N}\kappa_jx_j-\Big(x_\infty-t\sum_{j=1}^{N_1}x_j\Big)=\sum_{j=1}^{N}x_j+\sum_{j=1}^{m}\kappa_jc_{j,N}x_j-x_\infty.$$

Thus $(c_{j,N})$ satisfies $(\kappa_jx_j)_{j\ge1}$ if and only if $(\kappa_jc_{j,N})$ satisfies $x_\bullet$. This proves (4).

Step 6. In this step we prove (5).

Let $x_\bullet^{(1)},\dots,x_\bullet^{(p)}$ be moment rows for $D_\bullet$. For $q=1,\dots,p$, let $y_\bullet^{(q)}:=(\kappa_jx_j^{(q)})_{j\ge1}$, which is a moment row for $D'_\bullet$ by (4). Let $W'\subset H^0(X',\mathcal{O}_{X'}(-e(K_{X'}+B')))$ be a corrected system for $D'_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$, spanned by products $s_1\cdots s_k$ as in Definition 7.3, where $e=e_1+\cdots+e_k$, $-e_i(K_X+B)$ is Cartier, $s_i\in H^0(X',\mathcal{O}_{X'}(-e_i(K_{X'}+B')))$, and $\operatorname{div}(s_i)=e_iG'^{(i)}$.

We regard a section $s\in H^0(X',\mathcal{O}_{X'}(-e(K_{X'}+B')))\setminus\{0\}$ as a rational function ϕ with $\operatorname{div}_{X'}(\phi)-e(K_{X'}+B')\ge0$, so that $\operatorname{div}(s)=\operatorname{div}_{X'}(\phi)-e(K_{X'}+B')$. Since $\mu_*\operatorname{div}_{X'}(\phi)=\operatorname{div}_X(\phi)$ and $\mu_*(K_{X'}+B')=K_X+B+tD_{\le N_1}$,

$$\operatorname{div}_X(\phi)-e(K_X+B)=\mu_*\operatorname{div}(s)+etD_{\le N_1}\ge0.$$

Thus $\phi\in H^0(X,\mathcal{O}_X(-e(K_X+B)))$, and $G_s=\mu_*G'_s+tD_{\le N_1}$. Moreover, $\operatorname{div}_{X'}(\phi)=\mu^*\operatorname{div}_X(\phi)$ and $K_{X'}+B'=\mu^*(K_X+B)+t\mu^*D_{\le N_1}$, so $\mu^*G_s=G'_s+t\mu^*D_{\le N_1}$. For each i, we regard s_i as a rational function ϕ_i with $\operatorname{div}_{X'}(\phi_i)-e_i(K_{X'}+B')=e_iG'^{(i)}$. Pushing forward by μ, we obtain

$$\operatorname{div}_X(\phi_i)-e_i(K_X+B)=e_i\mu_*G'^{(i)}+e_itD_{\le N_1}\ge0.$$

Thus $s_i \in H^0(X, \mathcal{O}_X(-e_i(K_X+B)))$ and $\operatorname{div}_X(s_i) = e_i G^{(i)}$, where

$$G^{(i)} := \mu_* G'^{(i)} + tD_{\le N_1}.$$

It remains to show that $G^{(i)}$ is the corrected boundary of a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$, at an index N such that $1 + c_{j,N} \ge 0$ for any j. Then the products $s_1 \cdots s_k$ satisfy the conditions of Definition 7.3 for $D_\bullet$, and W' is a corrected system for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$.

Fix i. By Definition 7.3, there exist a correction family $(c'_{j,N})_{1\le j\le m, N\ge N_0}$ for $D'_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$ and an index $N' \ge N_0$ such that

$$G'^{(i)} = D'_{\le N'} + \sum_{j=1}^{m} c'_{j,N'} D'_j \quad \text{and} \quad 1 + c'_{j,N'} \ge 0 \text{ for any } j.$$

Suppose that $N' \ge N_1$. Restricting $(c'_{j,N})$ to the indices $N \ge \max\{N_0, N_1\}$, we obtain a correction family for $D'_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$ whose integer N_0 in Definition 7.1 is $\max\{N_0, N_1\} \ge \max\{m, N_1\}$, and $N' \ge \max\{N_0, N_1\}$. By (3) and (4), $(\kappa_j c'_{j,N})$ is a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$, and by (10.6), the corrected boundary of $(\kappa_j c'_{j,N})$ at N' is $\mu_* G'^{(i)} + tD_{\le N_1} = G^{(i)}$. Since $0 < \kappa_j \le 1$, $1 + \kappa_j c'_{j,N'} \ge \min\{1, 1 + c'_{j,N'}\} \ge 0$ for any j.

Suppose that $N' < N_1$. We extend $G'^{(i)}$ to a correction family whose indices are at least N_1. By (1), (X', B') is a projective lc pair, B' is a $\mathbb{Q}$-divisor, and $D'_\bullet$ is an lc anti-canonical series on (X', B'). By Lemma 7.2, there exists a correction family $(c^0_{j,N})_{1\le j\le m^0, N\ge N_0^0}$ for $D'_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$. Let $N_2 := \max\{N_1, m, m^0, N_0^0\}$. Define a family $(c^*_{j,N})_{1\le j\le N_2, N\ge N_2}$ by

$$c^*_{j,N_2} := \begin{cases} c'_{j,N'} & \text{if } j \le m, \\ 0 & \text{if } m < j \le N', \\ -1 & \text{if } N' < j \le N_2, \end{cases} \qquad c^*_{j,N} := \begin{cases} c^0_{j,N} & \text{if } j \le m^0, \\ 0 & \text{if } m^0 < j \le N_2, \end{cases} \quad \text{for } N > N_2.$$

Here $m \le N_0 \le N'$. The corrected boundary of $(c^*_{j,N})$ at N_2 is

$$D'_{\le N_2} + \sum_{j=1}^{N_2} c^*_{j,N_2} D'_j = D'_{\le N'} + \sum_{j=1}^{m} c'_{j,N'} D'_j = G'^{(i)},$$

and for $N > N_2$ the corrected boundary of $(c^*_{j,N})$ at N is that of $(c^0_{j,N})$ at N. Thus $K_{X'} + B' + G'^{(i)} \sim_{\mathbb{Q}} 0$, and the equations of Definition 7.1 for $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$ hold for $(c^*_{j,N})$ at N_2 because they hold for $(c'_{j,N})$ at N', and at any $N > N_2$ because they hold for $(c^0_{j,N})$. Moreover, $\lim_{N\to+\infty} c^*_{j,N} = 0$ for any j. Hence $(c^*_{j,N})$ is a correction family for $D'_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$, with $m = N_0 = N_2 \ge N_1$ in Definition 7.1. By (3) and (4), $(\kappa_j c^*_{j,N})$ is a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$. By (10.6), the corrected boundary of $(\kappa_j c^*_{j,N})$ at N_2 is

$$\mu_* G'^{(i)} + tD_{\le N_1} = D_{\le N'} + t \sum_{N' < j \le N_1} D_j + \sum_{j=1}^{m} (1-t) c'_{j,N'} D_j = G^{(i)}.$$

The coefficients $1 + \kappa_j c^*_{j,N_2}$ are $1 + (1-t)c'_{j,N'} \ge \min\{1, 1 + c'_{j,N'}\} \ge 0$ for $j \le m$, 1 for $m < j \le N'$, t for $N' < j \le N_1$, and 0 for $N_1 < j \le N_2$. They are all non-negative.

In both cases, $G^{(i)}$ is the corrected boundary of a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$, at an index N such that $1 + c_{j,N} \ge 0$ for any j. Together with the identity $\mu^* G_s = G'_s + t\mu^* D_{\le N_1}$ for any $s \in W' \backslash \{0\}$, this proves (5). □

The center of v on X' may be strictly smaller than $\mu^{-1}(\operatorname{center}_X v)$, so leading divisors are compared on a common carrier of X and X'. In the proof of Theorem 12.1, the following lemma

transfers the sub-lc property of $(S, B'_S + \mathrm{lead}_S(h'^*G'))$, obtained on X', to $(S, B_S + \mathrm{lead}_S(h^*G))$ on X.

Lemma 10.2. *Notation as in Construction-Proposition 10.1. Let (h, S) be an adapted carrier of v for (X, B) and a finite set containing $D_1, \dots, D_{N_1}$, with $h : X'' \to X$, such that $h = \mu \circ h'$ for a morphism $h' : X'' \to X'$. Then (h', S) is an adapted carrier of v for (X', B') and $\{D'_1, \dots, D'_{N_1}\}$. Moreover, let B_S and B'_S be the $\mathbb{Q}$-divisors on S defined in Construction-Proposition 8.1 for (h, S) and (X, B), and for (h', S) and (X', B'), respectively. Then*

$$B'_S = B_S + t\sum_{j=1}^{N_1} \mathrm{lead}_S(h^*D_j) \quad \text{and} \quad \mathrm{lead}_S(h'^*D'_j) = \kappa_j \,\mathrm{lead}_S(h^*D_j)$$

for any $j \geq 1$. In particular, $B'_S + \mathrm{lead}_S(h'^*G') = B_S + \mathrm{lead}_S(h^*G)$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors $G' \geq 0$ on X' and $G \geq 0$ on X such that $\mu^*G = G' + t\mu^*D_{\leq N_1}$.

Proof. We write $h : X'' \to X$ for the carrier, $h = \mu \circ h'$. Let E, $E_1, \dots, E_r$, and α be the data of (h, S) in Construction-Proposition 8.1. Define $\mathbb{Q}$-divisors $B_{X''}$ and $B'_{X''}$ on X'' by

$$K_{X''} + B_{X''} = h^*(K_X + B) \quad \text{and} \quad K_{X''} + B'_{X''} = h'^*(K_{X'} + B').$$

Since $h'^*\mu^* = h^*$ and $K_{X'} + B' = \mu^*(K_X + B + tD_{\leq N_1})$,

$$B'_{X''} = B_{X''} + th^*D_{\leq N_1}. \tag{10.7}$$

We show that (h', S) is an adapted carrier. The variety X'' is smooth, and h' is projective and birational since h is. Since $h = \mu \circ h'$, $\mathrm{Exc}(h') \subset \mathrm{Exc}(h)$. Since X' is $\mathbb{Q}$-factorial, $\mathrm{Exc}(h')$ is of pure codimension one [KM98, Corollary 2.63]. By Construction-Proposition 8.1(1) for (h, S), E contains $\mathrm{Exc}(h)$, $\mathrm{Supp}\, B_{X''}$, and $\mathrm{Supp}\, h^*D_j$ for $j \leq N_1$. By (10.7) and $h'^*D'_j = \kappa_j h^*D_j$ with $\kappa_j > 0$, E contains $\mathrm{Exc}(h')$, $\mathrm{Supp}\, B'_{X''}$, and $\mathrm{Supp}\, h'^*D'_j$ for $j \leq N_1$. In particular, h' is a log resolution of (X', B'), and Construction-Proposition 8.1(1) holds for (h', S), (X', B'), and $\{D'_1, \dots, D'_{N_1}\}$. Construction-Proposition 8.1(2) only involves $E_1, \dots, E_r$, S, and α. Thus (h', S) is an adapted carrier of v for (X', B') and $\{D'_1, \dots, D'_{N_1}\}$.

By Construction-Proposition 8.1(4), the leading exponent $b(\cdot)$ and the leading divisor $\mathrm{lead}_S(\cdot)$ of a $\mathbb{Q}$-divisor on X'' depend only on $E_1, \dots, E_r$, S, and α, so $b(\cdot)$ and $\mathrm{lead}_S(\cdot)$ defined by (h', S) coincide with those defined by (h, S). By Construction-Proposition 8.1(7) for (h, S), for any $j \leq N_1$,

$$\mathrm{lead}_S(h^*D_j) = \Big(h^*D_j - \sum_{i=1}^{r} (\mathrm{mult}_{E_i} h^*D_j)E_i\Big)\Big|_S.$$

Since

$$1 - a(E_i, X, B) = \mathrm{mult}_{E_i} B_{X''} \quad \text{and} \quad 1 - a(E_i, X', B') = \mathrm{mult}_{E_i} B'_{X''},$$

Construction-Proposition 8.1(6) and (10.7) imply that

$$\begin{aligned} B'_S - B_S &= \Big(B'_{X''} - B_{X''} - \sum_{i=1}^{r} \mathrm{mult}_{E_i}(B'_{X''} - B_{X''})E_i\Big)\Big|_S \\ &= t\sum_{j=1}^{N_1} \Big(h^*D_j - \sum_{i=1}^{r} (\mathrm{mult}_{E_i} h^*D_j)E_i\Big)\Big|_S \\ &= t\sum_{j=1}^{N_1} \mathrm{lead}_S(h^*D_j). \end{aligned}$$

This is the first identity. Since $h'^*D'_j = \kappa_j h^*D_j$ and κ_j is a positive rational number, Lemma 8.2 implies $\mathrm{lead}_S(h'^*D'_j) = \kappa_j \,\mathrm{lead}_S(h^*D_j)$. This is the second identity.

Finally, let $G' \geq 0$ and $G \geq 0$ be $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X' and X such that $\mu^*G = G' + t\mu^*D_{\leq N_1}$. Applying h'^*, we obtain $h^*G = h'^*G' + t\sum_{j=1}^{N_1} h^*D_j$, where $h'^*G' \geq 0$ and

$h^*D_j \geq 0$ for any j. By Lemma 8.2, applied to $h'^*G', h^*D_1, \ldots, h^*D_{N_1}$ with coefficients $1, t, \ldots, t$, and the first identity,

$$B_S + \mathrm{lead}_S(h^*G) = B_S + t\sum_{j=1}^{N_1} \mathrm{lead}_S(h^*D_j) + \mathrm{lead}_S(h'^*G') = B'_S + \mathrm{lead}_S(h'^*G'). \qquad \square$$

11. The residue model

The goal of this section is to prove Construction-Proposition 11.4, which appears in Subsection 11.2. It constructs a projective lc pair of dimension $d-r$ together with an lc anti-canonical series on it, such that, for the correction families for $D_\bullet$ and the indices N in Construction-Proposition 11.4(3), the sub-pairs $(S, B_S + \mathrm{lead}_S(h^*G_N))$ are crepant birational to pairs formed with corrected boundaries of the series. In Section 12, the induction hypothesis on the dimension is applied to the lc pair of dimension $d-r$ constructed here.

11.1. **Normalization of the center in the residue field.** In this subsection, we show that when the center V of v has codimension r, for any correction family for $D_\bullet$ satisfying the moment row $(b_j)_{j\geq 1}$ and any sufficiently large N, the sub-pair $(S, B_S + \mathrm{lead}_S(h^*G_N))$ is crepant birational to a pair $(V_v, B_{V_v,N})$ on the normalization V_v of V in k_v, and that V_v is pseudo-lc.

Set-up 11.1. Notation and conditions as in Set-up 8.4. Assume moreover the following.

(1) $r < d$.
(2) V is of codimension r in X.

The $\mathbb{Q}$-divisor B_S may have negative coefficients, which come from the intersections of S with the exceptional divisors of h whose log discrepancies are greater than one. Thus $(S, B_S + \mathrm{lead}_S(h^*G_N))$ is in general not a pair, and we cannot use the induction on the dimension for it directly. The pushforward $B_{V_v,N}$ of $B_S + \mathrm{lead}_S(h^*G_N)$ to V_v satisfies $B_{V_v,N} \geq 0$: the different Δ_{F^ν} on the normalization F^ν of the prime divisor F of Construction-Proposition 9.3 satisfies $\Delta_{F^\nu} \geq 0$, and since V is of codimension r, the normalization of F maps onto V_v.

Definition-Proposition 11.2. Notation and conditions as in Set-up 11.1. Then k_v is a finite extension of $\mathbb{C}(V)$. Let V_v be the normalization of V in k_v. Then $h|_S$ induces a projective birational morphism $\tau : S \to V_v$. Moreover, for any correction family $(c_{j,N})$ for $D_\bullet$ satisfying the moment row $(b_j)_{j\geq 1}$ and any sufficiently large N, the $\mathbb{Q}$-divisor

$$B_{V_v,N} := \tau_*(B_S + \mathrm{lead}_S(h^*G_N))$$

on V_v satisfies $B_{V_v,N} \geq 0$, $K_{V_v} + B_{V_v,N} \sim_{\mathbb{Q}} 0$, and

$$K_S + B_S + \mathrm{lead}_S(h^*G_N) = \tau^*(K_{V_v} + B_{V_v,N}).$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that k_v is finite over $\mathbb{C}(V)$ and construct τ.

By Construction-Proposition 8.1(3), S is a smooth projective variety with $\dim S = d-r$, $h(S) = V$, and $k_v = \mathbb{C}(S)$. By Set-up 11.1(2), $\dim V = d-r = \dim S$. Since $h|_S : S \to V$ is surjective, $\mathbb{C}(S)$ is a finitely generated extension of $\mathbb{C}(V)$ of transcendence degree zero. Thus $k_v = \mathbb{C}(S)$ is a finite extension of $\mathbb{C}(V)$. Hence V_v is a normal projective variety with $\mathbb{C}(V_v) = k_v$, and the morphism $V_v \to V$ is finite [Har77, Theorem I.3.9A] (cf. [Har77, Exercise II.3.8] for the gluing, integral closure being a local construction). Since S is normal and $h|_S$ is dominant with $\mathbb{C}(V) \subset k_v = \mathbb{C}(S)$, $h|_S$ factors through a morphism $\tau : S \to V_v$ which induces the identity of k_v (cf. [Har77, Exercise II.3.8]). Since S is projective, τ is projective, and τ is birational because $\mathbb{C}(S) = k_v = \mathbb{C}(V_v)$.

Step 2. In this step we prove that $K_{V_v} + B_{V_v,N} \sim_{\mathbb{Q}} 0$ and $K_S + B_S + \mathrm{lead}_S(h^*G_N) = \tau^*(K_{V_v} + B_{V_v,N})$.

By Lemma 8.5(1), $(b_j)_{j\geq 1}$ is a moment row for $D_\bullet$. Let $(c_{j,N})_{1\leq j\leq m, N\geq N_0}$ be a correction family for $D_\bullet$ satisfying $(b_j)_{j\geq 1}$. By Lemma 7.2, $G_N \geq 0$ for any sufficiently large N. Let $N \geq N_0$ be an integer such that $G_N \geq 0$. Then G_N is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, and Lemma 8.6(2) implies that

$$a(v, X, B + G_N) = 0 \quad \text{and} \quad K_S + B_S + \operatorname{lead}_S(h^*G_N) \sim_{\mathbb{Q}} 0.$$

Let l be a positive integer and φ a rational function on S such that

$$l(K_S + B_S + \operatorname{lead}_S(h^*G_N)) = \operatorname{div}(\varphi).$$

Since τ is birational, φ is a rational function on V_v, $\tau_* K_S = K_{V_v}$, and $\tau_* \operatorname{div}(\varphi) = \operatorname{div}(\varphi)$. Thus $l(K_{V_v} + B_{V_v,N}) = \operatorname{div}(\varphi)$. In particular, $K_{V_v} + B_{V_v,N} \sim_{\mathbb{Q}} 0$, and

$$\tau^*(K_{V_v} + B_{V_v,N}) = \frac{1}{l}\tau^* \operatorname{div}(\varphi) = \frac{1}{l}\operatorname{div}(\varphi) = K_S + B_S + \operatorname{lead}_S(h^*G_N).$$

Step 3. In this step we construct a surjective morphism $\phi : F^\nu \to V_v$ from the normalization of the prime divisor F of Construction-Proposition 9.3, and relate ϕ to τ.

Let N be as in **Step 2**. Since $G_N \geq 0$ and $a(v, X, B + G_N) = 0$, we may apply Construction-Proposition 9.3 to $G := G_N$. We obtain a primitive vector $\mathbf{e} \in \mathbb{Z}^r_{>0}$, a projective birational morphism $\rho : X_{\mathbf{e}} \to X$, a prime divisor F on $X_{\mathbf{e}}$, and the different $\Delta_{F^\nu} \geq 0$ on the normalization F^ν of F. Let $\nu : F^\nu \to X_{\mathbf{e}}$ be the composition of the normalization of F with the inclusion $F \subset X_{\mathbf{e}}$. By Construction-Proposition 9.3(1), the vector $\mathbf{e}$ satisfies the condition of Construction-Proposition 9.2 for G_N. Let $g : X'_{\mathbf{e}} \to X'$, $E_{\mathbf{e}}$, $p : E_{\mathbf{e}} \to S$, $D_{\mathbf{e}}$, and $\Delta_{E_{\mathbf{e}}}$ be as in Construction-Proposition 9.2 for G_N and $\mathbf{e}$, so that

$$\Delta_{E_{\mathbf{e}}} = D_{\mathbf{e}} + p^*(B_S + \operatorname{lead}_S(h^*G_N)).$$

By Construction-Proposition 9.3(6), (F^ν, Δ_{F^ν}) and $(E_{\mathbf{e}}, \Delta_{E_{\mathbf{e}}})$ are crepant birational over X. Hence there exist a normal variety W and projective birational morphisms $f : W \to F^\nu$ and $f' : W \to E_{\mathbf{e}}$ such that

$$f^*(K_{F^\nu} + \Delta_{F^\nu}) = f'^*(K_{E_{\mathbf{e}}} + \Delta_{E_{\mathbf{e}}}) \quad \text{and} \quad \rho \circ \nu \circ f = h \circ g \circ f'.$$

Since $g|_{E_{\mathbf{e}}} = p$ as morphisms to X', we have $h \circ g \circ f' = h|_S \circ p \circ f'$.

By Construction-Proposition 9.3(4), $\rho \circ \nu : F^\nu \to V$ is surjective. The birational morphisms f and f' induce an isomorphism $\mathbb{C}(F^\nu) \cong \mathbb{C}(E_{\mathbf{e}})$, and p induces an embedding $k_v = \mathbb{C}(S) \subset \mathbb{C}(E_{\mathbf{e}})$. Thus $\mathbb{C}(V) \subset k_v \subset \mathbb{C}(F^\nu)$, and by the equality $\rho \circ \nu \circ f = h|_S \circ p \circ f'$, the composite embedding $\mathbb{C}(V) \subset \mathbb{C}(F^\nu)$ is the one induced by $\rho \circ \nu$. Since F^ν is normal, $\rho \circ \nu$ factors through a morphism $\phi : F^\nu \to V_v$ which induces the embedding $k_v \subset \mathbb{C}(F^\nu)$ above (cf. [Har77, Exercise II.3.8]). Since F^ν is projective and ϕ is dominant, ϕ is surjective.

The morphisms $\phi \circ f$ and $\tau \circ p \circ f'$ from W to V_v are dominant, their composites with $V_v \to V$ are $\rho \circ \nu \circ f$ and $h|_S \circ p \circ f'$, which are equal, and the embeddings $k_v \subset \mathbb{C}(W)$ induced by them coincide. Since $V_v \to V$ is finite, hence affine, a dominant morphism from the integral variety W to V_v over V is determined by the induced embedding of k_v into $\mathbb{C}(W)$. Thus

$$\phi \circ f = \tau \circ p \circ f'.$$

Step 4. In this step we prove that $B_{V_v,N} \geq 0$.

Let N be as in **Step 2**, and let P be a prime divisor on V_v with generic point η_P. Since V_v is normal, $\mathcal{O}_{V_v,\eta_P}$ is a discrete valuation ring, so the birational morphism τ from the projective variety S is an isomorphism over a neighborhood of η_P. Let $V^\circ \subset V_v$ be an open neighborhood of η_P such that τ is an isomorphism over V°, both V° and $P \cap V^\circ$ are smooth, and no component of $B_{V_v,N}$ other than P intersects V°. Let $U := \tau^{-1}(V^\circ)$, $P_S := \tau_*^{-1}P$, and let c be the coefficient of P_S in $B_S + \operatorname{lead}_S(h^*G_N)$. Since $\tau|_U : U \to V^\circ$ is an isomorphism, c is also the coefficient of P in $B_{V_v,N}$,

$$(B_S + \operatorname{lead}_S(h^*G_N))|_U = cP_S|_U, \quad \text{and} \quad \tau^*P|_{V^\circ} = P_S|_U.$$

We show that $c \geq 0$.

Let $L := (1-c)P_S$. Then $(B_S + \mathrm{lead}_S(h^*G_N) + L)|_U = P_S|_U$ is a smooth prime divisor on the smooth variety U, so $(U, (B_S + \mathrm{lead}_S(h^*G_N) + L)|_U)$ is lc by [KM98, Corollary 2.31(3)]. By Construction-Proposition 9.2, the sub-pair

$$\big(p^{-1}(U), (\Delta_{E_\mathbf{e}} + (1-c)p^*P_S)|_{p^{-1}(U)}\big)$$

is sub-lc. Since P is Cartier on V°, $\phi^*P \geq 0$ is a Cartier divisor on $\phi^{-1}(V^\circ)$. Let $W^\circ := (\phi \circ f)^{-1}(V^\circ)$. By **Step 3**, $W^\circ = f^{-1}(\phi^{-1}(V^\circ)) = f'^{-1}(p^{-1}(U))$, and on W°,

$$f^*\phi^*P = (\tau \circ p \circ f')^*P = f'^*p^*P_S.$$

Hence, on W°,

$$f^*\big(K_{F^\nu} + \Delta_{F^\nu} + (1-c)\phi^*P\big) = f'^*\big(K_{E_\mathbf{e}} + \Delta_{E_\mathbf{e}} + (1-c)p^*P_S\big).$$

Since log discrepancies do not change under crepant projective birational morphisms [KM98, Lemma 2.30], the sub-pair

$$\big(\phi^{-1}(V^\circ), (\Delta_{F^\nu} + (1-c)\phi^*P)|_{\phi^{-1}(V^\circ)}\big)$$

is sub-lc.

Since ϕ is surjective, $\phi^{-1}(P \cap V^\circ) \to P \cap V^\circ$ is surjective. The closed subset $\phi^{-1}(P \cap V^\circ)$ is the support of the non-zero Cartier divisor $\phi^*P|_{\phi^{-1}(V^\circ)} \geq 0$, so each irreducible component of $\phi^{-1}(P \cap V^\circ)$ is of codimension one in $\phi^{-1}(V^\circ)$ [Har77, Theorem I.1.11A], and one of them dominates $P \cap V^\circ$. Let P' be the closure in F^ν of such a component. Then P' is a prime divisor on F^ν dominating P, and $\mathrm{mult}_{P'}\,\phi^*P$ is a positive integer. Since the sub-pair above is sub-lc and $\Delta_{F^\nu} \geq 0$,

$$(1-c)\,\mathrm{mult}_{P'}\,\phi^*P \leq \mathrm{mult}_{P'}\,\Delta_{F^\nu} + (1-c)\,\mathrm{mult}_{P'}\,\phi^*P \leq 1.$$

If $c < 0$, then $(1-c)\,\mathrm{mult}_{P'}\,\phi^*P \geq 1-c > 1$, a contradiction. Thus $c \geq 0$. Since P is an arbitrary prime divisor on V_v, $B_{V_v,N} \geq 0$. □

The log discrepancies of the pairs $(V_v, B_{V_v,N})$ have non-negative limits as $N \to +\infty$, so V_v is pseudo-lc, and we can use the small lc modification of Theorem 2.18 once more.

Lemma 11.3. *Notation and conditions as in Definition-Proposition 11.2. Then V_v is pseudo-lc. In particular, there exists a small birational morphism $\sigma : T_0 \to V_v$ from a normal projective variety T_0 such that K_{T_0} is $\mathbb{Q}$-Cartier and σ-ample and $(T_0, 0)$ is lc.*

Proof. Since $(b_j)_{j\geq 1}$ is a moment row for $D_\bullet$ (Lemma 8.5(1)), Lemma 7.2 implies that there exists a correction family $(c_{j,N})_{1\leq j\leq m, N\geq N_0}$ for $D_\bullet$ satisfying $(b_j)_{j\geq 1}$. By Lemma 7.2 and Definition-Proposition 11.2, for any sufficiently large N, we have $1 + c_{j,N} > 0$ for any $1 \leq j \leq m$, $G_N \geq 0$, $B_{V_v,N} \geq 0$, $K_{V_v} + B_{V_v,N} \sim_\mathbb{Q} 0$, and

$$K_S + B_S + \mathrm{lead}_S(h^*G_N) = \tau^*(K_{V_v} + B_{V_v,N}).$$

Let u be a divisorial valuation of $\mathbb{C}(V_v) = \mathbb{C}(S)$ given by a prime divisor over V_v, and write $\alpha(u, V_v, 0)$ and $a(u, V_v, B_{V_v,N})$ for the corresponding invariants of that prime divisor. We show that $\alpha(u, V_v, 0) \geq 0$.

Let N be sufficiently large. By [KM98, Lemma 2.30],

$$a(u, V_v, B_{V_v,N}) = a(u, S, B_S + \mathrm{lead}_S(h^*G_N)).$$

Since $N \geq N_0 \geq m$, we have $G_N = \sum_{j=1}^m (1 + c_{j,N})D_j + \sum_{j=m+1}^N D_j \geq 0$, so Lemma 8.2 implies

$$\mathrm{lead}_S(h^*G_N) = \sum_{j=1}^N \mathrm{lead}_S(h^*D_j) + \sum_{j=1}^m c_{j,N}\,\mathrm{lead}_S(h^*D_j).$$

Since S is smooth, $a(u,S,B_S+R)=a(u,S,B_S)-u(R)$ for any $\mathbb{Q}$-divisor R on S by Definition-Proposition 2.8. Hence

$$a(u,V_v,B_{V_v,N})=a(u,S,B_S)-\sum_{j=1}^{N}u(\mathrm{lead}_S(h^*D_j))-\sum_{j=1}^{m}c_{j,N}u(\mathrm{lead}_S(h^*D_j)).$$

Since $\mathrm{lead}_S(h^*D_j)\geq 0$, we have $u(\mathrm{lead}_S(h^*D_j))\geq 0$ for any j, and Lemma 8.6(1) implies that the series $\sum_{j\geq 1}u(\mathrm{lead}_S(h^*D_j))$ converges to a real number at most $a(u,S,B_S)$. Since $\lim_{N\to+\infty}c_{j,N}=0$ for any $1\leq j\leq m$,

$$\lim_{N\to+\infty}a(u,V_v,B_{V_v,N})=a(u,S,B_S)-\sum_{j\geq 1}u(\mathrm{lead}_S(h^*D_j))\geq 0.$$

Let N be sufficiently large, and let $U\subset V_v$ be an affine open subset such that $U\cap\mathrm{center}_{V_v}u\neq\emptyset$. Then $B_{V_v,N}|_U\geq 0$ and $K_U+B_{V_v,N}|_U$ is $\mathbb{Q}$-Cartier. By Definition 2.17,

$$\alpha(u,V_v,0)\geq a(u,U,B_{V_v,N}|_U)=a(u,V_v,B_{V_v,N}).$$

Letting $N\to+\infty$, we obtain $\alpha(u,V_v,0)\geq 0$. Since u is arbitrary, V_v is pseudo-lc.

Since V_v is projective, Theorem 2.18 implies that there exists a small birational morphism $\sigma:T_0\to V_v$ from a normal variety T_0 such that K_{T_0} is $\mathbb{R}$-Cartier and σ-ample and $(T_0,0)$ is lc. Since σ is projective by Theorem 2.18, T_0 is projective. By Lemma 2.5, K_{T_0} is $\mathbb{Q}$-Cartier. □

11.2. **Construction of the residue model.** In this subsection, we construct the residue model in the setting of Definition-Proposition 11.2. The proof of Construction-Proposition 11.4 first adds finitely many moment rows, which remove the prime divisors at which every partial sum has a non-positive coefficient. Afterwards the first m terms are rescaled by factors in $(0,1]$, and Lemma 5.1 provides a model on which the increments are $\mathbb{Q}$-Cartier. Construction-Proposition 11.4(4) is used in the proof of Theorem 12.1. There, an induction provides corrected systems on residue models whose general members are lc, and (4) lifts them to corrected systems whose general members G satisfy the hypotheses on G in Proposition 9.1.

Construction-Proposition 11.4. Notation and conditions as in Definition-Proposition 11.2. Then there exist a normal projective variety T of dimension $d-r$ with a small birational morphism $T\to V_v$, a $\mathbb{Q}$-divisor $B_T\geq 0$ on T such that K_T and B_T are $\mathbb{Q}$-Cartier, an lc anti-canonical series $D_{T,\bullet}$ on (T,B_T), integers $I\geq m\geq 1$, rational numbers $\lambda_1,\dots,\lambda_m$ with $0<\lambda_i\leq 1$, and finitely many moment rows $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$ for $D_\bullet$, one of which is $(b_j)_{j\geq 1}$, satisfying the following. For a moment row $x_\bullet$ for $D_\bullet$, let $\tilde{x}_\bullet$ be the moment row for $D_{T,\bullet}$ defined by $\tilde{x}_i:=\lambda_i x_i$ for $i\leq m$ and $\tilde{x}_{m+k}:=x_{I+k}$ for $k\geq 1$, whose sum is $x_\infty-\sum_{j=1}^{I}x_j+\sum_{j=1}^{m}\lambda_j x_j$. For a family of rational numbers $(c_{j,N})_{1\leq j\leq m,N\geq N_0}$ with $N_0\geq I$, let $\tilde{c}_{i,n}:=\lambda_i^{-1}c_{i,n+I-m}$ for $1\leq i\leq m$ and $n\geq N_0-I+m$.

(1) $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$ if and only if $(\tilde{c}_{i,n})$ is a correction family for $D_{T,\bullet}$ satisfying $\tilde{y}_\bullet^{(1)},\dots,\tilde{y}_\bullet^{(p)}$.

(2) If $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$, then for any moment row $x_\bullet$ for $D_\bullet$, $(c_{j,N})$ satisfies $x_\bullet$ if and only if $(\tilde{c}_{i,n})$ satisfies $\tilde{x}_\bullet$.

(3) If $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$, then for any $N\geq N_0$ such that $G_N\geq 0$, the sub-pair $(S,B_S+\mathrm{lead}_S(h^*G_N))$ is crepant birational to $(T,B_T+G_{T,N-I+m})$, where $G_{T,n}$ are the corrected boundaries of $(\tilde{c}_{i,n})$.

(4) For any positive integer e_0, any moment rows $x_\bullet^{(1)},\dots,x_\bullet^{(q)}$ for $D_\bullet$, and any corrected system W_T for $D_{T,\bullet}$ satisfying $\tilde{y}_\bullet^{(1)},\dots,\tilde{y}_\bullet^{(p)},\tilde{x}_\bullet^{(1)},\dots,\tilde{x}_\bullet^{(q)}$ such that (T,B_T+G_T) is lc for a general member G_T of W_T, there exists a corrected system W for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)},x_\bullet^{(1)},\dots,x_\bullet^{(q)}$ such that e_0 divides each of the positive integers e_i in Definition 7.3 defining W, and such that $a(v,X,B+G)=0$ and $(S,B_S+\mathrm{lead}_S(h^*G))$ is sub-lc for a general member G of W.

We call $(T, B_T; D_{T,\bullet})$ a *residue model* of $(X, B; D_\bullet)$ at v.

Proof. The proof is long, so we divide it into several steps. The integer m of the statement is fixed only in **Step 3**, so for an arbitrary correction family we write M for its integer m of Definition 7.1: $(c_{j,N})_{1\le j\le M, N\ge N_0}$ with $N_0 \ge M \ge 1$ and corrected boundaries $G_N = D_{\le N} + \sum_{j=1}^{M} c_{j,N} D_j$.

Step 1. In this step we define the first two moment rows, fix a common resolution of S and of the variety T_0 of Lemma 11.3, and prove (11.1)–(11.5).

Let $\mathcal{V} \subset \mathrm{Cl}(X)_{\mathbb{Q}}$ be the subspace spanned by $[D_j]$ for $j \ge 1$, and let m_0 and $a_{j,N}$ be the integer and the rational numbers in Definition 7.1(2) for $D_\bullet$. For $N > m_0$, the difference of the relations of Definition 7.1(2) at N and at $N-1$ is

$$D_N + \sum_{j=1}^{m_0} (a_{j,N} - a_{j,N-1}) D_j \sim_{\mathbb{Q}} 0.$$

Thus $\mathcal{V}$ is spanned by $[D_1], \dots, [D_{m_0}]$, and $\dim \mathcal{V} \le m_0$. The relation at $N = m_0$ shows $[K_X + B] \in \mathcal{V}$. Fix an injective $\mathbb{Q}$-linear map $\varphi : \mathcal{V} \to \mathbb{Q}^{m_0}$ and let $y_j^{(1)} := \varphi([D_j])$ for $j \ge 1$. By Definition 7.1(2),

$$\sum_{j=1}^{N} y_j^{(1)} = -\varphi([K_X + B]) - \sum_{j=1}^{m_0} a_{j,N} \varphi([D_j]) \to -\varphi([K_X + B]) \quad (N \to +\infty),$$

so $y_\bullet^{(1)}$ is a moment row for $D_\bullet$ with $y_\infty^{(1)} = -\varphi([K_X + B])$. Since φ is injective, a family of rational numbers $(c_{j,N})_{1\le j\le M, N\ge N_0}$ satisfies $y_\bullet^{(1)}$ if and only if $K_X + B + G_N \sim_{\mathbb{Q}} 0$ for any $N \ge N_0$. In particular, any correction family for $D_\bullet$ satisfies $y_\bullet^{(1)}$. Let $y_\bullet^{(2)} := (b_j)_{j\ge 1}$. By Lemma 8.5(1), this is a moment row for $D_\bullet$.

By Lemma 11.3, there exists a small birational morphism $\sigma : T_0 \to V_v$ from a normal projective variety T_0 such that K_{T_0} is $\mathbb{Q}$-Cartier and $(T_0, 0)$ is lc. Let S' be a smooth projective variety with birational morphisms $f : S' \to S$ and $g : S' \to T_0$ such that $\tau \circ f = \sigma \circ g$. Through f, g, τ, and σ, we identify the function fields of S, S', T_0, and V_v with $k_v = \mathbb{C}(S)$. Let

$$K_{S'} + B_{S'} := f^*(K_S + B_S), \qquad K_{S'} + \Gamma_{S'} := g^* K_{T_0}, \qquad F := B_{S'} - \Gamma_{S'}.$$

For $j \ge 1$, let $J_j := \mathrm{lead}_S(h^* D_j) \ge 0$ and $J_{S',j} := f^* J_j$, and for a prime divisor P on S', let $t_{P,j} := \mathrm{mult}_P J_{S',j} \in \mathbb{Q}_{\ge 0}$.

Let $(c_{j,N})_{1\le j\le M, N\ge N_0}$ be a correction family for $D_\bullet$ and $N \ge N_0$ with $G_N \ge 0$. By Lemma 8.2,

$$\mathrm{lead}_S(h^* G_N) = \sum_{j=1}^{N} J_j + \sum_{j=1}^{M} c_{j,N} J_j. \tag{11.1}$$

For a correction family $(c_{j,N})_{1\le j\le M, N\ge N_0}$ for $D_\bullet$ and $N \ge N_0$, let

$$B_{S,N} := B_S + \sum_{j=1}^{N} J_j + \sum_{j=1}^{M} c_{j,N} J_j, \qquad \Phi_N := F + \sum_{j=1}^{N} J_{S',j} + \sum_{j=1}^{M} c_{j,N} J_{S',j}.$$

By the definitions of $B_{S'}$, $\Gamma_{S'}$, and F,

$$f^*(K_S + B_{S,N}) = K_{S'} + \Gamma_{S'} + \Phi_N. \tag{11.2}$$

Next we show that if $(c_{j,N})$ satisfies $y_\bullet^{(2)}$, then

$$K_S + B_{S,N} \sim_{\mathbb{Q}} 0 \quad \text{for any } N \ge N_0. \tag{11.3}$$

For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D' on X', write $[D'|_S] \in \mathrm{Cl}(S)_{\mathbb{Q}}$ for the class of the restriction of D' to S. By Construction-Proposition 8.1(5), $[J_j] = [(h^* D_j)|_S] - \sum_{i=1}^{r} b_{j,i} [E_i|_S]$. Let $N, N' \ge N_0$.

Then

$$[B_{S,N}]-[B_{S,N'}] = [h^*(G_N-G_{N'})|_S]-\sum_{i=1}^{r}\Big(\sum_{j=1}^{N} b_{j,i}+\sum_{j=1}^{M} c_{j,N}b_{j,i}-\sum_{j=1}^{N'} b_{j,i}-\sum_{j=1}^{M} c_{j,N'}b_{j,i}\Big)[E_i|_S] = 0,$$

since $G_N - G_{N'} \sim_{\mathbb{Q}} 0$ and since $(c_{j,N})$ satisfies $y_\bullet^{(2)} = (b_j)_{j\geq 1}$. By Lemma 7.2, there exists $N' \geq N_0$ with $G_{N'} \geq 0$. Then $B_{S,N'} = B_S + \mathrm{lead}_S(h^*G_{N'})$ by (11.1), and $K_S + B_{S,N'} \sim_{\mathbb{Q}} 0$ by Lemma 8.6(2). Hence (11.3) holds.

Assume moreover that $(c_{j,N})$ satisfies $y_\bullet^{(2)}$ and that N is sufficiently large. By Lemma 7.2, $1+c_{j,N} > 0$ for any j and $G_N \geq 0$, and by Definition-Proposition 11.2, $B_{V_v,N} \geq 0$, $K_{V_v}+B_{V_v,N} \sim_{\mathbb{Q}} 0$, and $\tau^*(K_{V_v} + B_{V_v,N}) = K_S + B_S + \mathrm{lead}_S(h^*G_N) = K_S + B_{S,N}$, where the last equality is (11.1). Let $B_{T_0,N} := \sigma_*^{-1}B_{V_v,N} \geq 0$. Since σ is small, $\sigma^*(K_{V_v} + B_{V_v,N}) - (K_{T_0} + B_{T_0,N})$ is a σ-exceptional divisor, hence zero, and $B_{T_0,N}$ is $\mathbb{Q}$-Cartier. Pulling back to S' and using (11.2), we obtain

$$g^*B_{T_0,N} = \Phi_N \geq 0 \quad \text{for any sufficiently large } N. \tag{11.4}$$

Finally, let u be a divisorial valuation of k_v. By [KM98, Lemma 2.30],

$$a(u,T_0,0) = a(u,S',\Gamma_{S'}) = a(u,S',B_{S'}) + u(F) = a(u,S,B_S) + u(F).$$

By Lemma 8.6(1), $\sum_{j\geq 1} u(J_j) \leq a(u,S,B_S)$, and $u(J_{S',j}) = u(J_j)$. Hence the series below converges and

$$u(F) + \sum_{j\geq 1} u(J_{S',j}) \leq a(u,T_0,0). \tag{11.5}$$

Step 2. In this step we define the remaining moment rows, which remove the prime divisors P on S' with $\mathrm{mult}_P F < 0$ such that $\mathrm{mult}_P(F + \sum_{j\leq I} J_{S',j}) \leq 0$ for every $I \geq 1$.

By Lemma 7.2, there exists a correction family $(c_{j,N})_{1\leq j\leq M, N\geq N_0}$ for $D_\bullet$ satisfying $y_\bullet^{(2)}$. Let P be a prime divisor on S'. By (11.5) for $u = \mathrm{ord}_P$, the series $\sum_{j\geq 1} t_{P,j}$ converges. Let

$$L_P := \mathrm{mult}_P F + \sum_{j\geq 1} t_{P,j}.$$

For N sufficiently large, (11.4) implies

$$0 \leq \mathrm{mult}_P \Phi_N = \mathrm{mult}_P F + \sum_{j=1}^{N} t_{P,j} + \sum_{j=1}^{M} c_{j,N}t_{P,j}.$$

Since $c_{j,N} \to 0$ as $N \to +\infty$, the right-hand side converges to L_P, and so $L_P \geq 0$.

Let $\mathcal{P}$ be the set of prime divisors P on S' such that $\mathrm{mult}_P F < 0$ and $L_P = 0$. Then $\mathcal{P}$ is finite, as $\mathcal{P} \subset \mathrm{Supp}\, F$. For $P \in \mathcal{P}$, the numbers $t_{P,j}$ are rational and

$$\sum_{j\geq 1} t_{P,j} = -\mathrm{mult}_P F \in \mathbb{Q}, \tag{11.6}$$

so $(t_{P,j})_{j\geq 1}$ is a moment row for $D_\bullet$ with values in $\mathbb{Q}$. Let $y_\bullet^{(3)}, \ldots, y_\bullet^{(p)}$ be the moment rows $(t_{P,j})_{j\geq 1}$ for $P \in \mathcal{P}$, where $p = 2 + \#\mathcal{P}$. For $P \in \mathcal{P}$, (11.6) implies

$$\mathrm{mult}_P\Big(F + \sum_{j=1}^{I} J_{S',j}\Big) = -\sum_{j>I} t_{P,j} \leq 0 \quad \text{for any } I \geq 1,$$

so the uniform scaling in **Step 3** cannot control the coefficient at P.

Let

$$\overline{J}_j := J_{S',j} - \sum_{P\in\mathcal{P}} t_{P,j}P \geq 0, \qquad \overline{F} := F - \sum_{P\in\mathcal{P}} (\mathrm{mult}_P F)P.$$

No prime divisor in $\mathcal{P}$ is a component of $\overline{J}_j$ or of $\overline{F}$. Let $(c_{j,N})_{1\le j\le M, N\ge N_0}$ be a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$. For $P\in\mathcal{P}$, the row for P reads $\sum_{j=1}^N t_{P,j}+\sum_{j=1}^M c_{j,N}t_{P,j}=-\operatorname{mult}_P F$, that is, $\operatorname{mult}_P\Phi_N=0$. Hence

$$\Phi_N=\overline{F}+\sum_{j=1}^N\overline{J}_j+\sum_{j=1}^M c_{j,N}\overline{J}_j\quad\text{for any } N\ge N_0. \tag{11.7}$$

Moreover, for a divisorial valuation u of k_v, the series $\sum_j u(\overline{J}_j)$ and $\sum_j t_{P,j}u(P)$ have non-negative terms and converge, so (11.6) and (11.5) imply

$$u(\overline{F})+\sum_{j\ge1}u(\overline{J}_j)=u(F)+\sum_{j\ge1}u(J_{S',j})\le a(u,T_0,0). \tag{11.8}$$

Step 3. In this step we fix the integers m and I, and we prove that the boundaries $B_{T_0,N}$, scaled by factors tending to 1, are lc.

By Lemma 7.2, there exists a correction family $(c^0_{j,N})_{1\le j\le m,N\ge N_0^0}$ for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)}$. We fix it. The integer m of the statement is the integer m of Definition 7.1 for $(c^0_{j,N})$. We write Φ^0_N and $B^0_{T_0,N}$ for Φ_N and $B_{T_0,N}$ defined by $(c^0_{j,N})$.

Let P be a prime divisor on S' such that $P\notin\mathcal{P}$ and $\operatorname{mult}_P F<0$. Then $L_P>0$, since $L_P\ge0$ and $P\notin\mathcal{P}$. As $\operatorname{mult}_P(F+\sum_{j\le I}J_{S',j})\to L_P$ for $I\to+\infty$ and F has finitely many components, there exists an integer $I\ge m$ such that $\operatorname{mult}_P(F+\sum_{j\le I}J_{S',j})>0$ for any such P. Let

$$\overline{F}_{\le I}:=\overline{F}+\sum_{j=1}^I\overline{J}_j.$$

We have $\operatorname{mult}_P\overline{F}_{\le I}=0$ for $P\in\mathcal{P}$. For $P\notin\mathcal{P}$, we have $\operatorname{mult}_P\overline{F}_{\le I}=\operatorname{mult}_P F+\sum_{j\le I}t_{P,j}$. If $\operatorname{mult}_P F<0$, then $\operatorname{mult}_P\overline{F}_{\le I}>0$ by the choice of I. If $\operatorname{mult}_P F\ge0$, then $\operatorname{mult}_P\overline{F}_{\le I}\ge0$, and $\operatorname{mult}_P\overline{F}_{\le I}\ge t_{P,j}>0$ when $P\subset\operatorname{Supp}\overline{J}_j$ for some $j\le m\le I$. Hence $\overline{F}_{\le I}\ge0$, and $\operatorname{mult}_P\overline{F}_{\le I}>0$ for any prime divisor P in $\operatorname{Supp}\overline{J}_j$ with $j\le m$.

Choose rational numbers $\mu_j\ge0$ for $1\le j\le m$ such that $\overline{J}_j\le\mu_j\overline{F}_{\le I}$. For $N\ge N_0^0$, let $\epsilon_N:=\sum_{j=1}^m|c^0_{j,N}|\mu_j$, a rational number with $\epsilon_N\to0$. For $N\ge\max\{I,N_0^0\}$, (11.7) implies

$$\Phi^0_N=\overline{F}_{\le I}+\sum_{j=I+1}^N\overline{J}_j+\sum_{j=1}^m c^0_{j,N}\overline{J}_j\le(1+\epsilon_N)\Big(\overline{F}_{\le I}+\sum_{j=I+1}^N\overline{J}_j\Big).$$

By (11.8), $u(\overline{F}_{\le I})+\sum_{j>I}u(\overline{J}_j)\le a(u,T_0,0)$. Hence, for any divisorial valuation u of k_v,

$$u(\Phi^0_N)\le(1+\epsilon_N)a(u,T_0,0)\quad\text{for any } N\ge\max\{I,N_0^0\}. \tag{11.9}$$

For N sufficiently large, (11.4) and (11.9) imply

$$a\big(u,T_0,(1+\epsilon_N)^{-1}B^0_{T_0,N}\big)=a(u,T_0,0)-(1+\epsilon_N)^{-1}u(\Phi^0_N)\ge0.$$

Hence

$$(T_0,(1+\epsilon_N)^{-1}B^0_{T_0,N})\text{ is lc for any sufficiently large } N. \tag{11.10}$$

Step 4. In this step we construct T by Lemma 5.1.

For $j\ge1$, let $J_{T_0,j}:=g_*\overline{J}_j\ge0$, and let $\overline{F}_{T_0}:=g_*\overline{F}$. We show that no lc center of $(T_0,0)$ is contained in $\operatorname{Supp}J_{T_0,j}$. Let Q be a prime divisor in $\operatorname{Supp}J_{T_0,j}$ and P the strict transform of Q on S'. Then $P\subset\operatorname{Supp}\overline{J}_j$, so $P\notin\mathcal{P}$ and $t_{P,j}>0$. If $\operatorname{mult}_P F\ge0$, then $L_P\ge t_{P,j}>0$, and otherwise $L_P>0$ since $P\notin\mathcal{P}$. By (11.4), $\operatorname{mult}_Q B^0_{T_0,N}=\operatorname{mult}_P\Phi^0_N\to L_P>0$ as $N\to+\infty$. Thus $\operatorname{Supp}J_{T_0,j}\subset\operatorname{Supp}B^0_{T_0,N}$ for any sufficiently large N. Suppose that an lc center Z of $(T_0,0)$ is contained in $\operatorname{Supp}J_{T_0,j}$, and let u be a divisorial valuation with $a(u,T_0,0)=0$ and

$\operatorname{center}_{T_0} u = Z$. As $B^0_{T_0,N} \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor whose support contains Z, we have $u(B^0_{T_0,N}) > 0$, so $a(u, T_0, (1+\epsilon_N)^{-1}B^0_{T_0,N}) < 0$ for any sufficiently large N, contradicting (11.10).

We show that the classes $[J_{T_0,j}]$, $j \geq 1$, span a subspace of $\mathrm{Cl}(T_0)_{\mathbb{Q}}$ of finite dimension. The relations of Definition 7.1(2) are $\mathbb{Q}$-linear equivalences of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors, so the difference used in **Step 1** shows that each D_N is $\mathbb{Q}$-linearly equivalent to a rational combination of $D_1, \dots, D_{m_0}$. Pulling back by h and restricting to S, the classes $[(h^*D_j)|_S]$ are contained in the span of $[(h^*D_1)|_S], \dots, [(h^*D_{m_0})|_S]$. By $[J_j] = [(h^*D_j)|_S] - \sum_i b_{j,i}[E_i|_S]$, the classes $[J_j]$ are contained in a subspace of $\mathrm{Cl}(S)_{\mathbb{Q}}$ of dimension at most $m_0 + r$. Pulling back by f, subtracting the finitely many prime divisors in $\mathcal{P}$, and pushing forward by g, which preserves linear equivalence, the classes $[J_{T_0,j}]$ are contained in a subspace of dimension at most $m_0 + r + \#\mathcal{P}$.

Choose a finite set $\mathcal{J}$ of positive integers such that $[J_{T_0,j}]$, $j \in \mathcal{J}$, form a basis of the span of all $[J_{T_0,j}]$. By the last two paragraphs, Lemma 5.1 applied to T_0 and to $J_{T_0,j}$, $j \in \mathcal{J}$, implies that there exists a small birational morphism $\sigma_1 : T \to T_0$ from a normal projective variety T such that $K_T = \sigma_1^* K_{T_0}$, $(T, 0)$ is lc, and $\sigma_{1*}^{-1} J_{T_0,j}$ is $\mathbb{Q}$-Cartier for any $j \in \mathcal{J}$. For $j \geq 1$, let $J_{T,j} := \sigma_{1*}^{-1} J_{T_0,j} \geq 0$. Since σ_1 is small, strict transform induces an isomorphism $\mathrm{Cl}(T_0)_{\mathbb{Q}} \cong \mathrm{Cl}(T)_{\mathbb{Q}}$. Thus $[J_{T,j}]$ is a rational combination of the classes $[J_{T,j'}]$, $j' \in \mathcal{J}$, and $J_{T,j}$ is $\mathbb{Q}$-Cartier for any $j \geq 1$.

Let $\overline{F}_T := \sigma_{1*}^{-1} \overline{F}_{T_0}$. For N sufficiently large, let $B^0_{T,N} := \sigma_1^* B^0_{T_0,N}$. Since σ_1 is small, $B^0_{T,N}$ is the strict transform of $B^0_{T_0,N} = g_* g^* B^0_{T_0,N} = g_* \Phi^0_N$, so (11.4) and (11.7) imply

$$B^0_{T,N} = \overline{F}_T + \sum_{j=1}^{N} J_{T,j} + \sum_{j=1}^{m} c^0_{j,N} J_{T,j}.$$

Hence $\overline{F}_T$ is $\mathbb{Q}$-Cartier. Let

$$\overline{F}_{T,\leq I} := \overline{F}_T + \sum_{j=1}^{I} J_{T,j}.$$

Then $\overline{F}_{T,\leq I}$ is $\mathbb{Q}$-Cartier, and $\overline{F}_{T,\leq I} \geq 0$ since $\overline{F}_{T,\leq I}$ is the strict transform of $g_* \overline{F}_{\leq I} \geq 0$. Pushing forward and taking strict transforms preserve coefficientwise inequalities, so $J_{T,j} \leq \mu_j \overline{F}_{T,\leq I}$ for $1 \leq j \leq m$.

Since σ and σ_1 are small, $\sigma \circ \sigma_1$ contracts no divisor, and $\sigma \circ \sigma_1 : T \to V_v$ is a small birational morphism. Moreover, $\dim T = \dim V_v = \dim V = d - r$, since V_v is finite over V and V is of codimension r in X.

Step 5. In this step we prove that corrected boundaries for $D_\bullet$ satisfying $y^{(1)}_\bullet, \dots, y^{(p)}_\bullet$ correspond to crepant birational sub-pairs on S and T.

Let S'' be a smooth projective variety with birational morphisms $f_1 : S'' \to S$ and $g_1 : S'' \to T$ such that $\tau \circ f_1 = \sigma \circ \sigma_1 \circ g_1$. Let $(c_{j,N})_{1 \leq j \leq M, N \geq N_0}$ be a correction family for $D_\bullet$ satisfying $y^{(1)}_\bullet, \dots, y^{(p)}_\bullet$. Recall that $B_{S,N} = B_S + \sum_{j \leq N} J_j + \sum_{j \leq M} c_{j,N} J_j$ for $N \geq N_0$, and let

$$B_{T,N} := \overline{F}_T + \sum_{j=1}^{N} J_{T,j} + \sum_{j=1}^{M} c_{j,N} J_{T,j}.$$

For $(c^0_{j,N})$ and N sufficiently large, $B_{T,N}$ is the $\mathbb{Q}$-divisor $B^0_{T,N}$ of **Step 4**. We show that for any $N \geq N_0$,

$$K_T + B_{T,N} \sim_{\mathbb{Q}} 0 \quad \text{and} \quad f_1^*(K_S + B_{S,N}) = g_1^*(K_T + B_{T,N}). \tag{11.11}$$

By (11.3), there exist a positive integer ℓ and $\psi \in k_v \backslash \{0\}$ such that $\ell(K_S + B_{S,N}) = \mathrm{div}_S(\psi)$. By (11.2), $\ell(K_{S'} + \Gamma_{S'} + \Phi_N) = \mathrm{div}_{S'}(\psi)$. We push forward by g. We have $g_*(K_{S'} + \Gamma_{S'}) =$

$g_*g^*K_{T_0} = K_{T_0}$ and $g_*\operatorname{div}_{S'}(\psi) = \operatorname{div}_{T_0}(\psi)$, and (11.7) implies

$$g_*\Phi_N = \overline{F}_{T_0} + \sum_{j=1}^{N} J_{T_0,j} + \sum_{j=1}^{M} c_{j,N} J_{T_0,j}.$$

Thus

$$\ell\Big(K_{T_0} + \overline{F}_{T_0} + \sum_{j=1}^{N} J_{T_0,j} + \sum_{j=1}^{M} c_{j,N} J_{T_0,j}\Big) = \operatorname{div}_{T_0}(\psi).$$

Since σ_1 is small, the strict transform of $\operatorname{div}_{T_0}(\psi)$ is $\operatorname{div}_T(\psi)$, and the strict transform of K_{T_0} is $\sigma_1^*K_{T_0} = K_T$. Hence $\ell(K_T + B_{T,N}) = \operatorname{div}_T(\psi)$. In particular $K_T + B_{T,N} \sim_{\mathbb{Q}} 0$, and

$$f_1^*(K_S + B_{S,N}) = \ell^{-1}\operatorname{div}_{S''}(\psi) = g_1^*(K_T + B_{T,N}).$$

Step 6. In this step we define λ_i, B_T, and $D_{T,\bullet}$, we compare families on T with families on X, and we prove that $D_{T,\bullet}$ is an lc anti-canonical series on (T, B_T).

We first show that

$$\Big(T, \overline{F}_T + \sum_{j=1}^{I'} J_{T,j}\Big) \text{ is lc for any } I' \geq I. \tag{11.12}$$

The $\mathbb{Q}$-divisor $\overline{F}_T + \sum_{j\leq I'} J_{T,j} = \overline{F}_{T,\leq I} + \sum_{I<j\leq I'} J_{T,j} \geq 0$ is $\mathbb{Q}$-Cartier. Let u be a divisorial valuation of k_v and $N \geq I'$ sufficiently large. Since $K_T + B^0_{T,N} = \sigma_1^*(K_{T_0} + B^0_{T_0,N})$, [KM98, Lemma 2.30], (11.4), and (11.9) imply

$$a(u, T, B^0_{T,N}) = a(u, T_0, B^0_{T_0,N}) = a(u, T_0, 0) - u(\Phi^0_N) \geq -\epsilon_N a(u, T_0, 0).$$

By the formula for $B^0_{T,N}$ in **Step 4**,

$$\begin{aligned} a\Big(u, T, \overline{F}_T + \sum_{j=1}^{I'} J_{T,j}\Big) &= a(u, T, B^0_{T,N}) + \sum_{j=I'+1}^{N} u(J_{T,j}) + \sum_{j=1}^{m} c^0_{j,N} u(J_{T,j}) \\ &\geq -\epsilon_N a(u, T_0, 0) + \sum_{j=1}^{m} c^0_{j,N} u(J_{T,j}). \end{aligned}$$

The right-hand side tends to 0 as $N \to +\infty$, which proves (11.12).

For $1 \leq i \leq m$, let $\lambda_i := 1$ if $\mu_i = 0$, and $\lambda_i := \min\{1, (m\mu_i)^{-1}\}$ otherwise. Then λ_i is rational, $0 < \lambda_i \leq 1$, and $\sum_{i\leq m} \lambda_i\mu_i \leq 1$. Let

$$B_T := \overline{F}_{T,\leq I} - \sum_{i=1}^{m} \lambda_i J_{T,i}, \qquad D_{T,i} := \lambda_i J_{T,i}\ (1 \leq i \leq m), \qquad D_{T,m+k} := J_{T,I+k}\ (k \geq 1).$$

Since $J_{T,i} \leq \mu_i \overline{F}_{T,\leq I}$, we have $B_T \geq (1 - \sum_{i\leq m} \lambda_i\mu_i)\overline{F}_{T,\leq I} \geq 0$. The $\mathbb{Q}$-divisors B_T and $D_{T,j}$ are $\mathbb{Q}$-Cartier, and $D_{T,j} \geq 0$. For $0 \leq n \leq m$, we have $B_T + D_{T,\leq n} \geq B_T \geq 0$, and $\overline{F}_{T,\leq I} - (B_T + D_{T,\leq n}) = \sum_{n<i\leq m} \lambda_i J_{T,i} \geq 0$ is $\mathbb{Q}$-Cartier, so $a(u, T, B_T + D_{T,\leq n}) \geq a(u, T, \overline{F}_{T,\leq I})$ for any divisorial valuation u of k_v. For $n = m + k$ with $k \geq 1$,

$$B_T + D_{T,\leq n} = \overline{F}_{T,\leq I} + \sum_{j=I+1}^{I+k} J_{T,j} = \overline{F}_T + \sum_{j=1}^{I+k} J_{T,j}.$$

By (11.12) with $I' = I$ and $I' = I + k$, $(T, B_T + D_{T,\leq n})$ is lc for any $n \geq 0$. In particular (T, B_T) is a projective lc pair, and $D_{T,\bullet}$ satisfies Definition 7.1(1).

Next we compare families on T with families on X. Let $l \geq m$ and $n_0 \geq l$ be integers and $(\tilde{c}_{i,n})_{1\leq i\leq l, n\geq n_0}$ rational numbers. Let $M := I + l - m$ and $N_0 := n_0 + I - m$, so that $N_0 \geq M$. For $N \geq N_0$ and $n := N - I + m$, let

$$c_{i,N} := \lambda_i \tilde{c}_{i,n}\ (1 \leq i \leq m), \qquad c_{j,N} := 0\ (m < j \leq I), \qquad c_{I+k,N} := \tilde{c}_{m+k,n}\ (1 \leq k \leq l - m).$$

We call $(c_{j,N})_{1\le j\le M, N\ge N_0}$ the *family on X associated to* $(\tilde{c}_{i,n})$. Let $n \ge n_0$ and $N := n + I - m$. As $n \ge m$, we have $D_{T,\le n} = \sum_{i\le m} \lambda_i J_{T,i} + \sum_{I<j\le N} J_{T,j}$ and $B_T + \sum_{i\le m} \lambda_i J_{T,i} = \overline{F}_{T,\le I}$. Thus a direct calculation shows

$$B_T + D_{T,\le n} + \sum_{i=1}^{l} \tilde{c}_{i,n} D_{T,i} = \overline{F}_T + \sum_{j=1}^{N} J_{T,j} + \sum_{j=1}^{M} c_{j,N} J_{T,j}. \tag{11.13}$$

Let $x_\bullet$ be a moment row for $D_\bullet$. Then $\sum_{i\le n} \tilde{x}_i = \sum_{i\le m} \lambda_i x_i + \sum_{I<j\le N} x_j$, which converges to $\tilde{x}_\infty = x_\infty - \sum_{j\le I} x_j + \sum_{j\le m} \lambda_j x_j$. By the definitions of $\tilde{x}_\bullet$ and $(c_{j,N})$,

$$\sum_{j=1}^{N} x_j + \sum_{j=1}^{M} c_{j,N} x_j - x_\infty = \sum_{i=1}^{n} \tilde{x}_i + \sum_{i=1}^{l} \tilde{c}_{i,n} \tilde{x}_i - \tilde{x}_\infty. \tag{11.14}$$

Moreover, $c_{j,N} \to 0$ for any j if and only if $\tilde{c}_{i,n} \to 0$ for any i, and since $0 < \lambda_j \le 1$,

$$\begin{gathered} 1 + c_{j,N} \ge \min\{1, 1 + \tilde{c}_{j,n}\} \ (j \le m), \\ 1 + c_{j,N} = 1 \ (m < j \le I), \qquad 1 + c_{I+k,N} = 1 + \tilde{c}_{m+k,n}. \end{gathered} \tag{11.15}$$

Finally we verify Definition 7.1(2). If a correction family $(c_{j,N})_{1\le j\le M, N\ge N_0}$ satisfies a moment row $x_\bullet$, then for any $N_0' \ge N_0$, the family $(c_{j,N})_{1\le j\le M, N\ge N_0'}$ is a correction family satisfying $x_\bullet$. Possibly replacing N_0^0 with $\max\{N_0^0, I\}$, we may assume that $N_0^0 \ge I$. Let $n_0^0 := N_0^0 - I + m \ge m$ and $\tilde{c}^0_{i,n} := \lambda_i^{-1} c^0_{i,n+I-m}$ for $1 \le i \le m$ and $n \ge n_0^0$. The family on X associated to $(\tilde{c}^0_{i,n})$ is $(c^0_{j,N})$ extended by $c^0_{j,N} := 0$ for $m < j \le I$, which is a correction family for $D_\bullet$ whose integer m of Definition 7.1 is $I \le N_0^0$, satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$. By (11.13) and (11.11),

$$K_T + B_T + D_{T,\le n} + \sum_{i=1}^{m} \tilde{c}^0_{i,n} D_{T,i} \sim_{\mathbb{Q}} 0 \quad \text{for any } n \ge n_0^0,$$

and $\tilde{c}^0_{i,n} \to 0$. Taking n_0^0 in place of m_0, the numbers $\tilde{c}^0_{i,n}$ for $i \le m$, and 0 for $m < i \le n_0^0$, Definition 7.1(2) holds for $D_{T,\bullet}$. Hence $D_{T,\bullet}$ is an lc anti-canonical series on (T, B_T), and **Step 4** shows that T, B_T, and $\sigma \circ \sigma_1$ have the properties in the statement.

Step 7. In this step we prove (1), (2), and (3).

Let $(c_{j,N})_{1\le j\le m, N\ge N_0}$ be rational numbers with $N_0 \ge I$, and let $\tilde{c}_{i,n} := \lambda_i^{-1} c_{i,n+I-m}$ for $1 \le i \le m$ and $n \ge N_0 - I + m$. Then $(\tilde{c}_{i,n})_{1\le i\le m, n\ge N_0 - I + m}$ is a family as in **Step 6** with $l = m$ and $n_0 = N_0 - I + m \ge m$. The family on X associated to $(\tilde{c}_{i,n})$ is $(c_{j,N})$ extended by $c_{j,N} := 0$ for $m < j \le I$. Since $N_0 \ge I$, the extension has the corrected boundaries G_N of $(c_{j,N})$ for $N \ge N_0$, and it satisfies a moment row for $D_\bullet$ if and only if $(c_{j,N})$ does, so we may apply (11.13) and (11.14) to $(c_{j,N})$ and $(\tilde{c}_{i,n})$.

We prove (1). Assume that $(\tilde{c}_{i,n})$ is a correction family for $D_{T,\bullet}$ satisfying $\tilde{y}_\bullet^{(1)}, \dots, \tilde{y}_\bullet^{(p)}$. Then $c_{j,N} \to 0$, and $(c_{j,N})$ satisfies $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$ by (11.14). Since $(c_{j,N})$ satisfies $y_\bullet^{(1)}$, we have $K_X + B + G_N \sim_{\mathbb{Q}} 0$ for any $N \ge N_0$ by **Step 1**. Hence $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$. Conversely, assume that $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$. Then $\tilde{c}_{i,n} \to 0$, and $(\tilde{c}_{i,n})$ satisfies $\tilde{y}_\bullet^{(1)}, \dots, \tilde{y}_\bullet^{(p)}$ by (11.14). By (11.13) and (11.11), $K_T + B_T + G_{T,n} \sim_{\mathbb{Q}} 0$ for any $n \ge N_0 - I + m$. As $N_0 - I + m \ge m$, $(\tilde{c}_{i,n})$ is a correction family for $D_{T,\bullet}$ satisfying $\tilde{y}_\bullet^{(1)}, \dots, \tilde{y}_\bullet^{(p)}$. This proves (1).

The equivalence (2) follows from (11.14). We prove (3). Let $(c_{j,N})$ be a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}$, $N \ge N_0$ with $G_N \ge 0$, and $n := N - I + m$. By (11.1), $B_S + \mathrm{lead}_S(h^* G_N) = B_{S,N}$, and by (11.13), $B_T + G_{T,n} = B_{T,N}$. Thus (11.11) implies

$$f_1^*\big(K_S + B_S + \mathrm{lead}_S(h^* G_N)\big) = g_1^*(K_T + B_T + G_{T,n}),$$

and $(S, B_S + \mathrm{lead}_S(h^* G_N))$ is crepant birational to $(T, B_T + G_{T,n})$ by Definition 2.2. This proves (3).

Step 8. In this step we prove (4).

Fix e_0, $x_\bullet^{(1)},\dots,x_\bullet^{(q)}$, and W_T satisfying the hypotheses of (4), with $W_T \subset H^0(T,\mathcal{O}_T(-e_T(K_T+B_T)))$. By Definition 7.3, W_T is spanned by products $s_{T,\alpha}=s_{T,\alpha,1}\cdots s_{T,\alpha,k_\alpha}$ for $1\le\alpha\le A$, where $e_T=\sum_i e_{\alpha,i}$, $\operatorname{div}(s_{T,\alpha,i})=e_{\alpha,i}G_T^{(\alpha,i)}$, and $G_T^{(\alpha,i)}$ is the corrected boundary at an index $n_{\alpha,i}$ of a correction family $(\tilde c_{i',n}^{(\alpha,i)})$ for $D_{T,\bullet}$ satisfying $\tilde y_\bullet^{(1)},\dots,\tilde y_\bullet^{(p)},\tilde x_\bullet^{(1)},\dots,\tilde x_\bullet^{(q)}$, with $1+\tilde c_{i',n_{\alpha,i}}^{(\alpha,i)}\ge 0$ for every i'.

Fix (α,i), write $(\tilde c_{i',n})_{1\le i'\le l,n\ge n_0}$ for $(\tilde c_{i',n}^{(\alpha,i)})$, and let $n_1:=n_{\alpha,i}$. We show that we may assume $l\ge m$. Suppose that $l<m$, and set $\tilde c_{i',n}:=0$ for $l<i'\le m$. If $n_1\ge m$, the family $(\tilde c_{i',n})_{1\le i'\le m,n\ge\max\{n_0,m\}}$ has the corrected boundaries of $(\tilde c_{i',n})_{1\le i'\le l,n\ge n_0}$ at any index $n\ge\max\{n_0,m\}$, it satisfies any moment row for $D_{T,\bullet}$ that $(\tilde c_{i',n})_{1\le i'\le l,n\ge n_0}$ satisfies, and the index n_1 is kept. If $n_1<m$, define $(\tilde c'_{i',n})_{1\le i'\le m,n\ge\max\{n_0,m\}}$ by $\tilde c'_{i',n}:=\tilde c_{i',n}$ for $n\ne m$, and

$$\tilde c'_{i',m}:=\tilde c_{i',n_1}\ (i'\le n_1),\qquad \tilde c'_{i',m}:=-1\ (n_1<i'\le m).$$

Then the corrected boundary of $(\tilde c'_{i',n})$ at m is $D_{T,\le n_1}+\sum_{i'\le l}\tilde c_{i',n_1}D_{T,i'}=G_T^{(\alpha,i)}$, and for a moment row $z_\bullet$ for $D_{T,\bullet}$,

$$\sum_{i'=1}^{m} z_{i'}+\sum_{i'=1}^{m}\tilde c'_{i',m}z_{i'}=\sum_{i'=1}^{n_1} z_{i'}+\sum_{i'=1}^{l}\tilde c_{i',n_1}z_{i'}. \tag{11.16}$$

Moreover, $1+\tilde c'_{i',m}\ge 0$ for any i': for $i'\le l$ this is the hypothesis $1+\tilde c_{i',n_1}\ge 0$, for $l<i'\le n_1$ we have $\tilde c'_{i',m}=\tilde c_{i',n_1}=0$, and for $n_1<i'\le m$ we have $1+\tilde c'_{i',m}=0$. Since $n_0\le n_1<m$, the indices of $(\tilde c'_{i',n})$ start at $\max\{n_0,m\}=m$, so that the condition $N_0\ge m$ of Definition 7.1 holds for $(\tilde c'_{i',n})$. For $n>m$, $\tilde c'_{i',n}=\tilde c_{i',n}$ (with $\tilde c_{i',n}=0$ for $l<i'\le m$), and at $n=m$ its corrected boundary is $G_T^{(\alpha,i)}$ and its row sums are those of $(\tilde c_{i',n})$ at n_1 by (11.16). Hence $(\tilde c'_{i',n})$ is a correction family for $D_{T,\bullet}$ satisfying $\tilde y_\bullet^{(1)},\dots,\tilde y_\bullet^{(p)},\tilde x_\bullet^{(1)},\dots,\tilde x_\bullet^{(q)}$, and $G_T^{(\alpha,i)}$ is the corrected boundary of $(\tilde c'_{i',n})$ at m. Possibly replacing the family and the index, we may assume that $l\ge m$. Then $n\ge n_0\ge l\ge m$ for any index n of the family.

Let $(c_{j,N})_{1\le j\le M,N\ge N_0}$ be the family on X associated to $(\tilde c_{i',n})$ in **Step 6**, and $N_1:=n_1+I-m$. By (11.14) and the equivalence of $c_{j,N}\to 0$ and $\tilde c_{i',n}\to 0$, the family $(c_{j,N})$ tends to 0 and satisfies $y_\bullet^{(1)},\dots,y_\bullet^{(p)},x_\bullet^{(1)},\dots,x_\bullet^{(q)}$. By **Step 1**, $K_X+B+G_N\sim_{\mathbb{Q}}0$, so $(c_{j,N})$ is a correction family for $D_\bullet$ satisfying $y_\bullet^{(1)},\dots,y_\bullet^{(p)},x_\bullet^{(1)},\dots,x_\bullet^{(q)}$. By (11.15), $1+c_{j,N_1}\ge 0$ for any j. Let $G^{(\alpha,i)}:=G_{N_1}\ge 0$ be the corrected boundary of $(c_{j,N})$ at N_1. By (11.1), (11.13), and (11.11),

$$f_1^*\big(K_S+B_S+\operatorname{lead}_S(h^*G^{(\alpha,i)})\big)=g_1^*\big(K_T+B_T+G_T^{(\alpha,i)}\big), \tag{11.17}$$

where $f_1:S''\to S$ and $g_1:S''\to T$ are the morphisms of **Step 5**. By Lemma 8.6(2), $a(v,X,B+G^{(\alpha,i)})=0$.

Fix a positive integer e' such that $-e'(K_X+B)$ is Cartier and $e'\mathbf{a}\in\mathbb{Z}^r$, and let $\mathcal{L}$, $\mathcal{L}_S$, the subspaces $W_{l'}$ for positive integers l', and lead be the objects of Definition-Lemma 8.7 for e' in place of e. Since $K_X+B+G^{(\alpha,i)}\sim_{\mathbb{Q}}0$ for the finitely many pairs (α,i), there exists a positive integer κ divisible by e_0e' such that $\kappa e_{\alpha,i}(K_X+B+G^{(\alpha,i)})$ is a principal divisor for any (α,i). Thus there exists $s_{\alpha,i}\in H^0(X,\mathcal{O}_X(-\kappa e_{\alpha,i}(K_X+B)))$ with $\operatorname{div}(s_{\alpha,i})=\kappa e_{\alpha,i}G^{(\alpha,i)}$. As $a(v,X,B+G^{(\alpha,i)})=0$, Definition-Lemma 8.7(3) implies $s_{\alpha,i}\in W_{\kappa e_{\alpha,i}/e'}$ and $\operatorname{lead}(s_{\alpha,i})\ne 0$. Let $e:=\kappa e_T$. Since e' divides κ, e/e' is a positive integer. Let $s_\alpha:=s_{\alpha,1}\cdots s_{\alpha,k_\alpha}$. By Definition-Lemma 8.7(2), $s_\alpha\in W_{e/e'}$ and $\operatorname{lead}(s_\alpha)=\prod_i\operatorname{lead}(s_{\alpha,i})\ne 0$, as S is integral.

Let W'_T be the subspace spanned by $s_{T,\alpha}^{\kappa}$, $1\le\alpha\le A$. By Lemma 7.4, W'_T is a corrected system for $D_{T,\bullet}$ satisfying $\tilde y_\bullet^{(1)},\dots,\tilde y_\bullet^{(p)},\tilde x_\bullet^{(1)},\dots,\tilde x_\bullet^{(q)}$, and $\operatorname{Nlc}(T,B_T+G'_T)=\operatorname{Nlc}(T,B_T+G_T)=\emptyset$ for general members G'_T of W'_T and G_T of W_T. Thus there exists a non-empty Zariski open subset $U_T\subset W'_T$ such that $(T,B_T+G_{T,w})$ is lc for any $w\in U_T$, where $G_{T,w}:=e^{-1}\operatorname{div}(w)$.

Let $G_\alpha := e^{-1}\operatorname{div}(s_\alpha) = \sum_i (\kappa e_{\alpha,i}/e) G^{(\alpha,i)}$ and $G_{T,\alpha} := G_{T,s^\kappa_{T,\alpha}} = \sum_i (\kappa e_{\alpha,i}/e) G_T^{(\alpha,i)}$. We have $\kappa e_{\alpha,i}/e > 0$ and $\sum_i \kappa e_{\alpha,i}/e = 1$. By Lemma 8.2 and (11.17),

$$f_1^*\big(K_S + B_S + \operatorname{lead}_S(h^*G_\alpha)\big) = g_1^*\big(K_T + B_T + G_{T,\alpha}\big) \quad \text{for any } \alpha. \tag{11.18}$$

By Definition-Lemma 8.7(3), $\operatorname{div}(\operatorname{lead}(s_\alpha)) = e \operatorname{lead}_S(h^*G_\alpha)$.

We normalize the sections s_α. Let $\psi_\alpha := \operatorname{lead}(s_\alpha)/\operatorname{lead}(s_1) \in k_v$ and $\psi_{T,\alpha} := s^\kappa_{T,\alpha}/s^\kappa_{T,1} \in k_v$. Then $\operatorname{div}_S(\psi_\alpha) = e(\operatorname{lead}_S(h^*G_\alpha) - \operatorname{lead}_S(h^*G_1))$ and $\operatorname{div}_T(\psi_{T,\alpha}) = e(G_{T,\alpha} - G_{T,1})$. Subtracting (11.18) for 1 from (11.18) for α, we obtain $\operatorname{div}_{S''}(\psi_\alpha) = \operatorname{div}_{S''}(\psi_{T,\alpha})$. As S'' is a normal projective variety, $\psi_\alpha/\psi_{T,\alpha}$ is a non-zero constant. Replacing $s_{\alpha,1}$ by a non-zero scalar multiple, we may assume that $\psi_\alpha = \psi_{T,\alpha}$ for any α.

Let $W \subset H^0(X, \mathcal{O}_X(-e(K_X + B)))$ be the subspace spanned by $s_1, \dots, s_A$. As e' divides e, $-e(K_X+B)$ is Cartier. Thus W is a corrected system for $D_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p)}, x_\bullet^{(1)}, \dots, x_\bullet^{(q)}$, defined by the integers $\kappa e_{\alpha,i}$, which are divisible by e_0.

It remains to treat a general member of W. We have $W \subset W_{e/e'}$. Let

$$\iota_S : H^0(S, \mathcal{L}_S^{\otimes e/e'}) \to k_v,\ t \to t/\operatorname{lead}(s_1), \qquad \iota_T : W'_T \to k_v,\ w \to w/s^\kappa_{T,1},$$

which are injective and linear. By the normalization, $\iota_S(\operatorname{lead}(s_\alpha)) = \iota_T(s^\kappa_{T,\alpha})$ for any α. As lead is linear on $W_{e/e'}$ by Definition-Lemma 8.7(1), $\iota_S(\operatorname{lead}(W)) \subset \iota_T(W'_T)$, and

$$\Theta := \iota_T^{-1} \circ \iota_S \circ \operatorname{lead} : W \to W'_T$$

is a linear map with $\Theta(s_\alpha) = s^\kappa_{T,\alpha}$. Hence Θ is surjective, and $\Theta^{-1}(U_T)$ is a non-empty Zariski open subset of W. The subset of $s \in W$ with $\operatorname{lead}(s) \neq 0$ is Zariski open and contains s_1. Since W is irreducible, $U := \Theta^{-1}(U_T) \cap \{s \in W : \operatorname{lead}(s) \neq 0\}$ is a non-empty Zariski open subset of W.

Let $s \in U$, $G := G_s = e^{-1}\operatorname{div}(s)$, $w := \Theta(s) \in U_T$, and $\psi := \iota_S(\operatorname{lead}(s)) = \iota_T(w)$. As $\operatorname{lead}(s) \neq 0$, Definition-Lemma 8.7(3) implies $a(v, X, B+G) = 0$ and $\operatorname{div}(\operatorname{lead}(s)) = e \operatorname{lead}_S(h^*G)$. Thus

$$e \operatorname{lead}_S(h^*G) = e \operatorname{lead}_S(h^*G_1) + \operatorname{div}_S(\psi), \qquad eG_{T,w} = eG_{T,1} + \operatorname{div}_T(\psi).$$

Adding $e^{-1}\operatorname{div}_{S''}(\psi)$ to both sides of (11.18) for $\alpha = 1$, we obtain

$$f_1^*\big(K_S + B_S + \operatorname{lead}_S(h^*G)\big) = g_1^*\big(K_T + B_T + G_{T,w}\big).$$

Since $(T, B_T + G_{T,w})$ is lc, [KM98, Lemma 2.30] implies that $(S, B_S + \operatorname{lead}_S(h^*G))$ is sub-lc. Hence $a(v, X, B+G) = 0$ and $(S, B_S + \operatorname{lead}_S(h^*G))$ is sub-lc for a general member G of W. This proves (4). □

12. Finite attainment

In this section, we prove Theorem 12.1 by induction on the dimension. Lemma 12.3 finds a stationary valuation whose center contains a given irreducible closed subset near which the corrected boundaries are not klt, Sections 9, 10, and 11 reduce the lc property near the center of such a valuation to Theorem 12.1 for a residue model of lower dimension, and Lemma 12.4 merges finitely many corrected systems.

Theorem 12.1. *Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B), and $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ moment rows for $D_\bullet$, where $p \geq 0$. Then there exists a corrected system W for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that $(X, B+G)$ is lc for a general member G of W. In particular, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on X such that $(X, B+G)$ is lc and $K_X + B + G \sim_{\mathbb{Q}} 0$.*

Theorem 12.1 allows finitely many moment rows because its proof uses the theorem in lower dimension for a residue model $(T, B_T; D_{T,\bullet})$ (Construction-Proposition 11.4), with moment rows for $D_{T,\bullet}$ defined from the moment rows provided by Construction-Proposition 11.4 and from $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$. When the stationary valuation has rational rank $\dim X$, no residue model is used.

Lemma 12.2. *Let X be a smooth projective variety of dimension $d \geq 1$ and H a very ample Cartier divisor on X. Then there exists a real number $C > 0$ depending only on X and H such that*

$$v(D) \leq C(D \cdot H^{d-1})a(v, X, 0)$$

for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ on X and any real valuation v of $\mathbb{C}(X)$.

Proof. Since X is projective, any real valuation v of $\mathbb{C}(X)$ has a center on X by the valuative criterion of properness, so $v(D)$ is defined. Since X is smooth, $(X, 0)$ is klt [KM98, Corollary 2.31(3)], so $(X, 0)$ is a projective lc pair without lc centers. Thus [Yos26, Lemma 3.5], applied to $(X, 0)$ and H, implies the lemma. □

The next lemma shows that, near the generic point of an irreducible closed subset Z, the corrected boundaries G_N are klt for large N, unless Z is contained in the center of a quasi-monomial valuation stationary for $(X, B; D_\bullet)$ (Definition 7.1). The valuation in (1) is quasi-monomial, so that Construction-Proposition 8.1 can be used for it in the proof of Theorem 12.1. It is given by [Xu20, Theorem 1.1] when the generic point of Z is not in $\mathrm{Nklt}(X, B)$, and it is ord_E for a prime divisor E over X with $a(E, X, B) = 0$ otherwise.

Lemma 12.3. *Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B), and $Z \subset X$ an irreducible closed subset. Then at least one of the following holds.*

(1) There exists a quasi-monomial valuation v of $\mathbb{C}(X)$ that is stationary for $(X, B; D_\bullet)$ and satisfies $Z \subset \mathrm{center}_X v$.
(2) For any correction family $(c_{j,N})$ for $D_\bullet$ and any sufficiently large N, the pair $(X, B + G_N)$ is klt on a neighborhood of the generic point of Z.

Proof. The proof is long, so we divide it into several steps. Let $d := \dim X$ and let η_Z be the generic point of Z. If $d = 0$, then no prime divisor exists over X, and (2) holds. Thus we may assume that $d \geq 1$. Any real valuation u of $\mathbb{C}(X)$ has a center on X because X is projective, and we let

$$\sigma(u) := \sum_{j \geq 1} u(D_j) \in [0, +\infty].$$

Step 1. In this step, we prove two inequalities for real valuations, and we show that (1) holds if $\eta_Z \in \mathrm{Nklt}(X, B)$.

Let X_1 be a smooth variety, $\sum_F F$ a reduced simple normal crossing divisor on X_1, and u a real valuation of $\mathbb{C}(X_1)$ with a center on X_1. The retraction $r(u)$ of u to the monomial valuations in the components F satisfies $r(u)(F) = u(F)$ for any F, $A_{X_1}(r(u)) = \sum_F u(F)$, and $A_{X_1}(r(u)) \leq A_{X_1}(u)$ [JM12, Section 4.3, (5.1), and Corollary 5.4]. Thus

$$\sum_F u(F) \leq A_{X_1}(u). \tag{12.1}$$

Let u be a real valuation of $\mathbb{C}(X)$ and $N \geq 1$. The pair $(X, B + D_{\leq N})$ is lc by Definition 7.1(1). If $a(u, X, B) < +\infty$, then Definition-Proposition 2.8 implies $0 \leq a(u, X, B + D_{\leq N}) = a(u, X, B) - u(D_{\leq N})$, so $u(D_{\leq N}) \leq a(u, X, B)$. Letting $N \to +\infty$, we obtain

$$\sigma(u) \leq a(u, X, B) \tag{12.2}$$

for any real valuation u of $\mathbb{C}(X)$.

Suppose that $\eta_Z \in \mathrm{Nklt}(X, B)$. By Definition 2.4, there exists a prime divisor E over X such that $a(E, X, B) \leq 0$ and $\eta_Z \in \mathrm{center}_X E$. Since (X, B) is lc, $a(E, X, B) = 0$. For any $N \geq 1$, since $(X, B + D_{\leq N})$ is lc,

$$0 \leq a(E, X, B + D_{\leq N}) = -\sum_{j=1}^{N} \mathrm{mult}_E D_j.$$

Since $\mathrm{mult}_E D_j \geq 0$ for any j, we have $\mathrm{mult}_E D_j = 0$ for any $j \geq 1$. The valuation ord_E is the monomial valuation with weight 1 in E at the generic point of E on a log resolution of (X, B) on which E is a divisor, so ord_E is quasi-monomial. Since $\sum_{j\geq 1} \mathrm{mult}_E D_j = 0 = a(\mathrm{ord}_E, X, B)$, the valuation ord_E is stationary for $(X, B; D_\bullet)$. Moreover, $Z \subset \mathrm{center}_X E$. Thus (1) holds.

From now on, we assume that $\eta_Z \notin \mathrm{Nklt}(X, B)$. Let $U := X \setminus \mathrm{Nklt}(X, B)$, which is an open neighborhood of η_Z since $\mathrm{Nklt}(X, B)$ is closed: for a log resolution $f : X_2 \to X$ of (X, B) with $K_{X_2} + B_{X_2} = f^*(K_X + B)$, $\mathrm{Nklt}(X, B)$ is the image under the proper morphism f of the union of the prime divisors with coefficient ≥ 1 in B_{X_2} [KM98, Lemma 2.30, Corollary 2.31(3)].

Step 2. In this step, we bound the tails of σ uniformly for real valuations whose center on X is contained in U.

Let $g : Y \to X$ be a log resolution of (X, B), let $K_Y + B_Y := g^*(K_X + B)$, let H be a very ample divisor on Y, and let $d_j := (g^*D_j \cdot H^{d-1}) \geq 0$ for $j \geq 1$. Let m_0 and $a_{j,N}$ be the integer and the rational numbers in Definition 7.1(2). Then, for any $N \geq m_0$,

$$(g^*(K_X + B) \cdot H^{d-1}) + \sum_{j=1}^{N} d_j + \sum_{j=1}^{m_0} a_{j,N} d_j = 0.$$

Since $\lim_{N\to+\infty} a_{j,N} = 0$ for any j, we obtain

$$\sum_{j\geq 1} d_j = -(g^*(K_X + B) \cdot H^{d-1}) < +\infty.$$

For a prime divisor F on Y, let $b_F := \mathrm{mult}_F B_Y$. If $b_F > 0$ and $F \cap g^{-1}(U) \neq \emptyset$, then $\mathrm{center}_X F \cap U \neq \emptyset$, so $\mathrm{center}_X F \not\subset \mathrm{Nklt}(X, B)$ and $1 - b_F = a(F, X, B) > 0$. Since only finitely many F satisfy $b_F > 0$, the real number

$$\epsilon := \min\left(\{1\} \cup \{1 - b_F \mid b_F > 0 \text{ and } F \cap g^{-1}(U) \neq \emptyset\}\right)$$

is positive. Let u be a real valuation of $\mathbb{C}(X)$ whose center on X is contained in U. The center ξ of u on Y is contained in $g^{-1}(U)$. If $u(F) > 0$ for a prime divisor F on Y, then $\xi \in F$, so $F \cap g^{-1}(U) \neq \emptyset$. Hence $u(B_Y) \leq (1-\epsilon)\sum_F u(F)$, where F runs over the components of $\mathrm{Supp}\, B_Y$, and (12.1) implies

$$a(u, X, B) = A_Y(u) - u(B_Y) \geq \epsilon A_Y(u) = \epsilon a(u, Y, 0).$$

Let C be the constant of Lemma 12.2 for Y and H. Since $u(D_j) = u(g^*D_j)$, Lemma 12.2 implies $u(D_j) \leq C d_j a(u, Y, 0) \leq \frac{C}{\epsilon} d_j a(u, X, B)$ for any $j \geq 1$. Thus

$$\sum_{j>N} u(D_j) \leq \kappa_N a(u, X, B), \quad \text{where } \kappa_N := \frac{C}{\epsilon} \sum_{j>N} d_j, \tag{12.3}$$

for any integer $N \geq 0$ and any real valuation u of $\mathbb{C}(X)$ whose center on X is contained in U. Moreover, $\lim_{N\to+\infty} \kappa_N = 0$.

Step 3. In this step, we construct a compact space $\mathcal{V}$ of real valuations and the maximum τ of σ on $\mathcal{V}$.

Let $\mathcal{V}$ be the set of real valuations u of $\mathbb{C}(X)$ such that $u \geq 0$ on $\mathcal{O}_{X,\eta_Z}$ and $a(u, X, B) \leq 1$. For $u \in \mathcal{V}$, the prime ideal $\{\phi \in \mathcal{O}_{X,\eta_Z} \mid u(\phi) > 0\}$ corresponds to the center of u on X, so $\eta_Z \in \mathrm{center}_X u$. Since U is open and contains η_Z, the center of u on X is contained in U, and (12.3) holds for u.

Since $\mathbb{C}(X)$ is the fraction field of $\mathcal{O}_{X,\eta_Z}$, any $u \in \mathcal{V}$ is determined by the values of u on $\mathcal{O}_{X,\eta_Z}\setminus\{0\}$. For $\phi \in \mathcal{O}_{X,\eta_Z}\setminus\{0\}$, let $\mathrm{lct}(X, B; (\phi))$ be the log canonical threshold of the graded sequence $(\phi^l \mathcal{O}_{X,\eta_Z})_{l\geq 1}$ with respect to (X, B) (Definition 2.9). Since (X, B) is klt on a neighborhood of η_Z, $\mathrm{lct}(X, B; (\phi)) > 0$ [Xu20, Section 2.2.1], [KM98, Lemma 2.30,

Corollary 2.31(3)]. For $u \in \mathcal{V}$ with $u(\phi) > 0$, Definition 2.9 implies $\mathrm{lct}(X, B; (\phi)) \leq a(u, X, B)/u(\phi) \leq 1/u(\phi)$. Hence

$$0 \leq u(\phi) \leq \mathrm{lct}(X, B; (\phi))^{-1}$$

for any $u \in \mathcal{V}$, with the convention $(+\infty)^{-1} = 0$. We regard $\mathcal{V}$ as a subset of the product

$$\prod_{\phi \in \mathcal{O}_{X,\eta_Z} \setminus \{0\}} \left[0, \mathrm{lct}(X, B; (\phi))^{-1}\right],$$

which is compact by Tychonoff's theorem. In the product, the conditions $u(\phi\psi) = u(\phi) + u(\psi)$, $u(\phi + \psi) \geq \min\{u(\phi), u(\psi)\}$ for $\phi + \psi \neq 0$, and $u(c) = 0$ for $c \in \mathbb{C}^*$ define a closed subset $\mathcal{V}_0$, and any point of $\mathcal{V}_0$ extends uniquely to a real valuation of $\mathbb{C}(X)$ that is non-negative on $\mathcal{O}_{X,\eta_Z}$. By Definition-Proposition 2.8, $a(u, X, B) = A_Y(u) - u(B_Y)$ for the log resolution $g : Y \to X$ of **Step 2**. The function A_Y is lower semicontinuous [JM12, Lemma 5.7], and $u \to u(F)$ is continuous for any prime divisor F on Y [JM12, Lemma 4.1], for the topology of pointwise convergence on $\mathbb{C}(X)^*$. On $\mathcal{V}_0$, the topology of pointwise convergence on $\mathbb{C}(X)^*$ is the topology induced by the product, since $u(\phi/\psi) = u(\phi) - u(\psi)$. Hence $\mathcal{V} = \{u \in \mathcal{V}_0 \mid a(u, X, B) \leq 1\}$ is closed in $\mathcal{V}_0$, and so closed in the product. Thus $\mathcal{V}$ is compact.

For $j \geq 1$, let l_j be a positive integer such that $l_j D_j$ is Cartier near η_Z, and $\phi_j \in \mathcal{O}_{X,\eta_Z}$ a local equation of $l_j D_j$. For $u \in \mathcal{V}$, ϕ_j is also a local equation of $l_j D_j$ at the center of u, so $u(D_j) = \frac{1}{l_j} u(\phi_j)$, and $u \to u(D_j)$ is continuous on $\mathcal{V}$. By (12.3),

$$0 \leq \sigma(u) - \sum_{j=1}^{N} u(D_j) \leq \kappa_N$$

for any $u \in \mathcal{V}$ and $N \geq 0$. Hence σ is a uniform limit of continuous functions on $\mathcal{V}$, and σ is continuous on $\mathcal{V}$. Since $\mathcal{V}$ contains the trivial valuation, $\mathcal{V} \neq \emptyset$. Let

$$\tau := \max_{u \in \mathcal{V}} \sigma(u).$$

By (12.2), $\tau \leq 1$.

Step 4. In this step, we show that (2) holds if $\tau < 1$.

Let E be a prime divisor over X such that $\eta_Z \in \mathrm{center}_X E$. Since $\eta_Z \notin \mathrm{Nklt}(X, B)$, we have $a(E, X, B) > 0$. Since $a(cu, X, B) = ca(u, X, B)$ for any real valuation u and any real number $c > 0$ [JM12, Sections 5.1 and 5.2], we have $a(E, X, B)^{-1} \mathrm{ord}_E \in \mathcal{V}$. Hence

$$\sigma(\mathrm{ord}_E) \leq \tau a(E, X, B).$$

Let $(c_{j,N})_{1 \leq j \leq m, N \geq N_0}$ be a correction family for $D_\bullet$, and let $\theta_N := \max\{0, c_{1,N}, \ldots, c_{m,N}\}$ for $N \geq N_0$. Since $\mathrm{mult}_E D_j \geq 0$ for any j and $m \leq N_0$, for any $N \geq N_0$ we have

$$\begin{aligned} a(E, X, B + G_N) &= a(E, X, B) - \sum_{j=1}^{N} \mathrm{mult}_E D_j - \sum_{j=1}^{m} c_{j,N} \mathrm{mult}_E D_j \\ &\geq a(E, X, B) - (1 + \theta_N)\sigma(\mathrm{ord}_E) \geq (1 - \tau - \theta_N) a(E, X, B). \end{aligned}$$

Since $\lim_{N \to +\infty} \theta_N = 0$, Lemma 7.2 implies that there exists an integer $N_2 \geq N_0$ such that $G_N \geq 0$ and $\tau + \theta_N < 1$ for any $N \geq N_2$. For such N, $a(E, X, B + G_N) > 0$ for any prime divisor E over X with $\eta_Z \in \mathrm{center}_X E$, that is, $\eta_Z \notin \mathrm{Nklt}(X, B + G_N)$. Since $\mathrm{Nklt}(X, B + G_N)$ is closed (as in **Step 1**) [KM98, Lemma 2.30, Corollary 2.31(3)], $(X, B + G_N)$ is klt on the neighborhood $X \setminus \mathrm{Nklt}(X, B + G_N)$ of η_Z. Thus (2) holds.

Step 5. In this step, we show that (1) holds if $\tau = 1$.

Let $w \in \mathcal{V}$ be a valuation such that $\sigma(w) = 1$. By (12.2), $a(w, X, B) = 1$, so w is non-trivial. Let ζ be the center of w on X, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X,\zeta}$, and let $\beta := w(\mathfrak{m}) > 0$. Since $\eta_Z \in \mathrm{center}_X w$, we have $Z \subset \mathrm{center}_X w$ and $\zeta \in U$. For $l \geq 1$, let

$$\mathfrak{a}_l := \{\phi \in \mathcal{O}_{X,\zeta} \mid w(\phi) \geq l\}.$$

Then $\mathfrak{a}_l$ is an ideal of $\mathcal{O}_{X,\zeta}$, $\mathfrak{a}_l\mathfrak{a}_{l'} \subset \mathfrak{a}_{l+l'}$ for any $l, l' \geq 1$, and $\mathfrak{m}^{\lceil l/\beta \rceil} \subset \mathfrak{a}_l \subset \mathfrak{m}$. Hence $\mathfrak{a}_l \neq 0$.

We show that

$$u(\mathfrak{a}_\bullet) \leq \sigma(u) \leq a(u, X, B) \tag{12.4}$$

for any real valuation u of $\mathbb{C}(X)$ such that $\zeta \in \operatorname{center}_X u$. The second inequality is (12.2), so we may assume that $\sigma(u) < +\infty$. Since $\sigma(w) = 1$, there exists an integer $N_3 \geq 1$ such that $w(D_{\leq N}) > 0$ for any $N \geq N_3$. Fix $N \geq N_3$, a positive integer l_N such that $l_N D_{\leq N}$ is Cartier near ζ, and a local equation $\phi_N \in \mathcal{O}_{X,\zeta}$ of $l_N D_{\leq N}$. For $k \geq 1$, let $n_k := \lfloor k l_N w(D_{\leq N}) \rfloor$. Since $w(\phi_N^k) = k l_N w(D_{\leq N}) \geq n_k$, we have $\phi_N^k \in \mathfrak{a}_{n_k}$ when $n_k \geq 1$. By Definition 2.9,

$$u(\mathfrak{a}_\bullet) \leq \frac{1}{n_k} u(\mathfrak{a}_{n_k}) \leq \frac{k l_N}{n_k} u(D_{\leq N}).$$

Letting $k \to +\infty$, we obtain $u(\mathfrak{a}_\bullet) \leq u(D_{\leq N})/w(D_{\leq N})$. Letting $N \to +\infty$, we have $w(D_{\leq N}) \to \sigma(w) = 1$ and $u(D_{\leq N}) \to \sigma(u)$, so $u(\mathfrak{a}_\bullet) \leq \sigma(u)$. This proves (12.4).

By the definition of $\mathfrak{a}_l$, $w(\mathfrak{a}_l) \geq l$ for any l, so $w(\mathfrak{a}_\bullet) \geq 1$. By (12.4) for $u = w$, $w(\mathfrak{a}_\bullet) \leq \sigma(w) = 1$. Thus $w(\mathfrak{a}_\bullet) = 1 = a(w, X, B)$. Since (X, B) is klt on a neighborhood of $\zeta \in U$, the log canonical threshold $\operatorname{lct}(X, B; \mathfrak{a}_\bullet)$ of Definition 2.9 at ζ is defined, and (12.4) implies $a(u, X, B)/u(\mathfrak{a}_\bullet) \geq 1$ for any real valuation u of $\mathbb{C}(X)$ with $\zeta \in \operatorname{center}_X u$, $a(u, X, B) < +\infty$, and $u(\mathfrak{a}_\bullet) > 0$. Hence

$$\operatorname{lct}(X, B; \mathfrak{a}_\bullet) = 1,$$

and w computes $\operatorname{lct}(X, B; \mathfrak{a}_\bullet)$.

By the conventions of [Xu20, Section 2], $\operatorname{Spec} \mathcal{O}_{X,\zeta}$ is a germ, since $\mathcal{O}_{X,\zeta}$ is a local ring essentially of finite type, and (X, B) is klt on it. Since $\mathfrak{a}_\bullet$ is a graded sequence of non-zero ideals of $\mathcal{O}_{X,\zeta}$ with $\operatorname{lct}(X, B; \mathfrak{a}_\bullet) = 1 < +\infty$, [Xu20, Theorem 1.1] implies that there exists a quasi-monomial valuation v of $\mathbb{C}(X)$ such that $\zeta \in \operatorname{center}_X v$, $a(v, X, B) < +\infty$, $v(\mathfrak{a}_\bullet) > 0$, and $a(v, X, B) = v(\mathfrak{a}_\bullet)$. Since v is quasi-monomial on the germ, it is an Abhyankar valuation, that is, $\operatorname{tr.deg}_{\mathbb{C}} k_v + \operatorname{rat.rk} v = d$, where k_v is the residue field of v [Xu20, Section 2.1.1]. Let $X_1 \to X$ be a resolution [KM98, Theorem 0.2] and ξ the center of v on X_1. Since $\dim \mathcal{O}_{X_1,\xi} + \operatorname{tr.deg}_{\mathbb{C}} \kappa(\xi) = d$, v is an Abhyankar valuation over X_1 in the sense of [JM12, Section 3.2], so v is monomial on a log smooth pair over X_1 [JM12, Proposition 3.7]. By Chow's lemma [Har77, Exercise II.4.10], [KM98, Theorem 0.2], and [JM12, Lemma 3.6(i)], we may choose the log smooth pair to be projective over X. Thus v is quasi-monomial in the sense of Definition 2.7.

We show that ζ is the center of v on X. Otherwise, $\mathfrak{p} := \{\phi \in \mathcal{O}_{X,\zeta} \mid v(\phi) > 0\}$ is a prime ideal different from $\mathfrak{m}$, so $\mathfrak{m}^{\lceil l/\beta \rceil} \not\subset \mathfrak{p}$ and $\mathfrak{a}_l \not\subset \mathfrak{p}$. Hence $v(\mathfrak{a}_l) = 0$ for any l, and $v(\mathfrak{a}_\bullet) = 0$, a contradiction. Thus $\operatorname{center}_X v = \operatorname{center}_X w \supset Z$. By (12.4) for $u = v$,

$$a(v, X, B) = v(\mathfrak{a}_\bullet) \leq \sigma(v) \leq a(v, X, B).$$

Hence $\sigma(v) = a(v, X, B)$, that is, v is stationary for $(X, B; D_\bullet)$. Thus (1) holds. □

Lemma 12.4. *Let (X, B) be a projective lc pair such that B is a $\mathbb{Q}$-divisor, $D_\bullet$ an lc anti-canonical series on (X, B), $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ moment rows for $D_\bullet$, and W_1 and W_2 corrected systems for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$. Then there exists a corrected system W for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that $\operatorname{Nlc}(X, B + G) \subset \operatorname{Nlc}(X, B + G_1) \cap \operatorname{Nlc}(X, B + G_2)$ for general members G, G_1, and G_2 of W, W_1, and W_2, respectively.*

Proof. Let e_1 and e_2 be the positive integers such that $W_1 \subset H^0(X, \mathcal{O}_X(-e_1(K_X + B)))$ and $W_2 \subset H^0(X, \mathcal{O}_X(-e_2(K_X + B)))$, and let $e := e_1 e_2$. By Lemma 7.4, applied to W_1 with $k = e_2$ and to W_2 with $k = e_1$, there exist corrected systems $W_1', W_2' \subset H^0(X, \mathcal{O}_X(-e(K_X + B)))$ for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that $\operatorname{Nlc}(X, B + G_i) = \operatorname{Nlc}(X, B + G_i')$ for general members G_i of W_i and G_i' of W_i', where $i = 1, 2$. Let $W_0' := W_1' + W_2' \subset H^0(X, \mathcal{O}_X(-e(K_X + B)))$. Since W_0' is spanned by the products spanning W_1' and W_2', W_0' is a corrected system for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ (Definition 7.3). We show that $W := W_0'$ satisfies the lemma.

For $i = 0, 1, 2$, let $\mathfrak{b}_i \subset \mathcal{O}_X$ be the base ideal of W_i', that is, the image of the evaluation map $W_i' \otimes_{\mathbb{C}} \mathcal{O}_X(e(K_X + B)) \to \mathcal{O}_X$. Then $\mathfrak{b}_0 = \mathfrak{b}_1 + \mathfrak{b}_2$. Let $f : Y \to X$ be a log resolution of (X, B) such that $\mathfrak{b}_i\mathcal{O}_Y = \mathcal{O}_Y(-F_i)$ for divisors $F_i \geq 0$ on Y, where $i = 0, 1, 2$, and such that $\Gamma := \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B \cup \operatorname{Supp} F_1 \cup \operatorname{Supp} F_2$ is a simple normal crossing divisor. Let $K_Y + B_Y := f^*(K_X + B)$. Since $\mathfrak{b}_0\mathcal{O}_Y = \mathfrak{b}_1\mathcal{O}_Y + \mathfrak{b}_2\mathcal{O}_Y$, we have

$$\operatorname{mult}_P F_0 = \min\{\operatorname{mult}_P F_1, \operatorname{mult}_P F_2\} \tag{12.5}$$

for any prime divisor P on Y. In particular, $\operatorname{Supp} F_0 \subset \operatorname{Supp} F_1 \cap \operatorname{Supp} F_2 \subset \Gamma$.

Fix $i \in \{0, 1, 2\}$. For $s \in W_i' \backslash \{0\}$, we have $\operatorname{div}(f^*s) = F_i + M_s$, where $M_s \geq 0$ is a member of the base point free linear system defined by f^*W_i' and F_i, and

$$K_Y + B_Y + \frac{1}{e}(F_i + M_s) = f^*(K_X + B + G_s).$$

By Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], applied to Y and to the intersections of components of Γ, on which the restriction of the base point free linear system defined by f^*W_i' and F_i has no base points, for a general $s \in W_i'$, M_s is smooth and $\Gamma \cup M_s$ is a simple normal crossing divisor. Moreover, M_s has no common component with Γ, since the members through a given point form a hyperplane. For such s, f is a log resolution of $(X, B + G_s)$, and the coefficient of any component of M_s in $B_Y + e^{-1}(F_i + M_s)$ is $e^{-1} \leq 1$. Since $\operatorname{Supp} B_Y \cup \operatorname{Supp} F_i \subset \Gamma$, the components of $B_Y + e^{-1}(F_i + M_s)$ with coefficient > 1 are the prime divisors P on Y with $\operatorname{mult}_P B_Y + e^{-1} \operatorname{mult}_P F_i > 1$, and they have negative log discrepancy with respect to $(X, B + G_s)$. On the complement in Y of the components with coefficient > 1, $(Y, B_Y + e^{-1}(F_i + M_s))$ is a log smooth sub-pair with coefficients ≤ 1, hence sub-lc [KM98, Corollary 2.31(3)]. By [KM98, Lemma 2.30, Corollary 2.31(3)],

$$\operatorname{Nlc}(X, B + G_s) = \bigcup_P f(P), \tag{12.6}$$

where P runs over the prime divisors on Y with $\operatorname{mult}_P B_Y + e^{-1} \operatorname{mult}_P F_i > 1$.

By (12.5), a prime divisor P on Y satisfies $\operatorname{mult}_P B_Y + e^{-1} \operatorname{mult}_P F_0 > 1$ if and only if it satisfies $\operatorname{mult}_P B_Y + e^{-1} \operatorname{mult}_P F_i > 1$ for both $i = 1$ and $i = 2$. Thus (12.6) implies

$$\operatorname{Nlc}(X, B + G) \subset \operatorname{Nlc}(X, B + G_1') \cap \operatorname{Nlc}(X, B + G_2') = \operatorname{Nlc}(X, B + G_1) \cap \operatorname{Nlc}(X, B + G_2)$$

for general members G of W, G_i' of W_i', and G_i of W_i, where $i = 1, 2$. □

Proof of Theorem 12.1. The proof is long, so we divide it into several steps. We prove the theorem by induction on $d := \dim X$. Assume that the theorem holds in any dimension less than d, with any finitely many moment rows.

Step 1. In this step, we construct a closed subset $Z_0 \subset X$ and a corrected system W_0, and we reduce the theorem to constructing a corrected system W_Z for an irreducible component Z of Z_0.

Let $(c_{j,N})$ be a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$, and $N \geq N_0$ an index such that $G_N \geq 0$ and $1 + c_{j,N} > 0$ for any j. Let e be a positive integer such that $-e(K_X + B)$ is Cartier and $e(K_X + B + G_N) \sim 0$, which exists since $K_X + B + G_N \sim_{\mathbb{Q}} 0$. Then there exists $s \in H^0(X, \mathcal{O}_X(-e(K_X + B)))$ such that $\operatorname{div}(s) = eG_N$, and the line $\mathbb{C}s$ is a corrected system for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that $G_{s'} = G_N$ for any $s' \in \mathbb{C}s\backslash\{0\}$. By Lemma 7.2, such $(c_{j,N})$ and N exist.

Let $\mathcal{Z}$ be the set of closed subsets $Z' \subset X$ such that there exists a corrected system W for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ with $\operatorname{Nlc}(X, B + G) \subset Z'$ for a general member G of W. By the previous paragraph, $X \in \mathcal{Z}$. By Lemma 12.4, $Z_1' \cap Z_2' \in \mathcal{Z}$ for any $Z_1', Z_2' \in \mathcal{Z}$. Since X is Noetherian, $\mathcal{Z}$ has a minimal element Z_0 with respect to inclusion. Let W_0 be a corrected system for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that $\operatorname{Nlc}(X, B + G_0) \subset Z_0$ for a general member G_0 of W_0.

If $Z_0 = \emptyset$, then $(X, B + G_0)$ is lc for a general member G_0 of W_0. Moreover, let e_{W_0} be the positive integer with $W_0 \subset H^0(X, \mathcal{O}_X(-e_{W_0}(K_X + B)))$. For $s \in W_0\backslash\{0\}$, the $\mathbb{Q}$-divisor

$G_s = \frac{1}{e_{W_0}} \operatorname{div}(s)$ of Definition 7.3 is $\mathbb{Q}$-Cartier and $G_s \geq 0$, and $K_X + B + G_s \sim_{\mathbb{Q}} 0$ since $\operatorname{div}(s) \sim -e_{W_0}(K_X + B)$. Thus the theorem holds if $Z_0 = \emptyset$.

Suppose that $Z_0 \neq \emptyset$. Let Z be an irreducible component of Z_0 and η_Z the generic point of Z. Suppose that there exists a corrected system W_Z for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that $\eta_Z \notin \operatorname{Nlc}(X, B + G_Z)$ for a general member G_Z of W_Z. By Lemma 12.4, applied to W_0 and W_Z, there exists a corrected system W for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ such that

$$\operatorname{Nlc}(X, B + G) \subset \operatorname{Nlc}(X, B + G_0) \cap \operatorname{Nlc}(X, B + G_Z) \subset Z_0 \cap \operatorname{Nlc}(X, B + G_Z)$$

for general members G, G_0, and G_Z of W, W_0, and W_Z. Fix such a general G_Z. Since $\operatorname{Nlc}(X, B + G_Z)$ is closed, being the image under a log resolution of $(X, B + G_Z)$ of the union of the prime divisors with coefficient > 1 in the pullback boundary [KM98, Lemma 2.30, Corollary 2.31(3)], $Z_0 \cap \operatorname{Nlc}(X, B + G_Z)$ is an element of $\mathcal{Z}$ that does not contain η_Z, so it is strictly contained in Z_0. This contradicts the minimality of Z_0. Thus, to prove the theorem, it suffices to construct such a W_Z under the assumption $Z_0 \neq \emptyset$.

By Lemma 12.3, applied to (X, B), $D_\bullet$, and Z, either Lemma 12.3(1) or Lemma 12.3(2) holds. If Lemma 12.3(1) holds, we fix a quasi-monomial valuation v of $\mathbb{C}(X)$ that is stationary for $(X, B; D_\bullet)$ and satisfies $Z \subset \operatorname{center}_X v$, and we let r be the rational rank of v. Then $r \geq 1$, and $r \leq d$ by Construction-Proposition 8.1(3). We construct W_Z in **Step 2** if Lemma 12.3(2) holds, in **Step 3** if Lemma 12.3(1) holds with $r = d$, and in **Step 4** if Lemma 12.3(1) holds with $r < d$.

Step 2. In this step, we construct W_Z when Lemma 12.3(2) holds.

Let $(c_{j,N})$ be a correction family for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ (Lemma 7.2). By Lemma 12.3(2) and Lemma 7.2, there exists an index $N \geq N_0$ such that $(X, B + G_N)$ is klt on a neighborhood of η_Z, $G_N \geq 0$, and $1 + c_{j,N} > 0$ for any j. By the second paragraph of **Step 1**, applied to $(c_{j,N})$ and N, there exists a corrected system $W_Z = \mathbb{C}s$ for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ with $G_Z = G_N$ for any member G_Z of W_Z, so $\eta_Z \notin \operatorname{Nlc}(X, B + G_N)$.

Step 3. In this step, we construct W_Z when Lemma 12.3(1) holds with $r = d$.

By Construction-Proposition 8.1, applied to (X, B), v, $\Sigma = \emptyset$, and $\mu = \mathrm{id}_X$, there exists an adapted carrier (h, S) of v for (X, B) and $\emptyset$, with $h : X'' \to X$. Then the conditions of Set-up 8.4 hold for (X, B), $D_\bullet$, v, and (h, S), and we use the notation V and b_j of Set-up 8.4. By Construction-Proposition 8.1(3), $\dim S = d - r = 0$, so S is a point. Hence $B_S = 0$ and $\operatorname{lead}_S(h^*G) = 0$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on X.

By Lemma 8.5(1), $(b_j)_{j \geq 1}$ is a moment row for $D_\bullet$. By Lemma 7.2, there exist a correction family $(c_{j,N})$ for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ and $(b_j)_{j \geq 1}$, and an index $N \geq N_0$ such that $G_N \geq 0$ and $1 + c_{j,N} > 0$ for any j. By Lemma 8.6(2), $a(v, X, B + G_N) = 0$. The sub-pair $(S, B_S + \operatorname{lead}_S(h^*G_N)) = (S, 0)$ is sub-lc because S is a point. By Proposition 9.1, $(X, B + G_N)$ is lc on a neighborhood of $V = \operatorname{center}_X v$, and $Z \subset V$. By the second paragraph of **Step 1**, applied to $(c_{j,N})$ and N, there exists a corrected system $W_Z = \mathbb{C}s$ for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$ with $G_Z = G_N$ for any member G_Z of W_Z, so $\eta_Z \notin \operatorname{Nlc}(X, B + G_N)$.

Step 4. In this step, we construct W_Z when Lemma 12.3(1) holds with $r < d$, using the induction hypothesis in dimension $d - r$.

Construction-Proposition 10.1, applied to (X, B), $D_\bullet$, and v, yields an integer $N_1 \geq m_0$, a rational number $0 < t < 1$, a projective birational morphism $\mu : X' \to X$ from a $\mathbb{Q}$-factorial normal projective variety X', the $\mathbb{Q}$-divisor $B' \geq 0$ with $K_{X'} + B' = \mu^*(K_X + B + tD_{\leq N_1})$, the multipliers κ_j, and $D'_j = \kappa_j \mu^* D_j$ for $j \geq 1$. Thus (X', B') is a projective dlt pair of dimension d, $\operatorname{center}_{X'} v$ is of codimension r in X', $D'_\bullet$ is an lc anti-canonical series on (X', B'), and v is stationary for $(X', B'; D'_\bullet)$ (Construction-Proposition 10.1(1)).

By Construction-Proposition 8.1, applied to (X, B), v, $\Sigma := \{D_1, \dots, D_{N_1}\}$, and μ, there exists an adapted carrier (h, S) of v for (X, B) and Σ, with $h : X'' \to X$, such that $h = \mu \circ h'$ for a morphism $h' : X'' \to X'$. By Lemma 10.2, (h', S) is an adapted carrier of v for (X', B') and

$\{D'_1, \dots, D'_{N_1}\}$. Let B_S and B'_S denote the $\mathbb{Q}$-divisors on S defined in Construction-Proposition 8.1 for (h, S) and (X, B), and for (h', S) and (X', B'), respectively. Thus the conditions of Set-up 8.4 hold for (X', B'), $D'_\bullet$, v, and (h', S). Since $r < d$ and $\operatorname{center}_{X'} v$ is of codimension r in X', the conditions of Set-up 11.1 also hold.

Let $(T, B_T; D_{T,\bullet})$ be a residue model of $(X', B'; D'_\bullet)$ at v (Construction-Proposition 11.4), and let $y_\bullet^{(1)}, \dots, y_\bullet^{(p')}$ be the moment rows for $D'_\bullet$ of Construction-Proposition 11.4. For $1 \le k \le p$, let $x_\bullet'^{(k)} := (\kappa_j x_j^{(k)})_{j \ge 1}$. By Construction-Proposition 10.1(4), $x_\bullet'^{(k)}$ is a moment row for $D'_\bullet$ on (X', B'). For $1 \le k \le p$, let $\tilde{x}_\bullet'^{(k)}$ be the moment row for $D_{T,\bullet}$ on (T, B_T) defined from $x_\bullet'^{(k)}$ in Construction-Proposition 11.4. For $1 \le i \le p'$, let $\tilde{y}_\bullet^{(i)}$ be the moment row for $D_{T,\bullet}$ on (T, B_T) defined from $y_\bullet^{(i)}$ in Construction-Proposition 11.4. Since $\dim T = d - r < d$, the induction hypothesis, applied to (T, B_T), $D_{T,\bullet}$, and the moment rows $\tilde{y}_\bullet^{(1)}, \dots, \tilde{y}_\bullet^{(p')}, \tilde{x}_\bullet'^{(1)}, \dots, \tilde{x}_\bullet'^{(p)}$, yields a corrected system W_T for $D_{T,\bullet}$ satisfying $\tilde{y}_\bullet^{(1)}, \dots, \tilde{y}_\bullet^{(p')}, \tilde{x}_\bullet'^{(1)}, \dots, \tilde{x}_\bullet'^{(p)}$ such that $(T, B_T + G_T)$ is lc for a general member G_T of W_T.

Let e_0 be a positive integer such that $e_0(K_X + B)$ is Cartier. By Construction-Proposition 11.4(4), applied with $q = p$ and the moment rows $x_\bullet'^{(1)}, \dots, x_\bullet'^{(p)}$, there exists a corrected system W' for $D'_\bullet$ satisfying $y_\bullet^{(1)}, \dots, y_\bullet^{(p')}, x_\bullet'^{(1)}, \dots, x_\bullet'^{(p)}$ such that e_0 divides each of the positive integers e_i in Definition 7.3 defining W', and such that

$$a(v, X', B' + G') = 0 \quad \text{and} \quad (S, B'_S + \operatorname{lead}_S(h'^* G')) \text{ is sub-lc}$$

for a general member G' of W'. Since $-e_i(K_X + B)$ is Cartier for each e_i, Construction-Proposition 10.1(5) implies that the space W' of rational functions is a corrected system for $D_\bullet$ satisfying $x_\bullet^{(1)}, \dots, x_\bullet^{(p)}$, which we denote by W_Z, and that $\mu^* G_s = G'_s + t\mu^* D_{\le N_1}$ for any $s \in W_Z \backslash \{0\}$, where G'_s and G_s are the $\mathbb{Q}$-divisors of Definition 7.3 on X' and X, respectively.

Let $s \in W_Z \backslash \{0\}$ be general, so that $a(v, X', B' + G'_s) = 0$ and $(S, B'_S + \operatorname{lead}_S(h'^* G'_s))$ is sub-lc. Since

$$K_{X'} + B' + G'_s = \mu^*(K_X + B + tD_{\le N_1}) + G'_s = \mu^*(K_X + B + G_s),$$

computing log discrepancies on a common log resolution of $(X, B + G_s)$ and $(X', B' + G'_s)$ (Definition-Proposition 2.8), we obtain $a(v, X, B + G_s) = a(v, X', B' + G'_s) = 0$. Since $G_s \ge 0$ and $G'_s \ge 0$, Lemma 10.2 implies

$$B_S + \operatorname{lead}_S(h^* G_s) = B'_S + \operatorname{lead}_S(h'^* G'_s),$$

so $(S, B_S + \operatorname{lead}_S(h^* G_s))$ is sub-lc. The conditions of Set-up 8.4 hold for (X, B), $D_\bullet$, v, and (h, S). By Proposition 9.1, which we use on X rather than on X' since $\mu^{-1}(V)$ need not be contained in $\operatorname{center}_{X'} v$, $(X, B + G_s)$ is lc on a neighborhood of $V = \operatorname{center}_X v$, and $Z \subset V$. Thus $\eta_Z \notin \operatorname{Nlc}(X, B + G_Z)$ for a general member G_Z of W_Z.

Therefore W_Z exists in any case, which contradicts the minimality of Z_0 by **Step 1**. Hence $Z_0 = \emptyset$, and the theorem holds by **Step 1**. □

13. Proofs of the main theorems

In this section, we prove Theorems 1.3 and 1.5 and Corollaries 1.4 and 1.6.

Proof of Theorem 1.3. Let $h : X \to X$ be an int-amplified endomorphism. If $\dim X = 0$, then X is a point and $K_X = 0$, so $B = 0$ satisfies the conclusion. Thus we may assume that $\dim X = d \ge 1$. Let r be the degree of h, let $D_j := r^{-j}(h^j)_* R(h)$ for $j \ge 1$, and let ℓ and $a_{j,N}$ be as in Construction-Proposition 3.1. Let $\pi : X' \to X$ be the small birational morphism of Construction-Proposition 5.2, and let $D'_j := \pi_*^{-1} D_j$ for $j \ge 1$.

We show that $D'_\bullet := (D'_j)_{j \ge 1}$ is an lc anti-canonical series on $(X', 0)$ (Definition 7.1). By Construction-Proposition 5.2(1) and (2), $(X', 0)$ is a projective lc pair, D'_j is $\mathbb{Q}$-Cartier for any $j \ge 1$, and $(X', \sum_{j=1}^N D'_j)$ is lc for any $N \ge 1$. Since h is finite (Lemma 2.12) and surjective, $R(h) \ge 0$ by Definition-Proposition 2.10(1). Since the pushforward of cycles under the finite

surjective morphism h^j and the strict transform under π both preserve effectivity, we have $D_j \geq 0$ and $D'_j \geq 0$ for any $j \geq 1$. Finally, let $m_0 := \ell$. Then $\lim_{N\to+\infty} a_{j,N} = 0$ for any $1 \leq j \leq \ell$ by Construction-Proposition 3.1, and

$$K_{X'} + \sum_{j=1}^{N} D'_j + \sum_{j=1}^{\ell} a_{j,N} D'_j \sim_{\mathbb{Q}} 0$$

for any $N \geq \ell$ by Construction-Proposition 5.2(3). Thus $D'_\bullet$ is an lc anti-canonical series on $(X', 0)$.

By Theorem 12.1 with $p = 0$, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G' \geq 0$ on X' such that (X', G') is lc and $K_{X'} + G' \sim_{\mathbb{Q}} 0$. Since π is a small birational morphism between normal projective varieties, by Lemma 2.19, there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that $K_X + B \sim_{\mathbb{Q}} 0$ and $\pi_*^{-1}B = G'$. Moreover, (X, B) is lc by Lemma 2.19, since (X', G') is lc. □

Proof of Corollary 1.4. Let X be a normal projective variety and $f : X \to X$ a polarized endomorphism. Then f is q-polarized for some integer $q > 1$ (Definition 2.11), hence int-amplified by Lemma 2.12. By Theorem 1.3, X is of Calabi–Yau type. □

Proof of Theorem 1.5. If $\dim X = 0$, then X is a point, so $K_X = 0$ and $R(f^m) = 0$ for any positive integer m. Thus $s = 1$ satisfies the conclusion, since $B_m = 0$ for any m. Hence we may assume that $\dim X = d \geq 1$.

By Construction-Proposition 6.2, there exist a small birational morphism $\pi : X' \to X$ from a normal projective variety X' such that $K_{X'}$ is $\mathbb{Q}$-Cartier and $(X', 0)$ is lc, a positive integer a, and a q^a-polarized endomorphism $g : X' \to X'$ such that $\pi \circ g = f^a \circ \pi$. By Definition 2.11, g is surjective. Since X' is projective, π is a projective birational morphism. By Lemma 6.4 applied to f, π, a, and g, there exists a positive integer s such that for any positive integer m, (X, B_m) is lc and $K_X + B_m \sim_{\mathbb{Q}} 0$.

It remains to show that $B_m \geq 0$. By Lemma 2.12 and Definition 2.11, f^{ms} is a finite surjective q^{ms}-polarized endomorphism. Thus $R(f^{ms}) \geq 0$ by Definition-Proposition 2.10(1). Since $q > 1$ is an integer, $q^{ms} - 1 > 0$. Therefore, $B_m \geq 0$. □

Proof of Corollary 1.6. Let H be an ample Cartier divisor on X such that $f^*H \sim qH$ (Definition 2.11), and let s be the positive integer of Theorem 1.5. By Theorem 1.5 with $m = 1$,

$$K_X + \frac{1}{q^s - 1} R(f^s) \sim_{\mathbb{Q}} 0.$$

By Lemma 2.12, f^s is a finite q^s-polarized endomorphism. By Definition-Proposition 2.10(1) (cf. [KM98, Proposition 5.20]), $K_X \sim (f^s)^* K_X + R(f^s)$. Thus

$$(f^s)^* K_X \sim K_X - R(f^s) \sim_{\mathbb{Q}} K_X + (q^s - 1)K_X = q^s K_X.$$

By Lemma 2.12, f is int-amplified. By Construction-Proposition 4.2, the $\mathcal{O}_X$-algebra $\mathcal{R} := \bigoplus_{m\geq 0} \mathcal{O}_X(mK_X)$ is finitely generated, $X' = \operatorname{Proj}_X \mathcal{R}$ is a normal projective variety, the structure morphism $\pi : X' \to X$ is a small birational morphism, and $K_{X'}$ is $\mathbb{Q}$-Cartier and π-ample. Since π is small, there is no π-exceptional prime divisor on X', so $\pi_*^{-1} \operatorname{div}_X(u) = \operatorname{div}_{X'}(u)$ for any $u \in \mathbb{C}(X)^*$. Hence π_*^{-1} preserves $\mathbb{Q}$-linear equivalence. Since $\pi_*^{-1} K_X = K_{X'}$, we have

$$\pi_*^{-1}(f^s)^* K_X \sim_{\mathbb{Q}} q^s K_{X'}.$$

Since $q^s > 0$, both $\pi_*^{-1} K_X$ and $\pi_*^{-1}(f^s)^* K_X$ are $\mathbb{Q}$-Cartier and π-ample. By Lemma 6.1 applied to f^s and $D = K_X$, there exists a finite surjective endomorphism $g : X' \to X'$ such that $\pi \circ g = f^s \circ \pi$.

Finally, let $d := \dim X$. Since π is birational, $((\pi^*H)^d) = (H^d) > 0$, so the nef Cartier divisor π^*H is big [Laz04, Theorem 2.2.16]. Moreover,

$$g^*\pi^*H = \pi^*(f^s)^*H \sim q^s \pi^* H,$$

so $g^*\pi^*H \equiv q^s \pi^* H$. By Theorem 2.13 with $L = \pi^*H$, g is q^s-polarized. □

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